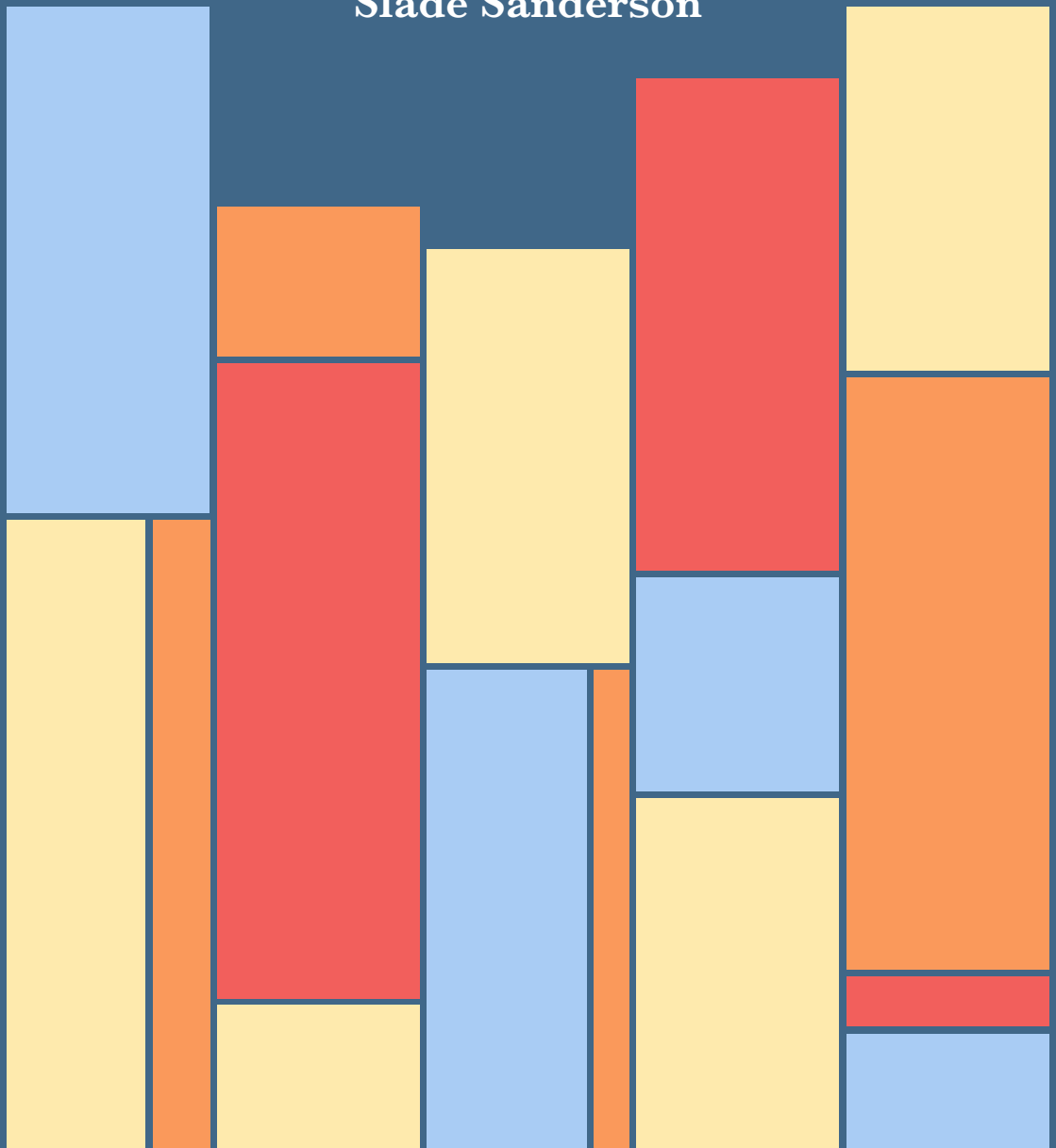
DYNAMICS OF NUMBER EXPANSIONS AND TRANSLATION SURFACES

Slade Sanderson



Dynamics of number expansions and translation surfaces

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Round and round we go, all of us,
and ever come back thither

Walt Whitman

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Introduction

Dynamical systems

Toward the end of the 19th century, one of the central problems in celestial mechanics was the *n-body problem*, which asks for a prediction of the motion of several objects interacting with each other gravitationally. Of particular concern was whether or not the orbit of the earth around the sun is stable, meaning that it returns arbitrarily close to some initial position over time. In honor of the 60th birthday in 1889 of Oscar II, King of Sweden and Norway, *Acta Mathematica* announced a prize for whoever could solve this problem. Although nobody at the time managed to solve it in its entirety, the prize was awarded to Henri Poincaré, who purportedly proved stability for the three-body problem but later demonstrated possible instability; for a detailed account of this history, see [61].

Poincaré's work [162] and subsequent monograph [163] laid the foundation for a measure-theoretic approach to an area of mathematics called *dynamical systems*, which—in broadest terms—is the study of time evolution in a given state space. Dynamical systems appear in a wide variety of disciplines, including biology ([130]), neuroscience ([94]), engineering ([113]) and economics ([75]). Even within mathematics itself, dynamical systems are ubiquitous in, e.g., number theory ([50]), combinatorics ([180]), probability theory ([152]) and automata theory ([40]).

Mathematically, a dynamical system may be described by a semigroup action (time evolution) on a set (the state space). The state space often carries some additional mathematical structure (e.g., it is a measure space or topological space), and the semigroup action describing time evolution preserves this structure (e.g., via measure-preserving transformations or continuous transformations). In what is called *discrete dynamics*, the semigroup consists of compositional powers $\{T^n\}_{n\in\mathbb{N}}$ of a transformation T on the state space; if T is invertible, $\mathbb{N}$ may be replaced by $\mathbb{Z}$. In *continuous dynamics*, the semigroup is a collection of transformations $\{T_t\}_{t\in\mathbb{R}_{\geq 0}}$ on the state space satisfying $T_t \circ T_s = T_{t+s}$ for all t and s with T_0 the identity; similarly here, if each T_t is invertible, $\mathbb{R}_{\geq 0}$ may be replaced by $\mathbb{R}$. The semigroup action of a dynamical system, however, could be more general than those considered in discrete and continuous dynamics. In this thesis, we use dynamics to study two types of mathematical objects: *number expansions* and *translation surfaces*.

Number expansions

The most familiar way—for modern humans—of representing real numbers is via *decimal expansions*, e.g., 51 or $3.14159\ldots$. However, there are several alternative number expansions. Some have been used by ancient civilisations (e.g., the *vigesimal*, or base-20, numeral system of the Mayans ([72])) and others find important applications in today's digital age: computers store data using bits of 0s and 1s coming from *binary expansions*; analog-to-digital conversion systems ([58]) and random number generators ([100]) use more general—possibly non-integer based—*β-expansions*; *generalised Lüroth series expansions* underlie the artificial neural network ChaosNet ([8]); and *continued fraction expansions* have been used in wireless networks ([2]).

In many cases—including each of those mentioned above—number expansions are generated by iterating some explicit function, i.e., via discrete dynamics. We briefly illustrate with the decimal

expansion. Suppose x_0 is some real number in the unit interval $[0,1)$ with decimal expansion $0.a_1a_2a_3\cdots$. Here, each a_n is a *digit* belonging to the set $\{0,1,\cdots,9\}$, and we assume that the decimal expansion does not end in an infinite sequence of digits equal to 9. Formally, we may write

$$x_0 = \sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{a_1}{10} + \frac{a_2}{10^2} + \frac{a_3}{10^3} + \cdots .$$

Now consider the interval map $T : [0,1) \to [0,1)$ defined by $T(x) \coloneqq 10x \pmod 1$ and the partition of $[0,1)$ into disjoint subintervals $[k/10,(k+1)/10)$, $k \in \{0,1,\ldots 9\}$; see the left-hand side of Figure 1. For $x \in [0,1)$, set $a(x) = k$ if and only if $x \in [k/10,(k+1)/10)$. Notice that with the decimal expansion of x_0 as above, $x_0 \in [a_1/10,(a_1+1)/10)$, i.e., $a_1 = a(x_0)$. Moreover,

$$T(x_0) = 10x_0 \pmod 1 = 0.a_2a_3a_4\cdots ,$$

and $a_2 = a(T(x_0))$. In general,

$$T^{n-1}(x_0) = 0.a_na_{n+1}a_{n+2}\cdots , \quad n \geq 1,$$

and $a_n = a(T^{n-1}(x_0))$. In words, to determine the digits a_n of the decimal expansion of x_0, it suffices to observe the subsequent subintervals $[k/10,(k+1)/10)$ entered by the T-orbit $(T^n(x_0))_{n\geq 0}$ of x_0. If $x_0 \geq 1$ with decimal expansion $x_0 = b_mb_{m-1}\cdots b_1.a_1a_2a_3\cdots$, the digits $b_n \in \{0,1,\ldots,9\}$ are obtained by applying the same technique to $x_0/10^m = 0.b_mb_{m-1}\cdots b_1a_1a_2a_3\cdots$, while if $x_0 < 0$, we determine the digits of $-x_0 > 0$.

In a similar fashion, digits of binary, β-, Lüroth, continued fraction, and many other number expansions are generated by some underlying dynamical system; see [50]. For instance, for each real number $\beta > 1$, the *β-transformation* defined by $x \mapsto \beta x \pmod 1$ generates β-expansions

$$x = \sum_{n=1}^{\infty} \frac{a_n}{\beta^n} = \frac{a_1}{\beta} + \frac{a_2}{\beta^2} + \frac{a_3}{\beta^3} + \cdots ,$$

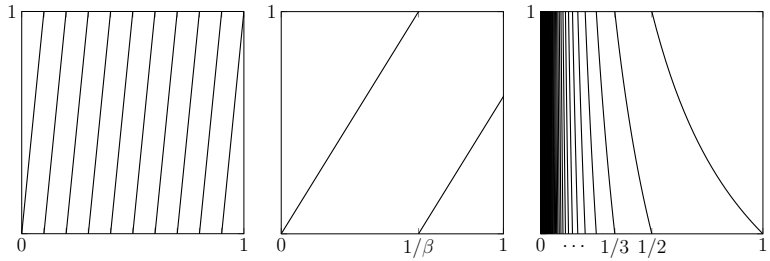


Figure 1: Left: The map $x \mapsto 10x \pmod 1$ generating decimal expansions. Center: The β-transformation $x \mapsto \beta x \pmod 1$ generating β-expansions; here, $\beta = (\sqrt{5}+1)/2$. Right: The Gauss map $0 \mapsto 0$ and $x \mapsto 1/x \pmod 1$, $x \neq 0$, generating continued fraction expansions.

while the *Gauss map* $G : [0,1) \to [0,1)$ given by $G(0) := 0$ and $G(x) := 1/x \pmod 1$ for $x \neq 0$ generates continued fraction expansions

$$x = \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots}}};$$

see Figure 1. (We remark that the partition of the domain needed to determine the digits of each expansion depends on the transformation at hand.) Statistical properties of the digits of these and other number expansions can be obtained by determining an *invariant measure* for the underlying transformation and using tools from *ergodic theory*.

Ergodicity

Definition. Let $(X, \mathcal{F}, \mu)$ be a measure space and $T : X \to X$ a measurable transformation. We say μ is T-*invariant* (equiva-

lently, T is *measure-preserving with respect to* μ) if for all $A \in \mathcal{F}$, $\mu(T^{-1}(A)) = \mu(A)$. The map T is *ergodic* if for every $A \in \mathcal{F}$ for which $T^{-1}(A) = A$, either $\mu(A) = 0$ or $\mu(X \setminus A) = 0$.

Ergodicity guarantees that—from the perspective of the measure μ—the dynamical system $(X, \mathcal{F}, \mu, T)$ cannot be decomposed into smaller, nontrivial components. The etymology of the word *ergodic* is disputed ([74]), but the commonly held belief is that it was introduced by Ludwig Boltzmann in the 19th century as a combination of the Greek words *ergon* and *odos*, meaning *work* and *path*, respectively ([192]). Around that time, a hypothesis was sought in statistical mechanics that would guarantee that the time average along a single trajectory equals the space average of the whole system. The correct hypothesis turned out to be ergodicity as defined above. Equality of time and space averages is made precise in the statement of G. D. Birkhoff's (pointwise) ergodic theorem:

Theorem (Birkhoff, 1931 [14])**.** *Let* $(X, \mathcal{F}, \mu)$ *be a probability space and* $T : X \to X$ *a measure-preserving transformation. Then for any integrable function* $f \in L^1(\mu)$*,*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} f(T^i(x)) = f^*(x)$$

exists μ*–almost everywhere. Moreover,* f^* *is* T*-invariant (i.e,* $f^* \circ T = f^*$*) and* $f^* \in L^1(\mu)$ *with* $\int_X f d\mu = \int_X f^* d\mu$*. If* T *is ergodic, then* $f^* = \int_X f d\mu$ *is constant.*

Birkhoff's ergodic theorem is perhaps best understood when f is taken to be an indicator function $f = \mathbf{1}_A$ for some $A \in \mathcal{F}$. In this case, $\frac{1}{n} \sum_{i=0}^{n-1} \mathbf{1}_A(T^i(x))$ gives the proportion of time the T-orbit of x spends in A in the first n steps. If T is ergodic, then as $n \to \infty$ this proportion almost surely tends to the μ-measure of A, $\int_X \mathbf{1}_A d\mu = \mu(A)$.

Besides Birkhoff's, there are several other ergodic theorems—

e.g., of von Neumann[1], Hopf and Hurewicz (see [46])—which guarantee equality of time and space averages in various senses and under various assumptions. There are also notions of invariance and ergodicity for more general dynamical systems (i.e., not necessarily in the setting of discrete dynamics) and corresponding ergodic theorems; see Chapter 8 of [65].

Elegant applications to number expansions

While Birkhoff's ergodic theorem is stated for a general probability space and measure-preserving transformation, it can be applied to obtain particularly elegant insights into number expansions. For instance, one can show that the *Gauss measure* ν_G with density $\frac{1}{\log 2(1+x)}$ is invariant and ergodic for the aforementioned Gauss map G generating continued fraction expansions. Moreover, the nth digit a_n in the continued fraction expansion of x equals 1 if and only if $G^{n-1}(x) \in (1/2, 1)$. By Birkhoff's ergodic theorem, we find that for Lebesgue–almost all[2] x,

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{n}\#\{1 \le i \le n \mid a_i = 1\} &= \lim_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} \mathbf{1}_{(1/2,1)}(G^i(x)) \\ &= \int_{[0,1)} \mathbf{1}_{(1/2,1)} d\nu_G \\ &= \nu_G((1/2,1)) \\ &\approx 0.415037\ldots.\end{aligned}$$

That is, $41.5037\ldots\%$ of the digits in the continued fraction expansion of a typical number x are equal to 1.

Similarly, Lebesgue measure is invariant and ergodic for each β-transformation with $\beta = n > 1$ an integer. Using this and Birkhoff's

[1] Although their theorems are different, there was some controversy regarding Birkhoff's and von Neumann's publication timelines; see [197].

[2] Note that ν_G is *equivalent* to Lebesgue measure, i.e., both measures are absolutely continuous with respect to one another and thus share the same null sets. Hence 'Lebesgue–almost all' is the same as 'ν_G–almost all.'

ergodic theorem, one can give a straightforward proof of a result of Borel from 1909 ([18])—originally proven with the Borel–Cantelli lemma—that Lebesgue–almost every real number is *normal*,[3] i.e., for any integer base $n > 1$, any finite string of k digits from the set $\{0, 1, \ldots, n-1\}$ occurs in the base n expansion of x with asymptotic frequency $1/n^k$. Normality can also be defined for other number expansions and is a subject of ongoing interest; see, e.g., [12, 51].

Invariant measures and natural extensions

As demonstrated above, invariant measures which are absolutely continuous with respect to Lebesgue can give immense insight into the typical behaviour of a dynamical system. Given a transformation, however, it is often a non-trivial task to find such measures—if they exist at all. In 1973, Lasota and Yorke proved that any piecewise C^2, monotone, expanding interval map admits an absolutely continuous invariant probability measure ([124]). Later, in 1990, Kopf gave expressions for the densities of these measures when the interval map is piecewise linear ([112]; see also [79]). We remark that these results may been viewed as generalisations of seminal papers from the 1950's and 1960's on β-transformations (existence of a unique, absolutely continuous invariant probability measure by Rényi ([165]) and an explicit expression for the density of this measure—independently—by Gelfond and Parry ([76, 159])). Though explicit, Kopf's expressions remain somewhat elusive in general, as the densities depend on the full forward orbits of the limits from the left and from the right at critical points.

For non-linear interval maps, the prospect of determining an absolutely continuous invariant measure is yet more challenging. It is thus quite mysterious that Gauss announced without derivation the discovery of his aforementioned namesake measure ν_G in an 1812 letter to Laplace—far before Poincaré's work and the subsequent

[3] Despite the fact that almost all numbers are known to be normal, proving or disproving the normality of common, specific numbers (e.g., $\pi, \sqrt{2}$ or e) is beyond our current reach.

development of ergodic theory. Several authors have since given methods of deriving the Gauss measure; see [106]. One such method arises from using a planar representation of the so-called *natural extension* of the Gauss map ([153, 155]).

In words, the natural extension of a non-invertible dynamical system $(X, \mathcal{F}, \mu, T)$ is the 'smallest' invertible dynamical system of which $(X, \mathcal{F}, \mu, T)$ is a *factor*, or subsystem; see [46] for a formal definition. Natural extensions possess several desirable properties: they always exist and—up to an appropriate notion of isomorphism for dynamical systems—are unique ([168]); a non-invertible dynamical system is ergodic (respectively, stronger notions called *weakly/strongly mixing*) if and only if the same is true of its natural extension ([168]); and a non-invertible dynamical system and its natural extension have the same measure-theoretic *entropy* ([30]), which is a quantitative notion of randomness. Natural extensions of the dynamical systems underlying various number expansions have proven to be extremely powerful. Besides allowing for the derivation of invariant measures, they have been used to give metric and arithmetic properties of several continued fraction algorithms, including regular ([21, 155]), Nakada's α- ([60, 129, 153]), Tanaka–Ito α- ([36, 157]), Katok–Ugarcovici (a, b)- ([103, 104, 105]) and Rosen continued fractions ([33, 117, 118, 119]). Natural extensions have also been used to relate the dynamics underlying various number expansions to one another, e.g., β- and generalised Lüroth series expansions ([54]) and Minkowski's diagonal continued fractions, Bosma's optimal continued fractions, and a subfamily of Nakada's α-continued fractions ([115]).

Translation surfaces

Translation surfaces can be motivated from several angles (e.g., billiards in rational polygons, electron transport on Fermi surfaces, and Teichmüller theory; see [172, 195]), but they also lay the foundation for a rich, fascinating, dynamical world of their own. Formally, a

translation surface is defined as a Riemann surface together with a non-zero, holomorphic one-form. Concretely, it may be represented by a collection of polygons in the plane together with an identification of pairs of parallel, equal length, and oppositely oriented edges. This representation is unique up to a 'cut-and-paste' equivalence respecting edge identifications (see Figure 2, which shows representations of the same translation surface after edges with the same labels are identified). Translation surfaces inherit a flat structure from the plane which is—at most points—locally Euclidean. At points where it fails to be locally Euclidean, the surface has a conical singularity whose angle is a positive, integral multiple of 2π.

When studying the dynamics of translation surfaces, there are two main perspectives: the dynamics of an individual surface, or the dynamics of families of surfaces. We briefly illustrate some of the questions that arise from both viewpoints; see also the surveys [71, 87, 131, 138, 190, 193, 195] for further background.

Dynamics on individual translation surfaces

One of the most immediate dynamical questions on fixed translation surfaces is that of geodesic flow. On the square torus, it is well-known that all linear trajectories (i.e., geodesics) in a fixed direction are either periodic or dense on the surface, depending on the rationality of the direction. Moreover, if the trajectory is dense, then it is in fact *equidistributed* on the torus, meaning that in the long run it will spend in a given subregion an amount of time proportional to the area of that subregion. In general, there are translation surfaces exhibiting dense linear trajectories that 'prefer' some regions of the surface to others (see [136]). Nevertheless, for any translation surface, geodesic flow in almost every direction is *uniquely ergodic* ([108]), meaning (normalised) Lebesgue measure is the only invariant probability measure for geodesic flow. Unique ergodicity implies that any linear trajectory in the corresponding direction which is not a *saddle connection* (a linear trajectory beginning and ending at a singularity, with no singularities in the interior; see, e.g., γ and

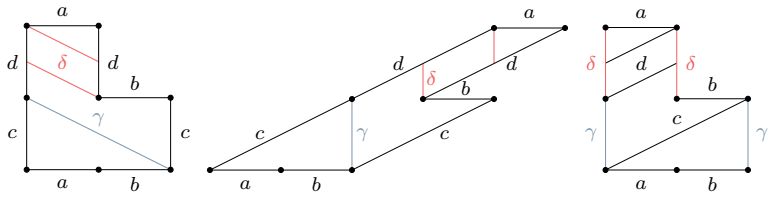


Figure 2: Left: A translation surface with one singularity of cone angle 6π along with two saddle connections, γ and δ. Center: A representation of the image of the translation surface on the left under the matrix $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$. Right: The translation surface in the center, when cut along γ and δ and pasted—via translation—along c and d, yields the translation surface on the left.

δ in Figure 2) is everywhere dense and equidistributed.

Several 'counting' questions on individual translation surfaces have also been considered. For instance, Masur proved in [133, 134] that for any translation surface, the number of closed geodesics[4] and saddle connections of length at most L grows quadratically in L, independent of the genus of the surface; more precise asymptotic growth rates for almost all surfaces were given by Eskin and Masur in [66]. Understanding the distribution of gaps between slopes of saddle connections is an area of ongoing research; see [13, 41, 122].

Dynamics on families of translation surfaces

Besides the dynamics on individual surfaces, there is an important $\mathrm{SL}_2\mathbb{R}$-action on families of (unit-area) translation surfaces, called *strata*, which share the same number and angles of singularities. Informally, this action may be thought of as deforming the polygonal representation of a translation surface by the linear transformation associated to a matrix in $\mathrm{SL}_2\mathbb{R}$. In general, this results in a polygonal

[4] Technically, maximal families of parallel, closed geodesics of the same length, called *cylinders*.

representation of a new translation surface. However, due to the cut-and-paste equivalence mentioned above, a non-trivial matrix may send a translation surface to itself (see Figure 2). This $\mathrm{SL}_2\mathbb{R}$-action on strata has been well-studied, as has the one-parameter action of the subgroup of diagonal matrices giving rise to what is called *Teichmüller geodesic flow*.

Each stratum of unit-area translation surfaces has local coordinates into $\mathbb{R}^{2n}$, allowing one to define a measure ν on the stratum as the pull back of Lebesgue measure.[5] In fact, this measure is finite ([132, 186]) and can be computed ([69]). Moreover, the $\mathrm{SL}_2\mathbb{R}$-action (and its restriction to diagonal matrices) preserves ν and is ergodic[6] on connected components of the stratum ([132, 186]). In particular, ergodicity implies that the $\mathrm{SL}_2\mathbb{R}$-orbit closure of ν–almost every surface is the whole connected component of the stratum to which the surface belongs. However, there are some rare translation surfaces, called *lattice surfaces* (or *Veech surfaces*), whose $\mathrm{SL}_2\mathbb{R}$-orbit is closed due to extra symmetry: for such surfaces, the stabilizer under this $\mathrm{SL}_2\mathbb{R}$-action—called the *Veech group* of the surface—is exceptionally large.

We also mention briefly the landmark 'Magic Wand Theorem' of Eskin, Mirzakhani and Mohammadi ([67, 68]), which roughly states that *all* (not just ν–almost all) $\mathrm{SL}_2\mathbb{R}$-orbit closures are as nice as one can hope: they are 'suborbifolds' of the stratum which—in local coordinates—are represented by affine subspaces. This result, its implications, and its striking similarity to unipotent flows on homogenous spaces are discussed in [196]. However, despite the immense power of the Magic Wand Theorem, a classification of $\mathrm{SL}_2\mathbb{R}$-orbit closures remains open in most genera except genus 2, where Calta and McMullen have provided a fairly complete treatment ([34, 139, 141, 142, 143, 144, 145]).

[5]Here, $n = 2g + \kappa - 1$, where g is the genus and κ the number of singularities corresponding to surfaces in the stratum.

[6]Here, ergodicity means that if A is a measurable subset of a connected component of a stratum for which $\nu(G \cdot A \,\Delta\, A) = 0$ for all $G \in \mathrm{SL}_2\mathbb{R}$, then A has either full or zero ν-measure in the connected component.

Interplay of both perspectives

These two dynamical perspectives—i.e., on fixed surfaces and on families of surfaces—often yield information about one another; see also the survey of Zorich ([195]) where this philosophy is stressed. We illustrate this phenomenon with two classical results in the theory of translation surfaces. The first is *Masur's criterion*, which states that if—for a fixed translation surface—geodesic flow in the vertical direction is not ergodic with respect to Lebesgue measure, then the orbit of the surface under Teichmüller geodesic flow (i.e., the subgroup of diagonal matrices) is *divergent*, meaning it eventually leaves any compact subset of the stratum ([132, 135]). Another result connecting these two dynamical perspectives is *Veech's dichotomy* ([187]): for lattice surfaces (i.e., those whose $\mathrm{SL}_2\mathbb{R}$-orbits are 'small'), geodesic flow in a fixed direction is either *periodic*, meaning the surface completely decomposes into parallel closed geodesics and saddle connections, or it is uniquely ergodic. Hence, geodesic flow on lattice surfaces satisfies the same dichotomy as that on the square torus (which is itself a lattice surface).

Surprising connections

At first glance, number expansions and translation surfaces may seem to share little in common besides the study of both benefiting from a dynamical perspective. The overlap, however, runs deeper.

Interval exchange transformations

Recall that the number expansions discussed above are generated using certain interval maps. Interval maps also arise naturally in the study of geodesic flow on fixed translation surfaces. Indeed, fixing a straight line segment on a translation surface and considering the first-return map of geodesic flow in a direction orthogonal to this segment gives rise to an *interval exchange transformation*

(IET), which is an interval map defined by permuting, via translation, the subintervals of a finite partition; see the surveys [190] and [195]. Moreover, there is an operator called *Rauzy-Veech induction* on the space of all IETs which sends an IET to a new IET given by an *induced map* (or *first-return map*) on an appropriate subinterval ([164, 186]). Zorich showed in [194] that accelerating a modified version of the Rauzy-Veech induction operator gives a new operator on the space of IETs which may be viewed as a multi-dimensional continued fraction map and which admits an ergodic, absolutely continuous invariant probability measure (see also [190, 195]).

Hyperbolic geometry

Continued fractions also arise in connection with Teichmüller geodesic flow. Up to isometry, the $\mathrm{SL}_2\mathbb{R}$-orbit of a translation surface in a stratum is parameterised by the quotient space $\Gamma\backslash\mathbb{H}$, where $\mathbb{H} \cong \mathrm{SL}_2\mathbb{R}/\mathrm{SO}_2\mathbb{R}$ is the upper-half plane and Γ denotes the Veech group (i.e., the stabilizer) of the surface. One can show that Γ is a discrete subgroup, so $\Gamma\backslash\mathbb{H}$ is a hyperbolic surface (see [102]), and Teichmüller geodesic flow corresponds to geodesic flow on $\Gamma\backslash\mathbb{H}$ under the hyperbolic metric. When $\Gamma = \mathrm{SL}_2\mathbb{Z}$ is the modular group (e.g., for the square torus, but also for the genus 3 'Eierlegende Wollmilchsau' and genus 4 'Ornithorynque'; see [71]), $\Gamma\backslash\mathbb{H}$ is the modular surface, and there is a well-studied connection between geodesics on this surface and continued fractions (see Series's lucid exposition in [174], though this connection was perhaps first observed by Artin ([6])). Further connections between the dynamics of translation surfaces and various continued fraction algorithms can be found in [3, 5, 35, 175, 176] and the references therein.

In this thesis

This thesis consists of three chapters. The first two deal with the dynamics of different types of number expansions, and the third

concerns the dynamics of the $\mathrm{SL}_2\mathbb{R}$-action on families of translation surfaces.

Chapter 1 – Inducing contractions and a unifying metrical theory for continued fractions

Our first chapter introduces a new, large class of continued fraction algorithms producing what are called *contracted Farey expansions* and studies their metrical properties. These algorithms are defined by coupling two acceleration techniques—*induced transformations* and *contraction*—in the setting of Shunji Ito's 1989 ([93]) natural extension of the Farey tent map. An induced transformation is an acceleration of a dynamical system via a first-return map to a given subregion of the domain; contraction is an old, seemingly overlooked arithmetic procedure which accelerates continued fractions by removing prescribed convergents; and the Farey tent map produces 'slow' semi-regular continued fraction expansions whose convergents consist of all regular continued fraction convergents and so-called *mediant convergents*. Succinctly, our algorithms use induced transformations of Ito's natural extension to govern contractions of Farey expansions.

Our construction is analogous to that defining Kraaikamp's S-expansions, which use induced transformations of the natural extension of the Gauss map to govern *singularisations*—special cases of contractions—of regular continued fractions. In [115], Kraaikamp shows that several continued fraction algorithms are realised as S-expansions, including Nakada's α-continued fractions for $1/2 \leq \alpha \leq 1$. However, the α-continued fractions are defined for all $0 \leq \alpha \leq 1$. These partially overlapping families are both found in our unified setting: that is, both S-expansions and α-continued fractions for all $0 < \alpha \leq 1$ are shown to be examples of contracted Farey expansions (§1.7.2 and §1.7.3). In particular, we give a new description of planar natural extensions of the α-continued fractions as explicit induced transformations of Ito's natural extension (cf. [120]).

In addition to defining new continued fraction algorithms (e.g.,

the *superoptimal continued fractions* of §1.9), we also develop a unifying metrical theory for subsequences of Farey convergents (i.e., regular continued fraction convergents and mediants). In §1.8, we prove a wealth of old and new results on these subsequences and their Diophantine properties. In particular, we extend to subsequences of Farey convergents a classical result of Lévy ([127]) on the growth rate of regular continued fraction convergent denominators; the Doeblin–Lenstra conjecture—proven by Bosma, Jager and Wiedijk ([21])—on the distribution of so-called *approximation coefficients*; Jager's dynamical system ([95, 96]) used to study consecutive approximation coefficients; and various analogues and refinements of a classical theorem of Legendre ([125]).

Chapter 1 is based on three articles. The theory of contracted Farey expansions appears in [52]. The metrical theory of §1.8 is largely reproduced from [53], which was first published in *Math. Comp.* (2025), published by the American Mathematical Society (©2025 American Mathematical Society). The superoptimal continued fractions introduced in §1.9 are based on [169].

Chapter 2 – Ergodic properties of a parameterised family of symmetric golden maps

In Chapter 2, we introduce a new, one-parameter family of piecewise linear interval maps and study properties of the number expansions they produce. Our *skewed symmetric golden maps* have slopes determined by the golden mean, $(\sqrt{5}+1)/2$. We show that all (but one) of our maps have a unique—hence ergodic—absolutely continuous invariant probability measure, and we obtain explicit densities for these measures on a countable collection of disjoint, open subintervals of the parameter space whose union has full Lebesgue measure. Analogous results are first proven for a simpler family of *symmetric golden maps* of constant slope equal to the golden mean. We use the fact that the original, skewed maps arise as *jump transformations* of the symmetric golden maps to transfer statements about the latter

family into statements about the former.

The main tool used to determine the densities of the invariant measures of the symmetric golden maps is a phenomenon called *matching*, which has gained much attention recently; see, e.g., [16, 31, 32, 37, 38, 39, 42, 44, 45, 47, 120, 156]. An interval map has matching if at each critical point, the orbits of the limits from the left and from the right agree after some finite number of steps. Kopf's ([112]) aforementioned expressions for densities of invariant measures imply that when matching occurs, the density is a step function depending only on points in the orbits before matching. With these explicit densities at hand, we use Birkhoff's ergodic theorem to compute the relative frequencies of digits typically occurring in number expansions produced by the symmetric—and hence the skewed symmetric—golden maps, and we study how these frequencies vary continuously with the parameters.

Up to minor changes, Chapter 2 is reproduced from [56].

Chapter 3 – Constructing lattice surfaces with prescribed Veech groups: an algorithm

In Chapter 3, we consider the dynamics of the $\mathrm{SL}_2\mathbb{R}$-action on families of translation surfaces. Recall that there is a natural, finite measure ν on each stratum of unit-area translation surfaces and that the $\mathrm{SL}_2\mathbb{R}$-action preserves ν and is ergodic on connected components of the stratum. Ergodicity implies that ν–almost every translation surface has a dense $\mathrm{SL}_2\mathbb{R}$-orbit in the connected component of the stratum to which it belongs. Among the exceptions are lattice surfaces whose stabilizers under this action are particularly large: they are lattice subgroups of $\mathrm{SL}_2\mathbb{R}$. Lattice surfaces are especially interesting because they have 'optimal dynamics' (they satisfy the aforementioned Veech dichotomy), and the projections of their $\mathrm{SL}_2\mathbb{R}$-orbits to the moduli space of Riemann surfaces are algebraic curves—called *Teichmüller curves*—isometrically immersed with respect to the Teichmüller metric (see, say, [193]). Despite much interest in these

remarkable surfaces, the list of known families of lattice surfaces is relatively short ([59]).

We develop a new, explicit algorithm which constructs all translation surfaces with a prescribed, lattice Veech group in any given stratum. In developing the algorithm, we present a new proof of a result of Smillie and Weiss ([179]) that there are at most finitely many unit-area lattice surfaces with a given Veech group in any given stratum. Moreover, our methods can be used to obtain obstructions for lattice subgroups of $\mathrm{SL}_2\mathbb{R}$ being realised as Veech groups: we show that the square torus is the only translation surface with exactly one singularity whose Veech group is $\mathrm{SL}_2\mathbb{Z}$.

Chapter 3 is—up to minor edits—reproduced from [170], which is available at: `http://dx.doi.org/10.1007/s10711-023-00818-7`.

Chapter 1

Inducing contractions and a unifying metrical theory for continued fractions

1.1 Introduction

In 1855, Seidel ([173]) introduced a seemingly overlooked,[1] arithmetic procedure, called *contraction*, which—under mild assumptions—allows one to produce from a given generalised continued fraction (GCF) a new GCF whose convergents are any prescribed subsequence of the original GCF-convergents. Nearly ninety years later, in 1943, Kakutani introduced in [101] *induced transformations*, which

[1] Contraction *is* used in the analytic theory of continued fractions, but usually only for subsequences of odd or even integers ([128]). See also [26], where the more general contraction procedure is used on the continued fraction expansion of the golden mean, $(\sqrt{5}+1)/2$.

accelerate a given dynamical system by only observing the dynamics within a subregion of the domain. In 1989, Shunji Ito ([93]) gave an explicit natural extension of what has been called[2] 'the mother of all continued fractions'—the *Farey tent map*—which generates 'slow' continued fraction expansions (*Farey expansions*) whose convergents (*Farey convergents*) consist of all regular continued fraction (RCF) convergents and so-called *mediant convergents* (see §1.2.3 below). In this chapter, we obtain a broad, unifying theory for various continued fraction expansions and their metrical properties by 'inducing contractions of the mother of all continued fractions.'

More formally, we use induced transformations of Ito's natural extension to govern contractions of Farey expansions. This coupling of inducing and contracting defines a large class of continued fraction algorithms—producing what we call *contracted Farey expansions*—which are parameterised by measurable subregions of the domain of Ito's natural extension. Within this collection of algorithms we find several well-studied examples. In particular, contracted Farey expansions contain the theory of Kraaikamp's S-expansions, which themselves contain the theory of regular continued fractions, Minkowski's diagonal continued fractions, Bosma's optimal continued fractions and more ([115]). Kraaikamp's S-expansions also partially contain Nakada's parameterised family of α-continued fractions: this latter family is defined for $0 \leq \alpha \leq 1$, but only those for which $\alpha \geq 1/2$ are realised as S-expansions. Our theory of contracted Farey expansions contains Nakada's α-continued fractions for all $0 < \alpha \leq 1$—thus providing a unifying framework within which to view these two partially overlapping families—and gives new descriptions of the natural extensions of the α-continued fractions as induced transformations of Ito's natural extension (cf. [120]). Moreover, we introduce a particular subclass of contracted Farey expansions, producing what we call *superoptimal continued fractions*, whose convergents have arbitrarily good approximation properties

[2]This is true 'up to isomorphism.' The maternal moniker was originally applied to the *Lehner map*, which is isomorphic to the Farey tent map ([49]); see also §1.4.2 below.

and converge arbitrarily fast.

In addition to defining a large, unifying class of continued fraction algorithms, we also study metrical properties of the Farey convergents, i.e., how well these RCF-convergents and mediants approximate a given irrational. For this, we exploit the fact that so-called *approximation coefficients* corresponding to an irrational x and its Farey convergents may be written explicitly in terms of the forward orbit of $(x, 1)$ under Ito's natural extension. This fact was essentially known and used in [93] and [29], where the orbit is put into one-to-one correspondence with the Farey convergents of x, but ordered with increasing denominators. By instead considering the sequence of Farey convergents 'as is' (i.e., not rearranged with increasing denominators), we obtain unified and simple proofs of results from, e.g., [20, 29, 93, 97], old and classical results of Legendre and Koksma, and various new results such as generalisations of Lévy's constant and of the Doeblin–Lenstra conjecture to subsequences of convergents and mediants.

This chapter is organised as follows. In §1.2 we set definitions and notation for generalised, semi-regular and regular continued fractions that are used throughout the chapter, and in §1.3 we recall several important continued fraction algorithms: the Gauss map and its natural extension, Nakada's α-continued fractions and Kraaikamp's S-expansions. We recall the Farey tent map, Farey expansions and Farey convergents in §1.4. In §1.5 we describe Ito's natural extension of the Farey tent map, and, moreover, define and set notation for induced transformations of it (§1.5.2). In §1.6.1 we describe contraction in the abstract setting of generalised continued fractions and in §1.6.2 use induced transformations of Ito's natural extension to govern contractions of Farey expansions. Furthermore, in §1.6.3 we define a dynamical system which acts essentially as a two-sided shift on contracted Farey expansions. Section 1.7 realises each of the examples from §1.3 within our theory of contracted Farey expansions. In §1.8 we explore the unifying metrical theory which arises by inducing Ito's natural extension, and we reobtain and extend several results (e.g., of Lévy ([127]), Bosma–Jager–Wiedijk ([21]), Vahlen ([185])

and Legendre ([125])) and their analogues for RCF-convergents and mediants. We end in §1.9 by defining and giving concrete examples of superoptimal continued fractions.

1.2 Generalised, semi-regular and regular continued fractions

1.2.1 Generalised continued fractions

A *generalised continued fraction* (GCF) is a formal (infinite or finite) expression of the form

$$[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots] = \cfrac{\alpha_{-1}}{\beta_{-1} + \cfrac{\alpha_0}{\beta_0 + \cfrac{\alpha_1}{\beta_1 + \cfrac{\alpha_2}{\beta_2 + \ddots}}}}, \tag{1.1}$$

where $(\alpha_{-1}, \beta_{-1}) \coloneqq (1, 0)$ and for $n \geq 0$, $\alpha_n, \beta_n \in \mathbb{C}$ with $\alpha_n \neq 0$.

Remark 1.2.1. Notice that for $\alpha_0, \beta_0, x \in \mathbb{C}$ with α_0 nonzero,

$$\cfrac{1}{0 + \cfrac{\alpha_0}{\beta_0 + x}} = \frac{1}{\alpha_0}(\beta_0 + x),$$

with the convention that $c/0 = \infty$ and $c/\infty = 0$ for $c \in \mathbb{C} \setminus \{0\}$. Thus—although at this point it is a strictly formal expression—a GCF should be thought of as $1/\alpha_0$ 'multiplied' with the expression

$$\beta_0 + \cfrac{\alpha_1}{\beta_1 + \cfrac{\alpha_2}{\beta_2 + \ddots}}$$

(hence the choice of notation β_0/α_0 rather than α_0/β_0 on the left-hand side of (1.1)). Besides allowing for this inversion of α_0, our inclusion of α_{-1} and β_{-1} in (1.1) also prevents us from needing to treat the index 0 as a special case in matrix notation introduced below.

The digits α_n and β_n are called *partial numerators* and *partial denominators*, respectively. When a GCF has only finitely many partial numerators and partial denominators, the expression on the right-hand side of (1.1) may be evaluated to give a number in $\widehat{\mathbb{C}} \coloneqq \mathbb{C} \cup \{\infty\}$. Define for each integer $n \geq -2$ (with the obvious restriction in the finite case) the nth *tail* of $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ to be the GCF

$$[0/1; \alpha_{n+1}/\beta_{n+1}, \alpha_{n+2}/\beta_{n+2}, \alpha_{n+3}/\beta_{n+3}, \dots].$$

For each integer $n \geq -1$, set

$$B_n = B_n([\beta_0/\alpha_0; \alpha_1/\beta_1, \dots]) \coloneqq \begin{pmatrix} 0 & \alpha_n \\ 1 & \beta_n \end{pmatrix},$$

and for integers $-1 \leq m \leq n$, let

$$B_{[m,n]} = B_{[m,n]}([\beta_0/\alpha_0; \alpha_1/\beta_1, \dots]) \coloneqq B_m B_{m+1} \cdots B_n.$$

Notice that $\det B_{[m,n]} = (-1)^{n-m+1} \alpha_m \alpha_{m+1} \dots \alpha_n \neq 0$. For a matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2\mathbb{C}$, denote by $A \cdot z \coloneqq \frac{az+b}{cz+d}$, $z \in \widehat{\mathbb{C}}$, the action of A as a Möbius transformation. (We remark that for any $r \in \mathbb{C} \backslash \{0\}$, $(rA) \cdot z = A \cdot z$; this fact will be used repeatedly throughout.) Writing the entries of $B_{[m,n]}$ as $\left(\begin{smallmatrix} R_{[m,n]} & P_{[m,n]} \\ S_{[m,n]} & Q_{[m,n]} \end{smallmatrix}\right)$, we have

$$\begin{aligned} \frac{P_{[m,n]}}{Q_{[m,n]}} =& B_{[m,n]} \cdot 0 = \cfrac{\alpha_m}{\beta_m + \cfrac{\alpha_{m+1}}{\beta_{m+1} + \cfrac{\alpha_{m+2}}{\ddots + \cfrac{\alpha_n}{\beta_n}}}} \\ =& [0/1; \alpha_m/\beta_m, \alpha_{m+1}/\beta_{m+1}, \dots, \alpha_n/\beta_n] \in \widehat{\mathbb{C}}. \end{aligned}$$

(Notice that if each $\alpha_j, \beta_j \in \mathbb{Z}$, then $P_{[m,n]}/Q_{[m,n]} \in \mathbb{Q} \cup \{\infty\}$, but in general, $\gcd(P_{[m,n]}, Q_{[m,n]}) \neq 1$.) When $m = -1$, we use the suppressed notation

$$\begin{pmatrix} R_n & P_n \\ S_n & Q_n \end{pmatrix} := \begin{pmatrix} R_{[-1,n]} & P_{[-1,n]} \\ S_{[-1,n]} & Q_{[-1,n]} \end{pmatrix} = B_{[-1,n]}$$

and call P_n/Q_n the nth *convergent* of[3] $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$. If a GCF is finite and evaluates to $x \in \mathbb{C}$, or if it is infinite and $x = \lim_{n\to\infty} P_n/Q_n \in \mathbb{C}$ exists, we call $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ a GCF-*expansion* of x, write $x = [\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ and—when the expansion $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ is understood—refer to the convergents P_n/Q_n as convergents of x.

Notice for any integers $-1 \leq m \leq n$ that

$$\begin{aligned} \begin{pmatrix} R_{[m,n+1]} & P_{[m,n+1]} \\ S_{[m,n+1]} & Q_{[m,n+1]} \end{pmatrix} &= B_{[m,n]} B_{n+1} \\ &= \begin{pmatrix} R_{[m,n]} & P_{[m,n]} \\ S_{[m,n]} & Q_{[m,n]} \end{pmatrix} \begin{pmatrix} 0 & \alpha_{n+1} \\ 1 & \beta_{n+1} \end{pmatrix} \\ &= \begin{pmatrix} P_{[m,n]} & \beta_{n+1} P_{[m,n]} + \alpha_{n+1} R_{[m,n]} \\ Q_{[m,n]} & \beta_{n+1} Q_{[m,n]} + \alpha_{n+1} S_{[m,n]} \end{pmatrix}. \end{aligned}$$

In particular, $R_{[m,n+1]} = P_{[m,n]}$ and $S_{[m,n+1]} = Q_{[m,n]}$. Setting $(P_{[m,m-1]}, Q_{[m,m-1]}) := (0, 1)$ for all $m \geq -1$, this gives

$$B_{[m,n]} = \begin{pmatrix} P_{[m,n-1]} & P_{[m,n]} \\ Q_{[m,n-1]} & Q_{[m,n]} \end{pmatrix}, \quad -1 \leq m \leq n,$$

and we obtain the following recurrence relations for all $-1 \leq m \leq n$:

$$\begin{aligned} P_{[m,n+1]} &= \beta_{n+1} P_{[m,n]} + \alpha_{n+1} P_{[m,n-1]}, \\ Q_{[m,n+1]} &= \beta_{n+1} Q_{[m,n]} + \alpha_{n+1} Q_{[m,n-1]}, \end{aligned} \tag{1.2}$$

where

$$\begin{aligned} P_{[m,m-1]} &= 0, & P_{[m,m]} &= \alpha_m, \\ Q_{[m,m-1]} &= 1, & Q_{[m,m]} &= \beta_m. \end{aligned} \tag{1.3}$$

[3]Note that $[0/1; \alpha_{-1}/\beta_{-1}, \alpha_0/\beta_0, \dots, \alpha_n/\beta_n] = [\beta_0/\alpha_0; \alpha_1/\beta_1, \dots, \alpha_n/\beta_n]$.

Let $(P_{-2}, Q_{-2}) \coloneqq (0, 1)$. When $m = -1$, the observations above give, for $n \geq -1$,

$$B_{[-1,n]} = \begin{pmatrix} P_{n-1} & P_n \\ Q_{n-1} & Q_n \end{pmatrix}$$

and the recurrence relations

$$\begin{aligned} P_{n+1} &= \beta_{n+1} P_n + \alpha_{n+1} P_{n-1}, & P_{-2} &= 0,\ P_{-1} = 1, \\ Q_{n+1} &= \beta_{n+1} Q_n + \alpha_{n+1} Q_{n-1}, & Q_{-2} &= 1,\ Q_{-1} = 0. \end{aligned} \tag{1.4}$$

Remark 1.2.2. The quantities P_n, Q_n are defined in terms of the partial numerators and partial denominators α_n, β_n of a GCF. Conversely, since $\det(B_{[-1,n]}) \neq 0$, the digits α_n, β_n are also determined by the quantities P_n, Q_n. In particular, the recurrence relations (1.4) imply

$$\begin{pmatrix} \alpha_{n+1} \\ \beta_{n+1} \end{pmatrix} = B_{[-1,n]}^{-1} \begin{pmatrix} P_{n+1} \\ Q_{n+1} \end{pmatrix}, \qquad n \geq -1.$$

Remark 1.2.3. It shall sometimes be useful to allow for infinite partial denominators $\beta_n = \infty$ for some $n \geq 1$ in a GCF. In this case, letting $n_0 \geq 0$ denote smallest index for which $\beta_{n_0+1} = \infty$, the GCF $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ is interpreted to be the finite GCF $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots, \alpha_{n_0}/\beta_{n_0}]$.

Letting $T_n \coloneqq [0/1; \alpha_{n+1}/\beta_{n+1}, \alpha_{n+2}/\beta_{n+2}, \dots]$, $n \geq 0$, denote the nth tail of the GCF-expansion $x = [\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$, one obtains

$$\begin{aligned} x =& \cfrac{\alpha_{-1}}{\beta_{-1} + \cfrac{\alpha_0}{\beta_0 + \cfrac{\alpha_1}{\ddots + \cfrac{\alpha_n}{\beta_n + T_n}}}} \\ =& \begin{pmatrix} 0 & \alpha_{-1} \\ 1 & \beta_{-1} \end{pmatrix} \begin{pmatrix} 0 & \alpha_0 \\ 1 & \beta_0 \end{pmatrix} \cdots \begin{pmatrix} 0 & \alpha_n \\ 1 & \beta_n \end{pmatrix} \cdot T_n \\ =& B_{[-1,n]} \cdot T_n. \end{aligned} \tag{1.5}$$

Notice also that for any $z \in \widehat{\mathbb{C}}$,

$$\begin{aligned} B^T_{[-1,n]} \cdot z &= \begin{pmatrix} 0 & 1 \\ \alpha_n & \beta_n \end{pmatrix} \cdots \begin{pmatrix} 0 & 1 \\ \alpha_0 & \beta_0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot z \\ &= \cfrac{1}{\beta_n + \cfrac{\alpha_n}{\beta_{n-1} + \cfrac{\alpha_{n-1}}{\ddots + \cfrac{\alpha_1}{\beta_0 + \cfrac{\alpha_0}{z}}}}}, \end{aligned} \tag{1.6}$$

or

$$B^T_{[-1,n]} \cdot z = [0/1; 1/\beta_n, \alpha_n/\beta_{n-1}, \ldots, \alpha_1/\beta_0, \alpha_0/z], \tag{1.7}$$

where, in the case that $z = \infty$, the right-hand side is interpreted as $[0/1; 1/\beta_n, \alpha_n/\beta_{n-1}, \ldots, \alpha_1/\beta_0]$.

1.2.2 Semi-regular continued fractions

A *semi-regular continued fraction* (SRCF) is a GCF as in (1.1) with integral partial numerators and partial denominators $\alpha_n, \beta_n \in \mathbb{Z}$ satisfying

(i) $\alpha_0 = 1$ and $\alpha_n = \pm 1$ for $n \geq 1$,

(ii) $\beta_n > 0$ for $n \geq 1$, and

(iii) $\alpha_{n+1} + \beta_n \geq 1$ for $n \geq 1$.

If there are infinitely many digits, we further require

(iv) $\alpha_{n+1} + \beta_n \geq 2$ infinitely often.

By Tietze's Convergence Theorem (see, say, [161]) the above conditions guarantee that $x = \lim_{n\to\infty} P_n/Q_n \in \mathbb{R}$ always exists, and

thus we call $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ a SRCF-expansion of x. Notice that the convergents P_n/Q_n of any SRCF-expansion of $x \in \mathbb{R}$ are reduced since

$$|P_{n-1}Q_n - P_nQ_{n-1}| = |\det(B_{[-1,n]})| = |\alpha_{-1}\alpha_0 \cdots \alpha_n| = 1.$$

1.2.3 Regular continued fractions

A *regular continued fraction* (RCF) is a SRCF with partial numerators $\alpha_n = 1$ for $n \geq 1$. (Note that with this assumption on partial numerators, conditions (iii) and (iv) of SRCFs are trivially satisfied for any choice of integral partial denominators satisfying condition (ii).) A RCF will also be denoted by

$$[a_0; a_1, a_2, \dots] \coloneqq [a_0/1; 1/a_1, 1/a_2, \dots].$$

For a RCF, we use the special notation $p_n \coloneqq P_n$ and $q_n \coloneqq Q_n$, $n \geq -2$, so the recurrence relations (1.4) become

$$\begin{aligned} p_{n+1} &= a_{n+1}p_n + p_{n-1}, \qquad p_{-2} = 0,\ p_{-1} = 1, \\ q_{n+1} &= a_{n+1}q_n + q_{n-1}, \qquad q_{-2} = 1,\ q_{-1} = 0. \end{aligned} \tag{1.8}$$

Since a RCF is a SRCF, the limit $x = \lim_{n\to\infty} p_n/q_n \in \mathbb{R}$ exists for any infinite choice of a_n, $n \geq 0$ (this can also be proven directly; see, e.g., [90]), and the odd-indexed and even-indexed RCF-convergents $(p_{2k-1}/q_{2k-1})_{k\geq 0}$ and $(p_{2k}/q_{2k})_{k\geq 0}$ form strictly decreasing and strictly increasing sequences, respectively (see, e.g., Theorem 4 of [109]). Conversely, every real number x has a RCF-expansion. Moreover, if x is irrational, its RCF-expansion is unique and has infinitely many partial denominators a_n; if x is rational, it has exactly two RCF-expansions,

$$[a_0; a_1, \dots, a_n] \quad \text{and} \quad [a_0; a_1, \dots, a_n - 1, 1],$$

where $a_n \geq 2$ if $n \geq 1$ ([90]). The *depth* of $x \in \mathbb{Q}$ is the number n occurring in the above two expansions.

Mediant convergents

The fractions

$$\frac{\lambda p_n + p_{n-1}}{\lambda q_n + q_{n-1}} \quad \text{for } \lambda \in \mathbb{N},\ 1 \le \lambda < a_{n+1},\ n \ge -1, \tag{1.9}$$

are called *mediants* (or *mediant convergents*) of $x = [a_0; a_1, a_2, \dots]$. Notice that if $\lambda = 0$, the expression in (1.9) gives p_{n-1}/q_{n-1}, while if $\lambda = a_{n+1}$, the expression gives p_{n+1}/q_{n+1} by the recurrence relations (1.8). Since the mediant $(a + b)/(c + d)$ of two fractions a/c and b/d with positive denominators lies between them in value, monotonicity of the odd-/even-indexed RCF-convergents gives the following relations for all $n \ge 0$ (see §1.4 of [109]):

$$\begin{aligned} x < \frac{p_{2n+1}}{q_{2n+1}} = \frac{a_{2n+1}p_{2n} + p_{2n-1}}{a_{2n+1}q_{2n} + q_{2n-1}} < \frac{(a_{2n+1}-1)p_{2n} + p_{2n-1}}{(a_{2n+1}-1)q_{2n} + q_{2n-1}} \\ < \cdots < \frac{p_{2n} + p_{2n-1}}{q_{2n} + q_{2n-1}} < \frac{p_{2n-1}}{q_{2n-1}} \end{aligned} \tag{1.10}$$

and

$$\begin{aligned} \frac{p_{2n}}{q_{2n}} < \frac{p_{2n+1} + p_{2n}}{q_{2n+1} + q_{2n}} < \cdots < \frac{(a_{2n+2}-1)p_{2n+1} + p_{2n}}{(a_{2n+2}-1)q_{2n+1} + q_{2n}} \\ < \frac{a_{2n+2}p_{2n+1} + p_{2n}}{a_{2n+2}q_{2n+1} + q_{2n}} = \frac{p_{2n+2}}{q_{2n+2}} < x. \end{aligned} \tag{1.11}$$

1.3 Some continued fraction algorithms

In this section we introduce some important continued fraction (CF) algorithms which shall be revisited throughout the chapter. In particular, the reader will find in §1.3.1 the Gauss map, which generates RCF-expansions; in §1.3.2 Nakada's parameterised family of α-CFs, which generate SRCFs including RCFs, Hurwitz's singular CFs, nearest integer CFs, and Rényi's backward CFs; and in §1.3.3 Kraaikamp's S-expansions, which generate SRCFs including Minkowski's diagonal CFs, Bosma's optimal CFs and (a strict subcollection of) Nakada's α-CFs.

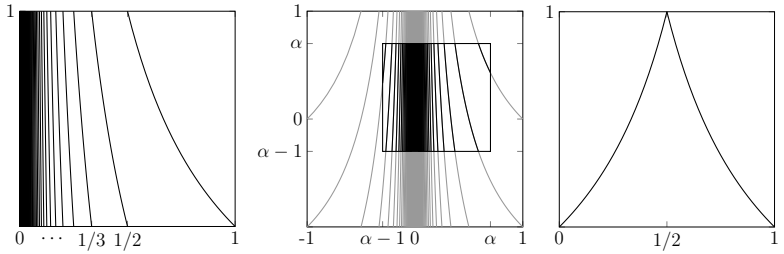


Figure 1.1: Graphs of the Gauss map G (left), Nakada's α-CF maps G_α (center), and the Farey tent map F (right).

1.3.1 The Gauss map

The Gauss map

The partial denominators a_n of RCF-expansions are generated by the *Gauss map* $G : [0,1] \to [0,1]$ defined by[4] $G(0) \coloneqq 0$ and $G(x) \coloneqq 1/x - \lfloor 1/x \rfloor$, $x > 0$; see Figure 1.1. Indeed, for $x \in \mathbb{R}$, set $a_0 = a_0(x) \coloneqq \lfloor x \rfloor$ and $x_0 \coloneqq x - a_0 \in [0,1)$. Define $a(0) \coloneqq \infty$, $a(x) \coloneqq \lfloor 1/x \rfloor$ for $x \neq 0$, and $a_n = a_n(x) \coloneqq a(G^{n-1}(x_0))$ for $n > 0$. Notice that for any integer $k \geq 1$, $a_n = k$ if and only if $G^{n-1}(x_0) \in (1/(k+1), 1/k]$. One finds that for $G^{n-1}(x_0) \neq 0$,

$$G^n(x_0) = \frac{1}{G^{n-1}(x_0)} - a_n.$$

Rearranging gives

$$G^{n-1}(x_0) = \frac{1}{a_n + G^n(x_0)},$$

[4]While G may be defined as a self-map of $[0,1)$, we choose to include the endpoint 1 for later notational purposes.

which holds for both $G^{n-1}(x_0) \neq 0$ and $G^{n-1}(x_0) = 0$, and which—with repeated applications—in turn gives

$$\begin{aligned} x =& a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_n + G^n(x_0)}}} \\ =& [a_0; a_1, \ldots, a_{n-1}, a_n + G^n(x_0)]. \end{aligned}$$

Symbolically, the Gauss map acts as a left-shift on RCF-expansions. That is, if $x = [0; a_1, a_2, \ldots] \in (0, 1)$, then $G(x) = [0; a_2, a_3, \ldots]$.

The dynamical system $([0, 1], \mathcal{B}, \nu_G, G)$ is exact (hence strongly mixing, weakly mixing and ergodic; see [90]), where $\mathcal{B}$ is the Borel σ-algebra[5] on $[0, 1]$ and ν_G is the *Gauss measure*, which is the absolutely continuous, G-invariant probability measure with density $1/(\log 2(1 + x))$.

The natural extension of the Gauss map

In the late 1970s and early 1980s, Nakada, Ito and Tanaka ([153, 155]) introduced an explicit, planar natural extension $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$ of the system $([0, 1], \mathcal{B}, \nu_G, G)$. Here, $\Omega \coloneqq [0, 1]^2$; the map $\mathcal{G} : \Omega \to \Omega$ is defined by $\mathcal{G}(0, y) \coloneqq (0, y)$ and for $z = (x, y) \in \Omega$ with $x \neq 0$,

$$\mathcal{G}(z) \coloneqq \left(G(x), \frac{1}{a(x) + y} \right); \tag{1.12}$$

and the $\mathcal{G}$-invariant probability measure $\bar{\nu}_G$ has density $1/(\log 2(1 + xy)^2)$. Since $([0, 1], \mathcal{B}, \nu_G, G)$ is strongly mixing, so is the natural extension $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$.

Symbolically, the map $\mathcal{G}$ acts as a two-sided shift on RCF-expansions. That is, if

$$(x, y) = ([0; a_1, a_2, \ldots], [0; b_1, b_2, \ldots]) \in \Omega$$

[5] Throughout, $\mathcal{B}$ represents the Borel σ-algebra on the appropriate domain.

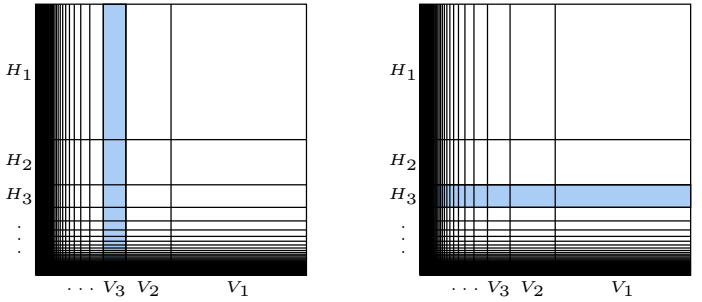


Figure 1.2: Up to null sets, the map $\mathcal{G}$ sends the vertical region V_k to the horizontal region H_k.

with $x \in (0,1)$, then

$$\mathcal{G}(x,y) = ([0; a_2, a_3, \dots], [0; a_1, b_1, b_2, \dots]). \tag{1.13}$$

The map $\mathcal{G}$ may also be understood geometrically: setting

$$V_k := \left(\frac{1}{k+1}, \frac{1}{k}\right] \times [0,1] \quad \text{and} \quad H_k := [0,1] \times \left(\frac{1}{k+1}, \frac{1}{k}\right] \tag{1.14}$$

for each integer $k \geq 1$, one finds that $\mathcal{G}(V_k) = H_k$, up to null sets; see Figure 1.2. We call V_k and H_k the kth *vertical region* and k^{th} *horizontal region*, respectively.

1.3.2 Nakada's α-continued fractions

In 1981, Nakada ([153]) introduced a one-parameter family of continued fraction algorithms, called α-CF *maps*, each of which generates—in a similar fashion as the Gauss map—SRCF-expansions. For each $\alpha \in [0,1]$, Nakada's α-CF map $G_\alpha : [\alpha - 1, \alpha] \to [\alpha - 1, \alpha]$ is defined by $G_\alpha(0) := 0$ and for $x \neq 0$,

$$G_\alpha(x) := \frac{1}{|x|} - \left\lfloor \frac{1}{|x|} + 1 - \alpha \right\rfloor;$$

see Figure 1.1. Notice that $G_1 = G$ is the Gauss map. In fact, Nakada's α-CFs contain several other well-studied continued fraction algorithms: when $\alpha = (\sqrt{5}-1)/2$, G_α generates Hurwitz's *singular* CFs ([88]); $\alpha = 1/2$ generates the *nearest integer* CFs introduced by Minnigerode in [148] and studied by Hurwitz in [88]; and $\alpha = 0$ generates Rényi's *backward* CFs ([166]). The latter map G_0 has an infinite, σ-finite, absolutely continuous invariant measure ρ_0 with density $1/(x+1)$, while for $\alpha \in (0,1]$, there is a unique, absolutely continuous invariant probability measure ρ_α ([129]). Moreover, for each $\alpha \in [0,1]$, the dynamical system $([\alpha-1,\alpha],\mathcal{B},\rho_\alpha,G_\alpha)$ is exact and, hence, ergodic ([129]).

Since Nakada's introduction of the α-CFs, much work has been done to determine explicitly the invariant measures ρ_α and to understand the metric entropy $h(G_\alpha) = h_{\rho_\alpha}(G_\alpha)$ as a function $\alpha \mapsto h(G_\alpha)$ of $\alpha \in (0,1]$. Nakada restricted his initial study in [153] to $\alpha \geq 1/2$ and derived ρ_α by constructing an explicit, planar natural extension of $([\alpha-1,\alpha],\mathcal{B},\rho_\alpha,G_\alpha)$. However, it was observed at the time that difficulties arose in extending these methods to $\alpha < 1/2$. Kraaikamp in 1991 ([115]) reobtained Nakada's natural extensions for $\alpha \geq 1/2$ in a simple fashion as special instances within his theory of S-expansions; see §1.3.3 below. In 1999, Moussa et al. ([150]) determined explicit, absolutely continuous invariant probability measures for a subset of a slightly different family of maps called *folded* α-CF*s*, which are factors of the α-CF maps. From their results one could obtain ρ_α for $\sqrt{2}-1 \leq \alpha < 1/2$ and—using Rohlin's formula—the entropy $h(G_\alpha)$ as a function of $\alpha \in [\sqrt{2}-1,1]$ (see [129]):

$$h(G_\alpha) = \begin{cases} \frac{\pi^2}{6\log(1+g)}, & \sqrt{2}-1 \leq \alpha \leq g, \\ \frac{\pi^2}{6\log(1+\alpha)}, & g < \alpha \leq 1, \end{cases} \tag{1.15}$$

where $g := (\sqrt{5}-1)/2$. Following this, the entropy function was conjectured to be monotone increasing and continuous on the remaining subinterval $(0,\sqrt{2}-1)$ ([24]).

It was thus quite surprising when, in 2008, Luzzi and Marmi ([129]) gave numerical evidence suggesting that $h(G_\alpha)$ possessed

a seemingly complicated, non-monotone, self-similar structure on $(0, \sqrt{2}-1)$. In the same year, Nakada and Natsui ([156]) confirmed this non-monotonicity by giving countably many non-empty intervals on which the function is increasing, decreasing and constant, respectively. These intervals are determined by a phenomenon called *matching*, where the G_α-orbits of α and $\alpha - 1$ coincide after some finite number of steps (see also Chapter 2 below). Further numerical data on these so-called *matching intervals* was given in [37], and the authors also exhibited points in the complement of the union of matching intervals at which the entropy function fails to be locally monotone. The matching intervals were completely classified in [38], and their union was shown to have full measure. (These intervals have surprising connections to unimodal maps, the real slice of the boundary of the Mandelbrot set, and the parameter space of a family of maps producing signed binary expansions ([16, 39, 45]).)

In 2012, Kraaikamp, Schmidt and Steiner ([120]) proved that the entropy function is indeed continuous on $(0, 1]$ (this fact had also been proven in a 2009 preprint of Tiozzo for $\alpha > 0.056\ldots$ and was later improved to Hölder continuity on $(0, 1]$ ([182, 183])). In [120], the authors construct a planar natural extension of $([\alpha - 1, \alpha], \mathcal{B}, \rho_\alpha, G_\alpha)$ for each $\alpha > 0$. The domain of this natural extension is first defined theoretically as an orbit closure of a certain planar map; a further detailed analysis of the G_α-orbits of α and $\alpha - 1$ allow for a more explicit description of this domain (see §7 of [120]). Moreover, the authors prove (Theorem 2 of [120]) a conjecture of Luzzi and Marmi ([129]) that the product of the entropy $h(G_\alpha)$ and the measure of the natural extension domain (using density $1/(1 + xy)^2$) is constant—in fact, equal to $\pi^2/6$—as a function of α, and they extend the constant branch of $h(G_\alpha)$ in (1.15) to the maximal interval $[g^2, g]$. However, even equipped with such machinery, a number of open questions are left at the end of [120]. In particular, the authors ask for explicit values of the entropy $h(G_\alpha)$ for $\alpha < g^2$, and they restate a conjecture of [37] on the explicit form of the density of ρ_α.

1.3.3 S-expansions

In 1991, Kraaikamp introduced in [115] a large class of new continued fraction algorithms by coupling two tools: singularisation and induced transformations of the natural extension $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$ of the Gauss map. Singularisation is an old, arithmetic procedure—tracing back as early as Lagrange's addendum ([123]) to Euler's *Vollständige Anleitung zur Algebra*—whereby one can sometimes manipulate a SRCF-expansion to produce a new, 'accelerated' SRCF-expansion of the same number.

Indeed, suppose that $x = [\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ is a SRCF-expansion with convergents $(P_n/Q_n)_{n\geq -1}$. Suppose, moreover, that $\beta_{n+1} = \alpha_{n+2} = 1$ for some $n \geq 0$. *Singularisation at position n* replaces this SRCF-expansion with the SRCF-expansion

$$x = [\beta_0/\alpha_0; \alpha_1/\beta_1, \dots, \alpha_{n-1}/\beta_{n-1}, \alpha_n/(\beta_n + \alpha_{n+1}), \\ -\alpha_{n+1}/(\beta_{n+2}+1), \alpha_{n+3}/\beta_{n+3}, \dots].$$

One can show that the convergents $(P'_n/Q'_n)_{n\geq -1}$ of this new expansion satisfy

$$\frac{P'_j}{Q'_j} = \begin{cases} \frac{P_j}{Q_j}, & j < n, \\ \frac{P_{j+1}}{Q_{j+1}}, & j \geq n, \end{cases}$$

i.e., singularisation at position n removes the nth convergent P_n/Q_n; see [115]. By iterating this procedure (possibly infinitely many times), one obtains a new SRCF-expansion of x whose convergents are a subsequence $(P_{n_k}/Q_{n_k})_{k\geq -1}$ of the original convergents.

Beginning from a RCF-expansion

$$[a_0; a_1, a_2, \dots] = [a_0/1; 1/a_1, 1/a_2, \dots]$$

with convergents p_n/q_n, acceleration via singularisation admits two major restrictions:

(i) the convergents p_n/q_n which are removed correspond to partial denominators $a_{n+1} = 1$, and

(ii) consecutive convergents p_n/q_n, p_{n+1}/q_{n+1} cannot be removed.

Restriction (ii) comes from the fact that in order to remove both p_n/q_n and p_{n+1}/q_{n+1}, one would need to either first singularise the original expansion at position n, then singularise the new expansion again at position n, or first singularise at position $n+1$ and then at position n. However, in either case, the partial denominator corresponding to the second singularisation is strictly greater than 1, contrary to the singularisation requirements. Nevertheless, one may singularise to remove non-consecutive convergents p_n/q_n with $a_{n+1} = 1$ independent of order and, thus, simultaneously; see [115].

In [115], Kraaikamp uses the natural extension $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$ of the Gauss map to govern the singularisation process, beginning from RCF-expansions. In particular, Kraaikamp fixes a measurable *singularisation area*[6] $S \subset \Omega$ satisfying $\bar{\nu}_G(\partial S) = 0$,

(a) $S \subset V_1$, and

(b) $S \cap \mathcal{G}(S) = \varnothing$,

and considers the $\mathcal{G}$-orbit of $(x, 0)$ in Ω with $x = [0; a_1, a_2, \dots]$ irrational. That $\bar{\nu}_G(\partial S) = 0$ is a technical condition, called $\bar{\nu}_G$-*continuity*, guaranteeing that for Lebesgue–a.e. x, the $\mathcal{G}$-orbit of $(x, 0)$ behaves like a '$\bar{\nu}_G$-generic' point; see Remark 4.6.i of [115]. Condition (a) guarantees that if $\mathcal{G}^n(x, 0) \in S$, then $a_{n+1} = 1$, and condition (b) guarantees that two consecutive points in the $\mathcal{G}$ orbit of $(x, 0)$ do not belong to S; cf. restrictions (i) and (ii) above. By simultaneously singularising the RCF-expansion of x at all positions n for which $\mathcal{G}^n(x, 0) \in S$, one obtains a SRCF-expansion of x, called an S-*expansion*, whose convergents are a subsequence of the RCF-convergents of x. Put a different way, Kraaikamp's S-expansions are SRCFs whose convergents are precisely the subse-

[6] Technically, condition (a) should be replaced by $S \subset \overline{V_1}$ and (b) by $S \cap \mathcal{G}(S) \subset \{(g, g)\}$ with $g = (\sqrt{5} - 1)/2$; see Definition 4.4 and Remark 4.6.ii of [115]. Moreover, in [115], $\mathcal{G}$ is defined on $[0, 1) \times [0, 1]$ rather than $\Omega = [0, 1]^2$. What follows in §1.7.2 below can be done with these minor adjustments, but for simplicity we shall omit these details.

quence of RCF-convergents p_n/q_n for which $\mathcal{G}^n(x,0) \in \Omega \setminus S$ for $n \geq 0$, i.e., S-expansions are governed by the induced transformation of $(\Omega, \mathcal{B}, \bar{\mu}_G, \mathcal{G})$ on $\Omega \setminus S$.

In addition to defining new CF-algorithms, Kraaikamp shows in [115] that many previously studied CF-algorithms are realised by certain singularisation areas S. Since these arise from induced transformations of a common dynamical system, ergodic properties of the underlying algorithms are easily comparable with one another. Kraaikamp's S-expansions include Minkowski's diagonal CFs ([147]), Bosma's optimal CFs ([19]) and (the natural extensions of) Nakada's α-CFs. However, the α-CFs realised as S-expansions are—somewhat curiously—only those for which $\alpha \geq 1/2$. This is explained[7] by Nakada and Natsui in [156] for $\alpha \in [\sqrt{2}-1, 1/2)$:

> Here we note that the natural extension of G_α cannot be obtained by a simple induced transformation, in the sense of [Kraaikamp's S-expansions], of the natural extension of G. This is related to the fact that a convergent of the continued fraction expansion of x by G_α may not be a convergent of the [regular] continued fraction expansion of x.

In §1.7.3 below, we exhibit the natural extension of each system $([\alpha-1, \alpha], \mathcal{B}, \rho_\alpha, G_\alpha)$, $\alpha \in (0,1]$, as an induced transformation of the natural extension of another, slower continued fraction map—the Farey tent map.

1.4 The Farey tent map and Farey expansions

In this section we introduce another CF-map—the Farey tent map—whose natural extension (see §1.5 below) shall be of central importance to us. The Farey tent map generates SRCF-expansions whose

[7] Notation is changed from the original to match that of the current chapter.

convergents consist of all RCF-convergents and mediant convergents; see (1.9) above and Proposition 1.4.1 below.

1.4.1 The Farey tent map

Define $\varepsilon : [0,1] \to \{0,1\}$ by

$$\varepsilon(x) \coloneqq \begin{cases} 0, & x \leq 1/2, \\ 1, & x > 1/2, \end{cases}$$

and for $\varepsilon \in \{0,1\}$, set

$$A_\varepsilon \coloneqq \begin{pmatrix} 1-\varepsilon & \varepsilon \\ 1 & 1 \end{pmatrix}. \tag{1.16}$$

The *Farey tent map* $F : [0,1] \to [0,1]$ is defined by

$$F(x) \coloneqq A^{-1}_{\varepsilon(x)} \cdot x = \begin{cases} x/(1-x), & x \leq 1/2, \\ (1-x)/x, & x > 1/2; \end{cases} \tag{1.17}$$

see Figure 1.1 above. The dynamical system $([0,1], \mathcal{B}, \mu, F)$ is ergodic, where μ is the infinite, σ-finite, absolutely continuous F-invariant measure with density $1/x$ ([57, 93, 160]). It follows from the definition of F that if x has RCF-expansion[8] $[0; a_1, a_2, a_3, \dots]$, then

$$F(x) = \begin{cases} [0; a_1 - 1, a_2, a_3, \dots], & a_1 > 1, \\ [0; a_2, a_3, a_4, \dots], & a_1 = 1. \end{cases} \tag{1.18}$$

From this, one finds that the Gauss map G is the *jump transformation* of F associated to the interval $(1/2, 1]$, meaning that for x as above with $x \neq 0$,

$$\min\{j \geq 0 \mid F^j(x) \in (1/2, 1]\} = a_1 - 1, \quad \text{and} \quad G(x) = F^{a_1}(x);$$

see, e.g., §11.4 of [46].

[8] If the expansion of x is finite, we set the remaining digits equal to ∞, e.g., $x = [0; a_1, \dots, a_n, \infty, \infty, \dots]$. This also holds for x equal to $0 = [0; \infty, \infty, \dots]$ and $1 = [0; 1, \infty, \infty, \dots]$, interpreting $\infty - 1 = \infty$.

1.4.2 Farey expansions and Farey convergents

The Farey tent map and RCF-convergents and mediants

In [93], Ito studied the ergodic properties of the dynamical system $([0,1], \mathcal{B}, \mu, F)$ and showed via matrix relations that F generates all convergents and mediant convergents of the RCF-expansion of any irrational $x \in (0,1)$. We reproduce this fact here, fixing notation[9] along the way.

Recall from (1.17) that $F(x) = A^{-1}_{\varepsilon(x)} \cdot x$, or $x = A_{\varepsilon(x)} \cdot F(x)$. Setting

$$x_n \coloneqq F^n(x) \qquad \text{and} \qquad \varepsilon_{n+1} = \varepsilon_{n+1}(x) \coloneqq \varepsilon(x_n), \tag{1.19}$$

we find for each $n \geq 0$ that $x_n = A_{\varepsilon(x_n)} \cdot F(x_n) = A_{\varepsilon_{n+1}} \cdot x_{n+1}$. Repeatedly applying this beginning from $x = x_0$, we have

$$x = (A_{\varepsilon_1} A_{\varepsilon_2} \cdots A_{\varepsilon_n}) \cdot x_n.$$

Let $A_{[0,0]} \coloneqq I_2$ be the identity matrix and for $n > 0$,

$$A_{[0,n]} = A_{[0,n]}(x) \coloneqq A_{\varepsilon_1} A_{\varepsilon_2} \cdots A_{\varepsilon_n}. \tag{1.20}$$

With this notation,

$$x_n = F^n(x) = A^{-1}_{[0,n]} \cdot x, \quad n \geq 0.$$

We wish to determine the entries of $A_{[0,n]}$. For $x = [0; a_1, a_2, \dots]$ irrational and $n \geq 0$, let $j_n = j_n(x)$ and $\lambda_n = \lambda_n(x)$ be the unique integers[10] satisfying

$$n = a_1 + a_2 + \cdots + a_{j_n} + \lambda_n, \qquad j_n \geq 0, \quad 0 \leq \lambda_n < a_{j_n+1}. \tag{1.21}$$

[9] Notation is largely recycled from [93] but with matrix entries permuted.

[10] These should be thought of in light of Euclid's division lemma: for integers $n, a \geq 0$, there exist unique integers j and λ such that $n = ja + \lambda$ with $0 \leq \lambda < a$. Instead of summing a fixed integer a with itself j times, we sum the first j RCF-digits $a_1, a_2, \dots, a_j$ of x.

From (1.18), we have

$$\varepsilon_1\varepsilon_2\varepsilon_3\cdots = 0^{a_1-1}10^{a_2-1}10^{a_3-1}1\cdots,$$

so

$$\varepsilon_1\varepsilon_2\cdots\varepsilon_n = 0^{a_1-1}10^{a_2-1}1\cdots 0^{a_{j_n}-1}10^{\lambda_n}. \tag{1.22}$$

Denote the entries of $A_{[0,n]},\ n \geq 0$, by

$$\begin{pmatrix} u_n & t_n \\ s_n & r_n \end{pmatrix} = \begin{pmatrix} u_n(x) & t_n(x) \\ s_n(x) & r_n(x) \end{pmatrix} \coloneqq A_{[0,n]},$$

and observe that for any $k \in \mathbb{Z}$,

$$A_0^k A_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^k \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & k+1 \end{pmatrix}. \tag{1.23}$$

From (1.22) and (1.23), it follows for $n > 0$ that

$$\begin{aligned} \begin{pmatrix} u_n & t_n \\ s_n & r_n \end{pmatrix} = A_{[0,n]} &= A_{\varepsilon_1}\cdots A_{\varepsilon_n} \\ &= A_0^{a_1-1}A_1\cdots A_0^{a_{j_n}-1}A_1A_0^{\lambda_n} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix}\cdots\begin{pmatrix} 0 & 1 \\ 1 & a_{j_n} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ \lambda_n & 1 \end{pmatrix} \\ &= \begin{pmatrix} p_{j_n-1} & p_{j_n} \\ q_{j_n-1} & q_{j_n} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ \lambda_n & 1 \end{pmatrix} \\ &= \begin{pmatrix} \lambda_n p_{j_n} + p_{j_n-1} & p_{j_n} \\ \lambda_n q_{j_n} + q_{j_n-1} & q_{j_n} \end{pmatrix}, \end{aligned} \tag{1.24}$$

where p_j/q_j is the jth RCF-convergent of x (see also Lemma 1.1 of [93]). Note that equality of the first and final expressions also holds for $n = j_n = \lambda_n = 0$, for in this case both matrices are the identity I_2. As a sequence, the quotients of the left-columns of the matrices

in (1.24) are

$$\begin{aligned}\left(\frac{u_n}{s_n}\right)_{n\geq 0} &= \left(\frac{\lambda_n p_{j_n}+p_{j_n-1}}{\lambda_n q_{j_n}+q_{j_n-1}}\right)_{n\geq 0} \\ &= \Bigg(\frac{p_{-1}}{q_{-1}}\,,\frac{p_0+p_{-1}}{q_0+q_{-1}}\,,\ldots,\frac{(a_1-1)p_0+p_{-1}}{(a_1-1)q_0+q_{-1}},\\ &\quad \frac{p_0}{q_0}\,,\frac{p_1+p_0}{q_1+q_0}\,,\ldots,\frac{(a_2-1)p_1+p_0}{(a_2-1)q_1+q_0},\ldots,\\ &\quad \frac{p_{j-1}}{q_{j-1}},\frac{p_j+p_{j-1}}{q_j+q_{j-1}},\ldots,\frac{(a_{j+1}-1)p_j+p_{j-1}}{(a_{j+1}-1)q_j+q_{j-1}},\ldots\Bigg),\end{aligned} \tag{1.25}$$

i.e., the map F generates all RCF-convergents and mediants. Notice that the denominators $(s_n)_{n\geq 0}$ do not form an increasing sequence. Supposedly to 'remedy' this, in [93] Ito instead considers the sequence $((u_n+t_n)/(s_n+r_n))_{n\geq 0}$ with increasing denominators. However, in light of Proposition 1.4.1 below and further results in the sequel, we find it more natural to study $(u_n/s_n)_{n\geq 0}$.

Recall that $x = A_{[0,n]}\cdot x_n$ and note that $\det(A_{[0,n]}) = \pm 1$. With these observations, we calculate the distance between x and u_n/s_n, which will prove to be useful later:

$$\left|x-\frac{u_n}{s_n}\right| = \left|\frac{u_n x_n+t_n}{s_n x_n+r_n}-\frac{u_n}{s_n}\right| = \frac{|t_n s_n-u_n r_n|}{s_n(s_n x_n+r_n)} = \frac{1}{s_n(s_n x_n+r_n)}. \tag{1.26}$$

Lehner and Farey expansions

Originally, there was no continued fraction expansion associated to the Farey tent map F. Such expansions do exist and can be obtained from a map introduced by Lehner in 1994; see [126]. The *Lehner map* (also referred to as 'the mother of all continued fractions' in [49]) is the map $L:[1,2]\to[1,2]$ defined by

$$L(x) \coloneqq \begin{cases} 1/(2-x), & x\leq 3/2,\\ 1/(x-1), & x>3/2.\end{cases}$$

For $x \in [1,2]$ and each $n \geq 0$, set

$$(b_n, e_{n+1}) = (b_n(x), e_{n+1}(x)) \coloneqq \begin{cases} (2,-1), & L^n(x) \leq 3/2, \\ (1,1), & L^n(x) > 3/2. \end{cases}$$

The digits (b_n, e_{n+1}) generate the so-called *Lehner expansion* of $x \in [1,2]$,

$$x = [b_0/1; e_1/b_1, e_2/b_2, \dots], \tag{1.27}$$

which is a SRCF-expansion (see [49, 126]).

Lehner studied expansions of the form (1.27) generated by L but no dynamical properties of this map. In [49] it is observed that the dynamical systems $([0,1], \mathcal{B}, \mu, F)$ and $([1,2], \mathcal{B}, \rho, L)$ are isomorphic via the translation $x \mapsto x+1$, where ρ is the absolutely continuous, L-invariant measure with density $1/(x-1)$. Via this isomorphism, the map F can be used to generate a *Farey expansion* for each $x \in [0,1]$ (see also [92]). Indeed, for $x \in [0,1]$, let $\varepsilon_{n+1} = \varepsilon_{n+1}(x)$, $n \geq 0$, be as in (1.19), and let $[b_0/1; e_1/b_1, e_2/b_2, \dots]$ be the Lehner expansion of $x+1$. Then $x = [b_0 - 1/1; e_1/b_1, e_2/b_2, \dots]$, and we find that

$$\begin{aligned} (b_n, e_{n+1}) &= \begin{cases} (2,-1), & L^n(x+1) \leq 3/2, \\ (1,1), & L^n(x+1) > 3/2, \end{cases} \\ &= \begin{cases} (2,-1), & F^n(x) \leq 1/2, \\ (1,1), & F^n(x) > 1/2, \end{cases} \\ &= \begin{cases} (2,-1), & \varepsilon_{n+1} = 0 \\ (1,1), & \varepsilon_{n+1} = 1 \end{cases} \\ &= (2 - \varepsilon_{n+1}, 2\varepsilon_{n+1} - 1). \end{aligned}$$

Hence F generates SRCF-expansions, called *Farey expansions*:

$$x = [(1-\varepsilon_1)/1; (2\varepsilon_1 - 1)/(2 - \varepsilon_2), (2\varepsilon_2 - 1)/(2 - \varepsilon_3), \dots]. \tag{1.28}$$

Farey convergents

The convergents

$$P_n/Q_n = [(1-\varepsilon_1)/1; (2\varepsilon_1 - 1)/(2-\varepsilon_2), (2\varepsilon_2 - 1)/(2-\varepsilon_3), \\ \ldots, (2\varepsilon_n - 1)/(2-\varepsilon_{n+1})]$$

of the Farey expansion of x are called the *Farey convergents* of x. It turns out that the sequence $(P_n/Q_n)_{n\geq -1}$ of Farey convergents is precisely the sequence $(u_n/s_n)_{n\geq 0}$ from (1.25) of RCF-convergents and mediants generated by F:

Proposition 1.4.1. *For each $n \geq 0$,*

$$\begin{pmatrix} u_n \\ s_n \end{pmatrix} = \begin{pmatrix} P_{n-1} \\ Q_{n-1} \end{pmatrix},$$

where P_n/Q_n is the nth Farey convergent of x.

Proof. The proof is by induction. For $n = 0$ and $n = 1$, we have

$$\begin{pmatrix} u_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} P_{-1} \\ Q_{-1} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} u_1 \\ s_1 \end{pmatrix} = \begin{pmatrix} 1-\varepsilon_1 \\ 1 \end{pmatrix} = \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix}.$$

Now suppose for some $n \geq 1$ it holds for each $0 \leq k \leq n$ that

$$\begin{pmatrix} u_k \\ s_k \end{pmatrix} = \begin{pmatrix} P_{k-1} \\ Q_{k-1} \end{pmatrix}.$$

We compute

$$\begin{aligned} \begin{pmatrix} u_n & t_n \\ s_n & r_n \end{pmatrix} &= \begin{pmatrix} u_{n-1} & t_{n-1} \\ s_{n-1} & r_{n-1} \end{pmatrix} \begin{pmatrix} 1-\varepsilon_n & \varepsilon_n \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} (1-\varepsilon_n)u_{n-1} + t_{n-1} & \varepsilon_n u_{n-1} + t_{n-1} \\ (1-\varepsilon_n)s_{n-1} + r_{n-1} & \varepsilon_n s_{n-1} + r_{n-1} \end{pmatrix}. \end{aligned} \tag{1.29}$$

The top-left entries give

$$t_{n-1} = u_n + (\varepsilon_n - 1)u_{n-1}.$$

Then the top-right entries—replacing t_{n-1} with the right-hand side of the previous line—give

$$t_n = u_n + (2\varepsilon_n - 1)u_{n-1}.$$

Since (1.29) holds for all $n \geq 1$, the top-left entries (replacing n with $n+1$) together with the previous line, our inductive hypothesis, (1.28) and (1.4) give

$$\begin{aligned} u_{n+1} =& (1 - \varepsilon_{n+1})u_n + t_n \\ =& (2 - \varepsilon_{n+1})u_n + (2\varepsilon_n - 1)u_{n-1} \\ =& (2 - \varepsilon_{n+1})P_{n-1} + (2\varepsilon_n - 1)P_{n-2} = P_n. \end{aligned}$$

With similar computations one finds $s_{n+1} = Q_n$. □

1.5 Ito's natural extension of the Farey tent map

We now come to one of the central tools of this chapter: the natural extension of the Farey tent map, originally introduced by Ito in 1989 ([93]). In §1.5.1, we recall the definition of Ito's natural extension and discuss a one-to-one correspondence between orbits under the natural extension map and Farey convergents. Via this correspondence, we find that certain subregions of the domain of Ito's natural extension give rise to certain types of Farey convergents. In §1.5.2, we discuss induced transformations of Ito's natural extension and their connection with subsequences of Farey convergents. Moreover, we revisit a theorem of Brown and Yin from [29] stating that the natural extension of the Gauss map is isomorphic to a certain induced transformation of Ito's natural extension, and—exploiting this fact—calculate the entropy of our induced systems (Theorem 1.5.6).

1.5.1 The natural extension of the Farey tent map

In [93], Ito determined a planar natural extension $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ of the dynamical system $([0,1], \mathcal{B}, \mu, F)$ associated to the Farey tent map. The map $\mathcal{F} : \Omega \to \Omega$ is defined for each $z = (x, y) \in \Omega$ by

$$\mathcal{F}(z) \coloneqq \left(A^{-1}_{\varepsilon(x)} \cdot x, A_{\varepsilon(x)} \cdot y\right) = \begin{cases} \left(\frac{x}{1-x}, \frac{y}{1+y}\right), & x \leq 1/2, \\ \left(\frac{1-x}{x}, \frac{1}{1+y}\right), & x > 1/2, \end{cases} \tag{1.30}$$

where again $\Omega = [0,1]^2$, and $\bar{\mu}$ is the infinite, σ-finite, absolutely continuous $\mathcal{F}$-invariant measure with density $1/(x+y-xy)^2$. Since $([0,1], \mathcal{B}, \mu, F)$ is ergodic, so is its natural extension $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$.

Notice that $\mathcal{F}$ is simply the Farey tent map F in the first coordinate. Setting $\varepsilon_{n+1} = \varepsilon_{n+1}(x) = \varepsilon(x_n)$ as in (1.19), we find that

$$z_n = (x_n, y_n) \coloneqq \mathcal{F}^n(z) = \left(A^{-1}_{[0,n]} \cdot x, A_{[n,0]} \cdot y\right), \quad n \geq 0, \tag{1.31}$$

where $A_{[0,n]}$ is defined as in (1.20), and

$$A_{[n,0]} = A_{[n,0]}(x) \coloneqq A_{\varepsilon_n} A_{\varepsilon_{n-1}} \cdots A_{\varepsilon_1}, \quad n \geq 1.$$

The entries of $A_{[n,0]}$ may be computed explicitly in terms of those of $A_{[0,n]}$ (recall (1.24)). Indeed, let $x = [0; a_1, a_2, \dots]$. Setting $j = j_n$ and $\lambda = \lambda_n$ (see (1.21)), we have for $n > 0$ that

$$\begin{aligned} A_{[n,0]} &= A_{\varepsilon_n} \cdots A_{\varepsilon_1} I_2 \\ &= A_0^{\lambda} A_1 A_0^{a_j - 1} \cdots A_1 A_0^{a_1 - 1} A_1 A_1^{-1} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & \lambda + 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & a_j \end{pmatrix} \cdots \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & \lambda + 1 \end{pmatrix} \left(\begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix} \cdots \begin{pmatrix} 0 & 1 \\ 1 & a_j \end{pmatrix} \right)^T \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & \lambda + 1 \end{pmatrix} \begin{pmatrix} p_{j-1} & q_{j-1} \\ p_j & q_j \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

$$= \begin{pmatrix} q_j - p_j & p_j \\ (\lambda+1)q_j + q_{j-1} - ((\lambda+1)p_j + p_{j-1}) & (\lambda+1)p_j + p_{j-1} \end{pmatrix}$$
$$= \begin{pmatrix} r_n - t_n & t_n \\ s_n + r_n - (u_n + t_n) & u_n + t_n \end{pmatrix}, \tag{1.32}$$

and the first and final expressions are also equal to I_2 for $n = 0$. Notice, furthermore, that $A^T_{[0,n]}$ and $A_{[n,0]}$ are conjugate under A_1:

$$A_1 A^T_{[0,n]} = \begin{pmatrix} t_n & r_n \\ u_n + t_n & s_n + r_n \end{pmatrix} = A_{[n,0]} A_1. \tag{1.33}$$

$\mathcal{F}$-orbits and Farey convergents

The map $\mathcal{F}$ admits particularly nice symbolic and geometric interpretations. Fix $z = (x, y) \in \Omega$ with (finite[11] or infinite) RCF-expansions

$$(x, y) = ([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]). \tag{1.34}$$

One verifies using (1.18) and (1.30) that

$$\mathcal{F}(z) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; 1 + b_1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3, \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases} \tag{1.35}$$

Recalling the vertical and horizontal regions from (1.14), the image of the rectangle $V_a \cap H_b$ for $a > 1$ is thus the rectangle $\mathcal{F}(V_a \cap H_b) = V_{a-1} \cap H_{b+1}$ immediately below and to the right of the original rectangle, and the image of the right-half V_1 of Ω is the top half $\mathcal{F}(V_1) = H_1$, up to a Lebesgue-null set. In particular, the iterates $\mathcal{F}^\lambda$, $0 \le \lambda < a$, 'slide' the rectangle $V_a \cap H_1$ 'down-and-right' through a rectangles, and the next image $\mathcal{F}^a(V_a \cap H_1)$ is mapped back as a subset of H_1 (see Figure 1.3).

[11] As in (1.18), if the expansion of x or y is finite, we set the remaining digits equal to ∞. If $x = 1/2$, we take the shorter of its two RCF-expansions, namely $x = [0; 2]$.

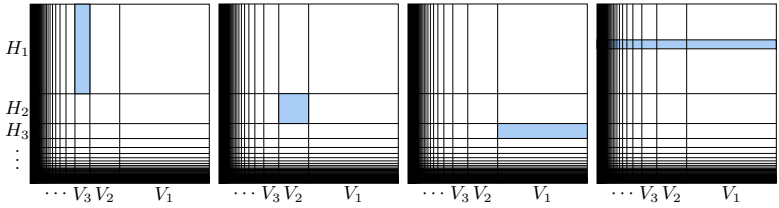


Figure 1.3: From left to right: The sets $V_3 \cap H_1$, $\mathcal{F}(V_3 \cap H_1)$, $\mathcal{F}^2(V_3 \cap H_1)$ and $\mathcal{F}^3(V_3 \cap H_1)$, respectively.

Now let $z = (x, y)$ be as in (1.34) with x irrational, and fix some $n \geq 0$. Recall from (1.21) that n may be written uniquely as

$$n = a_1 + a_2 + \cdots + a_{j_n} + \lambda_n,$$

where $0 \leq \lambda_n < a_{j_n+1}$. Repeatedly applying (1.35), one finds

$$\begin{aligned} z_n =& \mathcal{F}^n(z) \qquad (1.36)\\ =& \begin{cases} ([0; a_1 - \lambda_n, a_2, \dots], [0; \lambda_n + b_1, b_2, \dots, \dots]), & n < a_1, \\ ([0; a_{j_n+1} - \lambda_n, a_{j_n+2}, \dots], & \\ \quad [0; \lambda_n + 1, a_{j_n}, \dots, a_2, a_1 - 1 + b_1, b_2, \dots]), & n \geq a_1. \end{cases} \end{aligned}$$

In particular, if $z \in H_1$ so that $b_1 = 1$, then z_n belongs to (the closure of) $V_{a_{j_n+1}-\lambda_n} \cap H_{\lambda_n+1}$ for all $n \geq 0$.

Remark 1.5.1. The closure is needed in the previous statement if and only if $z = (x, 1)$ with $a_1 = 1$ and $a_1 \leq n < a_1 + a_2$. Indeed, in this case

$$z_n = (x_n, y_n) = ([0; a_2 - \lambda_n, a_3, \dots], [0; \lambda_n + 1, 1]).$$

Hence $y_n = 1/(\lambda_n + 2)$ which implies $z_n \notin H_{\lambda_n+1}$ lies on the lower boundary of H_{λ_n+1}. In all other cases for which $z \in H_1$, one has in fact $z_n \in V_{a_{j_n+1}-\lambda_n} \cap H_{\lambda_n+1}$ for all $n \geq 0$. In particular, this annoyance for $z = (x, 1)$ and $a_1 = 1$ is 'corrected' for $n \geq a_1 + a_2$,

and the closures are no longer needed. We shall frequently overlook this subtlety and make no mention of the special case $a_1 = 1$, and thus some claims should be understood up to this minor technicality. See also Remark 1.5.4 below.

Recall from Proposition 1.4.1 and (1.25) that the $(n-1)^{\text{st}}$ Farey convergent of x is

$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}}. \tag{1.37}$$

Identifying the nth point $z_n \in V_{a_{j_n+1}-\lambda_n} \cap H_{\lambda_n+1}$ of the $\mathcal{F}$-orbit of $z \in H_1$ with the $(n-1)^{\text{st}}$ Farey convergent u_n/s_n, we find that certain subregions $R \subset \Omega$ correspond to certain types of RCF-convergents or mediants. For instance, if $R = H_1$, then $z_n \in R$ if and only if $\lambda_n = 0$ in (1.37), i.e., u_n/s_n is a RCF-convergent. More generally, if $R = H_{\lambda+1}$, then $z_n \in R$ if and only if $\lambda_n = \lambda$, i.e., u_n/s_n is a RCF-convergent ($\lambda = 0$) or a 'λ^{th} mediant' convergent ($\lambda > 0$).

Moreover, setting $R = V_a$, we have $z_n \in R$ if and only if $a = a_{j_n+1} - \lambda_n$, or $\lambda_n = a_{j_n+1} - a$. Hence $z_n \in R$ if and only if u_n/s_n is a RCF-convergent ($a_{j_n+1} = a$) or '$(a-1)^{\text{st}}$-from-final' mediant convergent ($a_{j_n+1} > a$). Lastly, setting $R = V_{a-\lambda} \cap H_{\lambda+1}$, we have $z_n \in R$ if and only if $\lambda_n = \lambda$ and $a_{j_n+1} = a$, i.e., u_n/s_n is a RCF-convergent ($\lambda = 0$) or 'λ^{th} mediant' convergent ($\lambda > 0$) with partial denominator $a_{j_n+1} = a$.

These observations naturally lead us to consider the dynamics of Ito's natural extension $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ restricted to certain subregions of the domain in order to 'pick out' desired subsequences of Farey convergents.

1.5.2 Inducing Ito's natural extension

A $\bar{\mu}$-measurable set $R \subset \Omega$ is called *inducible*[12] if either $0 < \bar{\mu}(R) < \infty$ or $R = \Omega$. In the former case (i.e., $\bar{\mu}(R) < \infty$), we call R *proper*.

[12]We remark that the definition of inducible given here is broader than that in [53].

For R inducible, define the *hitting time to* R, denoted $N_R : \Omega \to \mathbb{N} \cup \{\infty\}$, by

$$N_R(z) \coloneqq \inf\{n \geq 1 \mid \mathcal{F}^n(z) \in R\}. \tag{1.38}$$

Since $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ is conservative and ergodic ([29]), $\bar{\mu}$–a.e. $z \in \Omega$ enters R infinitely often under iterates of $\mathcal{F}$ (see Remark 2.2.1 of [46]). Unless otherwise stated, we assume throughout that the null set of points from any $S \subset \Omega$ whose $\mathcal{F}$-orbits enter R at most finitely many times are removed from S, and—abusing notation—denote this new set again by S. Define $\mathcal{F}_R : \Omega \to R$ by

$$\mathcal{F}_R(z) \coloneqq \mathcal{F}^{N_R(z)}(z).$$

The *induced map of* $\mathcal{F}$ *on* R is the map $\mathcal{F}_R$ restricted to R, and the *induced measure* $\bar{\mu}_R$ is defined by

$$\bar{\mu}_R(S) \coloneqq \begin{cases} \frac{\bar{\mu}(S)}{\bar{\mu}(R)}, & R \neq \Omega, \\ \bar{\mu}(S), & R = \Omega, \end{cases} \qquad \text{for all} \quad S \in \mathcal{B},\ S \subset R.$$

Ergodicity of the *induced system* $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ follows from that of $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$. Notice that $\bar{\mu}_R$ is a probability measure if and only if R is proper.

Writing $z = (x, y)$ and setting $A_R(z) \coloneqq A_{[0, N_R(z)]}(x)$, $A_R^{-1}(z) \coloneqq (A_R(z))^{-1}$ and $A_R^T(z) \coloneqq (A_R(z))^T$, Equations (1.31) and (1.33) give

$$\mathcal{F}_R(z) = \left(A_R^{-1}(z) \cdot x, A_1 A_R^T(z) A_1^{-1} \cdot y\right). \tag{1.39}$$

We denote the entries of $A_R(z)$ by

$$\begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} \coloneqq A_R(z) = A_{\varepsilon_1} \cdots A_{\varepsilon_{N_R(z)}}. \tag{1.40}$$

For $n \geq 0$, set $z_n^R = (x_n^R, y_n^R) \coloneqq \mathcal{F}_R^n(z)$ and define $N_n^R(z)$ by $N_0^R(z) \coloneqq 0$ and

$$N_n^R(z) \coloneqq \sum_{\ell=0}^{n-1} N_R(z_\ell^R), \quad n \geq 1. \tag{1.41}$$

When the point z is understood, we use the suppressed notation $N_n^R \coloneqq N_n^R(z),\ n \geq 0$. We remark that when $R = \Omega$, $N_n^R = n$ for all $n \geq 0$. In general, the sequence $(N_n^R)_{n\geq 1}$ gives the indices $N \geq 1$ for which the forward orbit $(\mathcal{F}^N(z))_{N\geq 0}$ of z enters the region R, so $\mathcal{F}_R^n(z) = \mathcal{F}^{N_n^R}(z),\ n \geq 0$.

Let $A_0^R(z) \coloneqq I_2$ be the identity matrix, and for $n \geq 1$ set

$$A_n^R(z) \coloneqq A_R(z_{n-1}^R) = A_{\varepsilon_{N_{n-1}^R+1}} \cdots A_{\varepsilon_{N_n^R}}. \tag{1.42}$$

Notice that for $m \geq 1$ and $n \geq 0$,

$$A_m^R(z_n^R) = A_R(z_{n+m-1}^R) = A_{n+m}^R(z). \tag{1.43}$$

Moreover, for $0 \leq m \leq n$, set

$$A_{[m,n]}^R(z) \coloneqq A_m^R(z)A_{m+1}^R(z)\cdots A_n^R(z), \tag{1.44}$$

and denote the entries of $A_{[0,n]}^R(z)$ by

$$\begin{pmatrix} u_n^R(z) & t_n^R(z) \\ s_n^R(z) & r_n^R(z) \end{pmatrix} \coloneqq A_{[0,n]}^R(z) = A_{\varepsilon_1} \cdots A_{\varepsilon_{N_n^R}}. \tag{1.45}$$

When the point z is understood, we use the suppressed notation

$$A_{[0,n]}^R = \begin{pmatrix} u_n^R & t_n^R \\ s_n^R & r_n^R \end{pmatrix} \coloneqq A_{[0,n]}^R(z). \tag{1.46}$$

From (1.45) and (1.20), we have $A_{[0,n]}^R = A_{[0,N_n^R]}$ and thus, by (1.24),

$$\begin{pmatrix} u_n^R & t_n^R \\ s_n^R & r_n^R \end{pmatrix} = \begin{pmatrix} u_{N_n^R} & t_{N_n^R} \\ s_{N_n^R} & r_{N_n^R} \end{pmatrix} = \begin{pmatrix} \lambda_{N_n^R} p_{j_{N_n^R}} + p_{j_{N_n^R}-1} & p_{j_{N_n^R}} \\ \lambda_{N_n^R} q_{j_{N_n^R}} + q_{j_{N_n^R}-1} & q_{j_{N_n^R}} \end{pmatrix}. \tag{1.47}$$

The following lemma will be useful in §1.6.3 below.

Lemma 1.5.2. *For any $z = (x, y) \in \Omega$, $u_R(z), s_R(z) \in \mathbb{Z}$ satisfy $s_R(z) > 0$ and $0 \leq u_R(z) \leq s_R(z)$.*

Proof. It is clear from (1.40) that $u_R(z), s_R(z) \in \mathbb{Z}$. Moreover, $A_R(z) = A^R_{[0,1]}$, so setting $N = N^R_1$, (1.47) gives

$$\begin{pmatrix} u_R(z) \\ s_R(z) \end{pmatrix} = \begin{pmatrix} u_N \\ s_N \end{pmatrix} = \begin{pmatrix} \lambda_N p_{j_N} + p_{j_N - 1} \\ \lambda_N q_{j_N} + q_{j_N - 1} \end{pmatrix}.$$

Since $N > 0$, (1.21) implies either $j_N > 0$ or $\lambda_N > 0$. If $j_N = 0$, then $u_R(z) = \lambda_N p_0 + p_{-1} = p_{-1} = 1$ and $s_R(z) = \lambda_N q_0 + q_{-1} = \lambda_N > 0$, so the claim holds. If $j_N > 0$, then $s_R(z) \geq q_{j_N - 1} \geq q_0 = 1$, and $u_R(z)/s_R(z)$ lies between p_{j_N-1}/q_{j_N-1} and p_{j_N+1}/q_{j_N+1}, which are fractions between 0 and 1. Thus $0 \leq u_R(z) \leq s_R(z)$. □

Notice from (1.47) that $(u^R_n/s^R_n)_{n\geq 0}$ is a subsequence of the Farey convergents $(u_n/s_n)_{n\geq 0}$. In particular, the correspondence $z_n \leftrightarrow u_n/s_n$ between the $\mathcal{F}$-orbit of $z \in H_1$ and the Farey convergents of x gives a correspondence $z^R_n \leftrightarrow u^R_n/s^R_n$ between the subsequence $(z^R_n)_{n\geq 0} = (z_{N^R_n})_{n\geq 0}$ of points in the $\mathcal{F}$-orbit of z entering R and the subsequence $(u^R_n/s^R_n)_{n\geq 0} = (u_{N^R_n}/s_{N^R_n})_{n\geq 0}$ of Farey convergents. Hence a subregion R naturally determines a subsequence of Farey convergents. We illustrate with the examples discussed at the end of §1.5.1:

Example 1.5.3. The region $R = H_{\lambda+1}$ corresponds to RCF-convergents ($\lambda = 0$) or λ^{th} mediants ($\lambda > 0$):

$$\left\{ \frac{u^R_n}{s^R_n} \right\}_{n\geq 0} = \left\{ \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}} \;\middle|\; \lambda_n = \lambda \right\}_{n\geq 0}.$$

The vertical regions $R = V_a$ give—in addition to the RCF-convergents p_{j-1}/q_{j-1} for which $a_{j+1} = a$—final mediants, next-to-final mediants, and so on for $a = 1, 2, \ldots$, respectively:

$$\left\{ \frac{u^R_n}{s^R_n} \right\}_{n\geq 0} = \left\{ \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}} \;\middle|\; \lambda_n = a_{j_n+1} - a \right\}_{n\geq 0}.$$

The regions $R = V_{a-\lambda} \cap H_{\lambda+1}$ pick out RCF-convergents ($\lambda = 0$) or λ^{th}-mediants ($\lambda > 1$) corresponding to partial denominators a in the

RCF-expansion of x:

$$\left\{\frac{u_n^R}{s_n^R}\right\}_{n\geq 0} = \left\{\frac{\lambda_n p_{j_n}+p_{j_n-1}}{\lambda_n q_{j_n}+q_{j_n-1}} \;\middle|\; \lambda_n=\lambda,\ a_{j_n+1}=a\right\}_{n\geq 0}.$$

Remark 1.5.4. For the reasons explained in Remark 1.5.1, some of the statements in Example 1.5.3 are false for $z=(x,1)$, where $x=[0;a_1,a_2,\dots]$ with $a_1=a(x)=1$. For much of the metrical theory of §1.8 below, this technicality is innocuous. However, for some of the examples in §1.7 and §1.9, it will be advantageous to 'fix' this.

In such cases, we may 'adjust' the map $\mathcal{F}_R$ so that the corresponding statements on the subsets of Farey convergents are true for all $z=(x,1)$. For instance, when $R=H_1$ and $z=(x,1)$ with $x>1/2$, the sequence $(u_n^R/s_n^R)_{n\geq 0}$ skips the RCF-convergent p_0/q_0 of x. To 'catch' this convergent, we instead consider the map $\overline{\mathcal{F}}_{H_1}:\overline{H}_1\to\overline{H}_1$ where for $z=(x,y)\in\overline{H}_1$, $\overline{\mathcal{F}}_{H_1}(z)\coloneqq\mathcal{F}^{a(x)}(z)$ if $x\neq 0$ and $\overline{\mathcal{F}}_{H_1}(z)\coloneqq z$ if $x=0$. The maps $\overline{\mathcal{F}}_{H_1}$ and $\mathcal{F}_{H_1}$ agree on $H_1\backslash(A\cup B)$, where $A=(1/2,1]\times\{1\}$ and $B=\{0\}\times(1/2,1]$. The dynamical systems $(\overline{H}_1,\mathcal{B},\bar{\mu}_{\overline{H}_1},\overline{\mathcal{F}}_{H_1})$ and $(H_1,\mathcal{B},\bar{\mu}_{H_1},\mathcal{F}_{H_1})$ are isomorphic under the identity map and thus—from an ergodic-theoretic point of view—indistinguishable. Moreover, the map $\overline{\mathcal{F}}_{H_1}$ 'fixes' the subtlety in Remark 1.5.1 since, for $z\in A$, $\overline{\mathcal{F}}_{H_1}(z)=((1-x)/x,1/2)\in\overline{H}_1$. Thus we *do* include the convergent p_0/q_0 in $(u_n^R/s_n^R)_{n\geq 0}$. Throughout, we shall often consider such altered systems without mention, denoting them again by $(R,\mathcal{B},\bar{\mu}_R,\mathcal{F}_R)$.

Consider again $R=H_1$, which gives as a subsequence $(u_n^R/s_n^R)_{n\geq 0}$ of Farey convergents the RCF-convergents of x. Now R consists of all points $z=(x,y)$ as in (1.34) with $b_1=1$, so—after the alteration of Remark 1.5.4—we find from (1.36) that for $x\neq 0$, $N_R(z)=a_1=a(x)$ and

$$\begin{aligned}\mathcal{F}_R(z)=&\mathcal{F}^{a_1}([0;a_1,a_2,\dots],[0;1,b_2,b_3,\dots])\\ =&([0;a_2,a_3,\dots],[0;1,a_1,b_2,b_3,\dots]).\end{aligned}\tag{1.48}$$

Notice the similarity between this induced map and the map $\mathcal{G}$ from (1.13); they both act essentially as a two-sided shift on RCF-

expansions. In fact, Brown and Yin proved in 1996 that a copy of the Gauss natural extension is found sitting (inverted, scaled and 'suspended' from $y = 1$) within $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ as the induced system $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ with $R = H_1$ (Theorem 1 of [29]):

Theorem 1.5.5 (Brown–Yin, 1996 [29])**.** *When $R = H_1$, the induced system $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ is isomorphic to the Gauss natural extension $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$.*

Using Theorem 1.5.5, we can exploit knowledge about the Gauss natural extension $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$ to infer properties of other induced systems $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$. For instance, we find that the measure-theoretic entropy $h(\mathcal{F}_R) = h_{\bar{\mu}_R}(\mathcal{F}_R)$ of $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ is inversely proportional to the $\bar{\mu}$-measure of R:

Theorem 1.5.6. *For any proper, inducible subregion $R \subset \Omega$,*

$$h(\mathcal{F}_R) = \frac{\pi^2}{6\bar{\mu}(R)}.$$

Proof. We first note that for two proper, inducible subregions R_1 and R_2 with $R_1 \subset R_2$, the dynamical system $(R_1, \mathcal{B}, \bar{\mu}_{R_1}, \mathcal{F}_{R_1})$ is isomorphic to *the induced system of* $(R_2, \mathcal{B}, \bar{\mu}_{R_2}, \mathcal{F}_{R_2})$ *on* R_1. Hence, by Abramov's formula ([1]),

$$h(\mathcal{F}_{R_1}) = \frac{h(\mathcal{F}_{R_2})}{\bar{\mu}_{R_2}(R_1)} = \frac{\bar{\mu}(R_2)}{\bar{\mu}(R_1)} h(\mathcal{F}_{R_2}),$$

or

$$\bar{\mu}(R_1)h(\mathcal{F}_{R_1}) = \bar{\mu}(R_2)h(\mathcal{F}_{R_2}). \tag{1.49}$$

It is well-known that the entropy of the Gauss map G (and hence also its natural extension $\mathcal{G}$) is $\pi^2/6\log 2$, and thus $h(\mathcal{F}_{H_1}) = \frac{\pi^2}{6\log 2}$ by Theorem 1.5.5. Using this, (1.49) and a calculation of $\bar{\mu}(H_1) = \log 2$, we compute

$$\bar{\mu}(R)h(\mathcal{F}_R) = \bar{\mu}(R \cup H_1)h(\mathcal{F}_{R\cup H_1}) = \bar{\mu}(H_1)h(\mathcal{F}_{H_1}) = \frac{\pi^2}{6}. \qquad \square$$

Remark 1.5.7. We remark here the striking resemblance between Theorem 1.5.6, Remark 5.10 of [115] on the entropy of S-expansions and Theorem 2 of [120] (conjectured in Remark 2 of [129]) on the entropy of α-CFs. From the results of §1.7.2 and §1.7.3 below, Theorem 1.5.6 may be viewed as simultaneously extending these results from [115, 120].

1.6 Inducing contractions of the mother of all continued fractions

We have seen in §1.5.2 that inducible subregions $R \subset \Omega$ naturally determine subsequences $(u_n^R/s_n^R)_{n\geq 0} = (u_{N_n^R}/s_{N_n^R})_{n\geq 0}$ of the convergents of Farey expansions. In this section, we construct new GCF-expansions whose convergents are precisely the subsequences $(u_n^R/s_n^R)_{n\geq 0}$ (Corollary 1.6.11). These GCFs arise from a general procedure described in §1.6.1 called *contraction*, which—under very mild assumptions—allows one to produce from a given GCF a new GCF whose convergents are any desired subsequence of the original convergents. In §1.6.2, we use induced transformations of Ito's natural extension of the Farey tent map ('the mother of all continued fractions') to govern contractions of Farey expansions. In §1.6.3 we introduce a dynamical system—isomorphic to the induced system $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$—which acts essentially as a two-sided shift on contracted Farey expansions.

1.6.1 Contraction

Recall the singularisation procedure discussed in §1.3.3: beginning with a SRCF-expansion, one may (simultaneously) singularise at possibly countably many positions to produce a new SRCF-expansion whose convergents are a subsequence of the original convergents. However, singularisation at position n is subject to the condition that the partial numerator α_{n+2} and partial denominator β_{n+1} are both equal to 1. Moreover, beginning from a RCF-expansion, this

constraint on partial denominators implies that consecutive convergents cannot be removed via singularisation.

In this subsection, we recall an old acceleration technique of Seidel from 1855 ([173]; see also [161]), called *contraction*, which overcomes these obstacles. Although our main interest is in producing GCF-expansions whose convergents are subsequences of Farey convergents, we present contraction in the general, abstract setting of GCFs discussed in §1.2.1, as we feel this technique has been largely overlooked[13] and can be fruitfully applied to other continued fraction algorithms. For the same reason, we include a proof of Seidel's theorem (Theorem 1.6.6) below.

Definition 1.6.1. A GCF $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ is called *contractable* if $Q_{[m+1,n]} \neq 0$ for all $0 \leq m \leq n$. The *contracted continued fraction* (CCF) of a contractable GCF $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ with respect to a strictly increasing sequence of non-negative integers $(n_k)_{k\geq 0}$ is the GCF $[\beta'_0/\alpha'_0; \alpha'_1/\beta'_1, \alpha'_2/\beta'_2, \dots]$, where for $k \geq -1$,

$$\begin{pmatrix} \alpha'_{k+1} \\ \beta'_{k+1} \end{pmatrix} := \begin{pmatrix} -\det(B_{[n_{k-1}+2,n_k+1]})Q_{[n_{k-2}+2,n_{k-1}]}Q_{[n_k+2,n_{k+1}]} \\ Q_{[n_{k-1}+2,n_{k+1}]} \end{pmatrix},$$

with $n_k := k$ for $k < 0$.

Remark 1.6.2. The requirement that a contractable GCF satisfies $Q_{[m+1,n]} \neq 0$ for all $0 \leq m \leq n$ guarantees that the partial numerators α'_{k+1} are nonzero for all $k \geq -1$, i.e., that a CCF is in fact a GCF as defined in §1.2.1. Indeed, we have $\det(B_{[n_{k-1}+2,n_k+1]}) \neq 0$ for all $k \geq -1$ as noted in §1.2.1, and both $Q_{[n_{k-2}+2,n_{k-1}]} \neq 0,\ k \geq 1$, and $Q_{[n_k+2,n_{k+1}]} \neq 0,\ k \geq -1$, by assumption. Moreover, for $k = -1$ and $k = 0$, $Q_{[n_{k-2}+2,n_{k-1}]} = 1 \neq 0$; see (1.3).

This requirement also guarantees that the scalars c_k in Theorem 1.6.6 below are non-zero.

[13] As mentioned in the introduction (§1.1), contraction is used in the analytic theory of continued fractions, but usually only for subsequences of odd or even integers ([128]). See also [26].

Remark 1.6.3. Notice that the partial numerators α'_n of a CCF in general *do not* satisfy $|\alpha'_n| = 1$, even if this is true of the original GCF. Hence contraction does not necessarily send SRCFs to SRCFs.

Example 1.6.4. We compute the CCF with respect to $(n_k)_{k\geq 0} = (2k)_{k\geq 0}$ of $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots] = [1/1; 1/2, 2/3, 3/4, \dots]$, i.e., $\alpha_0 = 1$ and $\alpha_n = \beta_{n-1} = n$ for all $n > 0$. We first note that the recurrence relations (1.2) and the fact that all of the partial numerators and partial denominators are positive imply that this GCF is in fact contractable. From Definition 1.6.1,

$$\begin{pmatrix}\alpha'_0\\ \beta'_0\end{pmatrix} = \begin{pmatrix}-\det(B_{[0,0]})Q_{[-1,-2]}Q_{[1,0]}\\ Q_{[0,0]}\end{pmatrix} = \begin{pmatrix}\alpha_0\cdot 1\cdot 1\\ \beta_0\end{pmatrix} = \begin{pmatrix}1\\ 1\end{pmatrix},$$

$$\begin{aligned}\begin{pmatrix}\alpha'_1\\ \beta'_1\end{pmatrix} &= \begin{pmatrix}-\det(B_{[1,1]})Q_{[0,-1]}Q_{[2,2]}\\ Q_{[1,2]}\end{pmatrix}\\ &= \begin{pmatrix}\alpha_1\cdot 1\cdot \beta_2\\ \beta_2\beta_1+\alpha_2\end{pmatrix} = \begin{pmatrix}1\cdot 1\cdot 3\\ 3\cdot 2+2\end{pmatrix} = \begin{pmatrix}3\\ 8\end{pmatrix},\end{aligned}$$

$$\begin{aligned}\begin{pmatrix}\alpha'_2\\ \beta'_2\end{pmatrix} &= \begin{pmatrix}-\det(B_{[2,3]})Q_{[1,0]}Q_{[4,4]}\\ Q_{[2,4]}\end{pmatrix} = \begin{pmatrix}-\alpha_2\alpha_3\cdot 1\cdot \beta_4\\ \beta_4(\beta_3\beta_2+\alpha_3)+\alpha_4\beta_2\end{pmatrix}\\ &= \begin{pmatrix}-2\cdot 3\cdot 1\cdot 5\\ 5(4\cdot 3+3)+4\cdot 3\end{pmatrix} = \begin{pmatrix}-30\\ 87\end{pmatrix},\end{aligned}$$

and for $k > 1$,

$$\begin{aligned}\begin{pmatrix}\alpha'_{k+1}\\ \beta'_{k+1}\end{pmatrix} &= \begin{pmatrix}-\det(B_{[2k,2k+1]})Q_{[2k-2,2k-2]}Q_{[2k+2,2k+2]}\\ Q_{[2k,2k+2]}\end{pmatrix}\\ &= \begin{pmatrix}-\alpha_{2k}\alpha_{2k+1}\cdot \beta_{2k-2}\cdot \beta_{2k+2}\\ \beta_{2k+2}(\beta_{2k+1}\beta_{2k}+\alpha_{2k+1})+\alpha_{2k+2}\beta_{2k}\end{pmatrix}\\ &= \begin{pmatrix}-((2k)(2k+1))\cdot(2k-1)\cdot(2k+3)\\ (2k+3)((2k+2)(2k+1)+(2k+1))+(2k+2)(2k+1)\end{pmatrix}\\ &= \begin{pmatrix}-(2k-1)(2k)(2k+1)(2k+3)\\ (2k+1)((2k+3)^2+(2k+2))\end{pmatrix}.\end{aligned}$$

The first few terms of the CCF are thus

$$[1/1; 3/8, -30/87, -420/275, -1890/623, \\ -5544/1179, -12870/1991, -25740/3107, \dots].$$

Before proving that the convergents of a CCF are a subsequence of the original GCF-convergents, we need the following:

Lemma 1.6.5. *For any* GCF*,*

$$Q_{[m+1,n]} = \frac{Q_n P_{m-1} - P_n Q_{m-1}}{\det B_{[-1,m]}} \quad \textit{for all integers } -1 \le m \le n.$$

Proof. If $m = n$, then both the left- and right-hand sides of the expression equal 1. For $m < n$, $Q_{[m+1,n]}$ is the bottom-right entry of

$$\begin{aligned} B_{[m+1,n]} =& B^{-1}_{[-1,m]} B_{[-1,n]} \\ =& \frac{1}{\det B_{[-1,m]}} \begin{pmatrix} Q_m & -P_m \\ -Q_{m-1} & P_{m-1} \end{pmatrix} \begin{pmatrix} P_{n-1} & P_n \\ Q_{n-1} & Q_n \end{pmatrix}, \end{aligned}$$

which is evidently $\frac{Q_n P_{m-1} - P_n Q_{m-1}}{\det B_{[-1,m]}}$. □

Let $[\beta'_0/\alpha'_0; \alpha'_1/\beta'_1, \alpha'_2/\beta'_2, \dots]$ be the CCF of a contractable GCF $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ with respect to $(n_k)_{k\ge 0}$. For $-1 \le m \le n$, let

$$\begin{pmatrix} P'_{[m,n-1]} & P'_{[m,n]} \\ Q'_{[m,n-1]} & Q'_{[m,n]} \end{pmatrix} = B'_{[m,n]} \coloneqq B_{[m,n]}([\beta'_0/\alpha'_0; \alpha'_1/\beta'_1, \alpha'_2/\beta'_2, \dots]),$$

and when $m = -1$, set $P'_n \coloneqq P'_{[-1,n]}$ and $Q'_n \coloneqq Q'_{[-1,n]}$.

Theorem 1.6.6 (Seidel, 1855 [173])**.** *With notation as above,*

$$\begin{pmatrix} P'_k \\ Q'_k \end{pmatrix} = c_k \begin{pmatrix} P_{n_k} \\ Q_{n_k} \end{pmatrix}, \quad k \ge -2, \qquad \textit{where} \quad c_k = \prod_{j=0}^{k-1} Q_{[n_{j-1}+2, n_j]},$$

with $n_k \coloneqq k$, $k < 0$, *and the product defining* c_k *set equal to* 1 *for* $k < 1$. *In particular, the* CCF *with respect to* $(n_k)_{k\geq 0}$ *of a contractable* GCF*-expansion* $x = [\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots] \in \mathbb{C}$ *with convergents* $(P_k/Q_k)_{k\geq -1}$ *is a* GCF*-expansion of* x *with convergents* $(P_{n_k}/Q_{n_k})_{k\geq -1}$.

Proof. The proof of the first statement is by induction on k. The statement trivially holds for $k < 0$. Now let $k + 1 \geq 0$ and suppose the statement is true for k and $k - 1$. By the recurrence relations (1.4) and Definition 1.6.1—letting U represent either P or Q—we compute

$$\begin{aligned} U'_{k+1} &= \beta'_{k+1}U'_k + \alpha'_{k+1}U'_{k-1} \\ &= Q_{[n_{k-1}+2,n_{k+1}]}c_kU_{n_k} \\ &\quad - \det(B_{[n_{k-1}+2,n_k+1]})Q_{[n_{k-2}+2,n_{k-1}]}Q_{[n_k+2,n_{k+1}]}c_{k-1}U_{n_{k-1}} \\ &= c_k\big(Q_{[n_{k-1}+2,n_{k+1}]}U_{n_k} \\ &\quad - \det(B_{[n_{k-1}+2,n_k+1]})Q_{[n_k+2,n_{k+1}]}U_{n_{k-1}}\big). \end{aligned}$$

By Lemma 1.6.5,

$$Q_{[n_{k-1}+2,n_{k+1}]} = \frac{Q_{n_{k+1}}P_{n_{k-1}} - P_{n_{k+1}}Q_{n_{k-1}}}{\det B_{[-1,n_{k-1}+1]}}$$

and

$$Q_{[n_k+2,n_{k+1}]} = \frac{Q_{n_{k+1}}P_{n_k} - P_{n_{k+1}}Q_{n_k}}{\det B_{[-1,n_k+1]}},$$

so the above computation gives

$$\begin{aligned} U'_{k+1} = c_k\Bigg(&\frac{Q_{n_{k+1}}P_{n_{k-1}} - P_{n_{k+1}}Q_{n_{k-1}}}{\det B_{[-1,n_{k-1}+1]}}U_{n_k} \\ &- \frac{Q_{n_{k+1}}P_{n_k} - P_{n_{k+1}}Q_{n_k}}{\det B_{[-1,n_{k-1}+1]}}U_{n_{k-1}}\Bigg). \end{aligned}$$

For both $U = P$ and $U = Q$, the previous expression simplifies to

$$U'_{k+1} = c_k\frac{Q_{n_k}P_{n_{k-1}} - P_{n_k}Q_{n_{k-1}}}{\det B_{[-1,n_{k-1}+1]}}U_{n_{k+1}}.$$

Using Lemma 1.6.5 once more, we have

$$U'_{k+1} = c_k Q_{[n_{k-1}+2,n_k]} U_{n_{k+1}} = c_{k+1} U_{n_{k+1}}.$$

This proves the first statement. The second statement follows from the first, since for a contractable GCF, $Q_{[n_{j-1}+2,n_j]} \neq 0$ for all $j \geq 0$ implies $c_k \neq 0$ for all $k \geq -1$, and $x = \lim_{k\to\infty} \frac{P_k}{Q_k} = \lim_{k\to\infty} \frac{P_{n_k}}{Q_{n_k}}$. □

Example 1.6.7. Continuing with Example 1.6.4, for

$$[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots] = [1/1; 1/2, 2/3, 3/4, \dots],$$

one computes

$$\left(\begin{pmatrix} P_n \\ Q_n \end{pmatrix}\right)_{n\geq 0} = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 11 \\ 8 \end{pmatrix}, \begin{pmatrix} 53 \\ 38 \end{pmatrix}, \begin{pmatrix} 309 \\ 222 \end{pmatrix}, \begin{pmatrix} 2119 \\ 1522 \end{pmatrix}, \begin{pmatrix} 16687 \\ 11986 \end{pmatrix}, \begin{pmatrix} 148329 \\ 106542 \end{pmatrix}, \begin{pmatrix} 1468457 \\ 1054766 \end{pmatrix}, \begin{pmatrix} 16019531 \\ 11506538 \end{pmatrix}, \dots \right).$$

For the CCF

$$[\beta'_0/\alpha'_0; \alpha'_1/\beta'_1, \alpha'_2/\beta'_2, \dots] = [1/1; 3/8, -30/87, -420/275, -1890/623, -5544/1179, -12870/1991, -25740/3107, \dots]$$

of $[1/1; 1/2, 2/3, 3/4, \dots]$ with respect to $(2k)_{k\geq 0}$, we find

$$\left(\begin{pmatrix} P'_k \\ Q'_k \end{pmatrix}\right)_{k\geq 0} = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 11 \\ 8 \end{pmatrix}, 3 \begin{pmatrix} 309 \\ 222 \end{pmatrix}, 3 \cdot 5 \begin{pmatrix} 16687 \\ 11986 \end{pmatrix}, 3 \cdot 5 \cdot 7 \begin{pmatrix} 1468457 \\ 1054766 \end{pmatrix}, 3 \cdot 5 \cdot 7 \cdot 9 \begin{pmatrix} 190899411 \\ 137119578 \end{pmatrix}, \dots \right).$$

As fractions, $P'_k/Q'_k = P_{2k}/Q_{2k}$ for $k \geq 0$.

1.6.2 Contracted Farey expansions

Throughout this subsection, $R \subset \Omega$ is an inducible subregion and $z = (x, y) \in \Omega$ with x irrational. Using notation from §1.2.1 and §1.4.2, let $[\beta_0/\alpha_0; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots]$ denote the Farey expansion of x, i.e., $\alpha_0 = 1$, $\beta_0 = 1 - \varepsilon_1$, and for $n > 0$, $\alpha_n = 2\varepsilon_n - 1$ and $\beta_n = 2 - \varepsilon_{n+1}$; see (1.28). Below, we shall perform contraction on Farey expansions, but first we must show that Farey expansions are in fact contractable.

Proposition 1.6.8. *The Farey expansion of an irrational $x \in (0, 1)$ is contractable.*

Proof. We must show that $Q_{[m+1,n]} \neq 0$ for any $0 \leq m \leq n$. By Lemma 1.6.5, this is equivalent to $Q_n P_{m-1} \neq P_n Q_{m-1}$ for all $0 \leq m \leq n$. By Proposition 1.4.1, the Farey convergents $(P_j/Q_j)_{j \geq -1} = (u_j/s_j)_{j \geq 0}$ are all RCF-convergents and mediants, which are distinct by (1.10) and (1.11). □

Recall the definition of $N_n^R = N_n^R(z)$ from (1.41).

Definition 1.6.9. The *contracted Farey expansion* of x with respect to R and $z = (x, y)$, denoted

$$[\beta_0^R/\alpha_0^R; \alpha_1^R/\beta_1^R, \alpha_2^R/\beta_2^R, \dots] \\ = [\beta_0^R(z)/\alpha_0^R(z); \alpha_1^R(z)/\beta_1^R(z), \alpha_2^R(z)/\beta_2^R(z), \dots],$$

is the CCF of the Farey expansion of x with respect to $(n_k)_{k \geq 0}$, where $n_k := N_{k+1}^R - 1$, $k \geq 0$. If $z = (x, 1)$, we call this the *contracted Farey expansion* of x with respect to R.

In order to study contracted Farey expansions using the dynamics of $\mathcal{F}_R$, we wish to understand these expansions and their convergents in terms of entries from the matrices $A_R(z)$ and $A_{[m,n]}^R(z)$ from (1.40) and (1.44) rather than $B_{[m,n]}$. To this end, we begin with a lemma. Let $z_n^R = \mathcal{F}_R^n(z)$ and recall that $s_R(z)$ and $s_n^R(z)$ denote the bottom-left entries of the matrices $A_R(z)$ and $A_{[0,n]}^R(z)$, respectively; see (1.40) and (1.45).

Lemma 1.6.10. *For any $z \in \Omega$ and $0 \leq j < k$, one has*

$$Q_{[N_j^R+1, N_k^R-1]} = s_{k-j}^R(z_j^R).$$

In particular, if $k = j+1$, then $Q_{[N_j^R+1, N_{j+1}^R-1]} = s_R(z_j^R)$.

Proof. First, notice that for any $n > 0$,

$$\begin{aligned} \det A_{[0,n]} &= \det\left(A_{\varepsilon_1} \cdots A_{\varepsilon_n}\right) = \prod_{j=1}^{n}(1-2\varepsilon_j) \\ &= \det(B_{-1}B_0 \cdots B_n) = \det B_{[-1,n]}, \end{aligned} \tag{1.50}$$

and equality of the left- and right-hand sides also holds for $n = 0$ since both sides equal 1. Then by Lemma 1.6.5, Proposition 1.4.1, and Equations (1.46) and (1.47),

$$\begin{aligned} Q_{[N_j^R+1, N_k^R-1]} &= \frac{Q_{N_k^R-1}P_{N_j^R-1} - P_{N_k^R-1}Q_{N_j^R-1}}{\det B_{[-1,N_j^R]}} \\ &= \frac{s_{N_k^R}u_{N_j^R} - u_{N_k^R}s_{N_j^R}}{\det A_{[0,N_j^R]}} \\ &= \frac{s_k^R u_j^R - u_k^R s_j^R}{\det A_{[0,j]}^R}, \end{aligned}$$

where $A_{[0,j]}^R = A_{[0,j]}^R(z)$. For the first statement, it suffices to show that the right-hand side of the previous line equals the bottom-left entry of $A_{[0,k-j]}^R(z_j^R)$. From (1.43), (1.44) and (1.46), we have

$$\begin{aligned} A_{[0,k-j]}^R(z_j^R) &= A_1^R(z_j^R)A_2^R(z_j^R)\cdots A_{k-j}^R(z_j^R) \\ &= A_{j+1}^R(z)A_{j+2}^R(z)\cdots A_k^R(z) \\ &= \left(A_{[0,j]}^R\right)^{-1} A_{[0,k]}^R \\ &= \frac{1}{\det A_{[0,j]}^R}\begin{pmatrix} r_j^R & -t_j^R \\ -s_j^R & u_j^R \end{pmatrix}\begin{pmatrix} u_k^R & t_k^R \\ s_k^R & r_k^R \end{pmatrix}, \end{aligned}$$

and the first statement follows. The second statement follows immediately from the first and the fact that $A^R_{[0,1]}(z^R_j) = A^R_1(z^R_j) = A_R(z^R_j)$; see (1.43) and (1.44). □

Now let $[\beta^R_0/\alpha^R_0; \alpha^R_1/\beta^R_1, \alpha^R_2/\beta^R_2, \dots]$ be the contracted Farey expansion of x with respect to R and z, and for $-1 \le m \le n$, let

$$\begin{aligned}\begin{pmatrix} P^R_{[m,n-1]} & P^R_{[m,n]} \\ Q^R_{[m,n-1]} & Q^R_{[m,n]} \end{pmatrix} &= B^R_{[m,n]} \\ &:= B_{[m,n]}([\beta^R_0/\alpha^R_0; \alpha^R_1/\beta^R_1, \alpha^R_2/\beta^R_2, \dots]),\end{aligned} \tag{1.51}$$

and

$$P^R_n := P^R_{[-1,n]}, \quad Q^R_n := Q^R_{[-1,n]}. \tag{1.52}$$

Then Theorem 1.6.6, Proposition 1.4.1 and Lemma 1.6.10 imply:

Corollary 1.6.11. *With notation as above,*

$$\begin{pmatrix} P^R_k \\ Q^R_k \end{pmatrix} = c^R_k \begin{pmatrix} u^R_{k+1} \\ s^R_{k+1} \end{pmatrix}, \quad k \ge -1, \qquad \text{where} \quad c^R_k = \prod_{j=0}^{k-1} s_R(z^R_j),$$

with $c^R_k = 1$ for $k < 1$. In particular, the contracted Farey expansion of x with respect to R and $z = (x, y)$ has convergents $(u^R_k/s^R_k)_{k\ge 0}$.

Corollary 1.6.11 describes the convergents of a contracted Farey expansion in terms of the entries of $A^R_{[0,n]}$ (see (1.46)). Proposition 1.6.13 below gives an alternative description of the partial numerators and partial denominators α^R_k, β^R_k in terms of entries of the matrices $A_R(z)$ (see (1.40)). For this, we introduce three integer-valued maps on Ω: let $d_R, \alpha_R, \beta_R : \Omega \to \mathbb{Z}$ be defined for $z \in \Omega$ by

$$d_R(z) := \begin{cases} s_R(\mathcal{F}_R^{-1}(z)), & \text{if } \mathcal{F}_R^{-1}(z) \text{ is defined,} \\ 1, & \text{otherwise,} \end{cases}$$

$$\alpha_R(z) := -\det(A_R(z)) d_R(z) s_R(\mathcal{F}_R(z)) \tag{1.53}$$

and

$$\beta_R(z) := s_R(z) u_R(\mathcal{F}_R(z)) + r_R(z) s_R(\mathcal{F}_R(z)). \tag{1.54}$$

Remark 1.6.12. Notice that Lemma 1.5.2 implies $d_R(z)$ is a positive integer for any z. We claim that $d_R(z) = 1$ whenever $z = (x, 1)$. By definition of $\mathcal{F}$ (Equation (1.30)), $\mathcal{F}^{-1}(z) \in (1/2, 1] \times \{0\}$, and $\mathcal{F}^{-n}(z) \in [0, 1/2] \times \{0\}$ for $n > 1$. If $\mathcal{F}_R^{-1}(z)$ is not defined (e.g., if R does not intersect the line $[0, 1] \times \{0\}$), then $d_R(z) = 1$ by definition. Otherwise, $\mathcal{F}_R^{-1}(z) = \mathcal{F}^{-n}(z)$ for some $n \geq 1$, and

$$A_R(\mathcal{F}_R^{-1}(z)) = A_0^{n-1} A_1 = \begin{pmatrix} 0 & 1 \\ 1 & n \end{pmatrix}$$

implies $d_R(z) = s_R(\mathcal{F}_R^{-1}(z)) = 1$. In either case, $d_R(z) = 1$ for $z = (x, 1)$ as claimed.

One motivation for defining $d_R(z)$ as above lies in the second statement of the following proposition: when $d_R(z) = 1$, the notation simplifies and we need not consider the index 1 partial numerator α_1^R as a separate case.

Proposition 1.6.13. *The digits of the contracted Farey expansion of x with respect to R and $z = (x, y)$ are given by*

$$\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} s_R(z_0^R) \\ u_R(z_0^R) \end{pmatrix}, \qquad \begin{pmatrix} \alpha_1^R \\ \beta_1^R \end{pmatrix} = \begin{pmatrix} \alpha_R(z_0^R)/d_R(z_0^R) \\ \beta_R(z_0^R) \end{pmatrix}$$

and

$$\begin{pmatrix} \alpha_{k+1}^R \\ \beta_{k+1}^R \end{pmatrix} = \begin{pmatrix} \alpha_R(z_k^R) \\ \beta_R(z_k^R) \end{pmatrix}, \quad k > 0.$$

When $d_R(z) = 1$ (e.g., when $z = (x, 1)$ by Remark 1.6.12), this becomes

$$\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} s_R(z_0^R) \\ u_R(z_0^R) \end{pmatrix} \qquad \textit{and} \qquad \begin{pmatrix} \alpha_{k+1}^R \\ \beta_{k+1}^R \end{pmatrix} = \begin{pmatrix} \alpha_R(z_k^R) \\ \beta_R(z_k^R) \end{pmatrix}, \quad k \geq 0.$$

Proof. From Definitions 1.6.1 and 1.6.9, we have for $k \geq -1$,

$$\begin{pmatrix} \alpha_{k+1}^R \\ \beta_{k+1}^R \end{pmatrix} = \begin{pmatrix} -\det(B_{[N_k^R+1, N_{k+1}^R]}) Q_{[N_{k-1}^R+1, N_k^R-1]} Q_{[N_{k+1}^R+1, N_{k+2}^R-1]} \\ Q_{[N_k^R+1, N_{k+2}^R-1]} \end{pmatrix}, \tag{1.55}$$

where $N_k^R \coloneqq k$ for $k < 0$. For $k = -1$,

$$\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} -\det(B_{[0,0]})Q_{[-1,-2]}Q_{[N_0^R+1,N_1^R-1]} \\ Q_{[0,N_1^R-1]} \end{pmatrix} = \begin{pmatrix} Q_{[N_0^R+1,N_1^R-1]} \\ Q_{[0,N_1^R-1]} \end{pmatrix}.$$

Lemma 1.6.10 gives $Q_{[N_0^R+1,N_1^R-1]} = s_R(z_0^R)$, and Lemma 1.6.5 and Proposition 1.4.1 give

$$Q_{[0,N_1^R-1]} = \frac{Q_{N_1^R-1}P_{-2} - P_{N_1^R-1}Q_{-2}}{\det B_{[-1,-1]}} = P_{N_1^R-1} = u_1^R = u_R(z_0^R).$$

Thus the claim holds for α_0^R, β_0^R. Next, notice from Equation (1.50) that for any $0 \le m < n$,

$$\det B_{[m+1,n]} = \frac{\det B_{[-1,n]}}{\det B_{[-1,m]}} = \frac{\det A_{[0,n]}}{\det A_{[0,m]}} = \det(A_{\varepsilon_{m+1}} \cdots A_{\varepsilon_n}),$$

so by (1.42), we have for $k \ge 0$ that

$$\det(B_{[N_k^R+1,N_{k+1}^R]}) = \det(A_{\varepsilon_{N_k^R+1}} \cdots A_{\varepsilon_{N_{k+1}^R}}) = \det(A_R(z_k^R)).$$

This, Equation (1.55) and Lemma 1.6.10 give for $k \ge 0$

$$\begin{aligned}\begin{pmatrix} \alpha_{k+1}^R \\ \beta_{k+1}^R \end{pmatrix} &= \begin{pmatrix} -\det(B_{[N_k^R+1,N_{k+1}^R]})Q_{[N_{k-1}^R+1,N_k^R-1]}Q_{[N_{k+1}^R+1,N_{k+2}^R-1]} \\ Q_{[N_k^R+1,N_{k+2}^R-1]} \end{pmatrix} \\ &= \begin{cases} \begin{pmatrix} -\det(A_R(z_0^R))s_R(z_1^R) \\ s_2^R(z_0^R) \end{pmatrix}, & k = 0, \\ \begin{pmatrix} -\det(A_R(z_k^R))s_R(z_{k-1}^R)s_R(z_{k+1}^R) \\ s_2^R(z_k^R) \end{pmatrix}, & k > 0. \end{cases}\end{aligned}$$

When $k = 0$, this gives $\alpha_1^R = \alpha_R(z_0^R)/d_R(z_0^R)$. When $k > 0$, $s_R(z_{k-1}^R) = s_R(\mathcal{F}_R^{-1}(z_k^R)) = d_R(z_k)$, so $\alpha_{k+1}^R = \alpha_R(z_k^R)$. Moreover, for $k \ge 0$, $s_2^R(z_k^R)$ is the bottom-left entry of

$$\begin{aligned}A_{[0,2]}^R(z_k^R) =& A_1^R(z_k^R)A_2^R(z_k^R) = A_R(z_k^R)A_R(z_{k+1}^R) \\ &= \begin{pmatrix} u_R(z_k^R) & t_R(z_k^R) \\ s_R(z_k^R) & r_R(z_k^R) \end{pmatrix} \begin{pmatrix} u_R(z_{k+1}^R) & t_R(z_{k+1}^R) \\ s_R(z_{k+1}^R) & r_R(z_{k+1}^R) \end{pmatrix};\end{aligned}$$

see (1.45), (1.44), (1.43) and (1.40). Thus

$$s_2^R(z_k^R) = s_R(z_k^R)u_R(z_{k+1}^R) + r_R(z_k^R)s_R(z_{k+1}^R) = \beta_R(z_k^R).$$

This proves the first statement. The latter statement follows immediately from the first. □

We refer the reader to §1.7 and §1.9 below for examples using Proposition 1.6.13.

1.6.3 A two-sided shift for contracted Farey expansions

In this subsection, we associate to the induced system $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ an isomorphic dynamical system $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ acting essentially as a two-sided shift for contracted Farey expansions. This new system will serve several purposes in §1.7 below: we will see that $(\Omega_{H_1}, \mathcal{B}, \bar{\nu}_{H_1}, \tau_{H_1}) = (\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$ is the natural extension of the Gauss map; for certain subregions $R \subset H_1$, $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ will coincide with a two-sided shift system associated to S-expansions in [115]; and in §1.7.3, we describe the natural extension of each of Nakada's α-CFs, $0 < \alpha \leq 1$, as an induced system $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ by using the isomorphic system $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$.

To ease exposition, we impose some restrictions on our inducible subregion $R \subset \Omega$ throughout this subsection. First, we assume that R is bounded away from the origin and that for any $z = (x, y) \in R$, $y > 0$. Furthermore, we assume that $s_R(z) = 1$ for all $z \in R$, and hence—by Lemma 1.5.2—that $u_R(z) \in \{0, 1\}$. The regions R considered in §1.7 below shall satisfy these assumptions.

Remark 1.6.14. A two-sided shift space may be constructed without the restriction $s_R(z) = 1$, but in general the domain Ω_R consists of several planar 'sheets,' and the invariant measure $\bar{\nu}_R$ is a sum of measures which—restricted to each of these sheets—has density of the form in Theorem 1.6.15 below. However, this more general system is not needed for our purposes.

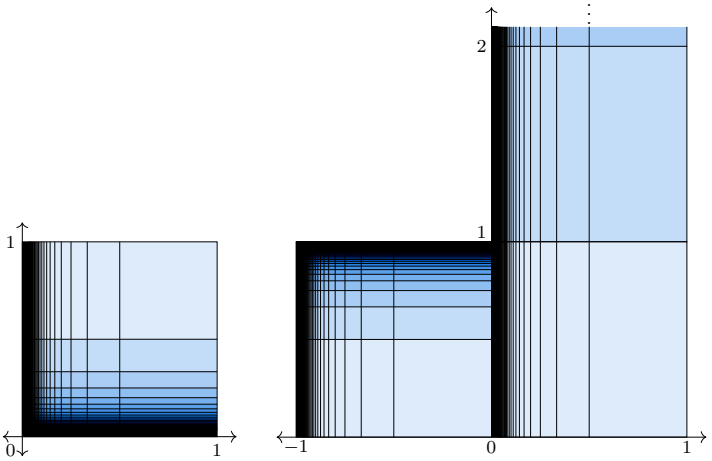


Figure 1.4: Left: The domain $\Omega = [0,1]^2$. Right: The first quadrant shows part of the image of $\Omega \backslash ([0,1] \times \{0\})$ under the map $(x,y) \mapsto (x,(1-y)/y)$; the second quadrant shows the image of Ω under the map $(x,y) \mapsto (x-1, 1-y)$.

Define $\varphi_R : R \to \mathbb{R}^2$, where for $z = (x,y) \in R$,

$$\begin{aligned} \varphi_R(z) =& (X(z), Y(z)) \\ :=& \left(\begin{pmatrix} 1 & -u_R(z) \\ 0 & 1 \end{pmatrix} \cdot x, \begin{pmatrix} -1 & 1 \\ 1-u_R(z) & u_R(z) \end{pmatrix} \cdot y \right) \\ =& \begin{cases} \left(x, \frac{1-y}{y} \right), & u_R(z) = 0, \\ (x-1, 1-y), & u_R(z) = 1. \end{cases} \end{aligned} \tag{1.56}$$

The map φ_R is injective, except possibly on the null-set of points $\{(x,y) \in R \mid x \in \{0,1\}\}$; see Figure 1.4. Setting $\Omega_R := \varphi_R(R)$, its inverse (off of the image of the aforementioned null-set) is the map

$\varphi_R^{-1} : \Omega_R \setminus (\{0\} \times [0,\infty)) \to R$, where

$$\varphi_R^{-1}(X,Y) = \begin{cases} \left(X, \frac{1}{Y+1}\right), & X > 0, \\ (X+1, 1-Y), & X < 0. \end{cases}$$

If $z = \varphi_R^{-1}(X,Y)$, this may also be written

$$\begin{aligned} \varphi_R^{-1}(X,Y) &= \left(\begin{pmatrix} 1 & u_R(z) \\ 0 & 1 \end{pmatrix} \cdot X, \begin{pmatrix} u_R(z) & -1 \\ u_R(z)-1 & -1 \end{pmatrix} \cdot Y \right) \\ &= \left(X + u_R(z), \frac{u_R(z)Y - 1}{(u_R(z)-1)Y - 1} \right). \end{aligned} \tag{1.57}$$

Define $\tau_R : \Omega_R \to \Omega_R$ by

$$\tau_R(X,Y) \coloneqq \begin{cases} (X,Y), & X = 0, \\ \varphi_R \circ \mathcal{F}_R \circ \varphi_R^{-1}, & X \neq 0. \end{cases}$$

We obtain a dynamical system $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$, where $\bar{\nu}_R \coloneqq \bar{\mu}_R \circ \varphi_R^{-1}$ denotes the pushforward measure of $\bar{\mu}_R$ under φ_R. By construction, $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ and $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ are isomorphic. Recall the definitions of α_R and β_R from (1.53) and (1.54).

Theorem 1.6.15. *The map $\tau_R : \Omega_R \to \Omega_R$ is given by $\tau_R(0,Y) = (0,Y)$ and for $X \neq 0$,*

$$\tau_R(X,Y) = \left(\frac{\alpha_R(z)}{X} - \beta_R(z), \frac{1}{\beta_R(z) + \alpha_R(z)Y} \right),$$

and the measure $\bar{\nu}_R$ has density

$$\frac{1}{\bar{\mu}(R)(1+XY)^2}.$$

Remark 1.6.16. We remark here the resemblance between the measures and maps from $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ and the natural extension of the Gauss map, $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$, from §1.3.1. We shall return to this point in §1.7.1 below.

Proof of Theorem 1.6.15. We begin with the statement about the map τ_R. By definition, $\tau_R(0,Y) = (0,Y)$, so let $(X,Y) \in \Omega_R$ with $X \neq 0$. Set $(X',Y') \coloneqq \tau_R(X,Y)$, $z = (x,y) \coloneqq \varphi_R^{-1}(X,Y)$ and $z' = (x',y') \coloneqq \mathcal{F}_R(z)$, and note that $(X',Y') = \varphi_R(z')$. Set $u = u_R(z)$ and $u' = u_R(z')$. Using Equations (1.56), (1.39), (1.57), and symmetry of A_1, respectively, (X',Y') equals

$$
\begin{aligned}
&\left(\begin{pmatrix} 1 & -u' \\ 0 & 1 \end{pmatrix} \cdot x', \begin{pmatrix} -1 & 1 \\ 1-u' & u' \end{pmatrix} \cdot y' \right) \\
&= \left(\begin{pmatrix} 1 & -u' \\ 0 & 1 \end{pmatrix} A_R^{-1}(z) \cdot x, \begin{pmatrix} -1 & 1 \\ 1-u' & u' \end{pmatrix} A_1 A_R^T(z) A_1^{-1} \cdot y \right) \\
&= \left(\begin{pmatrix} 1 & -u' \\ 0 & 1 \end{pmatrix} A_R^{-1}(z) \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \cdot X, \right. \\
&\qquad \left. \begin{pmatrix} -1 & 1 \\ 1-u' & u' \end{pmatrix} A_1 A_R^T(z) A_1^{-1} \begin{pmatrix} u & -1 \\ u-1 & -1 \end{pmatrix} \cdot Y \right) \\
&= \left(\begin{pmatrix} 1 & -u' \\ 0 & 1 \end{pmatrix} A_R^{-1}(z) \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \cdot X, \right. \\
&\qquad \left. \left(\begin{pmatrix} -1 & 1 \\ 1-u' & u' \end{pmatrix}^{-T} A_1^{-1} A_R^{-1}(z) A_1 \begin{pmatrix} u & -1 \\ u-1 & -1 \end{pmatrix}^{-T} \right)^{-T} \cdot Y \right).
\end{aligned}
$$

One easily computes

$$
\begin{pmatrix} -1 & 1 \\ 1-u' & u' \end{pmatrix}^{-T} A_1^{-1} = \begin{pmatrix} 1 & -u' \\ 0 & 1 \end{pmatrix}
$$

and

$$
A_1 \begin{pmatrix} u & -1 \\ u-1 & -1 \end{pmatrix}^{-T} = - \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix},
$$

so $(X',Y') = (M \cdot X, M^{-T} \cdot Y)$, where (recall $u = u_R(z)$, $u' = u_R(z')$

and $s_R(z) = s_R(z') = 1$)

$$\begin{aligned}
M =& \begin{pmatrix} 1 & -u' \\ 0 & 1 \end{pmatrix} A_R^{-1}(z) \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \\
=& \frac{1}{\det A_R(z)} \begin{pmatrix} 1 & -u' \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r_R(z) & -t_R(z) \\ -1 & u \end{pmatrix} \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \\
=& \frac{1}{\det A_R(z)} \begin{pmatrix} r_R(z) + u' & r_R(z)u - t_R(z) \\ -1 & 0 \end{pmatrix} \\
=& \frac{-1}{\det A_R(z)} \begin{pmatrix} -\left(r_R(z)s_R(z') + s_R(z)u_R(z')\right) & -\det(A_R(z)) \\ 1 & 0 \end{pmatrix} \\
=& \frac{-1}{\det A_R(z)} \begin{pmatrix} -\beta_R(z) & \alpha_R(z) \\ 1 & 0 \end{pmatrix}.
\end{aligned}$$

Thus

$$\begin{aligned}
\tau_R(X, Y) =& (X', Y') = (M \cdot X, M^{-T} \cdot Y) \\
=& \left(\begin{pmatrix} -\beta_R(z) & \alpha_R(z) \\ 1 & 0 \end{pmatrix} \cdot X, \begin{pmatrix} 0 & 1 \\ \alpha_R(z) & \beta_R(z) \end{pmatrix} \cdot Y \right),
\end{aligned}$$

proving the claim about τ_R.

Next we prove the statement about the density of $\bar{\nu}_R$. Let S be a measurable subset of Ω_R. Using a change of variables,

$$\begin{aligned}
\bar{\nu}_R(S) =& \bar{\mu}_R \circ \varphi_R^{-1}(S) = \int_{\varphi_R^{-1}(S)} d\bar{\mu}_R = \frac{1}{\bar{\mu}(R)} \iint_{\varphi_R^{-1}(S)} \rho(x, y) dx dy \\
=& \frac{1}{\bar{\mu}(R)} \iint_S \rho(\varphi_R^{-1}(X, Y)) |\det J| dX dY,
\end{aligned}$$

where

$$\rho(x, y) \coloneqq \frac{1}{(x + y - xy)^2}$$

is the density of $\bar{\mu}$ and J is the Jacobian of φ_R^{-1} at $(X, Y) \in S$. Let $u = u_R(z) \in \{0, 1\}$, where $z = \varphi_R^{-1}(X, Y)$. By Equation (1.57), the

Jacobian of φ_R^{-1} at (X, Y) is

$$J = \begin{pmatrix} 1 & 0 \\ 0 & \frac{u((u-1)Y-1)-(u-1)(uY-1)}{((u-1)Y-1)^2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{-1}{((u-1)Y-1)^2} \end{pmatrix}.$$

Moreover, $\rho(\varphi_R^{-1}(X, Y))$ equals

$$\begin{aligned} &\left((X+u) + \frac{uY-1}{(u-1)Y-1} - (X+u)\frac{uY-1}{(u-1)Y-1}\right)^{-2} \\ &= \left(\frac{(X+u)((u-1)Y-1) + (uY-1) - (X+u)(uY-1)}{((u-1)Y-1)}\right)^{-2} \\ &= \left(\frac{1+XY}{(u-1)Y-1}\right)^{-2} \end{aligned}$$

so that

$$\begin{aligned} \rho(\varphi_R^{-1}(X, Y))|\det J| &= \left(\frac{(u-1)Y-1}{1+XY}\right)^2 \frac{1}{((u-1)Y-1)^2} \\ &= \frac{1}{(1+XY)^2}. \end{aligned}$$

□

For given $(X, Y) = (X(z), Y(z)) \in \Omega_R$, set

$$(X_n^R, Y_n^R) = (X_n^R(z), Y_n^R(z)) \coloneqq \tau_R^n(X, Y), \quad n \geq 0. \tag{1.58}$$

Then, for $X_n^R \neq 0$,

$$z_n^R = \mathcal{F}_R^n(z) = \mathcal{F}_R^n \circ \varphi_R^{-1}(X, Y) = \varphi_R^{-1} \circ \tau_R^n(X, Y) = \varphi_R^{-1}(X_n^R, Y_n^R). \tag{1.59}$$

The next result states that the map τ_R acts essentially as a two-sided shift operator on contracted Farey expansions.

Proposition 1.6.17. *Let* $[\beta_0^R/\alpha_0^R; \alpha_1^R/\beta_1^R, \alpha_2^R/\beta_2^R, \dots]$ *denote the contracted Farey expansion of* $x \in (0,1)\backslash\mathbb{Q}$ *with respect to* R *and* $z = (x, y) \in R$. *Then for* $n \geq 0$,

$$\begin{aligned}(X_n^R, Y_n^R) = \big([0/1; \alpha_{n+1}^R/\beta_{n+1}^R, \alpha_{n+2}^R/\beta_{n+2}^R, \dots], \\ [0/1; 1/\beta_n^R, \alpha_n^R/\beta_{n-1}^R, \dots, \alpha_1^R/\beta_0^R, \alpha_0^R/(1/y-1)]\big).\end{aligned}$$

Proof. For each $n \geq 0$, set

$$\begin{aligned}(T_n, V_n) \coloneqq ([0/1; \alpha_{n+1}^R/\beta_{n+1}^R, \alpha_{n+2}^R/\beta_{n+2}^R, \dots], \\ [0/1; 1/\beta_n^R, \alpha_n^R/\beta_{n-1}^R, \dots, \alpha_1^R/\beta_0^R, \alpha_0^R/(1/y-1)]).\end{aligned}$$

Using (1.5) and (1.6), one finds that for each $n \geq 0$,

$$\begin{aligned}(T_{n+1}, V_{n+1}) &= \left(\begin{pmatrix} 0 & \alpha_{n+1}^R \\ 1 & \beta_{n+1}^R \end{pmatrix}^{-1} \cdot T_n, \begin{pmatrix} 0 & 1 \\ \alpha_{n+1}^R & \beta_{n+1}^R \end{pmatrix} \cdot V_n\right) \\ &= \left(\frac{\alpha_{n+1}^R}{T_n} - \beta_{n+1}^R, \frac{1}{\beta_{n+1}^R + \alpha_{n+1}^R V_n}\right).\end{aligned} \tag{1.60}$$

We will show by induction that $(X_n^R, Y_n^R) = (T_n, V_n)$ for all $n \geq 0$. By (1.56), Proposition 1.6.13 and the fact that $s_R(z) = 1$ for all z,

$$\begin{aligned}X_0^R &= \begin{pmatrix} 1 & -u_R(z) \\ 0 & 1 \end{pmatrix} \cdot x = \begin{pmatrix} \alpha_0^R & -\beta_0^R \\ 0 & 1 \end{pmatrix} \cdot x \\ &= \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \alpha_0^R \\ 1 & \beta_0^R \end{pmatrix}\right)^{-1} \cdot x.\end{aligned}$$

Setting $n = 0$ in (1.5) and multiplying both sides by $B_{[-1,0]}^{-1}$ reveals

that $X_0^R = T_0$. Similarly,

$$\begin{aligned} Y_0^R &= \begin{pmatrix} -1 & 1 \\ 1-u_R(z) & u_R(z) \end{pmatrix} \cdot y = \begin{pmatrix} -1 & 1 \\ \alpha_0^R - \beta_0^R & \beta_0^R \end{pmatrix} \cdot y \\ &= \begin{pmatrix} 0 & 1 \\ \alpha_0^R & \beta_0^R \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \cdot y \\ &= \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \alpha_0^R \\ 1 & \beta_0^R \end{pmatrix} \right)^T \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \cdot y. \end{aligned}$$

Since $\left(\begin{smallmatrix} -1 & 1 \\ 1 & 0 \end{smallmatrix}\right) \cdot y = 1/y - 1$, Equation (1.7) gives $Y_0^R = V_0$. Now suppose that $(X_n^R, Y_n^R) = (T_n, V_n)$ for some $n \geq 0$. By Theorem 1.6.15, Proposition 1.6.13, our inductive hypothesis and (1.60),

$$\begin{aligned} (X_{n+1}^R, Y_{n+1}^R) &= \left(\frac{\alpha_R(z_n^R)}{X_n^R} - \beta_R(z_n^R), \frac{1}{\beta_R(z_n^R) + \alpha_R(z_n^R) Y_n^R} \right) \\ &= \left(\frac{\alpha_{n+1}^R}{T_n} - \beta_{n+1}^R, \frac{1}{\beta_{n+1}^R + \alpha_{n+1}^R V_n} \right) \\ &= (T_{n+1}, V_{n+1}). \end{aligned}$$

□

For the remainder of this subsection, we restrict our attention to the full-measure subset of points $z \in R$ for which $z_n^R = (x_n^R, y_n^R) := \mathcal{F}_R^n(z) \in R$ is defined and $x_n^R \neq 0$ for all $n \in \mathbb{Z}$ (in particular, $x \notin \mathbb{Q}$). We remark that as $\mathcal{F}_R$ is totally invariant on this subset, the induced system $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ and its restriction to this full-measure subset are isomorphic. The same is true of the system $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ and its restriction to the image under φ_R of our full-measure, totally $\mathcal{F}_R$-invariant subset of R. Abusing notation, we denote these restricted, isomorphic systems again by $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ and $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$.

Now, let

$$\begin{aligned} &\Delta(0/1; \alpha_1/\beta_1, \alpha_2/\beta_2, \ldots, \alpha_n/\beta_n) \\ &\qquad \times \Delta(0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \alpha_{-1}/\beta_{-2}, \ldots, \alpha_{-(m-1)}/\beta_{-m}) \end{aligned}$$

be the (possibly empty) set of points $(X(z), Y(z)) \in \Omega_R$ satisfying

$$\alpha_R(z_j^R) = \alpha_{j+1} \quad \text{and} \quad \beta_R(z_k^R) = \beta_{k+1}$$

for all $-m \le j \le n-1$ and $-m-1 \le k \le n-1$. The following result is needed in §1.7.3 below when we realise the natural extensions of the α-CFs as induced systems $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$.

Proposition 1.6.18. *The Borel σ-algebra $\mathcal{B}$ restricted to Ω_R is generated by the sets*

$$\Delta(0/1; \alpha_1/\beta_1, \alpha_2/\beta_2, \ldots, \alpha_n/\beta_n) \\ \times \Delta(0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \alpha_{-1}/\beta_{-2}, \ldots, \alpha_{-(m-1)}/\beta_{-m}).$$

Proof. We first remark that each of these sets belongs to $\mathcal{B}$ since each may be written as an intersection of preimages of integers under compositions of the measurable functions α_R, β_R, $\mathcal{F}_R^{\pm 1}$ and φ_R^{-1}. Next, notice that there are only countably many such sets, so it suffices to show that any open set $U \in \Omega_R$ can be written as *some* union of these. It thus suffices to show that for any $(X, Y) = (X(z), Y(z)) \in U$, there exists some set

$$\begin{aligned} D_n = &\Delta(0/1; \alpha_1/\beta_1, \alpha_2/\beta_2, \ldots, \alpha_n/\beta_n) \\ &\times \Delta(0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \alpha_{-1}/\beta_{-2}, \ldots, \alpha_{-(n-2)}/\beta_{-(n-1)}) \end{aligned} \tag{1.61}$$

such that $(X, Y) \in D_n \subset U$. By definition, (X, Y) belongs to each D_n, $n \ge 1$, for which

$$\alpha_{j+1} = \alpha_R(z_j^R) \quad \text{and} \quad \beta_{k+1} = \beta_R(z_k^R) \tag{1.62}$$

for all $-(n-1) \le j \le n-1$ and $-n \le k \le n-1$. Thus, to prove that there is some n for which $D_n \subset U$, it suffices to show that the Euclidean diameters of the sets D_n tend to 0 uniformly in n. For this, it suffices to show that

$$|X - c_n| \to 0 \qquad \text{and} \qquad |Y - d_n| \to 0$$

uniformly in n, where—recycling notation—(X, Y) is an arbitrary point in D_n and

$$(c_n, d_n) \coloneqq ([0/1; \alpha_1/\beta_1, \alpha_2/\beta_2, \dots, \alpha_n/\beta_n], \\ [0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \alpha_{-1}/\beta_{-2}, \dots, \alpha_{-(n-2)}/\beta_{-(n-1)}]).$$

Fix D_n as in (1.61) and assume $(X, Y) = (X(z), Y(z)) \in D_n$ so that (1.62) holds. Proposition 1.6.13 and the fact that $s_R(z) = 1$ (and hence $d_R(z) = 1$) imply that the digits of the contracted Farey expansion of x with respect to R and $z = (x, y)$ are given by

$$\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} 1 \\ u_R(z) \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} \alpha_{j+1}^R \\ \beta_{j+1}^R \end{pmatrix} = \begin{pmatrix} \alpha_R(z_j^R) \\ \beta_R(z_j^R) \end{pmatrix}, \quad j \geq 0.$$

The previous line and (1.56) give $X = x - \beta_0^R$. Letting P_n^R, Q_n^R be as in (1.52), the previous line and (1.62) give

$$\frac{P_n^R}{Q_n^R} = [\beta_0^R/\alpha_0^R; \alpha_1^R/\beta_1^R, \dots, \alpha_n^R/\beta_n^R] = [\beta_0^R/1; \alpha_1/\beta_1, \dots, \alpha_n/\beta_n],$$

so also $c_n = \frac{P_n^R}{Q_n^R} - \beta_0^R$. By Corollary 1.6.11 and (1.47),

$$\frac{P_n^R}{Q_n^R} = \frac{u_{n+1}^R}{s_{n+1}^R} = \frac{\lambda_N p_{j_N} + p_{j_N - 1}}{\lambda_N q_{j_N} + q_{j_N - 1}},$$

where $N = N_{n+1}^R(z)$. By (1.10) and (1.11),

$$\begin{aligned} |X - c_n| &= \left| x - \frac{P_n^R}{Q_n^R} \right| = \left| x - \frac{\lambda_N p_{j_N} + p_{j_N - 1}}{\lambda_N q_{j_N} + q_{j_N - 1}} \right| \\ &\leq \left| x - \frac{p_{j_N - 1}}{q_{j_N - 1}} \right| \leq \frac{1}{q_{j_N - 1}^2}, \end{aligned}$$

where the final inequality follows classical arguments in the theory of RCFs. Since R is bounded away from the origin, there is some integer $M > 0$ such that for any integer $a \geq 1$, the number of rectangles

$V_{a-\lambda} \cap H_{\lambda+1}$, $0 \le \lambda < a$, intersecting R is no greater than M. By (1.21) and the fact that $z^R_{n+1} \in V_{a_{j_N+1}-\lambda_N} \cap H_{\lambda_N+1}$, this implies that $j_N = j_{N^R_{n+1}(z)}$ grows uniformly in n. Since the denominators q_j of RCF-convergents are strictly increasing, we have that $|X - c_n| \to 0$ uniformly in n.

It remains to show that $|Y - d_n| \to 0$ uniformly in n. Let $n \ge 1$, and consider $z^R_{-n} = (x^R_{-n}, y^R_{-n}) = \mathcal{F}_R^{-n}(z)$. By Proposition 1.6.13, the digits of the contracted Farey expansion of x^R_{-n} with respect to R and z^R_{-n} are given by

$$\begin{pmatrix} \alpha^R_0(z^R_{-n}) \\ \beta^R_0(z^R_{-n}) \end{pmatrix} = \begin{pmatrix} 1 \\ u_R(z^R_{-n}) \end{pmatrix}$$

and

$$\begin{pmatrix} \alpha^R_{k+1}(z^R_{-n}) \\ \beta^R_{k+1}(z^R_{-n}) \end{pmatrix} = \begin{pmatrix} \alpha_R(z^R_{-n+k}) \\ \beta_R(z^R_{-n+k}) \end{pmatrix}, \quad k \ge 0.$$

Since $(X^R_n(z^R_{-n}), Y^R_n(z^R_{-n})) = (X, Y) \in D_n$ (see (1.58)), we have by (1.62)

$$\alpha_{j+1} = \alpha_R(z^R_{-n+(n+j)}) = \alpha^R_{n+j+1}(z^R_{-n})$$

and

$$\beta_{k+1} = \beta_R(z^R_{-n+(n+k)}) = \beta^R_{n+k+1}(z^R_{-n})$$

for all $-(n-1) \le j \le n-1$ and $-n \le k \le n-1$. In particular, applying Proposition 1.6.17 to z^R_{-n}, we find

$$\begin{aligned} Y = Y^R_n(z^R_{-n}) = [0/1; 1/\beta^R_n(z^R_{-n}), \alpha^R_n(z^R_{-n})/\beta^R_{n-1}(z^R_{-n}), \dots, \\ \alpha^R_1(z^R_{-n})/\beta^R_0(z^R_{-n}), \alpha^R_0(z^R_{-n})/(1/y^R_{-n} - 1)]), \end{aligned}$$

while

$$\begin{aligned} d_n &= [0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \alpha_{-1}/\beta_{-2}, \dots, \alpha_{-(n-2)}/\beta_{-(n-1)}] \\ &= [0/1; 1/\beta^R_n(z^R_{-n}), \alpha^R_n(z^R_{-n})/\beta^R_{n-1}(z^R_{-n}), \dots, \alpha^R_2(z^R_{-n})/\beta^R_1(z^R_{-n})])]. \end{aligned}$$

Set

$$\begin{aligned} B^R_{[-1,n]}(z^R_{-n}) &\coloneqq B_{[-1,n]}\left([\beta^R_0(z^R_{-n})/\alpha^R_0(z^R_{-n}); \alpha^R_1(z^R_{-n})/\beta^R_1(z^R_{-n}), \right. \\ &\qquad \left. \ldots, \alpha^R_n(z^R_{-n})/\beta^R_n(z^R_{-n})]\right) \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \alpha^R_0(z^R_{-n}) \\ 1 & \beta^R_0(z^R_{-n}) \end{pmatrix} \begin{pmatrix} 0 & \alpha^R_1(z^R_{-n}) \\ 1 & \beta^R_1(z^R_{-n}) \end{pmatrix} \\ &\qquad \cdots \begin{pmatrix} 0 & \alpha^R_n(z^R_{-n}) \\ 1 & \beta^R_n(z^R_{-n}) \end{pmatrix}, \end{aligned}$$

and denote the entries by

$$B^R_{[-1,n]}(z^R_{-n}) = \begin{pmatrix} P^R_{n-1}(z^R_{-n}) & P^R_n(z^R_{-n}) \\ Q^R_{n-1}(z^R_{-n}) & Q^R_n(z^R_{-n}) \end{pmatrix}.$$

Then Equation (1.7) gives

$$\begin{aligned} Y =& (B^R_{[-1,n]}(z^R_{-n}))^T \cdot \left(\frac{1}{y^R_{-n}} - 1\right) \\ =& \frac{P^R_{n-1}(z^R_{-n})(1-y^R_{-n}) + Q^R_{n-1}(z^R_{-n})y^R_{-n}}{P^R_n(z^R_{-n})(1-y^R_{-n}) + Q^R_n(z^R_{-n})y^R_{-n}}, \end{aligned}$$

while

$$d_n = (B^R_{[-1,n]}(z^R_{-n}))^T \cdot 0 = \frac{Q^R_{n-1}(z^R_{-n})}{Q^R_n(z^R_{-n})}.$$

Notice by Proposition 1.6.13, Equation (1.53) and the fact that $s_R(z) = 1$ for all z, that

$$|\det(B^R_{[-1,n]}(z^R_{-n})| = |\alpha^R_0(z^R_{-n})\alpha^R_1(z^R_{-n}) \cdots \alpha^R_n(z^R_{-n})| = 1.$$

Moreover, recall that $y^R_{-n} \in [0,1]$. We thus compute

$$\begin{aligned}
|Y - d_n| &= \left| \frac{P^R_{n-1}(z^R_{-n})(1-y^R_{-n}) + Q^R_{n-1}(z^R_{-n})y^R_{-n}}{P^R_n(z^R_{-n})(1-y^R_{-n}) + Q^R_n(z^R_{-n})y^R_{-n}} - \frac{Q^R_{n-1}(z^R_{-n})}{Q^R_n(z^R_{-n})} \right| \\
&= \frac{|P^R_{n-1}(z^R_{-n})Q^R_n(z^R_{-n}) - P^R_n(z^R_{-n})Q^R_{n-1}(z^R_{-n})||1-y^R_{-n}|}{|P^R_n(z^R_{-n})(1-y^R_{-n}) + Q^R_n(z^R_{-n})y^R_{-n}||Q^R_n(z^R_{-n})|} \\
&\leq \frac{1}{|P^R_n(z^R_{-n}) + (Q^R_n(z^R_{-n}) - P^R_n(z^R_{-n}))y^R_{-n}|Q^R_n(z^R_{-n})} \\
&\leq \frac{1}{\min\left\{P^R_n(z^R_{-n}), Q^R_n(z^R_{-n})\right\} Q^R_n(z^R_{-n})}.
\end{aligned}$$

By Corollary 1.6.11,

$$\begin{aligned}
&\min\left\{P^R_n(z^R_{-n}), Q^R_n(z^R_{-n})\right\} Q^R_n(z^R_{-n}) \\
&\qquad = \min\left\{u^R_{n+1}(z^R_{-n}), s^R_{n+1}(z^R_{-n})\right\} s^R_{n+1}(z^R_{-n}),
\end{aligned}$$

so it suffices to show that

$$\min\left\{u^R_n(z), s^R_n(z)\right\} s^R_n(z) \to \infty$$

uniformly in n. Write $u^R_n(z) = \lambda_N p_{j_N} + p_{j_N-1} \geq p_{j_N-1}$ and $s^R_n(z) = \lambda_N q_{j_N} + q_{j_N-1} \geq q_{j_N-1}$, where $N = N^R_n(z)$. As before, since R is bounded away from the origin, $j_N = j_{N^R_n(z)}$ grows uniformly in n. Since $x \notin \mathbb{Q}$, there is some n large enough (independent of z) for which $u^R_n(z) \geq p_{j_N-1} \geq 1$. Since the RCF-convergent denominators q_j are strictly increasing for $j > 0$,

$$\min\left\{u^R_n(z), s^R_n(z)\right\} s^R_n(z) \geq \min\left\{p_{j_N-1}, q_{j_N-1}\right\} q_{j_N-1} \to \infty$$

uniformly in n. □

Notice from Propositions 1.6.13 and 1.6.17 that

$$X(z) = [0/1; \alpha_R(z^R_0)/\beta_R(z^R_0), \alpha_R(z^R_1)/\beta_R(z^R_1), \dots].$$

From the proof of Proposition 1.6.18, it is evident that the convergents

$$d_n = [0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \alpha_{-1}/\beta_{-2}, \dots, \alpha_{-(n-2)}/\beta_{-(n-1)}]$$

of the GCF

$$[0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \alpha_{-1}/\beta_{-2}, \dots]$$

with

$$\alpha_{j+1} = \alpha_R(z_j^R) \quad \text{and} \quad \beta_{j+1} = \beta_R(z_j^R), \quad j < 0,$$

also converge to $Y(z)$. We thus obtain GCF-expansions of both $X(z)$ and $Y(z)$ on which τ_R acts as a two-sided shift:

Corollary 1.6.19. *For $z \in R$ for which z_n^R is defined for all $n \in \mathbb{Z}$,*

$$\begin{aligned}(X(z), Y(z)) = \big([0/1; \alpha_R(z_0^R)/\beta_R(z_0^R), \alpha_R(z_1^R)/\beta_R(z_1^R), \dots],\\ [0/1; 1/\beta_R(z_{-1}^R), \alpha_R(z_{-1}^R)/\beta_R(z_{-2}^R), \dots]\big),\end{aligned}$$

and for any $n \in \mathbb{Z}$,

$$\begin{aligned}\tau_R^n\,(X(z), Y(z)) = \big([0/1; \alpha_R(z_n^R)/\beta_R(z_n^R), \alpha_R(z_{n+1}^R)/\beta_R(z_{n+1}^R), \dots],\\ [0/1; 1/\beta_R(z_{n-1}^R), \alpha_R(z_{n-1}^R)/\beta_R(z_{n-2}^R), \dots]\big).\end{aligned}$$

1.7 Examples of contracted Farey expansions

In this section we consider several examples of explicit, inducible regions R and the contracted Farey expansions they produce. We shall find in §1.7.1 RCFs, in §1.7.2 Kraaikamp's S-expansions, and in §1.7.3 Nakada's α-CFs for $\alpha \in (0, 1]$. Throughout this section, any reference to the induced system $(H_1, \mathcal{B}, \bar{\mu}_{H_1}, \mathcal{F}_{H_1})$ is to the 'altered' system from Remark 1.5.4.

1.7.1 Regular continued fractions, revisited

Set $R = H_1$, and recall from Theorem 1.5.5 above that the induced system $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ is isomorphic to the Gauss natural extension $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$. We re-obtain this fact here through the use of contracted Farey expansions and the two-sided shift of §1.6.3.

Proof of Theorem 1.5.5. Let $z = (x, y) \in R$ with $x \neq 0$ be as in (1.34). Using (1.35), we find that $N_R(z) = a_1 = a(x)$, and by (1.40),

$$\begin{aligned}\begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} &= A_R(z) = A_{\varepsilon_1} \dots A_{\varepsilon_{a(x)}} \\ &= A_0^{a(x)-1} A_1 = \begin{pmatrix} 0 & 1 \\ 1 & a(x) \end{pmatrix}.\end{aligned} \tag{1.63}$$

In particular, $s_R(z) = 1$ for all z, so we are in the setting of §1.6.3. We know that $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ is isomorphic to $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$; we shall show that this latter system is precisely $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$. Since $u_R(z) = 0$ for all z, the map $\varphi_R : R \to \mathbb{R}^2$ from (1.56) is

$$\varphi_R(z) = \left(x, \frac{1-y}{y} \right) \qquad \text{for all } z = (x, y) \in R,\ x \neq 0, \tag{1.64}$$

and thus $\Omega_R = \varphi_R(R) = \Omega$, up to a null set. Since $\bar{\mu}(R) = \log 2$, Theorem 1.6.15 gives that $\bar{\nu}_R = \bar{\nu}_G$. Moreover, from Equations (1.53), (1.54), and (1.63) we find

$$\begin{pmatrix} \alpha_R(z) \\ \beta_R(z) \end{pmatrix} = \begin{pmatrix} 1 \\ a(x) \end{pmatrix}. \tag{1.65}$$

But if $(X, Y) = \varphi_R(z)$, Equation (1.64) gives $X = x$, so by Theorem 1.6.15 and Equation (1.12),

$$\tau_R(X, Y) = \left(\frac{1}{X} - a(X), \frac{1}{a(X) + Y} \right) = \mathcal{G}(X, Y).$$

Thus $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R) = (\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$. □

Let $z = (x, y) \in R$ as in (1.34) (so $b_1 = 1$) with $x \notin \mathbb{Q}$, and notice that repeated use of (1.48) gives

$$\begin{aligned} z_k^R =& (x_k^R, y_k^R) = \mathcal{F}_R^k(x, y) \\ =& ([0; a_{k+1}, a_{k+2}, \dots], [0; 1, a_k, \dots, a_1, b_2, b_3, \dots]). \end{aligned}$$

Thus, by Proposition 1.6.13 and Equation (1.65), the digits of the contracted Farey expansion of x with respect to $R = H_1$ and $z = (x, y) \in R$ are

$$\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} s_R(z) \\ u_R(z) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \alpha_{k+1}^R \\ \beta_{k+1}^R \end{pmatrix} = \begin{pmatrix} 1 \\ a(x_k^R) \end{pmatrix} = \begin{pmatrix} 1 \\ a_{k+1} \end{pmatrix}.$$

That is, the contracted Farey expansion of x with respect to $R = H_1$ and $z = (x, y)$ recovers the RCF-expansion $[0/1; 1/a_1, 1/a_2, \dots] = [0; a_1, a_2, \dots]$ of x.

1.7.2 S-expansions, revisited

We also find Kraaikamp's S-expansions (and thus Minkowski's diagonal CFs, Bosma's optimal CFs, and Nakada's α-CFs for $\alpha \geq 1/2$; see [115] and §1.3.3 above) as special instances of contracted Farey expansions. Indeed, let $S \subset \Omega$ be a singularisation area, i.e., S is a $\bar{\nu}_G$-measurable set with $\bar{\nu}_G(\partial S) = 0$ satisfying both

(a) $S \subset V_1$ and

(b) $S \cap \mathcal{G}(S) = \varnothing$.

Moreover, let $[\beta_0^S/\alpha_0^S; \alpha_1^S/\beta_1^S, \alpha_2^S/\beta_2^S, \dots]$ be the S-expansion of $x = [0; a_1, a_2, \dots] \in (0, 1) \setminus \mathbb{Q}$ obtained by simultaneously singularising at all positions n for which $\mathcal{G}^n(x, 0) \in S$ (see Definitions 4.4, 4.5 of [115] and §1.3.3 above). For $n \geq -1$ let

$$B_{[-1,n]}^S = \begin{pmatrix} P_{n-1}^S & P_n^S \\ Q_{n-1}^S & Q_n^S \end{pmatrix} := B_{[-1,n]}([\beta_0^S/\alpha_0^S; \alpha_1^S/\beta_1^S, \alpha_2^S/\beta_2^S, \dots]).$$

From remarks preceding Theorem 4.13 and Theorem 5.3.i of [115], it follows that $P^S_{-2} = Q^S_{-1} = 0$, $P^S_{-1} = Q^S_{-2} = 1$, and for $k \geq 0$,

$$\begin{pmatrix} P^S_k \\ Q^S_k \end{pmatrix} = \begin{pmatrix} p_{j^S_k} \\ q_{j^S_k} \end{pmatrix},$$

where p_j/q_j is the jth RCF-convergent of x and $(j^S_k)_{k\geq 0}$ is the subsequence of powers $j \geq 0$ for which $\mathcal{G}^j(x,0) \in \Delta \coloneqq \Omega \backslash S$.

We wish to determine a proper, inducible subregion $R \subset \Omega$ for which the contracted Farey expansion $[\beta^R_0/\alpha^R_0; \alpha^R_1/\beta^R_1, \alpha^R_2/\beta^R_2, \dots]$ of x with respect to R coincides with the S-expansion of x. By Remark 1.2.2, it suffices to find R such that $P^R_k = P^S_k$ and $Q^R_k = Q^S_k$ for all $k \geq 0$, with

$$B^R_{[-1,k]} = \begin{pmatrix} P^R_{k-1} & P^R_k \\ Q^R_{k-1} & Q^R_k \end{pmatrix}$$

as in (1.51) and (1.52).

It seems natural to set $R \coloneqq \varphi^{-1}_{H_1}(\Delta) \subset H_1$, where $\varphi_{H_1} : H_1 \to \Omega$ is the isomorphism map between $(H_1, \mathcal{B}, \bar{\mu}_{H_1}, \mathcal{F}_{H_1})$ and $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$ from (1.64) above satisfying $\varphi_{H_1} \circ \mathcal{F}_{H_1}(z) = \mathcal{G} \circ \varphi_{H_1}(z)$ for all $z \in H_1$. However, in the classical setting of RCFs and, in particular, S-expansions, one uses the one-to-one correspondence between points in the $\mathcal{G}$-orbit of $(x,0)$ and RCF-convergents p_n/q_n, which come from the *right-hand column* of the matrix $\left(\begin{smallmatrix} p_{n-1} & p_n \\ q_{n-1} & q_n \end{smallmatrix}\right)$. On the other hand, for contracted Farey expansions we use the one-to-one correspondence between points in the $\mathcal{F}_R$-orbit of $(x,1)$ and contracted Farey convergents u^R_n/s^R_n coming from the *left-hand column* of the matrix $A^R_{[0,n]} = A_{[0,N^R_n]}$ from (1.46). When $R = \varphi^{-1}_{H_1}(\Omega) = H_1$, the matrix $A^{H_1}_{[0,n]}$ is of the form $\left(\begin{smallmatrix} p_{n-1} & p_n \\ q_{n-1} & q_n \end{smallmatrix}\right)$, so the one-to-one correspondence in this setting is between $\mathcal{F}^n_{H_1}(x,1) = \varphi^{-1}_{H_1} \circ \mathcal{G}^n(x,0)$ and p_{n-1}/q_{n-1}. This indexing discrepancy is fixed by instead considering the isomorphism map $\psi \coloneqq \mathcal{G}^{-1} \circ \varphi_{H_1}$ between $(H_1, \mathcal{B}, \bar{\mu}_{H_1}, \mathcal{F}_{H_1})$ and $(\Omega, \mathcal{B}, \bar{\nu}_G, \mathcal{G})$:

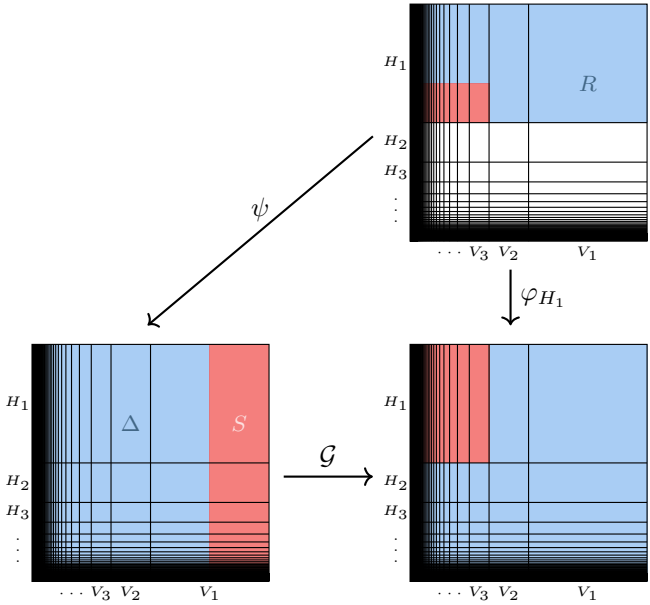


Figure 1.5: Bottom-left: A singularisation area S and its complement Δ in Ω. Bottom-right: The images of S and Δ under $\mathcal{G}$. Top-right: The region $R = \psi^{-1}(\Delta)$ and its complement in H_1.

$$\begin{array}{ccc} H_1 & \xrightarrow{\mathcal{F}_{H_1}} & H_1 \\ {\scriptstyle \varphi_{H_1}}\downarrow & \swarrow{\scriptstyle \psi} & \downarrow{\scriptstyle \varphi_{H_1}} \\ \Omega & \xrightarrow{\mathcal{G}} & \Omega \end{array}$$

Set

$$R := \psi^{-1}(\Delta) = H_1 \backslash \psi^{-1}(S);$$

see Figure 1.5. Notice that for any $z \in H_1$, either $\varphi_{H_1}(z) \in \Delta$ or $\mathcal{G} \circ \varphi_{H_1}(z) \in \Delta$; otherwise, both $\varphi_{H_1}(z)$ and $\mathcal{G} \circ \varphi_{H_1}(z)$ belong to

S, contrary to condition (ii) of a singularisation area. Thus, either $\psi^{-1} \circ \varphi_{H_1}(z) \in R$ or $\psi^{-1} \circ \mathcal{G} \circ \varphi_{H_1}(z) \in R$. But $\psi^{-1} \circ \varphi_{H_1} = \mathcal{F}_{H_1}$ and $\psi^{-1} \circ \mathcal{G} \circ \varphi_{H_1} = \mathcal{F}_{H_1}^2$, so for any $z \in H_1$, either $\mathcal{F}_{H_1}(z) \in R$ or $\mathcal{F}_{H_1}^2(z) \in R$. The entries of the matrices $A_R(z)$ depend on whether $\mathcal{F}_{H_1}(z) \in R$:

Lemma 1.7.1. *For any $z = (x, y) \in H_1$ with $x = [0; a_1, a_2, \dots]$,*

$$\begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} = A_R(z) = \begin{cases} \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix} & \text{if } \mathcal{F}_{H_1}(z) \in R, \\ \begin{pmatrix} 1 & a_2 \\ 1 & a_2 + 1 \end{pmatrix} & \text{if } \mathcal{F}_{H_1}(z) \notin R. \end{cases}$$

Proof. First suppose that $\mathcal{F}_{H_1}(z) \in R$. Now $\mathcal{F}_{H_1}(z) = \mathcal{F}^{a_1}(z)$, and for all $1 \le j < a_1$, $\mathcal{F}^j(z) \notin H_1$ implies $\mathcal{F}^j(z) \notin R \subset H_1$. Thus $N_R(z) = a_1$, and by (1.40) we have

$$A_R(z) = A_0^{a_1 - 1} A_1 = \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix}.$$

If $\mathcal{F}_{H_1}(z) = \mathcal{F}^{a_1}(z) \notin R$, then $\mathcal{F}_{H_1}^2(z) = \mathcal{F}^{a_1 + a_2}(z) \in R$. Since $\mathcal{F}^j(z) \notin H_1$ for $1 \le j < a_1 + a_2$ with $j \ne a_1$, we have $N_R(z) = a_1 + a_2$ and—by (1.40)—

$$A_R(z) = A_0^{a_1 - 1} A_1 A_0^{a_2 - 1} A_1 = \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & a_2 \end{pmatrix} = \begin{pmatrix} 1 & a_2 \\ a_1 & a_2 a_1 + 1 \end{pmatrix}.$$

But $\mathcal{F}_{H_1}(z) \notin R$ is equivalent to $\varphi_{H_1}(z) \notin \Delta$, or $\varphi_{H_1}(z) \in S$. Since φ_{H_1} acts as the identity on the first coordinate and $S \subset V_1$ by condition (a) of a singularisation area, this implies $a_1 = 1$. □

By Lemma 1.7.1, $s_R(z) = 1$ for all $z \in H_1$ and, in particular, for all $z = (x, 1)$. By Corollary 1.6.11, Equation (1.47), and the fact that $R \subset H_1$ (so $\lambda_{N_{k+1}^R} = 0$),

$$\begin{pmatrix} P_k^R \\ Q_k^R \end{pmatrix} = \begin{pmatrix} u_{k+1}^R \\ s_{k+1}^R \end{pmatrix} = \begin{pmatrix} p_{j_{N_{k+1}^R} - 1} \\ q_{j_{N_{k+1}^R} - 1} \end{pmatrix}, \qquad k \ge 0.$$

Writing $N^R_{k+1} = a_1 + \cdots + a_{j_{N^R_{k+1}}}$ (see (1.21)), we see that the indices $j_{N^R_{k+1}}$, $k \geq 0$, are precisely the powers $j > 0$ for which

$$\mathcal{F}^{a_1+\cdots+a_j}(x,1) = \mathcal{F}^j_{H_1}(x,1) \in R.$$

Equivalently, these are the powers $j > 0$ for which

$$\varphi^{-1}_{H_1} \circ \mathcal{G}^j \circ \varphi_{H_1}(x,1) \in R = \psi^{-1}(\Delta) = \varphi^{-1}_{H_1} \circ \mathcal{G}(\Delta),$$

or $\mathcal{G}^{j-1}(x,0) \in \Delta$. Thus $j_{N^R_{k+1}} - 1 = j^S_k$, and

$$\begin{pmatrix} P^R_k \\ Q^R_k \end{pmatrix} = \begin{pmatrix} p_{j_{N^R_{k+1}}-1} \\ q_{j_{N^R_{k+1}}-1} \end{pmatrix} = \begin{pmatrix} p_{j^S_k} \\ q_{j^S_k} \end{pmatrix} = \begin{pmatrix} P^S_k \\ Q^S_k \end{pmatrix}, \qquad k \geq 0.$$

By Remark 1.2.2, this proves:

Proposition 1.7.2. *The contracted Farey expansion of x with respect to $R = \psi^{-1}(\Delta)$ coincides with the S-expansion of x.*

In §5 of [115], a two-dimensional ergodic system[14] $(\Gamma_S, \mathcal{B}, \rho, \tau)$ is constructed corresponding to the two-sided shift operator for S-expansions. We briefly recall this system here and show that it coincides with $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ as defined in §1.6.3. (Note by Lemma 1.7.1 that $s_R(z) = 1$ for all $z \in R$, so we are in the setting of §1.6.3.) Set

$$\Delta^- := \mathcal{G}(S) \qquad \text{and} \qquad \Delta^+ := \Delta \backslash \Delta^-.$$

Define $M : \Delta \to \mathbb{R}^2$ for $z = (x,y)$ by

$$M(z) := \begin{cases} (x,y), & z \in \Delta^+, \\ \left(\frac{-x}{1+x}, 1-y\right), & z \in \Delta^-, \end{cases}$$

[14] We replace Kraaikamp's original notation Ω_S by Γ_S to avoid confusion with Ω_R defined §1.6.3. However, we shall see in Proposition 1.7.4 below that, in fact, $\Gamma_S = \Omega_R$.

and let $\Gamma_S \coloneqq M(\Delta)$. The map $\tau : \Gamma_S \to \Gamma_S$ is defined by $\tau \coloneqq M \circ \mathcal{G}_\Delta \circ M^{-1}$, where $\mathcal{G}_\Delta : \Delta \to \Delta$ is the map $\mathcal{G}$ induced on Δ, i.e., $\mathcal{G}_\Delta(z) = \mathcal{G}(z)$ if $\mathcal{G}(z) \in \Delta$ and $\mathcal{G}_\Delta(z) = \mathcal{G}^2(z)$ otherwise. The measure ρ is the probability measure on $(\Gamma_S, \mathcal{B})$ with density $1/(\log 2\bar{\nu}_G(\Delta)(1+XY)^2)$ (see Theorem 5.9 of [115]). Setting

$$X_k^S \coloneqq [0/1; \alpha_{k+1}^S/\beta_{k+1}^S, \alpha_{k+2}^S/\beta_{k+2}^S, \dots], \qquad k \geq 0,$$

$Y_0^S \coloneqq 0$ and

$$Y_k^S \coloneqq [0/1; 1/\beta_k^S, \alpha_k^S/\beta_{k-1}^S, \dots, \alpha_2^S/\beta_1^S], \qquad k \geq 1,$$

where $x = [\beta_0^S/\alpha_0^S; \alpha_1^S/\beta_1^S, \alpha_2^S/\beta_2^S, \dots]$ is the S-expansion of x, it is observed following Definition 5.8 of [115] that

$$(X_k^S, Y_k^S) = \tau^k(X_0^S, Y_0^S), \qquad k \geq 0.$$

Note that by Propositions 1.6.17 and 1.7.2, τ^n and τ_R^n agree for all $n \geq 0$ when evaluated at $(X_0^S, Y_0^S) = (X_0^S, 0)$. We claim that in fact $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R) = (\Gamma_S, \mathcal{B}, \rho, \tau)$. By (1.56) and Lemma 1.7.1,

$$\varphi_R(z) = \begin{cases} \left(x, \frac{1-y}{y}\right), & \mathcal{F}_{H_1}(z) \in R, \\ (x-1, 1-y), & \mathcal{F}_{H_1}(z) \notin R. \end{cases} \tag{1.66}$$

Lemma 1.7.3. *For any $z \in \Delta$, $M(z) = \varphi_R \circ \psi^{-1} \circ \mathcal{G}_\Delta^{-1}(z)$.*

Proof. Suppose first that $z = (x, y) \in \Delta^+$. Now, since $z \notin \Delta^- = \mathcal{G}(S)$, we have $\mathcal{G}^{-1}(z) \in \Omega \setminus S = \Delta$. Hence $\mathcal{G}_\Delta^{-1}(z) = \mathcal{G}^{-1}(z)$. Then

$$\varphi_R \circ \psi^{-1} \circ \mathcal{G}_\Delta^{-1}(z) = \varphi_R \circ \varphi_{H_1}^{-1}(z) = \varphi_R\left(x, \frac{1}{1+y}\right).$$

Notice that

$$\mathcal{F}_{H_1}\left(x, \frac{1}{1+y}\right) = \psi^{-1} \circ \varphi_{H_1}\left(x, \frac{1}{1+y}\right) = \psi^{-1}(z) \in \psi^{-1}(\Delta) = R,$$

so by (1.66), $\varphi_R(x, 1/(1+y)) = z$. Thus, for $z \in \Delta^+$,

$$\varphi_R \circ \psi^{-1} \circ \mathcal{G}_\Delta^{-1}(z) = z = M(z).$$

Next, suppose that $z \in \Delta^-$. Then $z \in \mathcal{G}(S)$, so $\mathcal{G}^{-1}(z) \in S = \Omega \setminus \Delta$ and $\mathcal{G}_\Delta^{-1}(z) = \mathcal{G}^{-2}(z)$. Moreover, since $\mathcal{G}^{-1}(z) \in S \subset V_1$, we have $\mathcal{G}^{-1}(z) = (1/(x+1), 1/y - 1)$. With these observations, we find

$$\begin{aligned}\varphi_R \circ \psi^{-1} \circ \mathcal{G}_\Delta^{-1}(z) =& \varphi_R \circ \varphi_{H_1}^{-1} \circ \mathcal{G}^{-1}(z) \\ =& \varphi_R \circ \varphi_{H_1}^{-1}\left(\frac{1}{x+1}, \frac{1}{y} - 1\right) \\ =& \varphi_R\left(\frac{1}{x+1}, y\right).\end{aligned}$$

We claim that $\mathcal{F}_{H_1}(1/(x+1), y) \notin R$. This is equivalent to

$$\psi^{-1} \circ \varphi_{H_1}\left(\frac{1}{x+1}, y\right) \notin \psi^{-1}(\Delta),$$

or $\varphi_{H_1}(1/(x+1), y) \in S$. But

$$\varphi_{H_1}\left(\frac{1}{x+1}, y\right) = \left(\frac{1}{x+1}, \frac{1}{y} - 1\right) = \mathcal{G}^{-1}(z) \in S$$

by assumption, so the claim holds. Thus, from (1.66), we have $\varphi_R(1/(x+1), y) = (-x/(x+1), 1-y)$ and

$$\varphi_R \circ \psi^{-1} \circ \mathcal{G}_\Delta^{-1}(z) = \left(-\frac{x}{x+1}, 1 - y\right) = M(z)$$

for $z \in \Delta^-$. □

Proposition 1.7.4. *With* $R = \psi^{-1}(\Delta)$,

$$(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R) = (\Gamma_S, \mathcal{B}, \rho, \tau).$$

Proof. By Lemma 1.7.3,

$$\Omega_R = \varphi_R(R) = \varphi_R \circ \psi^{-1}(\Delta) = \varphi_R \circ \psi^{-1} \circ \mathcal{G}_\Delta^{-1}(\Delta) = M(\Delta) = \Gamma_S.$$

Moreover,

$$\begin{aligned}
\psi \circ \mathcal{F}_R(z) &= \begin{cases} \psi \circ \mathcal{F}_{H_1}(z), & \mathcal{F}_{H_1}(z) \in R, \\ \psi \circ \mathcal{F}_{H_1}^2(z), & \mathcal{F}_{H_1}(z) \notin R, \end{cases} \\
&= \begin{cases} \varphi_{H_1}(z), & \varphi_{H_1}(z) \in \Delta, \\ \mathcal{G} \circ \varphi_{H_1}(z), & \varphi_{H_1}(z) \notin \Delta, \end{cases} \\
&= \begin{cases} \mathcal{G} \circ \psi(z), & \mathcal{G} \circ \psi(z) \in \Delta, \\ \mathcal{G}^2 \circ \psi(z), & \mathcal{G} \circ \psi(z) \notin \Delta, \end{cases} \\
&= \mathcal{G}_\Delta \circ \psi(z),
\end{aligned}$$

so

$$\tau_R = \varphi_R \circ \mathcal{F}_R \circ \varphi_R^{-1} = \varphi_R \circ \psi^{-1} \circ \mathcal{G}_\Delta \circ \psi \circ \varphi_R^{-1} = M \circ \mathcal{G}_\Delta \circ M^{-1} = \tau.$$

Lastly, $\bar{\nu}_R = \rho$ since these are both probability measures on $\Omega_R = \Gamma_S$ with densities of the form $C(1 + XY)^{-2}$, where C is a normalising constant. □

1.7.3 Nakada's α-continued fractions, revisited

Recall Nakada's parameterised family of α-CF maps from §1.3.2, which are defined for all $0 \leq \alpha \leq 1$. Moreover, recall from the end of §1.3.3 that the natural extensions of the α-CFs are realised as S-expansion systems, but only for $\alpha \geq 1/2$. Since, by §1.7.2, S-expansions are realised as contracted Farey expansions, so are Nakada's α-CFs for $\alpha \geq 1/2$. In this subsection we extend this fact to $\alpha > 0$, giving a new description of a planar natural extension of $([\alpha - 1, \alpha], \mathcal{B}, \rho_\alpha, G_\alpha)$ as an explicit induced transformation $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ of Ito's natural extension of the Farey tent map (Theorem 1.7.11 below; cf. [120]).

Remark 1.7.5. One finds that $G_\alpha([\alpha-1,\alpha]) = [\alpha-1,\alpha)$, so $([\alpha-1,\alpha],\mathcal{B},\rho_\alpha,G_\alpha)$ is isomorphic to the restriction of this system to $[\alpha-1,\alpha)$, which we denote by $([\alpha-1,\alpha),\mathcal{B},\rho_\alpha,G_\alpha)$. The endpoint α was included in the domain in §1.3.2 so that we could speak of matching, which depends on the G_α-orbits of α and $\alpha-1$. However, it shall be more convenient in this subsection to consider the isomorphic system $([\alpha-1,\alpha),\mathcal{B},\rho_\alpha,G_\alpha)$.

The domain R will be constructed in two steps: first, we define a subset $A \subset H_1$ via an integer-valued map k on H_1; second, R is defined by 'pushing' part of A down into $\Omega \setminus H_1$ with the map $\mathcal{F}$. Fix $\alpha \in (0,1]$ and define $k : H_1 \to \mathbb{N} \cup \{\infty\}$ by

$$k(z) \coloneqq \inf\{j > 0 \mid \mathcal{F}_{H_1}^{-j}(z) \in [0,\alpha) \times [1/2,1]\}, \qquad z \in H_1,$$

and let

$$A \coloneqq \{z \in H_1 \mid k(z) \text{ is odd}\};$$

see Figure 1.6. Recall the definition of hitting times N_R from (1.38). The restriction of N_R to R—also denoted N_R—is called the *return time* to R. We wish to determine the return times N_A under $\mathcal{F}$. For this, we use the following:

Lemma 1.7.6. *For any $z = (x,y) \in H_1$,*

$$k(\mathcal{F}_{H_1}(z)) = \begin{cases} 1, & x < \alpha, \\ k(z)+1, & x \geq \alpha. \end{cases}$$

Proof. First, notice that $\mathcal{F}_{H_1}^{-1}(\mathcal{F}_{H_1}(z)) = z$ belongs to $[0,\alpha)\times[1/2,1]$ if and only if $x < \alpha$. Thus, if $x < \alpha$, then $k(\mathcal{F}_{H_1}(z)) = 1$. If $x \geq \alpha$, then $k(z)$ is the infimum of powers $j > 0$ for which

$$\mathcal{F}_{H_1}^{-(j+1)}(\mathcal{F}_{H_1}(z)) = \mathcal{F}_{H_1}^{-j}(z) \in [0,\alpha) \times [1/2,1].$$

Hence $k(\mathcal{F}_{H_1}(z)) = k(z)+1$. □

Lemma 1.7.7. *The return times $N_A : A \to \mathbb{N}$ under $\mathcal{F}$ are*

$$N_A(z) = \begin{cases} a_1, & x < \alpha, \\ a_1 + a_2, & x \geq \alpha, \end{cases}$$

where $z = (x, y) \in A$ with $x = [0; a_1, a_2, \dots]$.

Proof. First, suppose that $x < \alpha$, and notice that for all $0 < j < a_1$, $\mathcal{F}^j(z) \in H_{j+1} \neq H_1$ implies $\mathcal{F}^j(z) \notin A \subset H_1$. On the other hand, by Lemma 1.7.6, $k(\mathcal{F}_{H_1}(z)) = 1$ is odd, so $\mathcal{F}_{H_1}(z) \in A$. Since $\mathcal{F}_{H_1}(z) = \mathcal{F}^{a_1}(z)$, we have $N_A(z) = a_1$.

Next, suppose that $x \geq \alpha$. As above, $\mathcal{F}^j(z) \notin H_1$ for all $0 < j < a_1 + a_2$ with $j \neq a_1$, so $\mathcal{F}^j(z) \notin A$ for such j. Moreover, $z \in A$ implies that $k(z)$ is odd, and thus by Lemma 1.7.6, $k(\mathcal{F}_{H_1}(z)) = k(z) + 1$ is even. Hence $\mathcal{F}^{a_1}(z) = \mathcal{F}_{H_1}(z) \notin A$. Write $z' = (x', y') \coloneqq \mathcal{F}_{H_1}(z)$. Again by Lemma 1.7.6,

$$k(\mathcal{F}_{H_1}(z')) = \begin{cases} 1, & x' < \alpha, \\ k(z') + 1, & x' \geq \alpha. \end{cases}$$

But $k(z') = k(\mathcal{F}_{H_1}(z))$ is even, so in either case $k(\mathcal{F}_{H_1}(z'))$ is odd. Hence $\mathcal{F}^{a_1+a_2}(z) = \mathcal{F}^2_{H_1}(z) = \mathcal{F}_{H_1}(z') \in A$, and $N_A(z) = a_1 + a_2$. □

We now define the subregion $R \subset \Omega$ in terms of the set $A \subset H_1$. For each integer $a > 1$, let

$$A_a \coloneqq A \cap V_a \cap ([\alpha, 1] \times [1/2, 1])$$

be the set of points $z = (x, y) \in A$ for which $x = [0; a_1, a_2, \dots]$ with $a_1 = a$ and $x \geq \alpha$. Next, define

$$R \coloneqq A \cup \bigcup_{a>1} \bigcup_{\lambda=1}^{a-1} \mathcal{F}^\lambda(A_a) \tag{1.67}$$

as the region $A \subset H_1$ together with each A_a 'pushed down' into $\Omega \backslash H_1$ under $\mathcal{F}$ a maximal number of times; see Figure 1.6. Notice that if $\alpha > 1/2$, then $A_a = \varnothing$ for $a > 1$ and hence $R = A$.

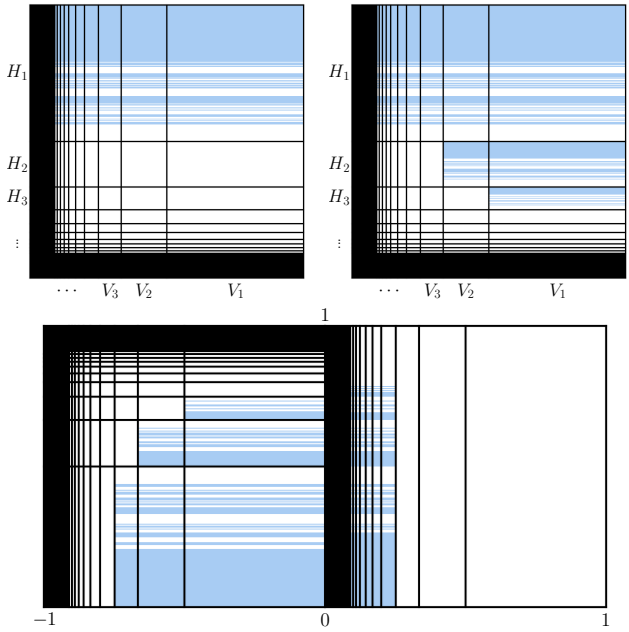


Figure 1.6: Approximations of the regions A (top-left), R (top-right), and Ω_R (bottom) for $\alpha = 1/4$.

Remark 1.7.8. In Figure 1.6, the region A consists of rectangles extending from $x = 0$ to $x = 1$, and the region R consists of A together with rectangles extending from $x = F(1/4) = 1/3$ to $x = 1$ and $x = F^2(1/4) = 1/2$ to $x = 1$. These 'full' rectangles are due to the fact that $\alpha = 1/4$ is of the form $\alpha = 1/n$ for some integer $n \geq 1$; see also [129], where the natural extension of the α-CF maps are constructed for such α. For general $\alpha > 0$, one can show that A consists of rectangles extending from various $x = x_0 \in [0, 1)$ to $x = 1$.

Lemma 1.7.7 and the definition of R give the following:

Corollary 1.7.9. *The return times $N_R : R \to \mathbb{N}$ under $\mathcal{F}$ are given by $N_R = N_A$ if $\alpha > 1/2$ and*

$$N_R(z) = \begin{cases} a_1, & x < \alpha, \\ 1, & \alpha \le x \le 1/2, \\ a_2 + 1, & 1/2 < x, \end{cases}$$

if $\alpha \le 1/2$, where $z = (x, y) \in A$ with $x = [0; a_1, a_2, \dots]$.

From this and Equation (1.40), we find that if $\alpha > 1/2$, then

$$\begin{aligned} A_R(z) = \begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} &= \begin{cases} A_0^{a_1 - 1} A_1, & x < \alpha \\ A_1 A_0^{a_2 - 1} A_1, & x \ge \alpha \end{cases} \\ &= \begin{cases} \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix}, & x < \alpha, \\ \begin{pmatrix} 1 & a_2 \\ 1 & a_2 + 1 \end{pmatrix}, & x \ge \alpha, \end{cases} \end{aligned} \tag{1.68}$$

while if $\alpha \le 1/2$,

$$\begin{aligned} A_R(z) = \begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} &= \begin{cases} A_0^{a_1 - 1} A_1, & x < \alpha \\ A_0, & \alpha \le x \le 1/2 \\ A_1 A_0^{a_2 - 1} A_1, & 1/2 < x \end{cases} \\ &= \begin{cases} \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix}, & x < \alpha, \\ \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, & \alpha \le x \le 1/2, \\ \begin{pmatrix} 1 & a_2 \\ 1 & a_2 + 1 \end{pmatrix}, & 1/2 < x. \end{cases} \end{aligned} \tag{1.69}$$

Notice, in particular, that $s_R(z) = 1$ for all z, and R satisfies the

assumptions of §1.6.3. Moreover,

$$u_R(z) = \begin{cases} 0, & x < \alpha, \\ 1 & x \geq \alpha, \end{cases} \tag{1.70}$$

so the map $\varphi_R : R \to \mathbb{R}^2$ from (1.56) is given by

$$\varphi_R(z) = \begin{cases} \left(x, \frac{1-y}{y}\right), & x < \alpha, \\ (x-1, 1-y), & x \geq \alpha. \end{cases} \tag{1.71}$$

The region $\Omega_R = \varphi_R(R)$ is shown in Figure 1.6.

Before proving that $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ is the natural extension of $([\alpha - 1, \alpha), \mathcal{B}, \rho_\alpha, G_\alpha)$, we determine the values of $\alpha_R(z)$ and $\beta_R(z)$ defined in (1.53) and (1.54).

Lemma 1.7.10. *Let $z = (x, y) \in R$ with $x \neq 0, 1$. Then*

$$\alpha_R(z) = \begin{cases} 1, & x < \alpha, \\ -1, & x \geq \alpha, \end{cases}$$

and

$$\beta_R(z) = \begin{cases} \left\lfloor \frac{1}{x} + 1 - \alpha \right\rfloor, & x < \alpha, \\ \left\lfloor \frac{1}{1-x} + 1 - \alpha \right\rfloor, & x \geq \alpha. \end{cases}$$

Proof. Let $z = (x, y) \in R$ with $x = [0; a_1, a_2, \dots]$, and set $z' = (x', y') = \mathcal{F}_R(z)$. From (1.53), the fact that $s_R(z) = 1$ for all z, and (1.68) and (1.69), we have

$$\alpha_R(z) = -\det(A_R(z)) = \begin{cases} 1, & x < \alpha, \\ -1, & x \geq \alpha, \end{cases}$$

as claimed.

Next, from (1.54) and the fact that $s_R(z) = 1$ for all z, we have $\beta_R(z) = r_R(z) + u_R(z')$. If $\alpha > 1/2$, then from (1.68) and (1.70), we

find that

$$\begin{aligned}\beta_R(z) &= \begin{cases} a_1 + u_R(z'), & x < \alpha \\ a_2 + 1 + u_R(z'), & x \geq \alpha \end{cases} \\ &= \begin{cases} a_1, & x < \alpha \text{ and } x' < \alpha, \\ a_1 + 1, & x < \alpha \text{ and } x' \geq \alpha, \\ a_2 + 1, & x \geq \alpha \text{ and } x' < \alpha, \\ a_2 + 2, & x \geq \alpha \text{ and } x' \geq \alpha. \end{cases} \end{aligned} \tag{1.72}$$

Now suppose $\alpha \leq 1/2$. Notice that if $\alpha \leq x \leq 1/2$, then by (1.39) and (1.69),

$$x' = A_R^{-1}(z) \cdot x = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \cdot x = \frac{x}{1-x} > x \geq \alpha \tag{1.73}$$

implies $u_R(z') = 1$. Hence, again from (1.69) and (1.70),

$$\begin{aligned}\beta_R(z) &= \begin{cases} a_1 + u_R(z'), & x < \alpha \\ 1 + u_R(z'), & \alpha \leq x \leq 1/2 \\ a_2 + 1 + u_R(z'), & 1/2 < x \end{cases} \\ &= \begin{cases} a_1, & x < \alpha \text{ and } x' < \alpha, \\ a_1 + 1, & x < \alpha \text{ and } x' \geq \alpha, \\ 2, & \alpha \leq x \leq 1/2, \\ a_2 + 1, & 1/2 < x \text{ and } x' < \alpha, \\ a_2 + 2, & 1/2 < x \text{ and } x' \geq \alpha. \end{cases} \end{aligned} \tag{1.74}$$

The remainder of the proof consists of cases. Throughout, we repeatedly use the two inequalities $\alpha \leq 1 + x'$ and $x' < 1 + \alpha$, which follow from $\alpha \in (0, 1]$ and $x' \in [0, 1]$.

(i) Suppose that $x < \alpha$. We must show $\beta_R(z) = \left\lfloor \frac{1}{x} + 1 - \alpha \right\rfloor$. By (1.39), (1.68) and (1.69),

$$x' = A_R^{-1}(z) \cdot x = \begin{pmatrix} a_1 & -1 \\ -1 & 0 \end{pmatrix} \cdot x = \frac{1}{x} - a_1,$$

so

$$\frac{1}{x} + 1 - \alpha = x' + a_1 + 1 - \alpha.$$

(a) If $x' < \alpha$, then

$$a_1 \le x' + a_1 + 1 - \alpha < a_1 + 1,$$

and by (1.72) and (1.74), $\beta_R(z) = a_1 = \left\lfloor \frac{1}{x} + 1 - \alpha \right\rfloor$.

(b) If $x' \ge \alpha$, then

$$a_1 + 1 \le x' + a_1 + 1 - \alpha < a_1 + 2,$$

so by (1.72) and (1.74), $\beta_R(z) = a_1 + 1 = \left\lfloor \frac{1}{x} + 1 - \alpha \right\rfloor$.

(ii) Now suppose $x \ge \alpha$. We must show $\beta_R(z) = \left\lfloor \frac{1}{1-x} + 1 - \alpha \right\rfloor$.

(a) If $x \le 1/2$, then from the computation in (1.73), $1 + x' = 1/(1-x)$. Hence

$$\frac{1}{1-x} + 1 - \alpha = 2 + x' - \alpha,$$

and (using $x' > x$)

$$2 \le 2 + x - \alpha < 2 + x' - \alpha < 3.$$

By (1.74), $\beta_R(z) = 2 = \left\lfloor \frac{1}{1-x} + 1 - \alpha \right\rfloor$.

(b) Now suppose that $x > 1/2$. By (1.39), (1.68) and (1.69),

$$x' = A_R^{-1}(z) \cdot x = \begin{pmatrix} a_2 + 1 & -a_2 \\ -1 & 1 \end{pmatrix} \cdot x = \frac{x}{1-x} - a_2,$$

and

$$\frac{1}{1-x} + 1 - \alpha = \frac{x}{1-x} + 2 - \alpha = a_2 + 2 + x' - \alpha.$$

(1) If $x' < \alpha$, then

$$a_2 + 1 \leq a_2 + 2 + x' - \alpha < a_2 + 2,$$

and by (1.72) and (1.74),

$$\beta_R(z) = a_2 + 1 = \left\lfloor \frac{1}{1-x} + 1 - \alpha \right\rfloor.$$

(2) Lastly, if $x' \geq \alpha$, then

$$a_2 + 2 \leq a_2 + 2 + x' - \alpha < a_2 + 3,$$

so by (1.72) and (1.74),

$$\beta_R(z) = a_2 + 2 = \left\lfloor \frac{1}{1-x} + 1 - \alpha \right\rfloor. \qquad \square$$

We are now in a position to prove:

Theorem 1.7.11. *The induced system* $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ *is the natural extension of* $([\alpha - 1, \alpha), \mathcal{B}, \rho_\alpha, G_\alpha)$.

Proof. When $\alpha = 1$, then $R = [0, 1) \times [1/2, 1)$, and $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ is isomorphic to $(H_1, \mathcal{B}, \bar{\mu}_{H_1}, \mathcal{F}_{H_1})$. The result follows from Theorem 1.5.5 and the fact that for $\alpha = 1$, $([\alpha-1, \alpha), \mathcal{B}, \rho_\alpha, G_\alpha)$ is (isomorphic to) $([0, 1], \mathcal{B}, \nu_G, G)$.

Now suppose $\alpha \in (0, 1)$. Since $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ and the system $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ from §1.6.3 are isomorphic, it suffices to show that the latter system is the natural extension of $([\alpha - 1, \alpha), \mathcal{B}, \rho_\alpha, G_\alpha)$. Throughout, we shall consider the restrictions of $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ and $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ to the full-measure subsets on which $\mathcal{F}_R^n$ and τ_R^n are defined for all $n \in \mathbb{Z}$, and such that for any $(x, y) \in R$ and any $(X, Y) \in \Omega_R$, both x and X are irrational; see the discussion preceding Proposition 1.6.18. Since $G_\alpha([\alpha - 1, \alpha)\backslash\mathbb{Q}) \subset [\alpha - 1, \alpha)\backslash\mathbb{Q}$, we shall in fact show that $(\Omega_R, \mathcal{B}, \bar{\nu}_R, \tau_R)$ is the natural extension of $([\alpha - 1, \alpha), \mathcal{B}, \rho_\alpha, G_\alpha)$ restricted to $[\alpha - 1, \alpha)\backslash\mathbb{Q}$, which we denote $([\alpha - 1, \alpha)\backslash\mathbb{Q}, \mathcal{B}, \rho_\alpha, G_\alpha)$

To distinguish the Borel σ-algebras restricted to Ω_R and $[\alpha - 1, \alpha)\backslash\mathbb{Q}$, we shall denote these by $\mathcal{C}$ and $\mathcal{D}$, respectively. Notice that $([\alpha-1,\alpha)\backslash\mathbb{Q}, \mathcal{D}, \rho_\alpha, G_\alpha)$ is non-invertible and $(\Omega_R, \mathcal{C}, \bar{\nu}_R, \tau_R)$ is invertible. We will show (i) that $([\alpha-1,\alpha)\backslash\mathbb{Q}, \mathcal{D}, \rho_\alpha, G_\alpha)$ is a factor of $(\Omega_R, \mathcal{C}, \bar{\nu}_R, \tau_R)$ with factor map $\pi_X : \Omega_R \to [\alpha-1,\alpha)\backslash\mathbb{Q}$ being the projection onto the first coordinate, and (ii) that the factor map π_X satisfies

$$\bigvee_{n=0}^{\infty} \tau_R^n \circ \pi_X^{-1}(\mathcal{D}) = \mathcal{C},$$

where $\bigvee_{n=0}^{\infty} \tau_R^n \circ \pi_X^{-1}(\mathcal{D})$ is the smallest σ-algebra containing each σ-algebra $\tau_R^n \circ \pi_X^{-1}(\mathcal{D}),\ n \geq 0$.

(i) We must show that $\pi_X : \Omega_R \to [\alpha - 1, \alpha)\backslash\mathbb{Q}$ is measurable, surjective, and satisfies $\pi_X \circ \tau_R = G_\alpha \circ \pi_X$ and $\bar{\nu}_R \circ \pi_X^{-1} = \rho_\alpha$. Certainly π_X is measurable, since for any Borel set $A \in \mathcal{D}$, $\pi_X^{-1}(A) = (A \times [0,1]) \cap \Omega_R \in \mathcal{C}$ is a Borel set in Ω_R. For surjectivity, suppose α has RCF-expansion $\alpha = [0; \alpha_1, \alpha_2, \dots]$, and let $z = (x, y) \in H_1$ with $x = [0; a_1, a_2, \dots] \notin \mathbb{Q}$ and $y = [0; 1, b, b, b, \dots]$ for some $b > \alpha_1$. Then

$$\mathcal{F}_{H_1}^{-1}(z) = ([0; b, a_1, a_2, \dots], [0; 1, b, b, \dots]) \in [0, \alpha) \times [1/2, 1],$$

so $k(z) = 1$ is odd and $z \in A$. Similarly, $k(\mathcal{F}_{H_1}^{-n}(z)) = 1$ for all $n \geq 0$, so $\mathcal{F}_{H_1}^{-n}(z) \in A$ for all $n \geq 0$. This—together with Corollary 1.7.9—implies that $\mathcal{F}_R^n(z) \in R$ is defined for all $n \in \mathbb{Z}$. Since $x \in [0,1]\backslash\mathbb{Q}$ was arbitrary, (1.71) gives $\pi_X(\Omega_R) = \pi_X(\varphi_R(R)) = [\alpha - 1, \alpha)\backslash\mathbb{Q}$, i.e., π_X is surjective.

Next, we show $\pi_X \circ \tau_R = G_\alpha \circ \pi_X$. Let $(X, Y) = (X(z), Y(z)) \in \Omega_R$, where $z = (x, y) \in R$, and notice from (1.71) that

$$X = \begin{cases} x, & x < \alpha, \\ x - 1, & x \geq \alpha. \end{cases}$$

Moreover, $x < \alpha$ if and only if $X > 0$, and $x \geq \alpha$ if and only if $X < 0$. These observations, together with Theorem 1.6.15

and Lemma 1.7.10, give

$$\begin{aligned}\pi_X \circ \tau_R(X,Y) =& \frac{\alpha_R(z)}{X} - \beta_R(z)\\ =& \begin{cases} \frac{1}{X} - \left\lfloor \frac{1}{x} + 1 - \alpha \right\rfloor, & x < \alpha, \\ -\frac{1}{X} - \left\lfloor \frac{1}{1-x} + 1 - \alpha \right\rfloor, & x \geq \alpha, \end{cases}\\ =& \begin{cases} \frac{1}{X} - \left\lfloor \frac{1}{X} + 1 - \alpha \right\rfloor, & X > 0, \\ -\frac{1}{X} - \left\lfloor -\frac{1}{X} + 1 - \alpha \right\rfloor, & X < 0, \end{cases}\\ =& \frac{1}{|X|} - \left\lfloor \frac{1}{|X|} + 1 - \alpha \right\rfloor\\ =& G_\alpha \circ \pi_X(X,Y)\end{aligned}$$

as desired. Lastly, notice that for any Borel set $A \in \mathcal{D}$, τ_R-invariance of $\bar{\nu}_R$ gives

$$\bar{\nu}_R \circ \pi_X^{-1}(G_\alpha^{-1}(A)) = \bar{\nu}_R \circ \tau_R^{-1}(\pi_X^{-1}(A)) = \bar{\nu}_R \circ \pi_X^{-1}(A),$$

so $\bar{\nu}_R \circ \pi_X^{-1}$ is an absolutely continuous, G_α-invariant probability measure. Uniqueness of ρ_α implies $\bar{\nu}_R \circ \pi_X^{-1} = \rho_\alpha$. Thus $([\alpha - 1, \alpha), \mathcal{D}, \rho_\alpha, G_\alpha)$ is a factor of $(\Omega_R, \mathcal{C}, \bar{\nu}_R, \tau_R)$.

(ii) We now show that

$$\bigvee_{n=0}^{\infty} \tau_R^n \circ \pi_X^{-1}(\mathcal{D}) = \mathcal{C}.$$

The forward inclusion follows from measurability of π_X and τ_R^{-1}, so it suffices to show the backward inclusion. For this, it suffices to show that every element of a generating set of the Borel σ-algebra $\mathcal{C}$ on Ω_R can be written as $\tau_R^k \circ \pi_X^{-1}(D)$ for some $D \in \mathcal{D}$ and $k \geq 0$. By Proposition 1.6.18, $\mathcal{C}$ is generated by the sets

$$\begin{aligned}C =& \Delta(0/1; \alpha_1/\beta_1, \alpha_2/\beta_2, \ldots, \alpha_n/\beta_n)\\ & \times \Delta(0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \alpha_{-1}/\beta_{-2}, \ldots, \alpha_{-(m-1)}/\beta_{-m})\end{aligned}$$

containing all points $(X(z), Y(z)) \in \Omega_R$ for which

$$\alpha_R(z_j^R) = \alpha_{j+1} \quad \text{and} \quad \beta_R(z_k^R) = \beta_{k+1}$$

for all $-m \le j \le n-1$ and $-m-1 \le k \le n-1$.

Let $D \in \mathcal{D}$ be the set of irrationals $X \in [\alpha-1, \alpha)$ for which

$$\operatorname{sgn}(G_\alpha^j(X)) = \alpha_{j-m} \quad \text{and} \quad \left\lfloor \frac{1}{|G_\alpha^k(X)|} + 1 - \alpha \right\rfloor = \beta_{k-m}$$

for all $1 \le j \le n+m$ and $0 \le k \le n+m$. Let $X \in [\alpha-1, \alpha) \setminus \mathbb{Q}$, $(X, Y) = (X(z), Y(z)) \in \pi_X^{-1}(\{X\})$, and $z_k^R = \mathcal{F}_R^k(z)$ for all $k \in \mathbb{Z}$. Using the fact that $G_\alpha \circ \pi_X = \pi_X \circ \tau_R$, Equations (1.58), (1.59) and Lemma 1.7.10 give

$$\operatorname{sgn}(G_\alpha^k(X)) = \alpha_R(z_k^R)$$

and

$$\left\lfloor \frac{1}{|G_\alpha^k(X)|} + 1 - \alpha \right\rfloor = \beta_R(z_k^R), \quad k \ge 0,$$

so $\pi_X^{-1}(D)$ is the set of points $(X(z), Y(z)) \in \Omega_R$ such that

$$\alpha_R(z_j^R) = \alpha_{j-m} \quad \text{and} \quad \beta_R(z_k^R) = \beta_{k-m}$$

for all $1 \le j \le n+m$ and $0 \le k \le n+m$. By Corollary 1.6.19, this is the set of points of the form

$$\begin{aligned} X = X(z) &= [0/1; \alpha_R(z_0^R)/\beta_R(z_0^R), \alpha_R(z_1^R)/\beta_R(z_1^R), \dots] \\ &= [0/1; \alpha_R(z_0^R)/\beta_{-m}, \alpha_{-(m-1)}/\beta_{-(m-1)}, \dots, \\ &\qquad \alpha_n/\beta_n, \alpha_R(z_{n+m+1}^R)/\beta_R(z_{n+m+1}^R), \dots] \end{aligned}$$

and

$$Y = Y(z) = [0/1; 1/\beta_R(z_{-1}^R), \alpha_R(z_{-1}^R)/\beta_R(z_{-2}^R), \dots].$$

Since $(X^R_{m+1}, Y^R_{m+1}) = \tau_R^{m+1}(X, Y)$ is of the form

$$X^R_{m+1} = [0/1; \alpha_1/\beta_1, \dots, \alpha_n/\beta_n, \\ \alpha_R(z^R_{n+m+1})/\beta_R(z^R_{n+m+1}), \dots]$$

and

$$Y^R_{m+1} = [0/1; 1/\beta_0, \alpha_0/\beta_{-1}, \dots, \alpha_{-(m-1)}/\beta_{-m}, \\ \alpha_R(z^R_0)/\beta_R(z^R_{-1}), \dots],$$

we have $\tau_R^{m+1} \circ \pi_X^{-1}(D) = C$. □

Remark 1.7.12. Recall from the end of §1.3.2 that there are several open questions about Nakada's α-CFs, including explicit descriptions of the values of the entropy $h(G_\alpha)$ for $\alpha < g^2$, $g = (\sqrt{5}-1)/2$, and of the densities of the invariant measures ρ_α ([120]). It is also open to explicitly compute the so-called *Legendre constant* for $\alpha < g^2$ ([60, 158]; see also Remark 1.8.14 below).

Each of these questions may be answered with an understanding of the domain of the natural extension of $([\alpha-1, \alpha), \mathcal{B}, \rho_\alpha, G_\alpha)$; see, e.g., Theorem 1.5.6 for the entropy. To date, however, the description of this domain has proven to be unmanageable for these tasks. Our new description of the natural extension $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ could bring many of these questions within reach. Indeed, by (1.67), in order to understand R is suffices to understand the set $A \subset H_1$. We hope to return to these questions in subsequent work and suspect that matching (see §1.3.2 above and Chapter 2 below) will play a crucial role in their resolution.

1.8 A unifying metrical theory

One important reason to study RCF-expansions is that their convergents yield rational approximations of irrationals of 'very high quality.' By this, we mean that the RCF-convergents $(p_n/q_n)_{n\geq 0}$ of an

irrational number x have the so-called *best approximation property*: for any $n \geq 1$, and any $p/q \in \mathbb{Q}$ in lowest terms with $0 < q \leq q_n$,

$$\left| x - \frac{p_n}{q_n} \right| \leq \left| x - \frac{p}{q} \right|;$$

see, e.g., [15]. This result was essentially already known to Christiaan Huygens when he constructed his planetarium in 1680 at the request of Jean-Baptiste Colbert (see Chapter IV in [167]). Later in this section (§1.8.6) we will revisit an old result by Legendre and some of its refinements and analogues for RCF-convergents and mediants. Legendre's result states that if p/q is a rational number in lowest terms with q positive, and

$$\left| x - \frac{p}{q} \right| < \frac{1}{2q^2}, \tag{1.75}$$

then there exists an $n \in \mathbb{N}$ such that $p = p_n$ and $q = q_n$. In other words, in order to approximate an irrational x 'well' by a rational p/q ('well' in the sense that (1.75) holds) one is bound to find a RCF-convergent of x. In view of this, for over a century so-called RCF-*approximation coefficients* θ_n, defined by

$$\theta_n = \theta_n(x) \coloneqq q_n \left| q_n x - p_n \right|, \qquad \text{for } n \in \mathbb{N}, \tag{1.76}$$

have been studied. Independently, Doeblin and Lenstra conjectured the distribution for almost all x of the sequence $(\theta_n(x))_{n \in \mathbb{N}}$. The proof of this so-called Doeblin–Lenstra conjecture by Bosma, Jager and Wiedijk in [21] (cf. the approach of Knuth in [110]; see also Corollary 1.8.13 below) lead to many new results in Diophantine approximation.

We also mention several classical results in the theory of RCFS; these statements hold for all irrational x and all $n \geq 1$:

$$\min\{\theta_{n-1}(x), \theta_n(x)\} < \frac{1}{2} \quad \text{(Vahlen, 1895 [185]; see also §1.8.5)}$$

and

$$\min\{\theta_{n-1}(x), \theta_n(x), \theta_{n+1}(x)\} < \frac{1}{\sqrt{5}} \quad \text{(Borel, 1903 [17])}. \quad (1.77)$$

Borel's result is a corollary of the following statement of Bagemihl and McLaughlin from 1966 ([7]):

$$\min\{\theta_{n-1}(x), \theta_n(x), \theta_{n+1}(x)\} < \frac{1}{\sqrt{a_{n+1}^2 + 4}}.$$

Related to this last result is the result by Tong from 1983 ([184]), which states that

$$\max\{\theta_{n-1}(x), \theta_n(x), \theta_{n+1}(x)\} > \frac{1}{\sqrt{a_{n+1}^2 + 4}}.$$

These results can easily be derived using the natural extension of the Gauss map (see, e.g., [50, 98]), and thus by Theorem 1.5.5 are attainable through the setup of this chapter.

Another important and classical result—proven in 1936 by Paul Lévy ([127])—concerns the exponential growth rate of the denominators of RCF convergents. It states that for almost all $x \in (0, 1)$,

$$\lim_{n\to\infty} \frac{1}{n} \log q_n = \frac{\pi^2}{12 \log 2}. \quad (1.78)$$

As a corollary of this and the fact that $1/2q_nq_{n+1} < |x - p_n/q_n| < 1/q_nq_{n+1}$, it follows that for almost every $x \in (0, 1)$,

$$\lim_{n\to\infty} \frac{1}{n} \log \left| x - \frac{p_n}{q_n} \right| = -\frac{\pi^2}{6 \log 2}. \quad (1.79)$$

In this section, we revisit and extend many of these classical metrical results, their refinements and their analogues to subsequences $(u_n^R/s_n^R)_{n\geq 0}$ of RCF-convergents and mediants. In light of the theory built in §1.6, these subsequences are in fact convergents of contracted Farey expansions, but this viewpoint will not be stressed

in this section. We simply consider $(u_n^R/s_n^R)_{n\geq 0}$ as subsequences of RCF-convergents and mediants and consider the approximation properties of these subsequences. Later, in §1.9, we shall combine the perspectives of §1.6 and the current section to define a new collection of GCF-algorithms producing contracted Farey expansions with arbitrarily good approximation and convergence properties.

1.8.1 On the (relative)-equidistribution of orbits

While Ito's natural extension $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ is two-dimensional, one of our main interests in this system is to understand properties of irrationals $x \in (0,1)$. In particular, we shall be interested in $\mathcal{F}$-orbits of points $z = (x,1)$ beginning from the 'top edge' of Ω. Since $[0,1] \times \{1\}$ is a $\bar{\mu}$-null set, it is not necessarily true that *any* point of the form $z = (x,1)$ ever enters a given inducible region R under iterates of $\mathcal{F}$. In order to make meaningful statements about such orbits, we must impose an extra condition on the inducible sets that we consider.

Definition 1.8.1. A Borel subset $R \subset \Omega$ is a *$\bar{\mu}$-continuity set* if $\bar{\mu}(\partial R) = 0$.

Proposition 1.8.2. *For any $\bar{\mu}$-continuity, inducible set R, the $\mathcal{F}$-orbit of $(x,1)$ enters R infinitely often for a.e.*[15] *irrational $x \in (0,1)$.*

Proof. Let $z = (x,y) \in \Omega$ with $x = [0; a_1, a_2, \dots] \notin \mathbb{Q}$, and set $z_n = (x_n, y_n) = \mathcal{F}^n(z)$. Equation (1.36) implies that for $n \geq a_1$, y_n belongs to the cylinder of points in $[0,1]$ whose RCF-expansions begin with $[0; \lambda_n + 1, a_{j_n}, \dots, a_2, \dots]$. The Euclidean diameter of this cylinder is no greater than the reciprocal of the j_n^{th} Fibonacci number squared and thus goes to 0 uniformly in y as n goes to infinity (see also [29]). Now let E be the set of irrationals $x \in (0,1)$ for which $z_n' := \mathcal{F}^n(x,1)$ enters R at most finitely often. Since R is a $\bar{\mu}$-continuity, inducible set, $\bar{\mu}(\text{int}(R)) > 0$ and there exists

[15] Unless otherwise stated, *almost every* (a.e.) statements are w.r.t. Lebesgue measure.

some point $c \in \mathrm{int}(R)$ and some $\delta > 0$ such that the Euclidean ball $B_\delta(c)$ of radius δ centered at c is contained in $\mathrm{int}(R)$. If E has positive Lebesgue measure, then also $\bar{\mu}(E \times [0,1]) > 0$. Since $\mathcal{F}$ is conservative ([29]), $\bar{\mu}$–a.e. point $z = (x,y) \in E \times [0,1]$ enters $B_{\delta/2}(c)$ infinitely often. The observation on the diameter of the cylinder above implies that for n large enough, $|z_n - z'_n| < \delta/2$, and hence z'_n enters $B_\delta(c) \subset \mathrm{int}(R)$ infinitely often—a contradiction. □

In 1996, Brown and Yin employed the Chacon–Ornstein ergodic theorem to derive metrical results on the system $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ (Theorem 2 of [29]):

Theorem 1.8.3 (Brown–Yin, 1996 [29])**.** *For any $f, g \in L^1(\bar{\mu})$ with $g \geq 0$ and $\int g d\bar{\mu} \neq 0$,*

$$\lim_{n\to\infty} \frac{\sum_{k=0}^{n-1} f(z_k)}{\sum_{k=0}^{n-1} g(z_k)} = \frac{\int f d\bar{\mu}}{\int g d\bar{\mu}} \quad \text{for } \bar{\mu}\text{–a.e. } z \in \Omega, \tag{1.80}$$

where $z_k = \mathcal{F}^k(z)$, $k \geq 0$.

Under certain Lipschitz conditions on f and g, Brown and Yin are able to replace $\bar{\mu}$–a.e. $z \in \Omega$ with $z = (x,1)$ for a.e. $x \in (0,1)\backslash\mathbb{Q}$ (see Theorem 3 of [29]). The following theorem—which does not require Lipschitz conditions but instead replaces f and g by indicator functions—may be seen as an analogue of an important result of Jager for the natural extension of the Gauss map, which states that for almost every irrational $x \in (0,1)$, the $\mathcal{G}$-orbit of $(x,0)$ is $\bar{\nu}_G$-equidistributed over Ω (see Theorem 3 of [95]). In fact, by Theorem 1.5.5 above and Corollary 1.8.5 below, the following statement may be read as a generalisation of Jager's result:

Theorem 1.8.4. *For almost every $x \in (0,1)\backslash\mathbb{Q}$, the $\mathcal{F}$-orbit of $z = (x,1)$ is $\bar{\mu}$-*relatively equidistributed[16] *over Ω. That is, for*

[16]This notion of $\bar{\mu}$-relative equidistribution differs slightly from that studied by Gerl in [77] and [78]. In particular, Gerl required a Radon measure, but $\bar{\mu}$ is not locally finite.

almost all $x \in (0,1)\backslash\mathbb{Q}$,

$$\lim_{n\to\infty} \frac{\sum_{k=0}^{n-1} \mathbf{1}_S(z_k)}{\sum_{k=0}^{n-1} \mathbf{1}_R(z_k)} = \frac{\bar{\mu}(S)}{\bar{\mu}(R)}$$

for any $\bar{\mu}$*-null or* $\bar{\mu}$*-continuity, proper, inducible set* $S \subset \Omega$ *and any* $\bar{\mu}$*-continuity, proper, inducible* $R \subset \Omega$*, where* $z_k = \mathcal{F}^k(z)$ *with* $z = (x, 1)$*.*

Proof. Let $\mathcal{U}$ be a countable base for the subspace topology on $\Omega \subset \mathbb{R}^2$, and let $\mathcal{C}$ denote the countable collection of all finite unions of finite $\bar{\mu}$-measure elements of $\mathcal{U}$, along with the empty set. Moreover, for $A, B \in \mathcal{C}$ with $\bar{\mu}(B) > 0$, let $E(A,B) \subset (0,1)\backslash\mathbb{Q}$ denote the set of irrational x for which there is *no* $y \in [0,1]$ satisfying

$$\lim_{n\to\infty} \frac{\sum_{k=0}^{n-1} \mathbf{1}_A(\mathcal{F}^k(x,y))}{\sum_{k=0}^{n-1} \mathbf{1}_B(\mathcal{F}^k(x,y))} = \frac{\bar{\mu}(A)}{\bar{\mu}(B)}. \tag{1.81}$$

Theorem 1.8.3 implies that $E(A,B)$ is a Lebesgue-null set: otherwise $E(A,B) \times [0,1]$ is a set of positive $\bar{\mu}$-measure for which (1.80) does not hold. Since $\mathcal{C}$ is countable, the union E of all such $E(A,B)$ is also a Lebesgue-null set. Hence, for every $x \in (0,1)\backslash(E \cup \mathbb{Q})$ it follows that for all $A, B \in \mathcal{C}$ with $\bar{\mu}(B) > 0$, there is some $y \in [0,1]$ for which (1.81) holds.

We claim that for any $\bar{\mu}$-null or $\bar{\mu}$-continuity, proper, inducible set $S \subset \Omega$ and any $\delta > 0$, there exist $S_{+\delta}, S_{-\delta} \in \mathcal{C}$ such that

(i) $S_{-\delta} \subset S \subset S_{+\delta}$,

(ii) $\bar{\mu}(S_{+\delta}\backslash S),\ \bar{\mu}(S\backslash S_{-\delta}) < \delta$, and

(iii) $d(S, \Omega\backslash S_{+\delta}) > 0$, and if $\bar{\mu}(S) > 0$, also $d(S_{-\delta}, \Omega\backslash S) > 0$,

where for $A, B \in \mathcal{B}$,

$$d(A,B) \coloneqq \inf\left\{|a-b| \;\middle|\; a \in A,\ b \in B\right\}$$

denotes the Euclidean distance between the sets A and B. Indeed, fix $S \subset \Omega$ and $\delta > 0$ as above. By the regularity of $\bar{\mu}$, there exists an open cover $\{U_i\}_{i\in I} \subset \mathcal{U}$ of the closure $\bar{S}$ of S for which $\bar{\mu}(\cup_{i\in I}U_i\backslash\bar{S}) < \delta$. Since $\bar{S}$ is compact, $\{U_i\}_{i\in I}$ has some finite subcover, the union of whose elements we denote by $S_{+\delta}$. Note, then, that $S \subset S_{+\delta} \subset \cup_{i\in I}U_i$, and since the boundary ∂S is a $\bar{\mu}$-null set, $\bar{\mu}(S_{+\delta}\backslash S) \leq \bar{\mu}(\cup_{i\in I}U_i\backslash\bar{S}) < \delta$. Moreover, since $\bar{S}$ and $\Omega\backslash S_{+\delta}$ are compact and disjoint, the distance between them is strictly positive, and thus the distance between $S \subset \bar{S}$ and $\Omega\backslash S_{+\delta}$ is strictly positive. Thus $S_{+\delta}$ satisfies each of the properties of the claim.

If $\bar{\mu}(S) = 0$, set $S_{-\delta} \coloneqq \varnothing \in \mathcal{C}$, which trivially satisfies the claim. Now suppose $\bar{\mu}(S) > 0$. Since $\bar{\mu}(\partial S) = 0$, we have $\bar{\mu}(\mathrm{int}(S)) > 0$. Again by regularity of $\bar{\mu}$, there exists some compact subset K of $\mathrm{int}(S)$ with $\bar{\mu}(\mathrm{int}(S)\backslash K) < \delta$. Since Ω is normal and $\Omega\backslash\mathrm{int}(S)$ and K are closed and disjoint, there exist open, disjoint sets $U,\ V \in \mathcal{B}$ containing $\Omega\backslash\mathrm{int}(S)$ and K, respectively. Let $\{V_j\}_{j\in J}$ be a collection of open sets from the countable base $\mathcal{U}$ whose union equals V. This collection of sets forms an open cover of the compact set K; let $S_{-\delta} \in \mathcal{C}$ be the union of elements of a finite subcover. We then have $S_{-\delta} \subset \mathrm{int}(S) \subset S$ and, since $\bar{\mu}(\partial S) = 0$, also $\bar{\mu}(S\backslash S_{-\delta}) \leq \bar{\mu}(\mathrm{int}(S)\backslash K) < \delta$. Furthermore, since $S_{-\delta} \subset \Omega\backslash U$ and $\Omega\backslash S \subset \Omega\backslash\mathrm{int}(S)$, we have $d(S_{-\delta}, \Omega\backslash S) \geq d(\Omega\backslash U, \Omega\backslash\mathrm{int}(S))$. But $\Omega\backslash U$ and $\Omega\backslash\mathrm{int}(S)$ are compact and disjoint, so the distance between them is again strictly positive. Thus $S_{-\delta}$ also satisfies the desired properties.

For $S \subset \Omega$ and $\delta > 0$ as above, let

$$d_S(\delta) \coloneqq \begin{cases} d(S, \Omega\backslash S_{+\delta}), & \bar{\mu}(S) = 0, \\ \min\{d(S, \Omega\backslash S_{+\delta}), d(S_{-\delta}, \Omega\backslash S)\}, & \bar{\mu}(S) > 0. \end{cases}$$

Recall from the proof of Proposition 1.8.2 that for any $x \in (0,1)\backslash\mathbb{Q}$, the distances between $\mathcal{F}^n(x,1)$ and $\mathcal{F}^n(x,y)$ approach zero uniformly in $y \in [0,1]$. Thus for any $x \in (0,1)\backslash\mathbb{Q}$ there exists some $n_S(\delta) \in \mathbb{N}$ such that for any $y \in [0,1]$ and $n \geq n_S(\delta)$,

$$|\mathcal{F}^n(x,1) - \mathcal{F}^n(x,y)| < d_S(\delta).$$

In particular, by definition of $d_S(\delta)$, for any $n \geq n_S(\delta)$,

$$\mathcal{F}^n(x,1) \in S \quad \text{implies} \quad \mathcal{F}^n(x,y) \in S_{+\delta},$$

and if $\bar{\mu}(S) > 0$,

$$\mathcal{F}^n(x,y) \in S_{-\delta} \quad \text{implies} \quad \mathcal{F}^n(x,1) \in S.$$

Now let $x \in (0,1)\backslash(E \cup \mathbb{Q})$, $S \subset \Omega$ a $\bar{\mu}$-null or $\bar{\mu}$-continuity, proper, inducible set and $R \subset \Omega$ a $\bar{\mu}$-continuity, proper, inducible set. For $\delta > 0$, let $S_{\pm\delta}, R_{\pm\delta} \in \mathcal{C}$ be the sets constructed above. When δ is sufficiently small, the observations at the beginning of the proof imply that there exist $y, y' \in [0,1]$ such that

$$\begin{aligned}
\frac{\bar{\mu}(S_{-\delta})}{\bar{\mu}(R_{+\delta})} &= \liminf_{n\to\infty} \frac{\sum_{k=0}^{n-1} \mathbf{1}_{S_{-\delta}}(\mathcal{F}^k(x,y))}{\sum_{k=0}^{n-1} \mathbf{1}_{R_{+\delta}}(\mathcal{F}^k(x,y))} \\
&\leq \liminf_{n\to\infty} \frac{\sum_{k=0}^{n-1} \mathbf{1}_{S}(\mathcal{F}^k(x,1))}{\sum_{k=0}^{n-1} \mathbf{1}_{R}(\mathcal{F}^k(x,1))} \\
&\leq \limsup_{n\to\infty} \frac{\sum_{k=0}^{n-1} \mathbf{1}_{S}(\mathcal{F}^k(x,1))}{\sum_{k=0}^{n-1} \mathbf{1}_{R}(\mathcal{F}^k(x,1))} \\
&\leq \limsup_{n\to\infty} \frac{\sum_{k=0}^{n-1} \mathbf{1}_{S_{+\delta}}(\mathcal{F}^k(x,y'))}{\sum_{k=0}^{n-1} \mathbf{1}_{R_{-\delta}}(\mathcal{F}^k(x,y'))} = \frac{\bar{\mu}(S_{+\delta})}{\bar{\mu}(R_{-\delta})}.
\end{aligned}$$

By properties (i) and (ii) above, taking $\delta \to 0$ gives the result. □

As a corollary, we obtain the following:

Corollary 1.8.5. *For almost every $x \in (0,1)\backslash\mathbb{Q}$, the $\mathcal{F}_R$-orbit of $z = (x,1)$ is $\bar{\mu}_R$-equidistributed for any $\bar{\mu}$-continuity, proper, inducible $R \subset \Omega$. That is, for almost every $x \in (0,1)\backslash\mathbb{Q}$,*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbf{1}_S(z_k^R) = \bar{\mu}_R(S)$$

for any $\bar{\mu}$-continuity, proper, inducible $R \subset \Omega$ and any $\bar{\mu}$-continuity $S \subset R$, where $z_k^R = \mathcal{F}_R^k(z)$ with $z = (x,1)$.

Proof. When $S \subset R$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbf{1}_S(\mathcal{F}_R^k(z)) = \lim_{n\to\infty} \frac{\sum_{k=0}^{n-1} \mathbf{1}_S(\mathcal{F}_R^k(z))}{\sum_{k=0}^{n-1} \mathbf{1}_R(\mathcal{F}_R^k(z))}$$
$$= \lim_{n\to\infty} \frac{\sum_{k=0}^{N_n^R-1} \mathbf{1}_S(\mathcal{F}^k(z))}{\sum_{k=0}^{N_n^R-1} \mathbf{1}_R(\mathcal{F}^k(z))},$$

where $N_n^R = N_n^R(z)$ and $z = (x, 1)$. The result follows from Theorem 1.8.4 and the fact that $\bar{\mu}_R(S) = \bar{\mu}(S)/\bar{\mu}(R)$. □

Remark 1.8.6. The $\bar{\mu}$-continuity condition on R, namely $\bar{\mu}(\partial R) = 0$, is necessary to avoid 'pathological' counter-examples to the above result. Indeed, since the second coordinate of $z_n = \mathcal{F}^n(x, 1)$, is rational for all n, if $R \subset [0, 1] \times ([0, 1] \backslash \mathbb{Q})$, then z_n^R is never defined. Using similar ideas, one can easily construct $R \in \mathcal{B}$ with $0 < \bar{\mu}(R) < \infty$ and $\bar{\mu}(\partial R)$, $\bar{\mu}(\text{int}(R)) > 0$ so that z_n almost surely enters R infinitely often but the conclusion of Corollary 1.8.5 is false.

1.8.2 Extending Lévy's Theorem

In this subsection, we extend Lévy's results (1.78) and (1.79) to subsequences of RCF-convergent and mediants. Let $R \subset \Omega$ be a $\bar{\mu}$-continuity, proper, inducible set and $z = (x, 1)$. Recall from (1.47) that $s_n^R = \lambda_N q_{j_N} + q_{j_N-1}$ where $N = N_n^R$. The recurrence relations (1.8) together with the fact that $0 \leq \lambda_N < a_{j_N+1}$ give

$$q_{j_N-1} \leq s_n^R < q_{j_N+1}, \qquad N = N_n^R. \tag{1.82}$$

The ratio $j_{N_n^R}/n$ thus measures the size of the denominators s_n^R relative to the denominators of RCF-convergents. As n tends toward ∞, this ratio converges a.s. to a number inversely proportional to the $\bar{\mu}$-measure of the region R:

Lemma 1.8.7. *For almost every $x \in (0, 1)$,*

$$\lim_{n\to\infty} \frac{j_{N_n^R}}{n} = \frac{\log 2}{\bar{\mu}(R)}$$

for any $\bar{\mu}$-continuity, proper, inducible $R \subset \Omega$.

Proof. Let $x \in (0,1) \setminus \mathbb{Q}$, and notice from (1.22) that $j_n = j_n(x)$ may be written as

$$j_n = \#\{1 \le k \le n \mid \varepsilon_k = 1\}.$$

That is, j_n counts the number of times that the F-orbit of $x \in (0,1)$ visits the region $(1/2, 1]$ in its first n steps $x, F(x), \ldots, F^{n-1}(x)$. Equivalently, j_n counts the number of times that the $\mathcal{F}$-orbit $(z_k)_{k\ge 0}$ of $z = (x, 1)$ visits $V_1 \subset \Omega$ in its first n steps:

$$j_n = \#\{1 \le k \le n \mid z_k \in V_1\} = \sum_{k=0}^{n-1} \mathbf{1}_{V_1}(z_k).$$

By Theorem 1.8.4, we have that for almost every $x \in (0,1)$,

$$\lim_{n\to\infty} \frac{j_{N_n^R}}{n} = \lim_{n\to\infty} \frac{\sum_{k=0}^{N_n^R - 1} \mathbf{1}_{V_1}(z_k)}{\sum_{k=0}^{N_n^R - 1} \mathbf{1}_{R}(z_k)} = \frac{\bar{\mu}(V_1)}{\bar{\mu}(R)} = \frac{\log 2}{\bar{\mu}(R)}$$

for any $\bar{\mu}$-continuity, proper, inducible $R \subset \Omega$. □

This lemma allows us to extend Lévy's results:

Theorem 1.8.8. *For almost every $x \in (0,1)\setminus\mathbb{Q}$,*

$$\textit{(i)} \quad \lim_{n\to\infty} \frac{1}{n} \log s_n^R = \frac{\pi^2}{12\bar{\mu}(R)}$$

and

$$\textit{(ii)} \quad \lim_{n\to\infty} \frac{1}{n} \log \left| x - \frac{u_n^R}{s_n^R} \right| = -\frac{\pi^2}{6\bar{\mu}(R)}$$

for any $\bar{\mu}$-continuity, proper, inducible $R \subset \Omega$.

Proof. Let x belong to the full-measure subset of $(0,1)\setminus\mathbb{Q}$ for which Lemma 1.8.7 and (1.78) both hold, and let $R \subset \Omega$ be a $\bar{\mu}$-continuity,

proper, inducible subregion. Taking logarithms and dividing by n in (1.82), we have

$$\begin{aligned}\frac{j_{N_n^R}-1}{n}\frac{1}{j_{N_n^R}-1}\log q_{j_{N_n^R}-1} \leq & \frac{1}{n}\log s_n^R \\ < & \frac{j_{N_n^R}+1}{n}\frac{1}{j_{N_n^R}+1}\log q_{j_{N_n^R}+1}.\end{aligned}$$

By Lemma 1.8.7 and Lévy's result (1.78), the limits as $n \to \infty$ of both the left- and right-hand sides of the previous line equal $\frac{\pi^2}{12\bar{\mu}(R)}$, proving (i).

For (ii), notice that (1.26) and (1.47) give

$$\left| x - \frac{u_n^R}{s_n^R} \right| = \frac{1}{s_n^R(s_n^R x_n^R + r_n^R)},$$

where $z_n^R = (x_n^R, y_n^R) = \mathcal{F}_R^n(x,1)$. Since $x_n^R \in (0,1)$, we have

$$\frac{1}{s_n^R(s_n^R + r_n^R)} < \left| x - \frac{u_n^R}{s_n^R} \right| < \frac{1}{s_n^R r_n^R}.$$

Using (1.82) and the facts that $r_n^R = q_{j_{N_n^R}}$ and $q_j < q_{j+1}$ for all j, one obtains from the previous line

$$\frac{1}{2q^2_{j_{N_n^R}+1}} < \left| x - \frac{u_n^R}{s_n^R} \right| < \frac{1}{q^2_{j_{N_n^R}-1}}.$$

Taking logarithms and dividing by n gives

$$\frac{q_{j_{N_n^R}+1}}{n}\frac{1}{q_{j_{N_n^R}+1}}\left(-\log 2 - 2\log q_{j_{N_n^R}+1}\right) < \frac{1}{n}\log\left| x - \frac{u_n^R}{s_n^R} \right|$$

and

$$\frac{1}{n}\log\left| x - \frac{u_n^R}{s_n^R} \right| < \frac{q_{j_{N_n^R}-1}}{n}\frac{1}{q_{j_{N_n^R}-1}}\left(-2\log q_{j_{N_n^R}-1}\right).$$

Again from Lemma 1.8.7 and Lévy's result (1.78), the limits as $n \to \infty$ of both the lower- and upper-bounds equal $-\frac{\pi^2}{6\bar{\mu}(R)}$, proving (ii). □

By specifying certain regions R, Theorem 1.8.8 realises a number of analogues of Lévy's results proven for particular subsequences of RCF-convergents and mediants. For instance, letting $S \subset \Omega$ be a singularisation area, $\Delta = \Omega \backslash S$ and $R = \psi^{-1}(\Delta) = \varphi_{H_1}^{-1} \circ \mathcal{G}(\Delta)$ as in §1.7.2, one computes

$$\begin{aligned}\bar{\mu}(R) =& \bar{\mu}(H_1)\bar{\mu}_{H_1}(R) = \log(2)\bar{\mu}_{H_1}(\varphi_{H_1}^{-1} \circ \mathcal{G}(\Delta)) \\ =& \log(2)\bar{\nu}_G(\Delta) = \log(2)(1 - \bar{\nu}_G(S)).\end{aligned}$$

Thus Theorem 1.8.8 realises Corollary 4.15 of [115] on S-expansions.

Similarly, letting

$$R = \bigcup_{\lambda=0}^{k} H_{\lambda+1} \cup \bigcup_{a=1}^{k} V_a$$

for some $k \geq 1$, one computes $\bar{\mu}(R) = \log(2k+2)$. For this R, $(u_n^R/s_n^R)_{n\geq 0}$ consists of RCF-convergents and the first k and final k mediant convergents, and Theorem 1.8.8 realises Theorem 4.i and 4.ii of Brown and Yin ([29]). When $k = 1$, this example also gives earlier results of Ito (Propositions 3.1 and 3.3(ii) of [93]) and Jager (Theorem 2.11 of [97]) concerning RCF-convergents and nearest mediants.

Remark 1.8.9. In this latter example, the statements of [29, 93, 97] are made assuming the subsequences of RCF-convergents and mediants are arranged with increasing denominators. However, recall from (1.25) that the denominators of the sequence $(u_n/s_n)_{n\geq 0}$ of Farey convergents are not necessarily increasing, and hence the same is true of the subsequence $(u_n^R/s_n^R)_{n\geq 0}$. Proposition 1.10.1 in the appendix (§1.10) implies that the original results (with increasing denominators) do indeed follow from Theorem 1.8.8.

Remark 1.8.10. It should be mentioned that originally Lévy's result (and similar other results by Lévy and Khintchine) also gave a 'speed of convergence,' as its proof relied on probability theory and was derived from the Gauss-Kuzmin-Lévy Theorem; see, e.g., [90] for more details. As we use ergodic theory, such speeds cannot be given.

1.8.3 Approximation coefficients

Recall the RCF-approximation coefficients θ_n from (1.76). We extend this notion as follows: given an irrational $x \in \mathbb{R}$ and a rational $p/q \in \mathbb{Q}$ with $\gcd(p,q) = 1$ and $q > 0$, the *approximation coefficient* corresponding to x and p/q is

$$\Theta(x, p/q) \coloneqq q|qx - p|. \tag{1.83}$$

As $|x - p/q| = \Theta(x, p/q)/q^2$, approximation coefficients give a measure of how well the rational p/q approximates x, relative to the size of the denominator q. When $p/q = u_n/s_n = P_{n-1}/Q_{n-1}$ is the $(n-1)^{\text{st}}$ Farey convergent of x, we use the special notation

$$\Theta_n(x) \coloneqq \Theta(x, u_n/s_n).$$

Similarly, for an inducible subregion $R \subset \Omega$, we set

$$\Theta_n^R(x) \coloneqq \Theta(x, u_n^R/s_n^R), \tag{1.84}$$

where $u_n^R/s_n^R = u_n^R(z)/s_n^R(z)$ with $z = (x, 1)$.

The following result—which is central to the remainder of this section and §1.9 below—states that the nth approximation coefficient $\Theta_n(x)$ is computed explicitly in terms of the nth point z_n in the $\mathcal{F}$-orbit of $z = (x, 1)$ and thus depends on both the 'future' and the 'past' of the F-orbit of x (cf. Propositions 1.2 and 2.2 of [93]). Hence—although statements of later results regarding approximations coefficients are about $x \in (0, 1)$—the proofs shall exploit the two-dimensional system $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$. Define $h : \Omega \backslash \{(0,0)\} \to [0, \infty)$ by

$$h(z) \coloneqq \frac{1-y}{x+y-xy}, \quad z = (x, y). \tag{1.85}$$

Proposition 1.8.11. *For any inducible $R \subset \Omega$ and $n \geq 0$,*

$$\Theta_n(x) = h(z_n) \quad \textit{and} \quad \Theta_n^R(x) = h(z_n^R),$$

where $z_n = \mathcal{F}^n(x, 1)$ and $z_n^R = \mathcal{F}_R^n(x, 1)$.

Proof. Set $z_n = (x_n, y_n)$. Notice from Equations (1.31) and (1.32) that

$$y_n = A_{[n,0]} \cdot 1 = \frac{r_n}{s_n + r_n},$$

so

$$h(z_n) = \frac{1 - y_n}{x_n + y_n - x_n y_n} = \frac{1 - y_n}{(1 - y_n)x_n + y_n} = \frac{s_n}{s_n x_n + r_n}.$$

On the other hand, Equation (1.26) also gives

$$\Theta_n(x) = s_n^2 \left| x - \frac{u_n}{s_n} \right| = \frac{s_n}{s_n x_n + r_n}.$$

The second equality of the proposition statement now follows from the first:

$$\begin{aligned} \Theta_n^R(x) =& \Theta(x, u_n^R/s_n^R) = \Theta(x, u_{N_n^R}/s_{N_n^R}) \\ =& \Theta_{N_n^R}(x) = h(z_{N_n^R}) = h(z_n^R). \end{aligned}$$

□

1.8.4 Generalised Doeblin–Lenstra

Let $R \subset \Omega$ be a $\bar{\mu}$-continuity, proper, inducible set. In this subsection we consider the limiting distributions of the approximation coefficients $\Theta_n^R(x)$. Such limiting distributions have a long history, dating back to a conjecture in 1940 by Doeblin ([62]) and—independently—in the 1980s by Lenstra (see [50]) for RCF-convergents. The Doeblin–Lenstra conjecture (Corollary 1.8.13 below) was first proven in 1983 by Bosma, Jager and Wiedijk ([21]) using the natural extension of the Gauss map.

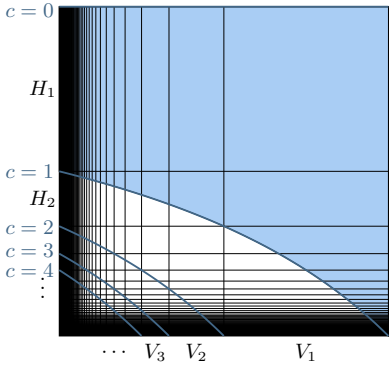


Figure 1.7: The curves $h(z) = c$ for $c \in \{0, 1, 2, 3, 4\}$. The region S_1 is the shaded region above the curve $h(z) = 1$.

For $c \in [0, \infty)$, let

$$S_c \coloneqq \{z \in \Omega \mid h(z) \leq c\}, \tag{1.86}$$

with $h(z)$ as in (1.85); see Figure 1.7. Corollary 1.8.5 and Proposition 1.8.11 allow us to calculate the limiting distributions of approximation coefficients $\Theta_n^R(x)$ in terms of the $\bar{\mu}_R$-measure of $S_c \cap R$:

Theorem 1.8.12. *For any $\bar{\mu}$-continuity, proper, inducible subregion $R \subset \Omega$, almost every $x \in (0,1)\backslash\mathbb{Q}$ satisfies*

$$\lim_{n\to\infty} \frac{1}{n} \#\left\{0 \leq k < n \mid \Theta_k^R(x) \leq c\right\} = \bar{\mu}_R(S_c \cap R)$$

for all $c \in [0, \infty)$.

Proof. With the stated assumptions, Proposition 1.8.11 and Corollary 1.8.5 give that for almost every irrational $x \in (0,1)$, it follows

for every $c \in [0,\infty)$ that

$$\begin{aligned}\lim_{n\to\infty}\frac{1}{n}\#\left\{0\le k<n \mid \Theta_k^R(x)\le c\right\} &= \lim_{n\to\infty}\frac{1}{n}\#\left\{0\le k<n \mid h(z_k^R)\le c\right\}\\ &= \lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1}\mathbf{1}_{S_c\cap R}(z_k^R)\\ &= \bar{\mu}_R(S_c\cap R),\end{aligned}$$

where $z_k^R = \mathcal{F}_R^k(x,1)$. □

Theorem 1.8.12 allows us to easily obtain a number of classical and new results on the limiting distributions of approximation coefficients corresponding to particular subsequences of RCF-convergents and mediants of generic $x \in (0,1)\backslash\mathbb{Q}$. We illustrate this with the Doeblin–Lenstra conjecture, first proven in [21].

Corollary 1.8.13 (Bosma–Jager–Wiedijk, 1983 [21])**.** *Set $R \coloneqq H_1$ so that for any $x \in (0,1)\backslash\mathbb{Q}$,*

$$\{u_n^R/s_n^R\}_{n\ge 0} = \{p_{j-1}/q_{j-1}\}_{j\ge 0}.$$

Then $0 < \Theta_n^R(x) < 1$, $n > 0$, with the upper and lower bounds optimal, and for almost every such x,

$$\lim_{n\to\infty}\frac{1}{n}\#\left\{0\le k<n \mid \Theta_k^R(x)\le c\right\} = \begin{cases}\frac{1}{D}c, & 0\le c\le 1/2,\\ \frac{1}{D}(1-c+\log(2c)), & 1/2\le c\le 1,\end{cases}$$

where $D \coloneqq \bar{\mu}(H_1) = \log 2$.

Proof. First, notice that 0 and 1 are the infimum and supremum values, respectively, of c for which $h(z) = c$ intersects $R = H_1$ (see Figure 1.7). By Proposition 1.8.11, these give bounds on $\Theta_n^R(x)$. That the bounds are optimal follows from Theorem 1.8.12: for any $c \in (0,1)$ one finds $\bar{\mu}_R(S_c \cap R) \in (0,1)$, and thus the limiting distribution of Theorem 1.8.12 implies that for almost every $x \in (0,1)\backslash\mathbb{Q}$, there exist n for which $\Theta_n^R(x) \le c$ and n for which $\Theta_n^R(x) > c$.

For the limiting distribution of approximation coefficients, it suffices to compute $\bar{\mu}(S_c \cap H_1)$ for each $c \in [0,1]$, and $D = \bar{\mu}(H_1)$. Fix $c \in [0,1]$. Rearranging terms, $h(z) = c$ with $z = (x,y)$ may be written as $y = f(x,c)$, where

$$f(x,c) = \frac{1-xc}{1-xc+c}.$$

Now $\frac{\partial f}{\partial x} \leq 0$, so f is monotone decreasing as a function of x. For $0 \leq c \leq 1/2$, both $f(0,c) = 1/(1+c)$ and $f(1,c) = 1-c$ are bounded between $1/2$ and 1, so the same is true of $f(x,c)$ for all $0 \leq x \leq 1$. Hence for $0 \leq c \leq 1/2$, one calculates

$$\bar{\mu}(S_c \cap H_1) = \int_0^1 \int_{f(x,c)}^1 \frac{dydx}{(x+y-xy)^2} = c.$$

On the other hand, if $1/2 \leq c \leq 1$, then $1/2 \leq f(0,c) \leq 1$ while $f(1,c) \leq 1/2$. One finds that $f((1-c)/c, c) = 1/2$ and thus computes

$$\begin{aligned}\bar{\mu}(S_c \cap H_1) = &\int_0^{\frac{1-c}{c}} \int_{f(x,c)}^1 \frac{dydx}{(x+y-xy)^2} + \int_{\frac{1-c}{c}}^1 \int_{1/2}^1 \frac{dydx}{(x+y-xy)^2} \\ =&1 - c + \log(2c).\end{aligned}$$

In particular, setting $c = 1$ gives

$$D = \bar{\mu}(H_1) = \bar{\mu}(S_1 \cap H_1) = \log 2. \qquad \square$$

Remark 1.8.14. Note that the limiting distribution of $\Theta_n^R(x)$ in Corollary 1.8.13 is linear on the (maximal) interval $[0,1/2]$; the supremum $1/2$ of this interval is called the *Lenstra constant*. The Lenstra constant coincides with the *Legendre constant*, which is defined as the infimum of real numbers $c > 0$ for which $\Theta(x, p/q) < c$ implies that p/q is a RCF-convergent of x (that is, $p/q = u_n^{H_1}/s_n^{H_1}$ for some n); see Theorem 1.8.25 below.

Analogues of Lenstra and Legendre constants can be defined for other continued fraction algorithms. In particular, for many other

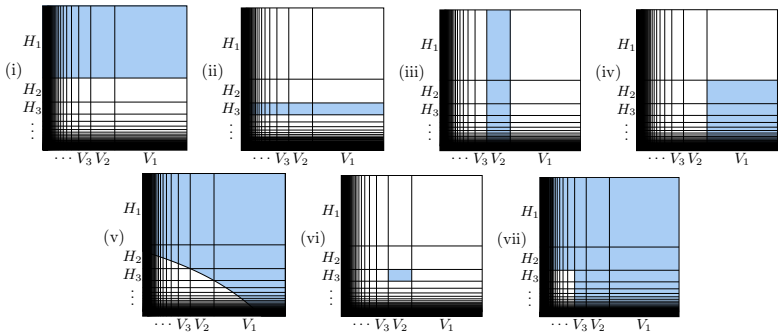


Figure 1.8: The regions R considered in Corollary 1.8.13 and Table 1.8.4.

regions R, there is a maximal interval on which the limiting distribution of approximation coefficients $\Theta_n^R(x)$ arising from Theorem 1.8.12 is linear, and a Lenstra constant may be defined as the supremum of the domain of this linear part (see [53]). In [154], Nakada shows that for a large class of algorithms for which the Legendre constant exists, the Lenstra constant also exists and equals the Legendre constant. However, the existence of a Lenstra constant does *not* necessarily imply the existence of a Legendre constant; in fact, results of Erdös and of Brown and Yin imply that for any $c > 1$, there is no $\bar{\mu}$-continuity, proper, inducible R for which $1 < \Theta(x, p/q) < c$ implies that $p/q = u_n^R/s_n^R$ for some n (see [29]).

By specifying certain subregions R, Theorem 1.8.12 realises a wealth of old and new analogues of the Doeblin–Lenstra conjecture for various subsequences of RCF-convergents and mediants. Table 1.8.4 summarises various regions we consider in [53], the subsequences of Farey convergents these regions 'pick out,' and—when applicable—the original theorems we recover as corollaries of Theorem 1.8.12; see also Figure 1.8. We remark that the authors in [20, 29, 93, 97] use independent methods to compute the corre-

Subregion R	Subsequence $(u_n^R/s_n^R)_{n\geq 0}$ of $\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{\lambda_n p_{j_n}+p_{j_n-1}}{\lambda_n q_{j_n}+q_{j_n-1}}\right)_{n\geq 0}$ specified by R	Original statement
$H_{\lambda+1}$, $\lambda > 0$	$\lambda_n = \lambda$	Theorem 1.9 of [20]
V_a, $a > 0$	$\lambda_n = a_{j_n+1} - a$	
$V_a \backslash H_1$, $a > 0$	$\lambda_n = a_{j_n+1} - a > 0$	Theorem 3.2 of [20], when $a = 1$
S_{c_0}, $c_0 \geq 0$	$\Theta(x, u_n/s_n) \leq c_0$	Theorem 2.2 of [20]
$V_{a-\lambda} \cap H_{\lambda+1}$, $0 \leq \lambda < a$	$\lambda_n = \lambda$ and $a_{j_n+1} = a$	Theorem 3.1 of [20], when $\lambda = a - 1 \geq 1$
$\bigcup_{\lambda=0}^{\Lambda} H_{\lambda+1} \cup \bigcup_{a=1}^{A} V_a$, $\Lambda \geq 0$, $A \geq 1$	$\lambda_n \leq \Lambda$ or $a_{j_n+1} - A \leq \lambda_n \leq a_{j_n+1} - 1$	Theorem 4.iii of [29], when $\Lambda = A$; Theorems 3.3, 3.1, 2.20 of [20, 93, 97], resp., when $\Lambda = A = 1$

Table 1.1: A summary of limiting distributions of approximation coefficients $\Theta_n^R(x)$ considered in [53], arranged by the subregion R, the subsequence $(u_n^R/s_n^R)_{n\geq 0}$ of Farey convergents specified by R, and—when applicable—the original theorem realised by applying Theorem 1.8.12.

sponding limiting distributions. In particular, as Bosma based his proofs in [20] on the natural extension of the Gauss map, which is a dynamical system only dealing with RCF-convergents and no mediant convergents, his proofs are quite involved. Using essentially the same approach but to a 'larger' and 'slower' dynamical system (induced transformations on the natural extension of the Farey tent map) makes the proofs of the results from [20] and their generalisations easier: as in the proof of Corollary 1.8.13 above, the measures $\bar{\mu}(S_c \cap R)$ (and hence the limiting distributions of the approximation coefficients) in each of these cases are computed via straightforward—though at times tedious—calculus; see Corollaries 5.6–5.9, 5.11 and 5.13 of [53] for explicit statements of these measures.

Remark 1.8.15. As in Remark 1.8.9, the original theorem statements on the limiting distributions of approximation coefficients corresponding to the subsequences from Table 1.8.4 assume that the subsequences are ordered with increasing denominators, but this is not necessarily the case for $(u_n^R/s_n^R)_{n\geq 0}$. Proposition 1.10.2—the statement and proof of which are found in the appendix (§1.10)—states that for the purposes of these limiting distributions, our reordering is innocuous and thus the original results do in fact follow from our methods.

Remark 1.8.16. Some of the theorems of [20] are obtained in a similar fashion in [91] using the natural extension of *Denjoy's canonical continued fraction map* $T_d : [0,\infty) \to [0,\infty)$, defined by

$$T_d(x) \coloneqq \begin{cases} 0 & x = 0, \\ \frac{1-x}{x} & x \leq 1, \\ \frac{1}{x} & x > 1. \end{cases}$$

In fact, the Farey tent map F is the first-return map of T_d to $[0,1]$, and the proofs of [91] and our application of Theorem 1.8.12 above are closely related. However, while the domain Ω of the natural extension of F is bounded, the domain of the natural extension of T_d

considered in [91] is the unbounded region $([0,1)\times[0,\infty))\cup([1,\infty)\times[0,1])$. We find the setting of $(\Omega,\mathcal{B},\bar{\mu},\mathcal{F})$ to be particularly insightful given the concrete geometric realisation of the regions $S_c\cap R$ to be integrated within Ω.

Consider again the region $R=V_{a-\lambda}\cap H_{\lambda+1}$ from Table 1.8.4; see also Figure 1.8.vi. The corresponding subsequence $(u_n^R/s_n^R)_{n\geq 0}$ consists precisely of the Farey convergents

$$\frac{u_n}{s_n}=\frac{\lambda_n p_{j_n}+p_{j_n-1}}{\lambda_n q_{j_n}+q_{j_n-1}}$$

for which $\lambda_n=\lambda$ and $a_{j_n+1}=a$, i.e., λ^{th} mediants corresponding to partial denominators a. In [53] we observe that when $1\leq\lambda<a$, the $\bar{\mu}_R$-measure of $S_c\cap R$ is left unchanged if λ is replaced by $a-\lambda$. Thus the λth and $(a-\lambda)^{\text{th}}$ mediants corresponding to partial denominators a give rational approximations of the same quality:

Corollary 1.8.17. *Let $R_1=V_{a-\lambda}\cap H_{\lambda+1}$ and $R_2=V_\lambda\cap H_{a-\lambda+1}$ with $1\leq\lambda<a$. Then for almost every $x\in(0,1)$, the limiting distributions*

$$\lim_{n\to\infty}\frac{1}{n}\#\Big\{0\leq k<n\,\Big|\,\Theta_k^{R_1}(x)\leq c\Big\}=\lim_{n\to\infty}\frac{1}{n}\#\Big\{0\leq k<n\,\Big|\,\Theta_k^{R_2}(x)\leq c\Big\}$$

both exist[17] *and are equal.*

1.8.5 Consecutive approximation coefficients

In [95, 96], Jager introduced a two-dimensional, ergodic dynamical system $(\Gamma,\mathcal{B},\nu,\mathcal{H})$ to study consecutive approximation coefficients $\theta_n(x)$ of RCF-convergents. The map $\mathcal{H}$ satisfies

$$\mathcal{H}(\theta_{n-1}(x),\theta_n(x))=(\theta_n(x),\theta_{n+1}(x)),\quad n>0, \tag{1.87}$$

and is conjugate to the natural extension map $\mathcal{G}$ of the Gauss map. This system has been used to obtain deep insights into metrical properties of approximation coefficients ([95, 96, 98, 114]). Maps which

[17] See Corollary 5.11 of [53] for these limits.

satisfy the property analogous to (1.87) have also been developed for S-expansions ([115]). In this subsection we extend these notions to the induced systems $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$.

We begin with a lemma that will be used in this section and in §1.9.2 below. Recall the definition of $A^R_{[m,n]}(z)$ from (1.44) and of h from (1.85).

Lemma 1.8.18. *Let $R \subset \Omega$ be an inducible subregion and $z = (x, y) \in \Omega \setminus \{(0, 0)\}$. Then for any integer $n \geq 0$,*

$$h(z_n^R) = \det(A) \frac{(u - sx)(u + (s - u)y)}{x + y - xy},$$

where $z_n^R = \mathcal{F}_R^n(z)$ and $A = \left(\begin{smallmatrix} u & t \\ s & r \end{smallmatrix}\right) := A^R_{[0,n]}(z)$.

Proof. First, notice that repeated use of Equation (1.39) gives

$$z_n^R = \mathcal{F}_R^n(z) = \left(A^{-1} \cdot x, A_1 A^T A_1^{-1} \cdot y\right).$$

Now

$$A_1 A^T A_1^{-1} = \begin{pmatrix} r - t & t \\ r - t + s - u & t + u \end{pmatrix},$$

so

$$z_n^R = \left(\frac{rx - t}{u - sx}, \frac{(r - t)y + t}{(r - t + s - u)y + t + u}\right).$$

Using the definition of h and simplifying, we compute

$$\begin{aligned} h(z_n^R) =& \frac{1 - \frac{(r-t)y+t}{(r-t+s-u)y+t+u}}{\frac{rx-t}{u-sx} + \frac{(r-t)y+t}{(r-t+s-u)y+t+u} - \frac{rx-t}{u-sx}\frac{(r-t)y+t}{(r-t+s-u)y+t+u}} \\ =& \frac{(u - sx)(u + (s - u)y)}{(rx - t)(u + (s - u)y) + (u - sx)((r - t)y + t)}. \end{aligned}$$

The denominator of this latter expression simplifies to $\det(A)(x+y-xy)$; the result follows from this and the fact that $\det(A) = \pm 1$. □

Recall Proposition 1.8.11, which states that for an inducible region R and an irrational x, the nth approximation coefficient $\Theta_n^R(x)$ may be written explicitly in terms of $z_n^R = (x_n^R, y_n^R) = \mathcal{F}_R^n(x,1)$. The next result, which follows from Proposition 1.8.11 and the previous lemma—replacing z with $z_n^R = \mathcal{F}_R^n(x,1)$ and n with 1—expresses the pair $(\Theta_n^R(x), \Theta_{n+1}^R(x))$ of consecutive approximation coefficients in terms of z_n^R:

Proposition 1.8.19. *For $R \subset \Omega$ an inducible subregion, define $\psi_R : R \setminus \{(0,0)\} \to \mathbb{R}^2$ for $z = (x,y) \in R \setminus \{(0,0)\}$ by*

$$\psi_R(z) \coloneqq \left(\frac{1-y}{x+y-xy}, \det(A) \frac{(u-sx)(u+(s-u)y)}{x+y-xy} \right),$$

where $A = \left(\begin{smallmatrix} u & t \\ s & r \end{smallmatrix}\right) \coloneqq A_R(z)$. Then for $x \in (0,1)\setminus\mathbb{Q}$,

$$\psi_R(z_n^R) = (\Theta_n^R(x), \Theta_{n+1}^R(x)), \quad n \geq 0,$$

where $z_n^R = \mathcal{F}_n^R(x,1)$.

We demonstrate how Proposition 1.8.19 can be used to obtain insights into consecutive approximation coefficients for three specific inducible subregions $R \subset \Omega$ below.

Consecutive RCF-convergents

Let $R = H_1$ so that $(u_n^R/s_n^R)_{n\geq 0}$ consists of RCF-convergents (see also Example 1.5.3 and §1.7.1). For any $a \geq 1$, if $z = (x,y) \in V_a \cap H_1$, then[18]

$$A_R(z) = A_0^{a-1} A_1 = \begin{pmatrix} 0 & 1 \\ 1 & a \end{pmatrix}.$$

The map ψ_R in Proposition 1.8.19 is thus

$$\psi_R(z) = \left(\frac{1-y}{x+y-xy}, \frac{xy}{x+y-xy} \right).$$

[18] See Remark 1.5.4.

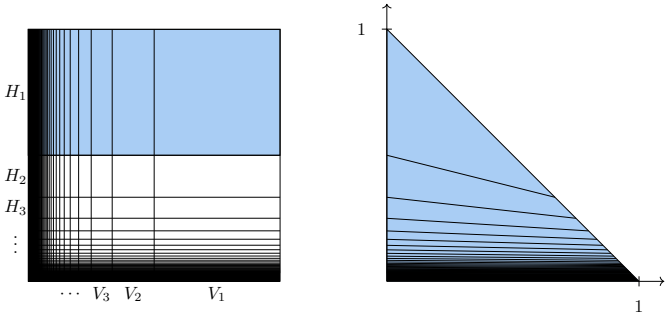


Figure 1.9: Left: The region $R = H_1$. Right: The image $\psi_R(R)$.

One finds that ψ_R is a diffeomorphism between the interior of R and the interior of the Euclidean triangle with vertices $(0,0)$, $(0,1)$ and $(1,0)$; see Figure 1.9. From this, one immediately obtains Vahlen's result that $\min\{\Theta_n^R(x), \Theta_{n+1}^R(x)\} < 1/2$ for all $n \geq 0$ ([185]; see also [96]). Up to the isomorphism of Theorem 1.5.5, in [96] it is shown that $(\Gamma, \mathcal{B}, \nu, \mathcal{H}) := (\psi_R(R), \mathcal{B}, \nu, \psi_R \circ \mathcal{F}_R \circ \psi_R^{-1})$ forms an ergodic system, where $d\nu = dxdy/(\log 2\sqrt{1-4xy})$. We refer the reader to [95, 96, 98, 114] for further metrical results which follow from a deeper analysis of this system.

Consecutive Farey convergents

Let $R = \Omega$. Then for any $z = (x, y) \in R$,

$$A_R(z) = A_{\varepsilon(x)} = \begin{pmatrix} 1 - \varepsilon(x) & \varepsilon(x) \\ 1 & 1 \end{pmatrix}.$$

The map ψ_R in Proposition 1.8.19 becomes

$$\psi_R(z) = \begin{cases} \left(\frac{1-y}{x+y-xy}, \frac{1-x}{x+y-xy} \right), & x \leq 1/2, \\ \left(\frac{1-y}{x+y-xy}, \frac{xy}{x+y-xy} \right), & x > 1/2; \end{cases} \tag{1.88}$$

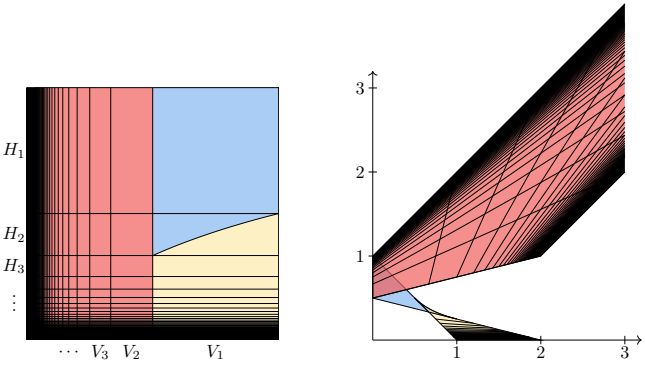


Figure 1.10: Left: The region $R = \Omega$. Right: The image $\psi_R(R\backslash\{(0,0)\})$.

see Figure 1.10. The map ψ_R allows one to easily compare consecutive approximation coefficients:

Proposition 1.8.20. *Let $x = [0; a_1, a_2, \dots] \in (0,1)\backslash\mathbb{Q}$. Then for any $j \geq 0$,*

$$1 < 2\Theta\left(x, \frac{p_j}{q_j}\right) + \Theta\left(x, \frac{(a_{j+1}-1)p_j + p_{j-1}}{(a_{j+1}-1)q_j + q_{j-1}}\right) < 2,$$

and for $a_{j+1} > 1$ and $0 \leq \lambda < a_{j+1} - 1$,

$$\left|\Theta\left(x, \frac{(\lambda+1)p_j + p_{j-1}}{(\lambda+1)q_j + q_{j-1}}\right) - \Theta\left(x, \frac{(\lambda p_j + p_{j-1}}{(\lambda q_j + q_{j-1}}\right)\right| < 1.$$

Proof. Up to non-strict inequalities, the first statement follows from (1.25), Proposition 1.8.19, Equation (1.88), and the fact that the image of V_1 under ψ_R is bounded[19] by the lines $y = -x/2 + 1$ and

[19]These lines do *not* describe the boundary of $\psi_R(V_1)$ (see Figure 1.10), but rather give upper and lower bounding lines of equal slope. More precise—though less elegant—statements can be made by analysing the boundary of the image.

$y = -x/2 + 1/2$; the second follows from (1.25), Proposition 1.8.19, Equation (1.88), and the fact that $\psi_R(\Omega\backslash(V_1 \cup \{(0,0)\}))$ is bounded by $y = x+1$ and $y = x-1$ (see Figure 1.10). The strict inequalities follow from the irrationality of x. □

The map ψ_R also gives information on the monotonicity of finite sequences of consecutive approximation coefficients. In particular, we find that the approximation coefficients of the mediant convergents $(\lambda p_j + p_{j-1})/(\lambda q_j + q_{j-1})$, $0 < \lambda < a_{j+1}$, between p_{j-1}/q_{j-1} and p_j/q_j first increase and then decrease monotonically in λ, with the maximum occurring in the middle. This statement is made precise in the following:

Theorem 1.8.21. *Let $x = [0; a_1, a_2, \dots] \in (0,1)\backslash\mathbb{Q}$, and suppose $a_{j+1} > 1$ for some fixed $j \geq 0$. Then for each $p/q \in \{p_j/q_j\} \cup \{(\lambda p_j + p_{j-1})/(\lambda q_j + q_{j-1})\}_{0 \leq \lambda < a_{j+1}}$, $p/q \neq 1/0$,*

$$0 < \Theta(x, p/q) < \begin{cases} \frac{a_{j+1}+2}{4}, & a_{j+1} \text{ even}, \\ \frac{(a_{j+1}+1)(a_{j+1}+3)}{4(a_{j+1}+2)}, & a_{j+1} \text{ odd}, \end{cases}$$

with the bounds optimal. Moreover,

$$\Theta\left(x, \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}}\right) < \Theta\left(x, \frac{\lambda' p_j + p_{j-1}}{\lambda' q_j + q_{j-1}}\right)$$

for all $0 \leq \lambda < \lambda' \leq \lfloor a_{j+1}/2 \rfloor$, and

$$\Theta\left(x, \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}}\right) > \Theta\left(x, \frac{\lambda' p_j + p_{j-1}}{\lambda' q_j + q_{j-1}}\right) > \Theta\left(x, \frac{p_j}{q_j}\right)$$

for all $\lceil a_{j+1}/2 \rceil \leq \lambda < \lambda' < a_{j+1}$.

Proof. Let x and a_{j+1} be as in the statement, $0 \leq \lambda \leq a_{j+1}$ and let n be such that $u_n/s_n = p_{j-1}/q_{j-1}$. We begin with the latter claims. We have (recall (1.25))

$$\Theta_{n+\lambda}(x) = \Theta\left(x, \frac{u_{n+\lambda}}{s_{n+\lambda}}\right) = \begin{cases} \Theta\left(x, \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}}\right), & \lambda < a_{j+1}, \\ \Theta\left(x, \frac{p_j}{q_j}\right), & \lambda = a_{j+1}, \end{cases}$$

so it suffices to show

$$\Theta_{n+\lambda}(x) < \Theta_{n+\lambda+1}(x), \quad 0 \leq \lambda < \lfloor a_{j+1}/2 \rfloor \tag{1.89}$$

and

$$\Theta_{n+\lambda}(x) > \Theta_{n+\lambda+1}(x), \quad \lceil a_{j+1}/2 \rceil \leq \lambda < a_{j+1}. \tag{1.90}$$

Set $z_{n+\lambda} = (x_{n+\lambda}, y_{n+\lambda}) = \mathcal{F}^{n+\lambda}(x, 1)$. For $\lambda < a_{j+1}$,

$$\begin{aligned} z_{n+\lambda} \in & V_{a_{j+1}-\lambda} \cap H_{\lambda+1} \\ &= \left(\frac{1}{a_{j+1}-\lambda+1}, \frac{1}{a_{j+1}-\lambda}\right] \times \left(\frac{1}{\lambda+2}, \frac{1}{\lambda+1}\right]. \end{aligned}$$

First, suppose $\lambda \neq a_{j+1}-1$. Then $x_{n+\lambda} \leq 1/2$, so by Equation (1.88) and Proposition 1.8.19, $\Theta_{n+\lambda}(x) < \Theta_{n+\lambda+1}(x)$ if $x_{n+\lambda} < y_{n+\lambda}$, and similarly with the reverse inequalities. If $\lambda < \lfloor a_{j+1}/2 \rfloor$, then $\lambda \leq a_{j+1}/2 - 1$, which implies that $\lambda + 2 \leq a_{j+1} - \lambda$. Thus in this case

$$x_{n+\lambda} \leq \frac{1}{a_{j+1}-\lambda} \leq \frac{1}{\lambda+2} < y_{n+\lambda},$$

proving (1.89). On the other hand, if $\lceil a_{j+1}/2 \rceil \leq \lambda$, then $a_{j+1}/2 \leq \lambda$. This implies $a_{j+1} - \lambda \leq \lambda$, so in this case

$$y_{n+\lambda} \leq \frac{1}{\lambda+1} \leq \frac{1}{a_{j+1}-\lambda+1} < x_{n+\lambda},$$

proving (1.90) for $\lambda < a_{j+1} - 1$.

If $\lambda = a_{j+1} - 1$, then since $a_{j+1} > 1$ by assumption, $z_{n+\lambda} \in V_1 \backslash H_1$. By Equation (1.88) and Proposition 1.8.19, $\Theta_{n+\lambda}(x) > \Theta_{n+\lambda+1}(x)$ if and only if $1 - y_{n+\lambda} > x_{n+\lambda} y_{n+\lambda}$, or, equivalently, $y_{n+\lambda} < 1/(1 + x_{n+\lambda})$, but this inequality holds since $x_{n+\lambda} < 1$ and $y_{n+\lambda} \leq 1/2$. Thus (1.90) is also true for $\lambda = a_{j+1} - 1$.

For the first statement, Corollary 5.11 of [53] implies that

$$\frac{(a_{j+1}-\lambda)\lambda}{a_{j+1}} < \Theta\left(x, \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}}\right) < \frac{(a_{j+1}-\lambda+1)(\lambda+1)}{a_{j+1}+2} \tag{1.91}$$

for $0 \leq \lambda < a_{j+1}$, with these bounds optimal. The optimal lower bound of 0 occurs when $\lambda = 0$. (This lower bound also applies for the rational p_j/q_j.) By the monotonicity statements just proven, the optimal upper bound is given when $\lambda = \lfloor a_{j+1}/2 \rfloor$ or $\lambda = \lceil a_{j+1}/2 \rceil$. If a_{j+1} is even, then in either case $\lambda = a_{j+1}/2$ and the right-hand side of (1.91) gives $(a_{j+1}+2)/4$. If a_{j+1} is odd, then either $\lambda = (a_{j+1}-1)/2$ or $\lambda = (a_{j+1}+1)/2$. In either case, the right-hand side of (1.91) gives $(a_{j+1}+1)(a_{j+1}+3)/4(a_{j+1}+2)$. □

As a corollary of the previous result, we find a lower bound on the maximum of the approximation coefficients of RCF-convergents and mediants corresponding to particular partial denominators.

Corollary 1.8.22. *Let $x = [0; a_1, a_2, \dots] \in (0,1)\backslash\mathbb{Q}$, and suppose $a_{j+1} > 1$ for some fixed $j \geq 0$. Then*

$$\max\left\{\Theta(x, p/q) \;\middle|\; \frac{p}{q} \in \left\{\frac{p_j}{q_j}\right\} \cup \left\{\frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}}\right\}_{0 \leq \lambda < a_{j+1}}, \; \frac{p}{q} \neq \frac{1}{0}\right\} > \begin{cases} \frac{a_{j+1}}{4}, & a_{j+1} \text{ even}, \\ \frac{a_{j+1}^2 - 1}{4a_{j+1}}, & a_{j+1} \text{ odd}, \end{cases}$$

with the lower bound optimal.

Proof. By Theorem 1.8.21, the maximum occurs when $p/q = (\lambda p_j + p_{j-1})/(\lambda q_j + q_{j-1})$ with $\lambda = \lfloor a_{j+1}/2 \rfloor$ or $\lambda = \lceil a_{j+1}/2 \rceil$. As in the final paragraph of the proof of Theorem 1.8.21, an analysis of the lower bound of (1.91) when a_{j+1} is even or odd gives the result. □

Remark 1.8.23. Since the partial denominators of a.e. $x \in (0,1)\backslash\mathbb{Q}$ are unbounded (see, say, Chapter V of [167]), the previous corollary implies that the approximation coefficients of the RCF-convergents and mediants of a.e. x are unbounded. This is also seen intuitively in Figure 1.10; when a partial denominator a_{j+1} is large, points in the $\mathcal{F}$-orbit of $(x, 1)$ will 'dip' close to the origin on the left-hand side of

the figure, and, consequently, the coordinates of the corresponding images under ψ_R on the right-hand side become large.

From Figure 1.10 it is clear that ψ_R is not injective,[20] and thus one cannot immediately conjugate $\mathcal{F}_R$ with ψ_R to study the dynamics of consecutive Farey convergents as done with consecutive RCF-convergents above. The same difficulty arises when studying consecutive convergents of nearest integer CFs ([98]). As in [98], this difficulty can be overcome by introducing a third coordinate to the image of the function ψ_R which 'flags' the color of the subregion that a point in the domain belongs to. This extra coordinate makes ψ_R invertible and could lead to the study of further metrical results on Farey convergents. We leave the details to future work.

Consecutive RCF-convergents and nearest mediants

Let $R = H_1 \cup H_2 \cup V_1$, which corresponds to RCF-convergents and nearest (i.e., first and final) mediants. For $z = (x, y) \in R$ with $x \neq 0$, we find

$$A_R(z) = \begin{cases} A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, & z \in V_1, \\ A_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, & z \in H_1 \backslash V_1, \\ A_0^{a-1} = \begin{pmatrix} 1 & 0 \\ a-1 & 1 \end{pmatrix}, & z \in V_a \cap H_2,\ a > 1, \end{cases}$$

and thus the map ψ_R in Proposition 1.8.19 is

$$\psi_R(z) = \begin{cases} \left(\frac{1-y}{x+y-xy}, \frac{xy}{x+y-xy} \right), & z \in V_1, \\ \left(\frac{1-y}{x+y-xy}, \frac{1-x}{x+y-xy} \right), & z \in H_1 \backslash V_1, \\ \left(\frac{1-y}{x+y-xy}, \frac{(1-(a-1)x)(1+(a-2)y)}{x+y-xy} \right), & z \in V_a \cap H_2,\ a > 1; \end{cases}$$

[20] The image of the red and blue regions clearly overlap, but ψ_R also 'folds' the blue and tan regions in $V_1 \cap H_2$ over one another. The boundary of $\psi_R(V_1 \cap H_2)$ is given by the curves $y = 1 - x$, $y = (2 - x)/4$, and $y = 1/4x$.

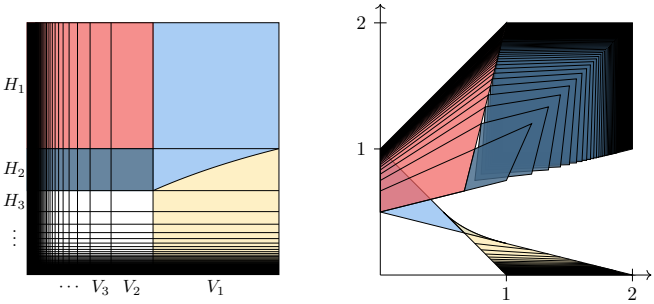


Figure 1.11: Left: The region $R = H_1 \cup H_2 \cup V_1$. Right: The image $\psi_R(R)$.

see Figure 1.11. As with Proposition 1.8.20, the images under ψ_R of subregions of R yield immediate information regarding consecutive RCF-convergents and nearest mediants. The first statement below—which is also a corollary of Proposition 1.8.20 and Theorem 1.8.21—corresponds to the image of $H_1 \backslash V_1$, and the second to the image of $H_2 \setminus V_1$; see Figure 1.11. These two images are bounded[21] by the pairs of lines $y = x+1$, $y = x$, and $y = x+1$, $y = x-1$, respectively.

Proposition 1.8.24. *Let* $x = [0; a_1, a_2, \dots] \in (0,1)\backslash\mathbb{Q}$. *For any* $j \geq 0$ *for which* $a_{j+1} > 1$,

$$0 < \Theta\left(x, \frac{p_j + p_{j-1}}{q_j + q_{j-1}}\right) - \Theta\left(x, \frac{p_{j-1}}{q_{j-1}}\right) < 1,$$

and

$$\left|\Theta\left(x, \frac{(a_{j+1}-1)p_j + p_{j-1}}{(a_{j+1}-1)q_j + q_{j-1}}\right) - \Theta\left(x, \frac{p_j + p_{j-1}}{q_j + q_{j-1}}\right)\right| < 1.$$

[21] As in the proof of Proposition 1.8.20, these lines are simply upper and lower bounding lines of equal slope and do *not* necessarily describe the boundaries of the images.

The map ψ_R is again not injective, but a similar process to that of [98] (as mentioned above for consecutive Farey convergents) may be used to overcome this difficulty and thus investigate further metrical results on RCF-convergents and nearest mediants. We again leave the details to future work.

1.8.6 Legendre-type theorems

Let $p/q \in \mathbb{Q} \cap [0,1]$ with $\gcd(p,q) = 1,\ q > 0$, and $x \in (0,1)\backslash\mathbb{Q}$. We recall here the important result of Legendre in the theory of RCFs:

Theorem 1.8.25 (Legendre, 1798 [125])**.** *If $\Theta(x, p/q) < 1/2$, then p/q is a* RCF*-convergent of x. Moreover, the constant $1/2$ is optimal.*

That the so-called *Legendre constant* $1/2$ is optimal means that for any $c > 1/2$, there exist p/q and x as above with $\Theta(x, p/q) < c$ but such that p/q is not a RCF-convergent of x. Legendre's Theorem implies that the 'excellent' rational approximations p/q to an irrational x are all RCF-convergents of x. Interpreted in a slightly different way, Legendre's Theorem gives a sufficient condition to verify that p/q is a RCF-convergent of x without computing the expansion of x.

A similar result, first stated by Fatou and proven by Grace (and later Koksma ([111])), gives a sufficient condition to verify that p/q is either a RCF-convergent or nearest mediant (i.e., a first or final mediant) of x:

Theorem 1.8.26 (Fatou–Grace, 1904–1918 [70, 80])**.** *If $\Theta(x, p/q) < 1$, then p/q is either a* RCF*-convergent or nearest mediant of x. The constant 1 is optimal.*

Later, Koksma formulated and proved a similar statement regarding RCF-convergents and first mediants:

Theorem 1.8.27 (Koksma, 1937 [111])**.** *If $\Theta(x, p/q) < 2/3$, then p/q is either a* RCF*-convergent or a first mediant of x. The constant $2/3$ is optimal.*

In [10], Barbolosi and Jager give refinements of the theorems of Legendre, Fatou–Grace and Koksma. The connections above between the map $\mathcal{F}$ and Farey convergents allow us to easily reobtain these refinements, *assuming Fatou–Grace*. We remark here that Barbolosi and Jager's proofs are in a sense more elementary, as they do not make this assumption. Nevertheless, we find it worthwhile to present a new approach to these results to further highlight the versatility of the natural extension map $\mathcal{F}$ for studying RCF-convergents and mediants.

We begin by setting necessary notation for the statements of the Barbolosi–Jager refinements. Following [10], for any nonzero rational[22] $p/q \in \mathbb{Q} \cap (0,1]$ with $\gcd(p,q) = 1$, set $\epsilon(p/q) \coloneqq (-1)^n$, where n is the depth of p/q (recall the definition of depth from §1.2.3). Furthermore, for $x \in (0,1)\backslash\mathbb{Q}$, set

$$\epsilon(x, p/q) \coloneqq \begin{cases} -1, & x < p/q, \\ 1, & p/q < x. \end{cases}$$

The *signature* of p/q with respect to x is defined as

$$\delta(x, p/q) \coloneqq \epsilon(p/q)\epsilon(x, p/q) \in \{\pm 1\}.$$

The following lemma classifies the signatures of the nonzero and finite Farey convergents of x:

Lemma 1.8.28. *Suppose $x \in (0,1)$ is irrational with* RCF*-expansion $[0; a_1, a_2, \dots]$, and let*

$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n q_{j_n} + q_{j_n - 1}} \in \mathbb{Q}\backslash\{0\}$$

be the $(n-1)^{st}$ Farey convergent of x, where $0 \leq \lambda_n < a_{j_n+1}$. Then $\delta(x, u_n/s_n) = -1$ if and only if u_n/s_n is

[22] Since for any $x \in (0,1)$ both $p_{-1}/q_{-1} = 1/0$ and $p_0/q_0 = 0/1$ are RCF-convergents of x, we only consider nonzero rationals p/q in what follows.

(i) a RCF*-convergent (i.e.,* $\lambda_n = 0$*) with* $a_{j_n-1} = 1,\ j_n > 1$*, or*

(ii) a first mediant (i.e., $\lambda_n = 1$*).*

Proof. From (1.10) and (1.11),

$$\epsilon(x, u_n/s_n) = \begin{cases} -1, & j_n \text{ is even}, \\ 1, & j_n \text{ is odd}. \end{cases}$$

We now consider the value of $\epsilon(u_n/s_n)$ in cases:

(a) Suppose that $\lambda_n = 0$ and $a_{j_n-1} = 1$ (the assumption that $u_n/s_n \in \mathbb{Q}\backslash\{0\}$ implies $j_n > 1$). Then

$$\frac{u_n}{s_n} = \frac{p_{j_n-1}}{q_{j_n-1}} = [0; a_1, \ldots, a_{j_n-1}] = [0; a_1, \ldots, a_{j_n-2} + 1],$$

so the depth of u_n/s_n is $j_n - 2$, and

$$\epsilon(u_n/s_n) = \begin{cases} -1, & j_n \text{ is odd}, \\ 1, & j_n \text{ is even}. \end{cases}$$

(b) Suppose that $\lambda_n = 0$ and $a_{j_n-1} > 1$ (here again $j_n > 1$). Then

$$\frac{u_n}{s_n} = \frac{p_{j_n-1}}{q_{j_n-1}} = [0; a_1, \ldots, a_{j_n-1}] = [0; a_1, \ldots, a_{j_n-1} - 1, 1],$$

so the depth of u_n/s_n is $j_n - 1$, and

$$\epsilon(u_n/s_n) = \begin{cases} -1, & j_n \text{ is even}, \\ 1, & j_n \text{ is odd}. \end{cases}$$

(c) Suppose that $\lambda_n = 1$. Then

$$\frac{u_n}{s_n} = \frac{p_{j_n} + p_{j_n-1}}{q_{j_n} + q_{j_n-1}} = [0; a_1, \ldots, a_{j_n}, 1] = [0; a_1, \ldots, a_{j_n} + 1],$$

so the depth of u_n/s_n is j_n, and

$$\epsilon(u_n/s_n) = \begin{cases} -1, & j_n \text{ is odd}, \\ 1, & j_n \text{ is even}. \end{cases}$$

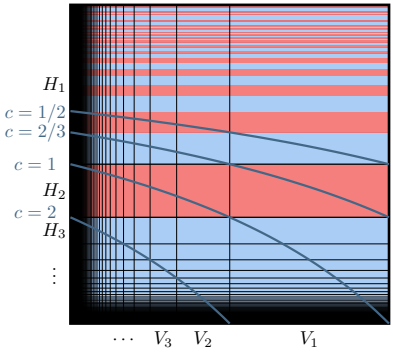


Figure 1.12: Red and blue regions correspond to negative and positive signatures $\delta(x, u_n/s_n) = -1, 1$, respectively. The curves $h(z) = c$ are shown in dark blue for $c \in \{1/2, 2/3, 1, 2\}$.

(d) Lastly, if $\lambda_n > 1$, then

$$\begin{aligned}\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} &= [0; a_1, \ldots, a_{j_n}, \lambda_n] \\ &= [0; a_1, \ldots, a_{j_n}, \lambda_n - 1, 1],\end{aligned}$$

so the depth of u_n/s_n is $j_n + 1$, and

$$\epsilon(u_n/s_n) = \begin{cases} -1, & j_n \text{ is even}, \\ 1, & j_n \text{ is odd}. \end{cases}$$

The result now follows by setting $\delta(x, u_n/s_n) = \epsilon(u_n/s_n)\epsilon(x, u_n/s_n)$. □

With Lemma 1.8.28, one may decompose the domain Ω of $\mathcal{F}$ according to the signatures of Farey convergents; see Figure 1.12. Recall (Example 1.5.3) that for fixed λ, the horizontal region $H_{\lambda+1} \subset \Omega$ corresponds to Farey convergents of x of the form $u_n/s_n = (\lambda_n p_{j_n} +$

$p_{j_n-1})/(\lambda_n q_{j_n} + q_{j_n-1})$ with $\lambda_n = \lambda$. Thus Lemma 1.8.28 implies that the region H_2 corresponds to Farey convergents with negative signature $\delta(x, u_n/s_n) = -1$, and the region $\cup_{\lambda>1} H_{\lambda+1}$ corresponds to Farey convergents with positive signature $\delta(x, u_n/s_n) = 1$. The region H_1 corresponding to RCF-convergents $u_n/s_n = p_{j_n-1}/q_{j_n-1}$ is further decomposed depending on the value of the partial denominator a_{j_n-1} in the RCF-expansion of x. Recall from (1.36) that if $z_n = \mathcal{F}^n(x, 1) \in H_1$, then

$$z_n = \big([0; a_{j_n+1}, a_{j_n+2}, \dots], [0; 1, a_{j_n}, a_{j_n-1}, \dots, a_1]\big).$$

The cylinder of points whose RCF-expansions begin as $[0; 1, a, b, \dots]$ is the interval

$$\big([0; 1, a, b+1], [0; 1, a, b]\big].$$

Thus, by Lemma 1.8.28, the regions

$$[0, 1] \times \big([0; 1, a, 2], [0; 1, a, 1]\big] = [0, 1] \times \left(\frac{2a+1}{2a+3}, \frac{a+1}{a+2}\right]$$

for $a \geq 1$ correspond to Farey convergents with negative signature $\delta(x, u_n/s_n) = -1$, while the regions

$$\begin{aligned}[0, 1] \times \bigcup_{b>1} \big([0; 1, a, b+1], [0; 1, a, b]\big] &= [0, 1] \times \big([0; 1, a], [0; 1, a, 2]\big] \\ &= [0, 1] \times \left(\frac{a}{a+1}, \frac{2a+1}{2a+3}\right]\end{aligned}$$

for $a \geq 1$ correspond to Farey convergents with positive signature $\delta(x, u_n/s_n) = 1$.

Recall that Theorem 1.8.26 implies that if $\Theta(x, p/q) < 1$, then p/q is *some* Farey convergent[23] u_n/s_n of x. With this in mind, Barbolosi and Jager's refinements (Theorems 2.2, 4.3 and 4.7, respectively, of [10]) now follow easily from Proposition 1.8.11 and Figure 1.12:

[23]Of course, Theorem 1.8.26 makes the stronger claim that p/q is a RCF-convergent or nearest mediant, but only this weaker implication is needed.

Corollary 1.8.29 (Barbolosi–Jager, 1994 [10]). *Let $p/q \in \mathbb{Q} \cap (0,1]$ with $\gcd(p,q) = 1$, $q > 0$, and $x \in (0,1)\backslash\mathbb{Q}$.*

- *When $\delta(x,p/q) = -1$, $\Theta(x,p/q) < 1/2$ implies that p/q is a* RCF*-convergent of x, while $\Theta(x,p/q) > 2/3$ implies that p/q is not a* RCF*-convergent of x.*
- *When $\delta(x,p/q) = 1$, $\Theta(x,p/q) < 2/3$ implies that p/q is a* RCF*-convergent of x, while $\Theta(x,p/q) > 1$ implies that p/q is not a* RCF*-convergent of x.*

All constants are optimal.

Corollary 1.8.30 (Barbolosi–Jager, 1994 [10]). *Let $p/q \in \mathbb{Q} \cap (0,1]$ with $\gcd(p,q) = 1$, $q > 0$, and $x \in (0,1)\backslash\mathbb{Q}$.*

- *When $\delta(x,p/q) = -1$, $\Theta(x,p/q) < 1$ implies p/q is a* RCF*-convergent or first mediant of x, while $\Theta(x,p/q) > 2$ implies p/q is neither a* RCF*-convergent nor first mediant of x.*
- *If $\delta(x,p/q) = 1$, then p/q is not a first mediant of x.*

All constants are optimal.

Notice that Corollary 1.8.29 implies Legendre's Theorem (Theorem 1.8.25), and Corollaries 1.8.29 and 1.8.30 imply Koksma's Theorem (Theorem 1.8.27).

Corollary 1.8.31 (Barbolosi–Jager, 1994 [10]). *Let $p/q \in \mathbb{Q} \cap (0,1]$ with $\gcd(p,q) = 1$, $q > 0$, and $x \in (0,1)\backslash\mathbb{Q}$.*

- *Suppose that $\delta(x,p/q) = -1$. If p/q is a final mediant of x, then it is also a first mediant of x. Moreover, $\Theta(x,p/q) < 2/3$ implies p/q is a* RCF*-convergent or a final mediant of x, while $\Theta(x,p/q) > 1$ implies p/q is neither a* RCF*-convergent nor a final mediant of x.*
- *When $\delta(x,p/q) = 1$, $\Theta(x,p/q) < 1$ implies that p/q is a* RCF*-convergent or a final mediant of x, while $\Theta(x,p/q) > 2$ implies that p/q is neither a* RCF*-convergent nor a nearest mediant of x.*

All constants are optimal.

The Barbolosi–Jager refinements of Legendre, Fatou–Grace and Koksma are refinements of the assumptions on the rational p/q (namely, its signature) which approximates x. One could instead ask for refinements of the bounds $1/2,\ 2/3,\ 1$ which occur in Theorems 1.8.25, 1.8.26 and 1.8.27. As the next theorem shows, such refinements give information about certain partial denominators occurring in the RCF-expansion of x.

Theorem 1.8.32. *Fix some positive integer k. If $p/q \in \mathbb{Q} \cap (0,1]$ with $\gcd(p,q) = 1$, $q > 0$, $x \in (0,1)\backslash\mathbb{Q}$ and*

$$\frac{k}{k+1} \leq \Theta(x, p/q) < \frac{k+1}{k+2},$$

then one of the following holds:

(i) p/q is a RCF-convergent of the form

$$\frac{p}{q} = \frac{p_{j-1}}{q_{j-1}}$$

for some j such that $a_j = 1$ and $a_{j+1} \geq k$,

(ii) p/q is a first mediant of the form

$$\frac{p}{q} = \frac{p_j + p_{j-1}}{q_j + q_{j-1}}$$

for some j such that $a_{j+1} \leq k+1$, or

(iii) p/q is a final mediant of the form

$$\frac{p}{q} = \frac{(a_{j+1}-1)p_j + p_{j-1}}{(a_{j+1}-1)q_j + q_{j-1}}$$

for some j such that $a_{j+1} \leq k+1$.

Proof. Since $\Theta(x, p/q) < 1$, Theorem 1.8.26 implies that there exists some n such that

$$\frac{p}{q} = \frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_{n-1}}}{\lambda_n q_{j_n} + q_{j_{n-1}}}$$

is a RCF-convergent or nearest mediant of x. Moreover, Proposition 1.8.11 implies that $z_n \in S$, where $z_n = (x_n, y_n) = \mathcal{F}^n(x, 1)$ and

$$S \coloneqq \left\{ z \in \Omega \;\middle|\; \frac{k}{k+1} \le h(z) < \frac{k+1}{k+2} \right\} \subset V_1 \cup H_1 \cup H_2.$$

Suppose $z_n \in S \cap H_1$, so that $u_n/s_n = p_{j_{n-1}}/q_{j_{n-1}}$. The curve $h(z) = k/(k+1)$ passes through the points $\left(0, \frac{k+1}{2k+1}\right)$ and $\left(\frac{1}{k}, \frac{1}{2}\right)$ (see, for instance, the curves determined by $c = 1/2$ and $c = 2/3$ in Figure 1.12 for $k = 1$ and $k = 2$, respectively). Using this, one finds $S \cap H_1 \subset [0, 1/k] \times (1/2, 2/3]$, which is contained in the union of $\{0, 1\} \times (1/2, 2/3]$ and

$$\mathcal{F}(V_1 \cap H_1) \cap \bigcup_{a \ge k} V_a.$$

Irrationality of x implies $x_n \notin \{0, 1\}$, so $z_n \in \mathcal{F}(V_1 \cap H_1) \cap \bigcup_{a \ge k} V_a$. That $z_n \in \mathcal{F}(V_1 \cap H_1)$ implies $a_{j_n} = 1$, and that $z_n \in \bigcup_{a \ge k} V_a$ implies $a_{j_n+1} \ge k$, proving case (i). The other two cases are proven similarly, considering instead when z_n belongs to $S \cap H_2$ and $S \cap V_1$, respectively. □

In 1952, Kuipers and Meulenbeld gave sufficient conditions to guarantee that the approximation coefficients corresponding to first and final mediants are less than 1 (Theorem 1 of [121]). This partial converse of Theorem 1.8.26 is also easily obtained via $\mathcal{F}$:

Corollary 1.8.33 (Kuipers–Meulenbeld, 1952 [121])**.** *Let $x \in (0, 1)$ be irrational with* RCF*-expansion $[0; a_1, a_2, \dots]$, and suppose that $a_{j+1} \ge 2$ for some $j \ge 0$. If $a_{j+1} \le a_j + 1$, then*

$$\Theta\left(x, \frac{p_j + p_{j-1}}{q_j + q_{j-1}}\right) < 1,$$

while if $a_{j+1} \leq a_{j+2} + 1$, *then*

$$\Theta\left(x, \frac{(a_{j+1}-1)p_j + p_{j-1}}{(a_{j+1}-1)q_j + q_{j-1}}\right) < 1.$$

Proof. With notation as in the statement, let n be such that $u_n/s_n = (p_j + p_{j-1})/(q_j + q_{j-1})$, and suppose $a_{j+1} \leq a_j + 1$. Then, setting $z_n = (x_n, y_n) = \mathcal{F}^n(x, 1)$,

$$z_n = ([0; a_{j+1} - 1, a_{j+2}, a_{j+3}, \dots], [0; 2, a_j, \dots, a_1]) \in V_{a_{j+1}-1} \cap H_2,$$

and we have both $x_n > 1/a_{j+1}$ and—since $a_j \geq a_{j+1} - 1$—also $y_n \geq [0; 2, a_{j+1} - 1] = (a_{j+1}-1)/(2a_{j+1}-1)$. By Proposition 1.8.11,

$$\Theta\left(x, \frac{p_j + p_{j-1}}{q_j + q_{j-1}}\right) = h(z_n) < h\left(\frac{1}{a_{j+1}}, \frac{a_{j+1}-1}{2a_{j+1}-1}\right) = 1.$$

A similar argument proves the claim for final mediants. □

1.9 Superoptimal continued fractions

In this section, we combine the theory from §1.6 and §1.8 to introduce a collection of so-called *superoptimal* CF-algorithms producing GCF-expansions with arbitrarily good approximation and convergence properties. We give two explicit families of such algorithms in §1.9.1 and §1.9.2, producing what we call *linear recurrence* and *Legendre* CFs, and give explicit examples of each of these. To motivate our superoptimal CFs, we first recall some properties of Bosma's optimal CFs, introduced in [19].

Letting P_k/Q_k with $\gcd(P_k, Q_k) = 1$ and $Q_k > 0$ denote the kth convergent of a SRCF-expansion of an irrational x and $\Theta(x, P_k/Q_k)$ the approximation coefficient corresponding to x and P_k/Q_k (recall (1.83)), one has that $\sup_{k\geq 1}\Theta(x, P_k/Q_k) \geq 1/2$ a.s. (see Remark 2.19 of [19]). On the other hand, for Bosma's optimal CFs, $\Theta(x, P_k/Q_k) < 1/2$ for all $k \geq 1$ (Theorem 4.28 of [19]), i.e., the optimal CF-expansion provides the 'best approximations' to x among

all SRCF-expansions. For a general SRCF-expansion of x, let $n(k) \in \mathbb{N}$ be defined by $q_{n(k)} \leq Q_k < q_{n(k)+1}$, where p_n/q_n denotes the nth RCF-convergent of x. A SRCF-expansion is considered to be 'fast' if the subsequence $n(k)$ grows quickly. For SRCF-expansions, $\limsup_{k\to\infty} n(k)/k \leq \log 2/\log G$ where $G = (\sqrt{5}+1)/2$, while Bosma's optimal CFs satisfy $\lim_{k\to\infty} n(k)/k = \log 2/\log G$ a.s. (see Remarks 3.16 (i) and (ii) and Theorem 4.28 of [19]). Thus the optimal CFs are also 'fastest' among all SRCF-expansions.

We remark that as optimal CFs are examples of S-expansions, they are also examples of contracted Farey expansions (see §1.3.3 and §1.7.2). By extending our considerations from SRCFs to GCFs, we can obtain CFs which simultaneously provide *arbitrarily good* approximations and *arbitrarily fast* expansions.

Definition 1.9.1. Let $\varepsilon,\ C > 0$. A GCF-expansion of x with convergents P_k/Q_k, $\gcd(P_k, Q_k) = 1$, $Q_k > 0$, is called (ε, C)-*superoptimal* if both

(i) $\Theta(x, P_k/Q_k) \leq \varepsilon$ for all $k \geq 1$, and

(ii) $\limsup_{k\to\infty} n(k)/k \geq C$, where $n(k)$ satisfies $q_{n(k)} \leq Q_k < q_{n(k)+1}$, with p_n/q_n the nth RCF-convergent of x.

A $\bar{\mu}$-continuity, proper, inducible subregion $R \subset \Omega$ is (ε, C)-*superoptimal* if, for a.e. $x \in (0,1)$, the contracted Farey expansion of x with respect to R is (ε, C)-superoptimal.

From the preceding discussion, optimal CF-expansions satisfy the above definition with $\varepsilon \geq 1/2$ and $C \leq \log 2/\log G \approx 1.440420\ldots$. Recall the definition of $S_c \subset \Omega$ from (1.86).

Theorem 1.9.2. *Let $\varepsilon,\ C > 0$, and let $R \subset \Omega$ be a $\bar{\mu}$-continuity, proper, inducible subregion satisfying both*

$$(a)\quad R \subset S_\varepsilon \qquad \text{and} \qquad (b)\quad \bar{\mu}(R) \leq \frac{\log 2}{C}.$$

Then R is (ε, C)-superoptimal.

Proof. Let x belong to the full-measure subset of $(0,1)$ for which Lemma 1.8.7 holds, and let $R \subset \Omega$ satisfy conditions (a) and (b). By Corollary 1.6.11, the kth convergent of the contracted Farey expansion of x with respect to R is u^R_{k+1}/s^R_{k+1}. We have $s^R_{k+1} > 0$ for $k \geq 1$, and $\gcd(u^R_{k+1}, s^R_{k+1}) = 1$ since $\det(A^R_{[0,k+1]}) = \det(A_{[0,N^R_{k+1}]}) = \pm 1$. By Proposition 1.8.11, condition (a) guarantees $\Theta(x, u^R_{k+1}/s^R_{k+1}) \leq \varepsilon$ for all $k \geq 1$. Moreover, Equation (1.82) implies $q_{n(k)} \leq s^R_{k+1} < q_{n(k)+1}$, where $n(k) = j_{N^R_{k+1}} + \delta(k)$ and $\delta(k) \in \{0, -1\}$. By Lemma 1.8.7 and condition (b),

$$\lim_{k\to\infty} \frac{n(k)}{k} = \lim_{k\to\infty} \frac{j_{N^R_{k+1}} + \delta(k)}{k} = \frac{\log 2}{\bar{\mu}(R)} \geq C.$$

Thus the contracted Farey expansion of x with respect to R is (ε, C)-superoptimal. □

Below, we introduce two explicit families of superoptimal CFs: *linear recurrence* and *Legendre* CFs.

1.9.1 Linear recurrence continued fractions

We begin by introducing some symbolic notation. For $W = w_1 w_2 \ldots$ an infinite or finite word in the alphabet of positive integers, let $[0; W] := [0; w_1, w_2, \ldots]$. If $W = w_1 \cdots w_n$ is finite, denote the *transpose* of W by ${}^tW := w_n \cdots w_1$, and, moreover, let

$$\frac{p(W)}{q(W)} := [0; W] = [0; w_1, \ldots, w_n], \tag{1.92}$$

where $p(W), q(W) \in \mathbb{Z}$ are non-negative and relatively prime. We extend these notions to the empty word ϵ by setting ${}^t\epsilon := \epsilon$, $p(\epsilon) := 0$ and $q(\epsilon) := 1$.

Now let $R = [0,1] \times [b/(b+1), 1] \subset \Omega$ where b is a positive integer. One verifies that

$$\text{(a)} \quad R \subset S_{1/b} \qquad \text{and} \qquad \text{(b)} \quad \bar{\mu}(R) = \log(1 + 1/b),$$

so R is (ε, C)-superoptimal for any $\varepsilon \geq 1/b$ and $C \leq \log 2/\log(1 + 1/b)$. Let

$$\mathcal{W}_b = \{1, \dots, b-1\}^* \coloneqq \{\epsilon\} \cup \bigcup_{n \geq 1} \{1, \dots, b-1\}^n$$

be the set of all finite words $W = w_1 \cdots w_n$ in the alphabet $\{1, \dots, b-1\}$, including the empty word. Almost every $z = (x, 1) \in R$ may be written

$$z = ([0; W_1 a_1 W_2 a_2 \cdots], [0; 1]),$$

where for each $k \geq 1$, $W_k \in \mathcal{W}_b$ and $a_k \geq b$. Notice that $b/(b+1) = [0; 1, b]$. With z as above, we have by (1.36) that for all $k \geq 0$,

$$z_k^R = (x_k^R, y_k^R) = ([0; W_{k+1} a_{k+1} W_{k+2} a_{k+2} \cdots], \\ [0; 1 a_k \, {}^t W_k \cdots a_1 \, {}^t W_1]).$$

Furthermore, if $W_{k+1} \neq \epsilon$, writing $W_{k+1} = w_1^{(k+1)} \cdots w_{n_{k+1}}^{(k+1)}$ with each $w_j^{(k+1)} \in \{1, \dots, b-1\}$, (1.40) gives

$$\begin{aligned} \begin{pmatrix} u_R(z_k^R) & t_R(z_k^R) \\ s_R(z_k^R) & r_R(z_k^R) \end{pmatrix} &= A_R(z_k^R) \\ &= A_0^{w_1^{(k+1)}-1} A_1 \cdots A_0^{w_{n_{k+1}}^{(k+1)}-1} A_1 A_0^{a_{k+1}-1} A_1 \\ &= \begin{pmatrix} 0 & 1 \\ 1 & w_1^{(k+1)} \end{pmatrix} \cdots \begin{pmatrix} 0 & 1 \\ 1 & w_{n_{k+1}}^{(k+1)} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & a_{k+1} \end{pmatrix} \\ &= \begin{pmatrix} p(W_{k+1}) & p(W_{k+1} a_{k+1}) \\ q(W_{k+1}) & q(W_{k+1} a_{k+1}) \end{pmatrix}. \end{aligned}$$

Equality of the first and final matrices also holds for $W_{k+1} = \epsilon$. From the above computation, one finds that $\det(A_R(z_k^R)) = (-1)^{n_{k+1}+1}$, where $n_{k+1} = 0$ if $W_{k+1} = \epsilon$. Proposition 1.6.13 and Equations (1.53) and (1.54) give the digits of the contracted Farey expansion of x with respect to R:

$$\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} s_R(z_0^R) \\ u_R(z_0^R) \end{pmatrix} = \begin{pmatrix} q(W_1) \\ p(W_1) \end{pmatrix},$$

$$\begin{aligned}\begin{pmatrix}\alpha_1^R\\ \beta_1^R\end{pmatrix} &= \begin{pmatrix}\alpha_R(z_0^R)/d_R(z_0^R)\\ \beta_R(z_0^R)\end{pmatrix} = \begin{pmatrix}-\det(A_R(z_0^R))s_R(z_1^R))\\ s_R(z_0^R)u_R(z_1^R) + r_R(z_0^R)s_R(z_1^R)\end{pmatrix}\\ &= \begin{pmatrix}(-1)^{n_1}q(W_2)\\ q(W_1)p(W_2) + q(W_1a_1)q(W_2)\end{pmatrix},\end{aligned}$$

and for $k > 0$, since $d_R(z_k^R) = s_R(z_{k-1}^R) = q(W_k)$,

$$\begin{aligned}\begin{pmatrix}\alpha_{k+1}^R\\ \beta_{k+1}^R\end{pmatrix} &= \begin{pmatrix}\alpha_R(z_k^R)\\ \beta_R(z_k^R)\end{pmatrix} = \begin{pmatrix}-\det(A_R(z_k^R))d_R(z_k^R)s_R(z_{k+1}^R)\\ s_R(z_k^R)u_R(z_{k+1}^R) + r_R(z_k^R)s_R(z_{k+1}^R)\end{pmatrix}\\ &= \begin{pmatrix}(-1)^{n_{k+1}}q(W_k)q(W_{k+2})\\ q(W_{k+1})p(W_{k+2}) + q(W_{k+1}a_{k+1})q(W_{k+2})\end{pmatrix}.\end{aligned}$$

As the partial numerators α_k^R and partial denominators β_k^R are expressed in terms of (1.92)—and hence the relations (1.8)—we call this a *linear recurrence* CF-expansion.

Fibonacci continued fractions

Continuing notation from above, consider $R = [0,1] \times [b/(b+1), 1]$ with $b = 2$, i.e., $R = [0,1] \times [2/3, 1]$. In this case, R is (ε, C)-superoptimal for any $\varepsilon \geq 1/2$ and $C \leq \log 2/\log(3/2) \approx 1.709511\ldots$. Now $\mathcal{W}_b$ consists of the empty word and all finite words consisting only of the letter 1, so we find

$$\begin{aligned}\begin{pmatrix}p(W_{k+1}) & p(W_{k+1}a_{k+1})\\ q(W_{k+1}) & q(W_{k+1}a_{k+1})\end{pmatrix} &= \begin{pmatrix}0 & 1\\ 1 & 1\end{pmatrix}^{n_{k+1}}\begin{pmatrix}0 & 1\\ 1 & a_{k+1}\end{pmatrix}\\ &= \begin{pmatrix}F_{n_{k+1}-1} & F_{n_{k+1}}\\ F_{n_{k+1}} & F_{n_{k+1}+1}\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & a_{k+1}\end{pmatrix}\\ &= \begin{pmatrix}F_{n_{k+1}} & a_{k+1}F_{n_{k+1}} + F_{n_{k+1}-1}\\ F_{n_{k+1}+1} & a_{k+1}F_{n_{k+1}+1} + F_{n_{k+1}}\end{pmatrix},\end{aligned}$$

where F_n is the nth Fibonacci number, i.e., $F_n = F_{n-1}+F_{n-2}$ for $n \geq 1$, with $F_{-1} \coloneqq 1$ and $F_0 \coloneqq 0$. Thus the contracted Farey expansion of x with respect to R, which we call the *Fibonacci continued fraction*

expansion[24] of x, is given by $[\beta_0^R/\alpha_0^R; \alpha_1^R/\beta_1^R, \alpha_2^R/\beta_2^R, \dots]$, where

$$\begin{pmatrix}\alpha_0^R\\ \beta_0^R\end{pmatrix} = \begin{pmatrix}q(W_1)\\ p(W_1)\end{pmatrix} = \begin{pmatrix}F_{n_1+1}\\ F_{n_1}\end{pmatrix},$$

$$\begin{aligned}\begin{pmatrix}\alpha_1^R\\ \beta_1^R\end{pmatrix} &= \begin{pmatrix}(-1)^{n_1}q(W_2)\\ q(W_1)p(W_2)+q(W_1a_1)q(W_2)\end{pmatrix}\\ &= \begin{pmatrix}(-1)^{n_1}F_{n_2+1}\\ F_{n_1+1}F_{n_2}+(a_1F_{n_1+1}+F_{n_1})F_{n_2+1}\end{pmatrix},\end{aligned}$$

and for $k > 0$,

$$\begin{aligned}\begin{pmatrix}\alpha_{k+1}^R\\ \beta_{k+1}^R\end{pmatrix} &= \begin{pmatrix}(-1)^{n_{k+1}}q(W_k)q(W_{k+2})\\ q(W_{k+1})p(W_{k+2})+q(W_{k+1}a_{k+1})q(W_{k+2})\end{pmatrix}\\ &= \begin{pmatrix}(-1)^{n_{k+1}}F_{n_k+1}F_{n_{k+2}+1}\\ F_{n_{k+1}+1}F_{n_{k+2}}+(a_{k+1}F_{n_{k+1}+1}+F_{n_{k+1}})F_{n_{k+2}+1}\end{pmatrix}.\end{aligned}$$

Example 1.9.3. Let

$$x = \frac{\sqrt{4226}-51}{25} = [0; \overline{1,1,3,1,1,1,4}],$$

where $[0; \overline{a_1, \dots, a_n}] := [0; a_1, \dots, a_n, a_1, \dots, a_n, a_1, \dots]$ denotes a periodic RCF-expansion with periodic part $a_1, \dots, a_n$. We may write $x = [0; W_1a_1W_2a_2\cdots]$, where

$$W_j = \left\{\begin{array}{ll}11 & \text{if } j \text{ is odd}\\ 111 & \text{if } j \text{ is even}\end{array}\right\} \quad\Longrightarrow\quad n_j = \left\{\begin{array}{ll}2 & \text{if } j \text{ is odd}\\ 3 & \text{if } j \text{ is even}\end{array}\right\},$$

and

$$a_j = \begin{cases}3 & \text{if } j \text{ is odd},\\ 4 & \text{if } j \text{ is even}.\end{cases}$$

[24] See also [26], where contraction is applied to obtain a GCF-expansion of the golden mean, $(\sqrt{5}+1)/2$, in terms of Fibonacci numbers.

We thus compute the Fibonacci CF-expansion of x as

$$\begin{pmatrix}\alpha_0^R\\ \beta_0^R\end{pmatrix} = \begin{pmatrix}F_{n_1+1}\\ F_{n_1}\end{pmatrix} = \begin{pmatrix}F_3\\ F_2\end{pmatrix} = \begin{pmatrix}2\\ 1\end{pmatrix},$$

$$\begin{aligned}\begin{pmatrix}\alpha_1^R\\ \beta_1^R\end{pmatrix} &= \begin{pmatrix}(-1)^{n_1}F_{n_2+1}\\ F_{n_1+1}F_{n_2} + (a_1F_{n_1+1} + F_{n_1})F_{n_2+1}\end{pmatrix}\\ &= \begin{pmatrix}(-1)^2F_4\\ F_3^2 + (3F_3 + F_2)F_4\end{pmatrix} = \begin{pmatrix}3\\ 25\end{pmatrix},\end{aligned}$$

and for $k > 0$,

$$\begin{aligned}\begin{pmatrix}\alpha_{k+1}^R\\ \beta_{k+1}^R\end{pmatrix} &= \begin{pmatrix}(-1)^{n_{k+1}}F_{n_k+1}F_{n_{k+2}+1}\\ F_{n_{k+1}+1}F_{n_{k+2}} + (a_{k+1}F_{n_{k+1}+1} + F_{n_{k+1}})F_{n_{k+2}+1}\end{pmatrix}\\ &= \begin{cases}\begin{pmatrix}(-1)^3F_3^2\\ F_4F_2 + (4F_4 + F_3)F_3\end{pmatrix} & \text{if } k \text{ is odd},\\ \begin{pmatrix}(-1)^2F_4^2\\ F_3^2 + (3F_3 + F_2)F_4\end{pmatrix} & \text{if } k \text{ is even},\end{cases}\\ &= \begin{cases}\begin{pmatrix}-4\\ 31\end{pmatrix} & \text{if } k \text{ is odd},\\ \begin{pmatrix}9\\ 25\end{pmatrix} & \text{if } k \text{ is even}.\end{cases}\end{aligned}$$

More succinctly, we have

$$\begin{aligned}x =& [1/2; 3/25, \overline{-4/31, 9/25}]\\ :=& [1/2; 3/25, -4/31, 9/25, -4/31, 9/25, -4/31, 9/25, \dots].\end{aligned}$$

The convergents of the Fibonacci CF-expansion of x are

$$\left(\frac{u_n^R}{s_n^R}\right)_{n\geq 0} = \left(\frac{1}{0}, \frac{1}{2}, \frac{14}{25}, \frac{144}{257}, \frac{1821}{3250}, \dots\right)$$

and form a subsequence of the RCF-convergents

$$\left(\frac{p_n}{q_n}\right)_{n\geq -1} = \left(\frac{1}{0}, \frac{0}{1}, \frac{1}{1}, \frac{1}{2}, \frac{4}{7}, \frac{5}{9}, \frac{9}{16}, \frac{14}{25}, \frac{65}{116}, \frac{79}{141}, \frac{144}{257}, \frac{511}{912}, \frac{655}{1169}, \frac{1166}{2081}, \frac{1821}{3250}, \frac{8450}{15081}, \frac{10271}{18331}, \dots\right).$$

1.9.2 Legendre continued fractions

Let $\varepsilon_0 \leq 1/2$ and let R be the set S_{ε_0} with the lower boundary curve removed, i.e.,

$$R = \{z \in \Omega \mid h(z) < \varepsilon_0\};$$

see (1.85) and (1.86). One computes that $\bar{\mu}(R) = \varepsilon_0$, so by Theorem 1.9.2, R is (ε, C)-superoptimal for any $\varepsilon \geq \varepsilon_0$ and any $C \leq \log 2/\varepsilon_0$. Note that if $z = (x, 1)$ with x irrational and the $\mathcal{F}$-orbit of z enters R infinitely often, then Proposition 1.8.11 implies that $(u_n^R/s_n^R)_{n\geq 0}$ consists of those Farey convergents of x satisfying $\Theta(x, u_n/s_n) < \varepsilon_0$. By Legendre's Theorem (Theorem 1.8.25), our assumption that $\varepsilon_0 \leq 1/2$ and the fact that the sequence of Farey convergents of x contains the RCF-convergents as a subsequence, $(u_n^R/s_n^R)_{n\geq 1}$ is *precisely* the sequence of all rationals p/q with $\gcd(p, q) = 1$, $q > 0$, satisfying $\Theta(x, p/q) < \varepsilon_0$. We call the contracted Farey expansion of x with respect to R the ε_0–*Legendre* CF-expansion of x. We analyse ε_0–Legendre expansions and the region R in further detail.

Suppose $z = (x, y) \in R$ with $x = [0; a_1, a_2, \dots]$, and consider the hitting time $N_R(z)$ to R from (1.38). Since $R \subset H_1$, (1.21) implies that $N_R(z)$ may be written as a sum

$$N_R(z) = a_1 + a_2 + \cdots + a_{j(z)}$$

of partial denominators of the RCF-expansion of x, where—to ease

notation—we set $j(z) = j_{N_R(z)}$. By (1.40),

$$\begin{aligned}\begin{pmatrix} u_R(z) & t_R(z) \\ s_R(z) & r_R(z) \end{pmatrix} &= A_R(z) = A_0^{a_1-1}A_1 \cdots A_0^{a_{j(z)}-1}A_1 \\ &= \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & a_2 \end{pmatrix}\cdots\begin{pmatrix} 0 & 1 \\ 1 & a_{j(z)} \end{pmatrix} \\ &= \begin{pmatrix} p_{j(z)-1}(x) & p_{j(z)}(x) \\ q_{j(z)-1}(x) & q_{j(z)}(x) \end{pmatrix},\end{aligned} \tag{1.93}$$

where $p_j(x)/q_j(x) = p_j/q_j$ is the jth RCF-convergent of x. (The dependence on x in the notation will be useful below.) Beginning with $z = (x, 1)$ and setting $z_k^R = (x_k^R, y_k^R) = \mathcal{F}_R^k(z)$, $k \geq 0$, Proposition 1.6.13 and Equations (1.53) and (1.54) give the digits of the ε_0–Legendre expansion as

$$\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} s_R(z_0^R) \\ u_R(z_0^R) \end{pmatrix} = \begin{pmatrix} q_{j(z_0^R)-1}(x_0^R) \\ p_{j(z_0^R)-1}(x_0^R) \end{pmatrix}, \tag{1.94}$$

$$\begin{aligned}\begin{pmatrix} \alpha_1^R \\ \beta_1^R \end{pmatrix} &= \begin{pmatrix} \alpha_R(z_0^R)/d_R(z_0^R) \\ \beta_R(z_0^R) \end{pmatrix} \\ &= \begin{pmatrix} (-1)^{j(z_0^R)+1}q_{j(z_1^R)-1}(x_1^R) \\ q_{j(z_0^R)-1}(x_0^R)p_{j(z_1^R)-1}(x_1^R) + q_{j(z_0^R)}(x_0^R)q_{j(z_1^R)-1}(x_1^R) \end{pmatrix},\end{aligned} \tag{1.95}$$

and for $k > 0$,

$$\begin{aligned}\begin{pmatrix} \alpha_{k+1}^R \\ \beta_{k+1}^R \end{pmatrix} &= \begin{pmatrix} \alpha_R(z_k^R) \\ \beta_R(z_k^R) \end{pmatrix} \\ &= \begin{pmatrix} (-1)^{j(z_k^R)+1}q_{j(z_{k-1}^R)-1}(x_{k-1}^R)q_{j(z_{k+1}^R)-1}(x_{k+1}^R) \\ q_{j(z_k^R)-1}(x_k^R)p_{j(z_{k+1}^R)-1}(x_{k+1}^R) + q_{j(z_k^R)}(x_k^R)q_{j(z_{k+1}^R)-1}(x_{k+1}^R) \end{pmatrix}.\end{aligned} \tag{1.96}$$

Note that the digits depend heavily on the integers $j(z_k^R) = j_{N_R(z_k^R)}$. To aid our computation of these digits, we decompose the region R into subregions

$$R_j \coloneqq \{z \in R \mid j(z) = j\}$$

on which $j(z)$ is constant. Using the 'altered' map $\mathcal{F}_{H_1}$ from Remark 1.5.4, $j(z)$ may be thought of as the return time of z to R under $\mathcal{F}_{H_1}$,

$$j(z) = \inf\left\{ j \geq 1 \;\middle|\; \mathcal{F}_{H_1}^j(z) \in R \right\}.$$

Let $z = (x, y) \in H_1$ with $x = [0; a_1, a_2, \dots]$, and set $z_j^{H_1} = \mathcal{F}_{H_1}^j(z)$, $j \geq 0$. Note that (1.42), (1.44) and (1.63) give

$$\begin{aligned} A_{[0,j]}^{H_1}(z) =& A_{H_1}(z_0^{H_1}) A_{H_1}(z_1^{H_1}) \cdots A_{H_1}(z_{j-1}^{H_1}) \\ =& \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & a_2 \end{pmatrix} \cdots \begin{pmatrix} 0 & 1 \\ 1 & a_j \end{pmatrix} = \begin{pmatrix} p_{j-1} & p_j \\ q_{j-1} & q_j \end{pmatrix}, \end{aligned} \tag{1.97}$$

so by (1.93), $A_R(z) = A_{[0,j(z)]}^{H_1}(z)$. The sets R_j may be written recursively as

$$R_1 = \left\{ z \in R \;\middle|\; z_1^{H_1} \in R \right\}$$

and

$$R_j = \left\{ z \in R \setminus \bigcup_{\ell=1}^{j-1} R_\ell \;\middle|\; z_j^{H_1} \in R \right\}, \quad j > 1.$$

By definition of R, $z_j^{H_1} \in R$ if and only if $h(z_j^{H_1}) < \varepsilon_0$, and by Lemma 1.8.18 and (1.97),

$$h(z_j^{H_1}) = (-1)^j \frac{(p_{j-1} - q_{j-1}x)(p_{j-1} + (q_{j-1} - p_{j-1})y)}{x + y - xy}, \tag{1.98}$$

where $z = (x, y)$ and p_j/q_j is the jth RCF-convergent of x. Since $p_0 = 0$ and $q_0 = 1$ for any $x \in (0, 1)$,

$$R_1 = \left\{ z \in R \;\middle|\; \frac{xy}{x + y - xy} < \varepsilon_0 \right\}, \tag{1.99}$$

and for $j > 1$,

$$R_j = \left\{ z \in R \setminus \bigcup_{\ell=1}^{j-1} R_\ell \;\middle|\; h(z_j^{H_1}) < \varepsilon_0 \right\}, \tag{1.100}$$

with $h(z_j^{H_1})$ as in (1.98).

In general, the ε_0–Legendre CF is not defined for a null-set of points in $(0,1)$. However, for ε_0 large enough, the ε_0–Legendre expansion is defined *for all* irrationals $x \in (0,1)$. We illustrate with the smallest value of ε_0 for which this is true.

Hurwitz–Borel continued fractions

In 1891, Hurwitz proved that for any irrational number x, there are infinitely many pairs of relatively prime integers p and q, $q > 0$, such that

$$\Theta(x, p/q) < 1/\sqrt{5}, \tag{1.101}$$

and, moreover, the constant $1/\sqrt{5}$ is optimal ([89]). In fact, the first part of Hurwitz's result is a direct consequence of the aforementioned 1903 result of Borel on three consecutive RCF-approximation coefficients; see (1.77). It follows that for any irrational x, the $1/\sqrt{5}$–Legendre CF-expansion of x exists, and the convergents $\{u_n^R/s_n^R\}_{n\geq 1}$ are precisely the set of *all* rationals satisfying (1.101). We call the $1/\sqrt{5}$–Legendre CF-expansion of x the *Hurwitz–Borel* CF-expansion of x. Hurwitz–Borel CFs are (ε, C)-superoptimal for any $\varepsilon \geq 1/\sqrt{5} \approx 0.447213\ldots$ and $C \leq \sqrt{5}\log 2 \approx 1.549924\ldots$.

Let us consider the regions R_j. By (1.99),

$$R_1 = \left\{ z \in R \;\middle|\; \frac{xy}{x+y-xy} < 1/\sqrt{5} \right\};$$

see the blue region of Figure 1.13. Now $R_2 \subset R \setminus R_1 \subset V_1 \cup V_2$. Hence, if $z = (x,y) \in R_2$ with $x = [0; a_1, a_2, \ldots]$, either $z \in R_2 \cap V_1$ and $a_1 = 1$, or $z \in R_2 \cap V_2$ and $a_1 = 2$. The first RCF-convergent of x is thus $p_1/q_1 = 1/1$ if $z \in R_2 \cap V_1$ or $p_1/q_1 = 1/2$ if $z \in R_2 \cap V_2$, and by (1.100),

$$R_2 = \begin{cases} \left\{ z \in R \setminus R_1 \;\middle|\; \frac{1-x}{x+y-xy} < 1/\sqrt{5} \right\}, & z \in R_2 \cap V_1, \\ \left\{ z \in R \setminus R_1 \;\middle|\; \frac{(1-2x)(1+y)}{x+y-xy} < 1/\sqrt{5} \right\}, & z \in R_2 \cap V_2; \end{cases}$$

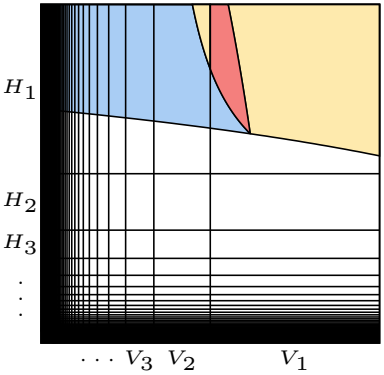


Figure 1.13: The region R (union of colored regions) defining Hurwitz–Borel continued fractions and its partition into subregions R_1 (blue), R_2 (tan) and R_3 (red).

see the tan regions[25] in Figure 1.13. Next, $R_3 \subset R \setminus (R_1 \cup R_2) \subset V_1 \cap ((1/2, 2/3] \times (1/2, 1])$. This implies that if $z = (x, y) \in R_3$ with $x = [0; a_1, a_2, \dots]$, then $a_1 = a_2 = 1$. The second RCF-convergent of x is thus $p_2/q_2 = 1/2$, so by (1.100),

$$R_3 = \left\{ z \in R \setminus (R_1 \cup R_2) \;\middle|\; \frac{(2x-1)(1+y)}{x+y-xy} < 1/\sqrt{5} \right\};$$

see the red region[26] in Figure 1.13. Since $R = R_1 \cup R_2 \cup R_3$, we have $R_j = \varnothing$ for $j > 3$.

Remark 1.9.4. That $j(z) \le 3$ for any $z = (x, y)$ with x irrational implies Borel's result (1.77), and hence the first statement of Hurwitz's result (1.101). Optimality of $1/\sqrt{5}$ can also be obtained by showing that $\mathcal{F}^n(g, 1) \to (g, g)$ and $h(g, g) = 1/\sqrt{5}$, where $g = (\sqrt{5} - 1)/2$.

[25] The curve $(1-2x)(1+y)/(x+y-xy) = 1/\sqrt{5}$ is not shown, as its intersection with R is contained in R_1.

[26] The curve $(2x-1)(1+y)/(x+y-xy) = 1/\sqrt{5}$ is not shown, as its intersection with R is contained in $R_2 \cap V_1$.

We illustrate the use of this partition for computing Hurwitz–Borel expansions with an example.

Example 1.9.5. Let $x = \pi - 3 = [0; 7, 15, 1, 292, 1, 1, \dots]$. We compute

$$\begin{aligned}
z_0^R =& ([0; 7, 15, 1, 292, 1, 1, \dots], [0; 1]) \\
\approx& (0.1415926\dots, 1) \in R_1, \\
z_1^R =& ([0; 15, 1, 292, 1, 1, 1, \dots], [0; 1, 7]) \\
\approx& (0.062513\dots, 0.875) \in R_1, \\
z_2^R =& ([0; 1, 292, 1, 1, 1, 2, \dots], [0; 1, 15, 7]) \\
\approx& (0.996594\dots, 0.938053\dots) \in R_2 \cap V_1, \\
z_3^R =& ([0; 1, 1, 1, 2, 1, 3, \dots], [0; 1, 292, 1, 15, 7]) \\
\approx& (0.634591\dots, 0.996597\dots) \in R_2 \cap V_1, \\
z_4^R =& ([0; 1, 2, 1, 3, 1, 14, \dots], [0; 1, 1, 1, 292, 1, 15, 7]) \\
\approx& (0.736659\dots, 0.666288\dots) \in R_2 \cap V_1, \\
z_5^R =& ([0; 1, 3, 1, 14, 2, 1, \dots], [0; 1, 2, 1, 1, 1, 292, 1, 15, 7]) \\
\approx& (0.797366\dots, 0.727244\dots) \in R_2 \cap V_1, \\
z_6^R =& ([0; 1, 14, 2, 1, 1, 2, \dots], [0; 1, 3, 1, 2, 1, 1, 1, 292, 1, 15, 7]) \\
\approx& (0.935008\dots, 0.788460\dots) \in R_2 \cap V_1, \\
z_7^R =& ([0; 2, 1, 1, 2, 2, 2, \dots], [0; 1, 14, 1, 3, 1, 2, 1, 1, 1, 292, 1, 15, 7]) \\
\approx& (0.386718\dots, 0.936662\dots) \in R_1, \\
z_8^R =& ([0; 1, 1, 2, 2, 2, 2, \dots], [0; 1, 2, 14, 1, 3, 1, 2, 1, 1, 1, 292, 1, 15, 7]) \\
\approx& (0.585857\dots, 0.674014\dots) \in R_3, \\
& \vdots
\end{aligned}$$

Using (1.94)–(1.96), we compute

$$\begin{pmatrix} \alpha_0^R \\ \beta_0^R \end{pmatrix} = \begin{pmatrix} q_0(x_0^R) \\ p_0(x_0^R) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$$\begin{aligned}
\begin{pmatrix}\alpha_1^R\\ \beta_1^R\end{pmatrix} &= \begin{pmatrix}(-1)^2 q_0(x_1^R)\\ q_0(x_0^R)p_0(x_1^R)+q_1(x_0^R)q_0(x_1^R)\end{pmatrix} = \begin{pmatrix}1\cdot 1\\ 1\cdot 0+7\cdot 1\end{pmatrix} = \begin{pmatrix}1\\ 7\end{pmatrix},\\
\begin{pmatrix}\alpha_2^R\\ \beta_2^R\end{pmatrix} &= \begin{pmatrix}(-1)^2 q_0(x_0^R)q_1(x_2^R)\\ q_0(x_1^R)p_1(x_2^R)+q_1(x_1^R)q_1(x_2^R)\end{pmatrix} = \begin{pmatrix}1\cdot 1\cdot 1\\ 1\cdot 1+15\cdot 1\end{pmatrix} = \begin{pmatrix}1\\ 16\end{pmatrix},\\
\begin{pmatrix}\alpha_3^R\\ \beta_3^R\end{pmatrix} &= \begin{pmatrix}(-1)^3 q_0(x_1^R)q_1(x_3^R)\\ q_1(x_2^R)p_1(x_3^R)+q_2(x_2^R)q_1(x_3^R)\end{pmatrix} = \begin{pmatrix}-1\cdot 1\cdot 1\\ 1\cdot 1+293\cdot 1\end{pmatrix} = \begin{pmatrix}-1\\ 294\end{pmatrix},\\
\begin{pmatrix}\alpha_4^R\\ \beta_4^R\end{pmatrix} &= \begin{pmatrix}(-1)^3 q_1(x_2^R)q_1(x_4^R)\\ q_1(x_3^R)p_1(x_4^R)+q_2(x_3^R)q_1(x_4^R)\end{pmatrix} = \begin{pmatrix}-1\cdot 1\cdot 1\\ 1\cdot 1+2\cdot 1\end{pmatrix} = \begin{pmatrix}-1\\ 3\end{pmatrix},\\
\begin{pmatrix}\alpha_5^R\\ \beta_5^R\end{pmatrix} &= \begin{pmatrix}(-1)^3 q_1(x_3^R)q_1(x_5^R)\\ q_1(x_4^R)p_1(x_5^R)+q_2(x_4^R)q_1(x_5^R)\end{pmatrix} = \begin{pmatrix}-1\cdot 1\cdot 1\\ 1\cdot 1+3\cdot 1\end{pmatrix} = \begin{pmatrix}-1\\ 4\end{pmatrix},\\
\begin{pmatrix}\alpha_6^R\\ \beta_6^R\end{pmatrix} &= \begin{pmatrix}(-1)^3 q_1(x_4^R)q_1(x_6^R)\\ q_1(x_5^R)p_1(x_6^R)+q_2(x_5^R)q_1(x_6^R)\end{pmatrix} = \begin{pmatrix}-1\cdot 1\cdot 1\\ 1\cdot 1+4\cdot 1\end{pmatrix} = \begin{pmatrix}-1\\ 5\end{pmatrix},\\
\begin{pmatrix}\alpha_7^R\\ \beta_7^R\end{pmatrix} &= \begin{pmatrix}(-1)^3 q_1(x_5^R)q_0(x_7^R)\\ q_1(x_6^R)p_0(x_7^R)+q_2(x_6^R)q_0(x_7^R)\end{pmatrix} = \begin{pmatrix}-1\cdot 1\cdot 1\\ 1\cdot 0+15\cdot 1\end{pmatrix} = \begin{pmatrix}-1\\ 15\end{pmatrix},\\
\begin{pmatrix}\alpha_8^R\\ \beta_8^R\end{pmatrix} &= \begin{pmatrix}(-1)^2 q_1(x_6^R)q_2(x_8^R)\\ q_0(x_7^R)p_2(x_8^R)+q_1(x_7^R)q_2(x_8^R)\end{pmatrix} = \begin{pmatrix}1\cdot 1\cdot 2\\ 1\cdot 1+2\cdot 2\end{pmatrix} = \begin{pmatrix}2\\ 5\end{pmatrix},\\
&\vdots
\end{aligned}$$

Hence, the first few terms of the Hurwitz–Borel expansion are

$$\begin{aligned}
\pi-3 =&[\beta_0^R/\alpha_0^R;\alpha_1^R/\beta_1^R,\alpha_2^R/\beta_2^R,\dots]\\
=&[0/1;1/7,1/16,-1/294,-1/3,-1/4,-1/5,-1/15,2/5,\dots].
\end{aligned}$$

The convergents are

$$\left(\frac{u_n^R}{s_n^R}\right)_{n\ge 1} = \left(\frac{0}{1},\frac{1}{7},\frac{16}{113},\frac{4703}{33215},\frac{14093}{99532},\frac{51669}{364913},\frac{244252}{1725033},\frac{3612111}{25510582},\frac{18549059}{131002976},\dots\right),$$

forming the subsequence of RCF-convergents

$$\left(\frac{p_n}{q_n}\right)_{n\geq 0} = \left(\frac{0}{1}, \frac{1}{7}, \frac{15}{106}, \frac{16}{113}, \frac{4687}{33102}, \frac{4703}{33215}, \frac{9390}{66317}, \frac{14093}{99532}, \frac{37576}{265381}, \frac{51669}{364913}, \frac{192583}{1360120}, \dots\right)$$

which satisfy (1.101).

1.10 Appendix: Rearranging sequences of Farey convergents

Let $x \in (0,1)\backslash\mathbb{Q}$ with RCF-expansion $x = [0; a_1, a_2, \dots]$. Recall from (1.25) that the denominators of the sequence of Farey convergents $(u_n/s_n)_{n\geq 0}$ of x are given by

$$\begin{aligned}(s_n)_{n\geq 0} = (q_{-1}\ &, q_0 + q_{-1}\ , \dots, (a_1-1)q_0 + q_{-1}, \\ q_0\ &, q_1 + q_0\ \ , \dots, (a_2-1)q_1 + q_0, \dots, \\ &q_{j-1}, q_j + q_{j-1}, \dots, (a_{j+1}-1)q_j + q_{j-1}, \dots).\end{aligned} \tag{1.102}$$

Let ρ be the bijection[27] of non-negative integers for which these denominators are arranged in increasing order:

$$\begin{aligned}(s_{\rho(n)})_{n\geq 0} = (q_{-1}, q_0, q_0 + q_{-1}\ &, \dots, (a_1-1)q_0 + q_{-1}, \\ q_1, q_1 + q_0\ \ &, \dots, (a_2-1)q_1 + q_0, \dots, \\ &q_j, q_j + q_{j-1}, \dots, (a_{j+1}-1)q_j + q_{j-1}, \dots).\end{aligned} \tag{1.103}$$

For an inducible subregion $R \subset \Omega$, let ρ_R be the bijection of non-negative integers for which $(u^R_{\rho_R(n)}/s^R_{\rho_R(n)})_{n\geq 0}$ forms a subsequence of $(u_{\rho(n)}/s_{\rho(n)})_{n\geq 0}$. That is, ρ_R permutes elements of the sequence $(u^R_n/s^R_n)_{n\geq 0}$ of Farey convergents determined by R so as to have increasing denominators. We aim to prove the following two results:

[27] For $a_1 > 1$ there are in fact two such bijections since $s_n = s_m$ if and only if $n = m$ or (since $q_{-1} = 0$) $\{n, m\} = \{1, a_1\}$. However, in the limits considered below, the choice between these two bijections becomes irrelevant.

Proposition 1.10.1. *Let $R \subset \Omega$ be a $\bar{\mu}$-continuity, proper, inducible subregion. Then for any $x \in (0,1)\backslash\mathbb{Q}$,*

(i) $\lim\limits_{n\to\infty} \frac{1}{n} \log s^R_{\rho_R(n)} = \lim\limits_{n\to\infty} \frac{1}{n} \log s^R_n$ *and*

(ii) $\lim\limits_{n\to\infty} \frac{1}{n} \log \left| x - \frac{u^R_{\rho_R(n)}}{s^R_{\rho_R(n)}} \right| = \lim\limits_{n\to\infty} \frac{1}{n} \log \left| x - \frac{u^R_n}{s^R_n} \right|$

when any of the limits exist.

Proposition 1.10.2. *Let $R \subset \Omega$ be a $\bar{\mu}$-continuity, proper, inducible subregion and $c \in [0,\infty)$. Then for any $x \in (0,1)\backslash\mathbb{Q}$,*

$$\lim_{n\to\infty} \frac{1}{n} \#\big\{0 \le k < n \mid \Theta^R_{\rho_R(k)}(x) \le c\big\} = \lim_{n\to\infty} \frac{1}{n} \#\big\{0 \le k < n \mid \Theta^R_k(x) \le c\big\}$$

when either limit exists.

To prove these, we need the following:

Lemma 1.10.3. *For any $x = [0; a_1, a_2, \dots] \in (0,1)\backslash\mathbb{Q}$, any inducible $R \subset \Omega$ and any $n \ge 0$, the cardinality of the symmetric difference between the sets $\{N^R_k\}_{k=0}^n$ and $\{N^R_{\rho_R(k)}\}_{k=0}^n$ is at most two, and $|j_{N^R_{\rho_R(n)}} - j_{N^R_n}| \le 1$.*

Proof. We begin with preliminary notation and observations. Set $A_0 := 0$ and for $j \ge 1$ set $A_j := \sum_{k=1}^j a_k$. From (1.102) and (1.103), one finds that ρ fixes $A_0 = 0$ and, for $j \ge 0$, acts as a 'cyclic permutation' on subsequent blocks of length a_{j+1}, namely

$$\begin{aligned}(A_j + 1, A_j + 2, \dots, A_j + a_{j+1} - 1, A_{j+1}) \\ \overset{\rho}{\mapsto} (A_{j+1}, A_j + 1, A_j + 2, \dots, A_j + a_{j+1} - 1),\end{aligned} \tag{1.104}$$

where ρ is applied entry-wise. In particular, for $N > 0$ with $A_j < N \le A_{j+1}$,

$$\rho(\{0, 1, \dots, N\}) = \{0, 1, \dots, N-1\} \cup \{A_{j+1}\}. \tag{1.105}$$

Also observe that since $s^R_k = s_{N^R_k}$ and $s^R_{\rho_R(k)} = s_{N^R_{\rho_R(k)}}$, the numbers N^R_k and $N^R_{\rho_R(k)}$ are the indices of the $(k+1)^{\text{st}}$ elements of the set $\{s^R_n\}_{n\geq 0} = \{s^R_{\rho_R(n)}\}_{n\geq 0}$ to appear in the sequences $(s_n)_{n\geq 0}$ and $(s_{\rho(n)})_{n\geq 0}$, respectively.

If $n = 0$, then since $N^R_0 = N^R_{\rho_R(0)} = 0$, the statement of the lemma holds. Now fix $n > 0$ and let j be such that $A_j < N^R_n \leq A_{j+1}$. We consider two cases:

(i) If $s_{A_{j+1}} \notin \{s^R_k\}_{k\geq 0}$, then (1.105) implies that $\{s_{N^R_{\rho_R(k)}}\}^n_{k=0} = \{s^R_{N_k}\}^n_{k=0}$, so[28] $\{N^R_k\}^n_{k=0} = \{N^R_{\rho_R(k)}\}^n_{k=0}$. Moreover, (1.104) implies that $N^R_{\rho_R(n)} = N^R_n$, and thus the claim holds in this case.

(ii) Suppose $s_{A_{j+1}} \in \{s^R_k\}_{k\geq 0}$. Then (1.105) gives $\{s_{N^R_{\rho_R(k)}}\}^n_{k=0} = \{s_{N^R_k}\}^{n-1}_{k=0} \cup \{s_{A_{j+1}}\}$. In particular, the symmetric difference between $\{N^R_k\}^n_{k=0}$ and $\{N^R_{\rho_R(k)}\}^n_{k=0}$ is zero if $N^R_n = A_{j+1}$ and two if $N^R_n < A_{j+1}$. Moreover, (1.104) implies that

$$N^R_{\rho_R(n)} = \begin{cases} N^R_{n-1} & \text{if } A_j < N^R_{n-1}, \\ A_{j+1} & \text{otherwise.} \end{cases}$$

If $N^R_{\rho_R(n)} = N^R_{n-1}$, then $A_j < N^R_{n-1} < N^R_n \leq A_{j+1}$ implies $j_{N^R_{\rho_R(n)}} = j$, while if $N^R_{\rho_R(n)} = A_{j+1}$ we have $j_{N^R_{\rho_R(n)}} = j+1$. But also $A_j < N^R_n \leq A_{j+1}$ implies $j_{N^R_n} = j$ or $j_{N^R_n} = j+1$ (with the latter occurring if and only if $N^R_n = A_{j+1}$). Thus $|j_{N^R_{\rho_R(n)}} - j_{N^R_n}| \leq 1$. □

Propositions 1.10.1 and 1.10.2 now follow almost immediately from Lemma 1.10.3:

[28] Recall that $s_n = s_m$ if and only if $n = m$ or $\{n, m\} = \{1, a_1\}$. Throughout, we consider s_1 and s_{a_1} for $a_1 > 1$ as *distinct* elements of $\{s_n\}_{n\geq 0}$, even though as integers these both equal $q_0 = 1$.

Proof of Proposition 1.10.1. By Lemma 1.10.3,

$$\lim_{n\to\infty} \frac{j_{N^R_{\rho_R(n)}}}{n} = \lim_{n\to\infty} \frac{j_{N^R_n}}{n}$$

whenever either limit exists. Using this, the proof of Theorem 1.8.8 goes through with each of s^R_n, r^R_n, x^R_n and N^R_n replaced by $s^R_{\rho_R(n)}$, $r^R_{\rho_R(n)}$, $x^R_{\rho_R(n)}$ and $N^R_{\rho_R(n)}$, respectively. □

Proof of Proposition 1.10.2. By definition (see (1.47) and (1.84)), $\Theta^R_{\rho_R(k)}(x) = \Theta(x, u_{N^R_{\rho_R(k)}}/s_{N^R_{\rho_R(k)}})$ and $\Theta^R_k(x) = \Theta(x, u_{N^R_k}/s_{N^R_k})$, so Lemma 1.10.3 implies that for any $n > 0$,

$$\left|\#\left\{0 \le k < n \mid \Theta^R_{\rho_R(k)}(x) \le c\right\} - \#\left\{0 \le k < n \mid \Theta^R_k(x) \le c\right\}\right| \le 2.$$

Dividing by n and taking $n \to \infty$ gives the result. □

Chapter 2

Ergodic properties of a parameterised family of symmetric golden maps

2.1 Introduction

Dynamical systems given by piecewise monotone maps $T : I \to I$ of an interval have a rich history: besides having applications in various fields—including population ecology ([22]) and controlled switching circuits ([9])—these systems are often used to produce expansions of numbers from the underlying interval I. Examples include decimal, n-ary, continued fraction, (generalised) Lüroth and β-expansions, though this list is far from exhaustive. A common theme in the study of these expansions is the investigation of asymptotic relative frequencies of digits occurring in typical (i.e., Lebesgue–almost all) expansions. To this end, the standard procedure is the construction of an ergodic, T-invariant probability measure μ equivalent to Lebesgue measure λ and a calculation of the μ-measure of the subin-

terval of I corresponding to the digit(s) in question. Birkhoff's ergodic theorem asserts that the measure of this subinterval equals the desired asymptotic frequency.

In [45], Dajani and Kalle study invariant measures and frequencies of digits for a family of *symmetric doubling maps* $\{D_\eta\}_{\eta\in[1,2]}$ defined on $[-1,1]$ by $D_\eta(x) \coloneqq 2x - d(x)\eta$ with $d(x) \in \{-1,0,1\}$. These maps produce *signed binary expansions* of numbers $x \in [-1,1]$ of the form $x = \eta\sum_{n\geq 1} d_n/2^n$ with each $d_n \in \{-1,0,1\}$. It is shown that each D_η, $\eta > 1$, admits an ergodic, invariant probability measure equivalent to Lebesgue measure. The authors use a curious property called *matching*—defined in the sequel—to prove that there is a countable collection of disjoint, open subintervals of $[1,2]$ whose union has full measure, and such that on each such subinterval, the densities of the corresponding invariant measures are step functions with at most the same, finite number of jumps. These explicitly constructed measures are then used to study the asymptotic frequency of the digit 0 in generic expansions. This frequency is shown to be continuous as a function of η and attains a maximum value of $2/3$ on the maximal interval $[6/5, 3/2]$. Moreover, the frequency function is either constant, strictly increasing or strictly decreasing on each of the aforementioned subintervals of $[1,2]$.

The present chapter continues these themes of inquiry with a parameterised family of *skewed symmetric golden maps* $\{T_\alpha\}_{\alpha\in[1,\beta]}$, with $\beta = (\sqrt{5}+1)/2$ the golden mean, i.e., the positive real solution to $\beta^2 = \beta + 1$. Each $T_\alpha : [-1,1] \to [-1,1]$ is defined by

$$T_\alpha(x) \coloneqq \begin{cases} \beta^2 x + \beta\alpha, & x \in [-1, -1/\beta), \\ \beta x, & x \in [-1/\beta, 1/\beta], \\ \beta^2 x - \beta\alpha, & x \in (1/\beta, 1]; \end{cases}$$

see Figure 2.1. Note that the interval $[1,\beta]$ is the maximal parameter space for α which guarantees that the image of each branch of T_α is contained in $[-1,1]$. Setting $J_{-1} \coloneqq [-1,-1/\beta)$, $J_0 \coloneqq [-1/\beta, 1/\beta]$ and $J_1 \coloneqq (1/\beta, 1]$, the map T_α may be written more succinctly as

$$T_\alpha(x) = \beta^{1+|t(x)|}x - t(x)\beta\alpha, \tag{2.1}$$

where $t(x) \in \{-1, 0, 1\}$ is the unique index for which $x \in J_{t(x)}$. For $j \geq 1$, set $t_{\alpha,j}(x) \coloneqq t(T_\alpha^{j-1}(x))$; the sequence of digits $(t_{\alpha,j}(x))_{j\geq 1} \in \{-1, 0, 1\}^{\mathbb{N}}$ records indices of the subsequent subintervals J_{-1}, J_0 or J_1 entered by the forward orbit of x. With this notation, Equation (2.1) gives for each $j \geq 1$

$$T_\alpha^j(x) = \beta^{1+|t_{\alpha,j}(x)|} T_\alpha^{j-1}(x) - t_{\alpha,j}(x)\beta\alpha.$$

Solving this for $T_\alpha^{j-1}(x)$, induction shows that for any $n \geq 1$,

$$x = \alpha \sum_{j=1}^{n} \frac{t_{\alpha,j}(x)}{\beta^{j-1+\sum_{k=1}^{j} |t_{\alpha,k}(x)|}} + \frac{T_\alpha^n(x)}{\beta^{n+\sum_{k=1}^{n} |t_{\alpha,k}(x)|}}.$$

Taking the limit $n \to \infty$ and recalling that $|T_\alpha^n(x)| \leq 1$ gives

$$x = \alpha \sum_{j\geq 1} \frac{t_{\alpha,j}(x)}{\beta^{j-1+\sum_{k=1}^{j} |t_{\alpha,k}(x)|}}.$$

Note that for fixed α, this process determines a unique expansion for each $x \in [-1, 1]$. We refer to both this expansion and the corresponding sequence of digits $(t_{\alpha,j}(x))_{j\geq 1}$ as the T_α-*expansion* of x. The above expressions imply that the collection of x whose T_α-expansion is eventually the constant sequence 0—or, equivalently, for which there is some $j \geq 0$ such that $T_\alpha^j(x) = 0$—is countable.

Phenomena analogous to those observed in [45] are found to occur for the skewed symmetric binary maps T_α. In particular, we prove:

Theorem 2.1.1. *For each $\alpha \in (1, \beta]$, the map T_α has a unique—hence ergodic—absolutely continuous invariant probability measure μ_α. Moreover, μ_α is equivalent to Lebesgue measure λ, and there is a countable collection $\{I_{\mathbf{d}}\}_{\mathbf{d}\in\mathcal{M}}$ of disjoint open subintervals of $[1, \beta]$ of full Lebesgue measure, such that for fixed $\mathbf{d} \in \mathcal{M}$ the density of each μ_α with $\alpha \in I_{\mathbf{d}}$ is a step function with at most the same, finite number of jumps.*

When $\alpha = 1$, T_1 has two ergodic, absolutely continuous invariant probability measures; we treat this case separately in §2.3. Via Birkhoff's ergodic theorem, the measures above are employed to show the following:

Theorem 2.1.2. *The asymptotic relative frequency of the digit* 0 *in Lebesgue–a.e. T_α-expansion depends continuously on $\alpha \in [1, \beta]$ and attains a maximum value of* $3/4$ *on the maximal interval* $[1/2 + 1/\beta, 1 + 1/\beta^2]$. *Furthermore, the frequency function is either constant, strictly increasing or strictly decreasing on each $I_{\mathbf{d}}$.*

As in [45], the main tool used to construct the T_α-invariant measures is a property called *matching*. An interval map $T : I \to I$ is said to have matching if for each critical point $c \in I$, the orbits of the left and right limits $y_\pm \coloneqq \lim_{x \to c^\pm} T(x)$ agree after some finite number of steps.[1] That is, for each critical point $c \in I$ there are integers $M, N \geq 0$ for which $T^M(y_-) = T^N(y_+)$.

Matching has gained considerable attention in recent years. Intricacies of the metric entropy function of Nakada's α-continued fraction maps have been studied using matching in [16, 37, 38, 39, 120, 156]. In particular, matching is used in [120] to determine the natural extension for each α-continued fraction transformation, and it is shown that the set of $\alpha \in [0, 1]$ for which matching does not occur has zero Lebesgue measure. The Lebesgue measure of this set of non-matching parameters—in addition to the fact that its Hausdorff dimension is 1—is also shown in [38]. Matching is used in [55] to determine invariant measures for the related family of α-Rosen continued fraction transformations. A parameterised family of linear maps with one increasing and one decreasing branch are considered in [23], and matching is used to show that in some parameter regions, the Lyapunov exponent and topological entropy are constant. A geometric explanation of matching for a similar family of maps is given in [44], and further implications of matching for these

[1] Some authors require that the one-sided derivatives also agree at these times, in which case the map may be said to have *strong matching* ([47]). This extra condition is not needed for our purposes.

maps—including smoothness of entropy on an open dense subset of parameters—is considered in [32]. The notion of matching is extended to random dynamical systems in [47] and is used to study the asymptotic frequency of the digit 0 in typical signed binary expansions arising from a family of random interval maps. Matching has also been investigated for generalised β-transformations, a certain class of continued fraction expansions with finite digit sets, and Lorenz maps (see [31, 42, 43], respectively).

The present chapter exploits the phenomenon of matching in a fashion similar to that of [45]. There the authors use results of Kopf ([112]) which give formulas for densities of the absolutely continuous invariant measures of piecewise linear expanding interval maps. These densities are—in general—infinite sums of (finite) step functions which are determined by the orbits of the left and right limits at critical points of the underlying interval map. However, when matching occurs the infinite sum becomes finite, and the density itself is a finite step function depending only on these orbits before matching. In [45], it is shown that matching occurs for the symmetric doubling maps D_η on a set of parameters η in $[1, 2]$ of full Lebesgue measure. For these matching parameters, the orbits of the left and right limits at the critical points before matching are studied in detail, and this information is used to provide an explicit formula for the density of the (unique) absolutely continuous invariant probability measure for each D_η with matching. The parameter space $[1, 2]$ is decomposed into a countable union of (maximal) open intervals—called *matching intervals*—where each D_η has matching, and a Lebesgue-null set of non-matching parameters with Hausdorff dimension[2] 1. On each matching interval, matching occurs after the same number of steps, and for each left/right limit at a critical point, the digits of the corresponding signed binary expansions agree before matching.

While the results of the present chapter imply that the same direct approach of understanding matching for the skewed symmet-

[2]Hausdorff dimensions are not treated in the current chapter.

ric golden maps T_α can be applied to construct the invariant measures asserted in Theorem 2.1.1, we find that the unequal slopes of the different branches present difficulties. To circumvent these, we instead study matching for a family of *symmetric golden maps* $\{S_\alpha\}_{\alpha\in[1,\beta]}$ of constant slope for which the skewed symmetric golden maps $\{T_\alpha\}_{\alpha\in[1,\beta]}$ are jump transformations, and it is subsequently shown that the parameters α for which the maps T_α and S_α have matching coincide (Proposition 2.2.10). Equipped with this result, one could then use the formulas from [112] to determine invariant densities for the T_α with matching; however, we proceed in the simpler setting of the symmetric maps S_α—determining invariant densities and the frequencies of digits for these—and finally use the fact that T_α is the jump transformation of S_α to determine invariant measures and frequencies of digits for the original skewed symmetric golden maps.

This chapter is organised as follows. In §2.2 the symmetric golden maps $\{S_\alpha\}_{\alpha\in[1,\beta]}$ are introduced. These are shown in §2.2.1 to have matching for Lebesgue–a.e. $\alpha \in [1,\beta]$, and we also prove here that the matching parameters of both families $\{S_\alpha\}_{\alpha\in[1,\beta]}$ and $\{T_\alpha\}_{\alpha\in[1,\beta]}$ coincide. Subsections 2.2.2 and 2.2.3 are devoted to understanding the finer structure of the set of matching parameters. The former provides a classification of all matching intervals and of the orbits of all left and right limits at critical points before matching occurs. In the latter, it is shown that all (but two) of the matching intervals generate in a natural fashion a whole 'cascade' of countably many matching intervals with adjacent endpoints. In §2.3 we use the results of the preceding section to prove Theorems 2.1.1 and 2.1.2. In particular, explicit formulas for densities of the unique, absolutely continuous invariant probability measures of the symmetric golden maps S_α are provided in §2.3.1, and the invariant measures of the skewed maps T_α are expressed in terms of these. These measures are used in §2.3.2 to determine expressions for the asymptotic frequencies of the digit 0 in typical S_α- and T_α-expansions. The maximal frequencies of the digit 0 as functions of α are considered in §2.3.3. Proofs of some technical results are provided in an appendix (§2.4).

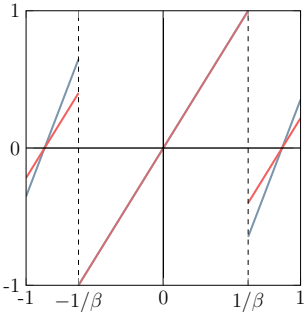


Figure 2.1: The maps T_α (blue) and S_α (red) with $\alpha = 1.4$. Note that $T_\alpha = S_\alpha$ on the middle interval $J_0 = [-1/\beta, 1/\beta]$.

2.2 Symmetric golden maps S_α

As mentioned in §2.1, we determine invariant measures and the frequencies of digits for a family of *symmetric golden maps* $\{S_\alpha\}_{\alpha\in[1,\beta]}$ for which the $\{T_\alpha\}_{\alpha\in[1,\beta]}$ are jump transformations. These invariant measures and frequencies are then used to determine the invariant measures and frequencies of digits for the original T_α. The maps S_α are defined as follows: for $\alpha \in [1, \beta]$, let $S_\alpha : [-1, 1] \to [-1, 1]$ be given by

$$S_\alpha(x) \coloneqq \beta x - t(x)\alpha,$$

with $t(x) \in \{-1, 0, 1\}$ as in §2.1; see Figure 2.1. Note that $S_\alpha(x) \in J_0$ for each $x \in J_{-1} \cup J_1$. Using this, one readily verifies that

$$T_\alpha(x) = \begin{cases} S_\alpha(x), & x \in J_0, \\ S_\alpha^2(x), & x \in J_{-1} \cup J_1, \end{cases} \tag{2.2}$$

i.e., T_α is the jump transformation for S_α with respect to the sweep-out set $J_0 = [-1/\beta, 1/\beta]$ (see, e.g., §11.4 of [46]). For each $j \geq 1$, let

$s_{\alpha,j}(x) \coloneqq t(S_\alpha^{j-1}(x))$. With induction one finds that for each $k \geq 0$,

$$S_\alpha^k(x) = \beta^k \left(x - \alpha \sum_{j=1}^{k} s_{\alpha,j}(x)/\beta^j \right) \tag{2.3}$$

(with the summation for $k = 0$ understood to be 0). Since $|S_\alpha^k| \leq 1$, dividing both sides by β^k and taking the limit as k approaches infinity gives

$$x = \alpha \sum_{j \geq 1} s_{\alpha,j}(x)/\beta^j. \tag{2.4}$$

Following our convention from §2.1, we refer to both the right-hand side of Equation (2.4) and the corresponding sequence $(s_{\alpha,j}(x))_{j\geq 1}$ of digits in $\{0, \pm 1\}^{\mathbb{N}}$ as the S_α-*expansion* of x. Again this process determines—for fixed α—a unique expansion for each $x \in [-1, 1]$; moreover, if $x, y \in [-1, 1]$ have the same S_α-expansion, then $x = y$ by Equation (2.4). Also note that not every sequence in $\{0, \pm 1\}^{\mathbb{N}}$ is an S_α-expansion; in particular, a 1 or -1 is necessarily followed by a 0.

As the orbits of 1 and $1 - \alpha$ will be studied in detail below, we fix special notation for their S_α-expansions: let $d_{\alpha,j} \coloneqq s_{\alpha,j}(1)$ and $e_{\alpha,j} \coloneqq s_{\alpha,j}(1 - \alpha)$ for each $\alpha \in [1, \beta]$ and $j \geq 1$. When α is understood, it is suppressed from the notation, and we simply write $d_j \coloneqq d_{\alpha,j}$ and $e_j \coloneqq e_{\alpha,j}$.

2.2.1 Matching almost everywhere

In this section, we show that the maps S_α (and T_α) have matching on a set of full Lebesgue measure.[3] The map S_α has two critical points $\pm 1/\beta$. Due to symmetry, it suffices to consider the matching criteria

[3]The general approach to proving this result largely follows that of §2.2 of [45]; however, we shall see that the dynamics of the symmetric golden maps S_α are—in a sense—more delicate than those of the previously studied symmetric binary maps (compare, e.g., Proposition 2.2.1 below with Proposition 2.1 of [45]).

only for the positive critical point $1/\beta$. Note that $\lim_{x\to 1/\beta^-} S_\alpha(x) = 1$ and $\lim_{x\to 1/\beta^+} S_\alpha(x) = 1-\alpha$. Hence S_α has matching if and only if there are integers $M, N \geq 1$ for which $S_\alpha^M(1) = S_\alpha^N(1-\alpha)$.

We begin by investigating matching in a number of specific cases. First, note that $1 \in J_1$ and $1-\alpha \in J_0$ for all $\alpha \in [1, \beta]$.

(i) If $\alpha \in (1+1/\beta^2, \beta]$, then

$$S_\alpha(1) = \beta - \alpha \in [0, 1/\beta^3) \subset J_0,$$
$$S_\alpha^2(1) = \beta^2 - \beta\alpha \in J_0,$$

and

$$S_\alpha(1-\alpha) = \beta - \beta\alpha \in [-1, -1/\beta) \subset J_{-1},$$
$$S_\alpha^2(1-\alpha) = \beta^2 - \beta^2\alpha + \alpha = \beta^2 - \beta\alpha \in J_0$$

shows that S_α has matching with $M = N = 2$.

(ii) If $\alpha = 1 + 1/\beta^2$, then

$$S_\alpha(1) = \beta - \alpha = 1/\beta^3 \in J_0,$$
$$S_\alpha^2(1) = 1/\beta^2 \in J_0,$$
$$S_\alpha^3(1) = 1/\beta \in J_0,$$
$$S_\alpha^4(1) = 1 \in J_1,$$

and

$$S_\alpha(1-\alpha) = \beta - \beta\alpha = -1/\beta \in J_0,$$
$$S_\alpha^2(1-\alpha) = -1 \in J_{-1},$$
$$S_\alpha^3(1-\alpha) = -1/\beta^3 \in J_0,$$
$$S_\alpha^4(1-\alpha) = -1/\beta^2 = 1-\alpha \in J_0,$$

so S_α has a Markov partition, namely

$$\left\{[-1/\beta^3, 1/\beta^3],\ \pm(1/\beta^3, 1/\beta^2],\ \pm(1/\beta^2, 1/\beta],\ \pm(1/\beta, 1]\right\},$$

and no matching. (As this Markov partition is not needed explicitly for our purposes, we do not explore it any further.)

(iii) If $\alpha \in (1+1/\beta^3, 1+1/\beta^2)$,

$$\begin{aligned} S_\alpha(1) &= \beta - \alpha \in (1/\beta^3, 1/\beta^2) \subset J_0, \\ S_\alpha^2(1) &= \beta^2 - \beta\alpha \in (1/\beta^2, 1/\beta) \subset J_0, \\ S_\alpha^3(1) &= \beta^3 - \beta^2\alpha \in (1/\beta, 1) \subset J_1, \\ S_\alpha^4(1) &= \beta^4 - (\beta^3+1)\alpha \in J_0, \end{aligned}$$

and

$$\begin{aligned} S_\alpha(1-\alpha) &= \beta - \beta\alpha \in (-1/\beta, -1/\beta^2) \subset J_0, \\ S_\alpha^2(1-\alpha) &= \beta^2 - \beta^2\alpha \in (-1, -1/\beta) \subset J_{-1}, \\ S_\alpha^3(1-\alpha) &= \beta^3 - (\beta^3-1)\alpha \in (-1/\beta^3, 1/\beta^3) \subset J_0, \\ S_\alpha^4(1-\alpha) &= \beta^4 - (\beta^4-\beta)\alpha \in J_0. \end{aligned}$$

Since $\beta^4 - \beta^3 = \beta^2 = \beta + 1$, we find that $S^4(1) = S^4(1-\alpha)$, so S_α has matching with $M = N = 4$.

(iv) If $\alpha = 1 + 1/\beta^3$,

$$\begin{aligned} S_\alpha(1) &= \beta - \alpha = 1/\beta^2 \in J_0, \\ S_\alpha^2(1) &= 1/\beta \in J_0, \\ S_\alpha^3(1) &= 1 \in J_1, \end{aligned}$$

and

$$\begin{aligned} S_\alpha(1-\alpha) &= \beta - \beta\alpha = -1/\beta^2 \in J_0, \\ S_\alpha^2(1-\alpha) &= -1/\beta \in J_0, \\ S_\alpha^3(1-\alpha) &= -1 \in J_{-1}, \\ S_\alpha^4(1-\alpha) &= -1/\beta^2 \in J_0, \end{aligned}$$

so S_α has a Markov partition and no matching.

(v) If $\alpha \in (1, 1 + 1/\beta^3)$, then

$$\begin{aligned} S_\alpha(1) &= \beta - \alpha \in (1/\beta^2, 1/\beta) \subset J_0, \\ S_\alpha^2(1) &= \beta^2 - \beta\alpha \in (1/\beta, 1) \subset J_1, \\ S_\alpha^3(1) &= \beta^3 - (\beta^2 + 1)\alpha \in (-1/\beta^3, 1/\beta) \subset J_0, \end{aligned}$$

and

$$\begin{aligned} S_\alpha(1-\alpha) &= \beta - \beta\alpha \in (-1/\beta^2, 0) \subset J_0, \\ S_\alpha^2(1-\alpha) &= \beta^2 - \beta^2\alpha \in (-1/\beta, 0) \subset J_0, \\ S_\alpha^3(1-\alpha) &= \beta^3 - \beta^3\alpha \in (-1, 0) \subset J_{-1} \cup J_0. \end{aligned}$$

This case will be considered more closely in what follows.

(vi) If $\alpha = 1$, then $S_\alpha(1) = 1/\beta \in J_0$, $S_\alpha^2(1) = 1 \in J_1$ and $S_\alpha(1 - \alpha) = 0 = 1 - \alpha \in J_0$. Thus there is a Markov partition and no matching.

Note that in the cases above in which there is matching—namely (i) and (iii)—we have $M = N$ (a property called *neutral matching* in [32]). We shall see below that this is always the case, i.e., S_α has matching if and only if there is some $m \geq 1$ for which $S_\alpha^m(1) = S_\alpha^m(1 - \alpha)$. For this we need the following proposition—key to a number of arguments throughout—which states that the difference between subsequent points in the orbits of 1 and $1 - \alpha$ can take on at most four values. Recall that $(d_j)_{j\geq 1}$ and $(e_j)_{j\geq 1}$ denote the S_α-expansions of 1 and $1 - \alpha$, respectively.

Proposition 2.2.1. *For every $\alpha \in [1, \beta]$ and $j \geq 0$,*

$$S_\alpha^j(1) - S_\alpha^j(1 - \alpha) \in \{0, \alpha/\beta, \alpha, \beta\alpha\}.$$

Proof. For $\alpha \notin (1, 1+1/\beta^3)$, the statement is verified with the cases above, so assume $\alpha \in (1, 1 + 1/\beta^3)$. We use induction on j. The result clearly holds for $j = 0$; assume for some $j = k - 1 \geq 0$ that

$$S_\alpha^{k-1}(1) - S_\alpha^{k-1}(1 - \alpha) = y$$

for some $y \in \{0, \alpha/\beta, \alpha, \beta\alpha\}$. If $y = 0$, then also $S_\alpha^j(1) - S_\alpha^j(1-\alpha) = 0$ for all $j \geq k-1$. Suppose $y \neq 0$, and note that

$$\begin{aligned} S_\alpha^k(1) - S_\alpha^k(1-\alpha) =& (\beta S_\alpha^{k-1}(1) - d_k\alpha) - (\beta S_\alpha^{k-1}(1-\alpha) - e_k\alpha) \\ =& \beta y - (d_k - e_k)\alpha. \end{aligned}$$

We determine the difference above for each $y \in \{\alpha/\beta, \alpha, \beta\alpha\}$:

(i) $y = \alpha/\beta$: Since $1/\beta < y < 2/\beta$, we have $(d_k, e_k) = (1,0)$, $(0,-1)$ or $(0,0)$. In the first two cases

$$S_\alpha^k(1) - S_\alpha^k(1-\alpha) = 0,$$

and in the third

$$S_\alpha^k(1) - S_\alpha^k(1-\alpha) = \alpha.$$

(ii) $y = \alpha$: Since $1/\beta < y < 1 + 1/\beta^3 = 2/\beta$, we again have $(d_k, e_k) = (1,0), (0,-1)$ or $(0,0)$. In the first two cases

$$S_\alpha^k(1) - S_\alpha^k(1-\alpha) = \beta\alpha - \alpha = \alpha/\beta,$$

and in the third

$$S_\alpha^k(1) - S_\alpha^k(1-\alpha) = \beta\alpha.$$

(iii) $y = \beta\alpha$: Since $y > 2/\beta$, we must have $(d_k, e_k) = (1,-1)$, and hence

$$S_\alpha^k(1) - S_\alpha^k(1-\alpha) = \beta^2\alpha - 2\alpha = \alpha/\beta. \qquad \square$$

The previous proposition can be used to give an alternative characterisation of matching in our setting:

Proposition 2.2.2. *The map S_α has matching if and only if there is some $m \geq 1$ for which $S_\alpha^m(1) = S_\alpha^m(1-\alpha)$.*

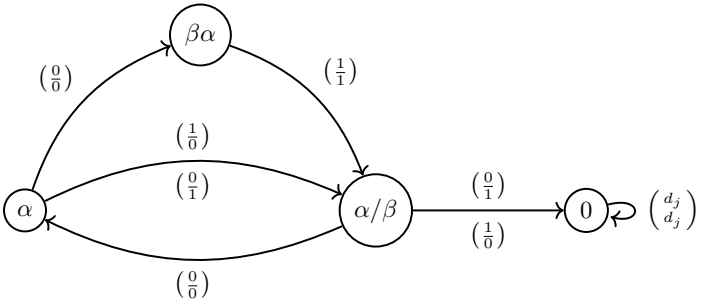


Figure 2.2: A graphical representation of the interdependence of the orbits of 1 and $1-\alpha$ for $\alpha \in [1, \beta]$. Vertices represent the differences $S_\alpha^{j-1}(1) - S_\alpha^{j-1}(1-\alpha)$ for $j \geq 1$, and the edges are marked $\binom{d_j}{e_j}$, where $\overline{w} \coloneqq -w$ for $w \in \{0, \pm 1\}$.

Proof. One direction is immediate; for the other, suppose there are $M, N \geq 1$ for which $S_\alpha^M(1) = S_\alpha^N(1-\alpha)$. If $S_\alpha^M(1) = S_\alpha^N(1-\alpha) = 0$, then since 0 is fixed by S_α, setting $m \coloneqq \max\{M, N\}$ gives $S_\alpha^m(1) = 0 = S_\alpha^m(1-\alpha)$. Now suppose $S_\alpha^M(1) = S_\alpha^N(1-\alpha) > 0$. Since the middle branch $S_\alpha|_{J_0}$ is the expanding function βx, we may increase M and N if necessary to assume that $S_\alpha^M(1) = S_\alpha^N(1-\alpha) \in J_1$. In particular, $S_\alpha^N(1-\alpha) > 1/\beta$. By Proposition 2.2.1, $S_\alpha^N(1) = S_\alpha^N(1-\alpha) + y > 1/\beta + y$ for some $y \in \{0, \alpha/\beta, \alpha, \beta\alpha\}$. But $S_\alpha^N(1) \leq 1$, and since $\alpha \geq 1$, the only value of $y \in \{0, \alpha/\beta, \alpha, \beta\alpha\}$ for which $1/\beta + y \leq 1$ is $y = 0$. Thus $S_\alpha^N(1) = S_\alpha^N(1-\alpha)$. The case when $S_\alpha^M(1) = S_\alpha^N(1-\alpha) < 0$ is similar. □

We can now define a canonical index to describe when matching occurs:

Definition 2.2.3. The *matching index* of S_α is

$$m(\alpha) \coloneqq \inf\{m \geq 1 \mid S_\alpha^m(1) = S_\alpha^m(1-\alpha)\} \in \mathbb{N} \cup \{\infty\}.$$

The cases above together with the proof of Proposition 2.2.1 reveal a strong interdependence between the orbits of 1 and $1-\alpha$,

which is summarised in the graph of Figure 2.2. In particular, note that if matching occurs with matching index $m := m(\alpha)$, then $S_\alpha^{m-1}(1) - S_\alpha^{m-1}(1-\alpha) = \alpha/\beta$ and $(d_m, e_m) \in \{(1,0),(0,-1)\}$. Since S_α-expansions cannot contain consecutive non-zero digits, this implies $S_\alpha^{m-2}(1) - S_\alpha^{m-2}(1-\alpha) = \alpha$ and $(d_{m-1}, e_{m-1}) \in \{(1,0),(0,-1)\}$. For $m > 2$, this further implies $S_\alpha^{m-3}(1) - S_\alpha^{m-3}(1-\alpha) = \alpha/\beta$ and $(d_{m-2}, e_{m-2}) = (0,0)$. Thus if S_α has matching with index $m > 2$, then the final three digits of the S_α-expansions of 1 and $1-\alpha$ before matching are given by

$$\begin{pmatrix} d_{m-2}d_{m-1}d_m \\ e_{m-2}e_{m-1}e_m \end{pmatrix} \in \left\{ \begin{pmatrix} 010 \\ \overline{001} \end{pmatrix}, \begin{pmatrix} 001 \\ \overline{010} \end{pmatrix} \right\}, \tag{2.5}$$

where $\overline{w} := -w$ for $w \in \{0, \pm 1\}$. Conversely, if for some $m > 2$, three consecutive digits of the S_α-expansions of 1 and $1-\alpha$ are given by (2.5), then the proof implies that S_α has matching with index m.

A number of further characterisations of matching for S_α can be derived from Propositions 2.2.1, 2.2.2 and Figure 2.2. We shall see below that matching occurs for $\alpha \neq 1$ if and only if the S_α-orbit of $1-\alpha$ ever becomes non-negative or the S_α-orbit of 1 ever becomes non-positive. We introduce notation to capture the indices prior to such a change in sign: for each $x \in [-1,1]$ and $\alpha \neq 1$, let

$$\ell_\alpha(x) := \inf_{j \geq 0} \{ S_\alpha^j(|x|) \leq 0 \} - 1 \quad \text{and} \quad \ell_\alpha := \min\{\ell_\alpha(1), \ell_\alpha(1-\alpha)\}. \tag{2.6}$$

The next result gives a characterisation of matching in terms of ℓ_α and also states that matching occurs soon after one of the orbits of 1 or $1-\alpha$ changes sign:

Lemma 2.2.4. *For $\alpha \neq 1$, S_α has matching if and only if $\ell_\alpha < \infty$. Moreover, if $\ell_\alpha < \infty$, then $m(\alpha) \in \{\ell_\alpha + 1, \ell_\alpha + 2\}$.*

Proof. Let $\ell := \ell_\alpha$. If S_α has matching, then there is necessarily some $j \geq 0$ for which $S_\alpha^j(1-\alpha) \geq 0$ or $S_\alpha^j(1) \leq 0$, and thus $\ell < \infty$. Now suppose $\ell < \infty$, and assume that $\ell = \ell_\alpha(1-\alpha)$ and hence

$S_\alpha^{\ell+1}(1-\alpha) \geq 0$. The definitions of ℓ and $m(\alpha)$ give $\ell+1 \leq m(\alpha)$. By Proposition 2.2.1, $S_\alpha^{\ell+1}(1-\alpha) \geq 0$ and $\alpha > 1$ imply

$$S_\alpha^{\ell+1}(1) - S_\alpha^{\ell+1}(1-\alpha) \in \{0, \alpha/\beta\}.$$

The result holds if the difference is 0. If the difference is α/β, we must have $(d_{\ell+2}, e_{\ell+2}) = (1,0)$. From Figure 2.2, this implies

$$S_\alpha^{\ell+2}(1) - S_\alpha^{\ell+2}(1-\alpha) = 0.$$

The case when $\ell = \ell_\alpha(1)$ is similar. □

Corollary 2.2.5. *For $\alpha \neq 1$, S_α has matching if and only if there exists some $j \geq 1$ such that*

$$S_\alpha^j(1) \in (1/\beta, \alpha/\beta] \qquad \text{or} \qquad S_\alpha^j(1-\alpha) \in [-\alpha/\beta, -1/\beta).$$

Moreover, $\ell_\alpha(1)$ and $\ell_\alpha(1-\alpha)$, respectively, are the infimums over all j for which the above inclusions hold.

Proof. This follows from Lemma 2.2.4 and the facts that

$$S_\alpha^{-1}([-1,0]) \cap (0,1] = (1/\beta, \alpha/\beta]$$

and

$$S_\alpha^{-1}([0,1]) \cap [-1,0) = [-\alpha/\beta, -1/\beta).$$ □

Due to symmetry, the above corollary states that S_α has matching if and only if the orbit of either 1 or of $\alpha - 1$ enters the region $(1/\beta, \alpha/\beta]$. We shall see that this occurs for Lebesgue–a.e. α by relating the beginnings of these orbits to the beginnings of certain orbits of the (ergodic) *β-transformation* $B : [0,1] \to [0,1]$ defined by $B(x) \coloneqq \beta x \pmod 1$. Set

$$b(x) \coloneqq \begin{cases} 0, & x < 1/\beta, \\ 1, & x \geq 1/\beta, \end{cases}$$

and for each $j \geq 1$, let

$$b_j(x) \coloneqq b(B^{j-1}(x)).$$

We call the sequence $(b_j(x))_{j\geq 1}$ the *β-expansion* (also referred to as the *greedy-expansion*) of x. Via induction, one finds that for each $k \geq 0$,

$$B^k(x) = \beta^k \left(x - \sum_{j=1}^{k} b_j(x)/\beta^j \right). \tag{2.7}$$

Lemma 2.2.6. *Let $x \in \{1, \alpha - 1\}$, $\alpha \neq 1$. Then*

(i) $S_\alpha^j(x) = \alpha B^j(x/\alpha)$ for each $0 \leq j \leq \ell_\alpha(x)$,

(ii) $s_{\alpha,j}(x) = b_j(x/\alpha)$ for each $1 \leq j \leq \ell_\alpha(x)$ and

(iii) $\ell_\alpha(x)$ is the infimum of all j for which $B^j(x/\alpha) \in (1/\beta\alpha, 1/\beta]$.

Proof. Claim (iii) will follow from claim (i), Corollary 2.2.5 and the fact that $\ell_\alpha(x) = \ell_\alpha(-x)$. We prove claim (i) via induction on j. Certainly $S_\alpha^j(x) = \alpha B^j(x/\alpha)$ for $j = 0$. Now suppose this equality holds for some $j = k-1$ with $0 \leq k-1 < \ell_\alpha(x)$. By Corollary 2.2.5, $S_\alpha^{k-1}(x) \in [0,1]\backslash(1/\beta, \alpha/\beta]$, and we find

$$\begin{aligned} S_\alpha^k(x) &= \begin{cases} \beta S_\alpha^{k-1}(x), & S_\alpha^{k-1}(x) \in [0, 1/\beta], \\ \beta S_\alpha^{k-1}(x) - \alpha, & S_\alpha^{k-1}(x) \in (\alpha/\beta, 1], \end{cases} \\ &= \begin{cases} \beta\alpha B^{k-1}(x/\alpha), & B^{k-1}(x/\alpha) \in [0, 1/\beta\alpha], \\ \beta\alpha B^{k-1}(x/\alpha) - \alpha, & B^{k-1}(x/\alpha) \in (1/\beta, 1/\alpha], \end{cases} \\ &= \alpha B^k(x/\alpha), \end{aligned}$$

so claim (i) holds. Moreover, the equality in (i) gives for each $1 \leq j \leq \ell_\alpha(x)$ that $S_\alpha^{j-1}(x) \in [0, 1/\beta]$ if and only if $B^{j-1}(x/\alpha) \in [0, 1/\beta\alpha]$, and $S_\alpha^{j-1}(x) \in (\alpha/\beta, 1]$ if and only if $B^{j-1}(x/\alpha) \in (1/\beta, 1/\alpha]$. Thus $s_{\alpha,j}(x) = b_j(x/\alpha)$ for such j, proving claim (ii). $\square$

Corollary 2.2.5, Lemma 2.2.6 and symmetry of S_α give yet another characterisation of matching in terms of the map B:

Corollary 2.2.7. *For $\alpha \neq 1$, S_α has matching if and only if there exists some $j \geq 1$ such that*

$$B^j(1/\alpha) \in (1/\beta\alpha, 1/\beta] \qquad \text{or} \qquad B^j(1-1/\alpha) \in (1/\beta\alpha, 1/\beta].$$

Moreover, $\ell_\alpha(1)$ and $\ell_\alpha(1-\alpha)$, respectively, are the infimums over all j for which the above inclusions hold.

The previous results together with ergodicity of B can now be used to prove that S_α has matching for a set of parameters α of full Lebesgue measure. The proof is nearly identical to that of Proposition 2.3 of [45] but is included here for the ease of the reader.

Proposition 2.2.8. *The map S_α has matching for Lebesgue–a.e. $\alpha \in [1, \beta]$.*

Proof. Let $\alpha \in (1, \beta]$ and $k \in \mathbb{N}$ with $k > \beta^3$. By ergodicity of B with respect to Lebesgue measure (§4 of [165]), for Lebesgue–a.e. $x \in [0, 1]$ there exists some $j \geq 1$ such that $B^j(x) \in (1/\beta - 1/k, 1/\beta]$. Note that $1/\beta\alpha < 1/\beta - 1/k$ if and only if $\alpha > k/(k-\beta)$. Thus for Lebesgue–a.e. $\alpha \in (k/(k-\beta), \beta]$, there exists some $j \geq 1$ such that

$$B^j(1/\alpha) \in (1/\beta - 1/k, 1/\beta] \subset (1/\beta\alpha, 1/\beta].$$

By Corollary 2.2.7, S_α has matching for Lebesgue–a.e. $\alpha \in (k/(k-\beta), \beta]$. Let A_k denote the set of $\alpha \in (k/(k-\beta), \beta]$ for which S_α does not have matching. Then $\cup_{k>\beta^3} A_k$ has Lebesgue measure 0 and equals the set of all $\alpha \in (1, \beta]$ for which S_α does not have matching. □

The finer structure of the set of matching parameters $\alpha \in [1, \beta]$ is considered in §2.2.2 and §2.2.3 below. Before investigating this structure, we show that matching occurs for S_α if and only if it occurs for the corresponding jump transformation T_α. The following lemma may be deduced from the general theory of jump transformations, but a proof is included for completeness.

Lemma 2.2.9. *Fix $x \in [-1,1]$ and let $j_1 < j_2 < j_3 < \dots$ be an enumeration of the set*

$$\{j \geq 0 \mid S_\alpha^j(x) \in J_0\}.$$

Then $T_\alpha^k(x) = S_\alpha^{j_k+1}(x)$ for all $k \geq 1$.

Proof. The claim is immediate for $k = 1$ by (2.2) and the fact that $S_\alpha(J_{-1} \cup J_1) \subset J_0$. Now suppose the result holds for some $k \geq 1$, and let $i \in \{0,1\}$ be minimal such that $S_\alpha^i(S_\alpha^{j_k+1}(x)) \in J_0$. By definition, then, $j_{k+1} = j_k + i + 1$, and

$$T_\alpha^{k+1}(x) = T_\alpha(S_\alpha^{j_k+1}(x)) = S_\alpha^{i+1}(S_\alpha^{j_k+1}(x)) = S_\alpha^{j_{k+1}+1}(x). \qquad \square$$

Proposition 2.2.10. *The matching parameters $\alpha \in [1,\beta]$ for T_α and for S_α coincide.*

Proof. Recall that T_α has critical points at $\pm 1/\beta$, and note that $\lim_{x\to 1/\beta^-} T_\alpha(x) = 1$ while $\lim_{x\to 1/\beta^+} T_\alpha(x) = \beta(1-\alpha)$. Due to symmetry, T_α has matching if and only if there are integers $M, N > 0$ for which $T_\alpha^M(1) = T_\alpha^N(\beta(1-\alpha))$.

Suppose first that T_α has matching. Then $T_\alpha^M(1) = T_\alpha^N(\beta(1-\alpha))$ for some $M, N > 0$. By (2.2) and the fact that $S_\alpha(1-\alpha) = \beta(1-\alpha)$, this implies the existence of some $M', N' > 0$ for which $S_\alpha^{M'}(1) = S_\alpha^{N'}(1-\alpha)$.

Conversely, suppose S_α has matching with matching index $m := m(\alpha)$. From the proof of Proposition 2.2.1 it is clear that $S_\alpha^m(1) = S_\alpha^m(1-\alpha) \in J_0$. By Lemma 2.2.9, there are $M, N > 0$ for which

$$T_\alpha^M(1) = S_\alpha^{m+1}(1) = S_\alpha^{m+1}(1-\alpha) = S_\alpha^m(\beta(1-\alpha)) = T_\alpha^N(\beta(1-\alpha)).$$

$\square$

2.2.2 Matching words and matching intervals

When S_α has matching, we call the first $m(\alpha) < \infty$ digits of the S_α-expansion of 1 the *matching word* corresponding to α. A maximal subinterval of $[1,\beta]$ on which matching words coincide is called

a *matching interval* corresponding to the common matching word. Here, we classify matching words and matching intervals (Corollary 2.2.23); as all matching parameters belong to some matching interval, this gives a complete classification of matching parameters $\alpha \in [1, \beta]$. (Propositions 2.2.16, 2.2.21 and 2.2.22 imply that this also classifies the first $m(\alpha) < \infty$ digits of the S_α-expansions of $1-\alpha$ for S_α with matching and the maximal subintervals of parameters α on which these digits coincide.) Note that matching words and matching intervals for $\alpha \in [1, \beta]\backslash(1, 1 + 1/\beta^3)$ have been implicitly determined via the cases considered in §2.2.1. For instance, $(1+1/\beta^2, \beta]$ is the matching interval corresponding to the matching word 10, and the S_α-expansion of $1-\alpha$ for each $\alpha \in (1+1/\beta^2, \beta]$ begins with $0(-1)$. Similarly, $(1+1/\beta^3, 1+1/\beta^2)$ is the matching interval corresponding to the matching word 1001, and the S_α-expansion of $1-\alpha$ for each α in this interval begins with $00(-1)0$.

Denote by $\prec$ the lexicographical ordering on $\{0, \pm 1\}^{\mathbb{N}}$. Note that $\prec$ may also be defined on the set $\{0, \pm 1\}^*$ of finite words with alphabet $-1, 0, 1$ by first sending $\mathbf{w} \in \{0, \pm 1\}^*$ to $\mathbf{w}0^\infty$.

Definition 2.2.11. Let

$$\mathbf{w}_0 \coloneqq 00 \prec \mathbf{w}_1 \coloneqq 001 \prec \mathbf{w}_2 \coloneqq 01.$$

We say that $\mathbf{d} \in \{0, 1\}^*$ is in *admissible block form* if $\mathbf{d} = 10$ or

$$\mathbf{d} = 1\mathbf{w}_{i_1}\mathbf{w}_{i_2}\cdots\mathbf{w}_{i_n}(1 - i_n/2)$$

for some $i_1, \ldots, i_n \in \{0, 1, 2\}$, $n \geq 1$ with $i_n \neq 1$, and, when $n \geq 2$, $i_1 = 2$. The collection of all words in admissible block form is denoted $\mathcal{B}$.

The condition that a word in admissible block form ends in $\mathbf{w}_{i_n}(1 - i_n/2)$, $i_n \neq 1$, guarantees that the final three digits are either 001 or 010 (recall (2.5)); however, not every word ending this way belongs to $\mathcal{B}$:

Example 2.2.12. One verifies that

$$\mathbf{d} \coloneqq 1\mathbf{w}_2\mathbf{w}_0\mathbf{w}_1\mathbf{w}_0 1 = 10100001001 \in \mathcal{B},$$

whereas

$$\mathbf{d}' \coloneqq 1010001 \notin \mathcal{B}.$$

Note that the indices i_j for $\mathbf{d} \in \mathcal{B}$ are uniquely determined; that is, if

$$1\mathbf{w}_{i_1}\mathbf{w}_{i_2}\cdots\mathbf{w}_{i_n}(1 - i_n/2) = 1\mathbf{w}_{j_1}\mathbf{w}_{j_2}\cdots\mathbf{w}_{j_m}(1 - j_m/2),$$

then $m = n$ and $i_k = j_k$ for each $1 \leq k \leq n$. Define $\varphi : \mathcal{B} \to \{0, -1\}^*$ by $\varphi(10) \coloneqq \overline{01}$ and for each $\mathbf{d} \in \mathcal{B}$ of the form

$$\mathbf{d} = 1\mathbf{w}_{i_1}\mathbf{w}_{i_2}\cdots\mathbf{w}_{i_n}(1 - i_n/2)$$

by

$$\varphi(\mathbf{d}) \coloneqq \overline{0\mathbf{w}_{2-i_1}\mathbf{w}_{2-i_2}\cdots\mathbf{w}_{2-i_n}(i_n/2)},$$

where $\overline{\mathbf{w}} \coloneqq -\mathbf{w}$ for each $\mathbf{w} \in \{0, \pm 1\}^*$.

Let $\sigma : \{0, \pm 1\}^{\mathbb{N}} \to \{0, \pm 1\}^{\mathbb{N}}$ denote the left shift defined by $\sigma((w_j)_{j\geq 1}) \coloneqq (w_{j+1})_{j\geq 1}$ for each $(w_j)_{j\geq 1} \in \{0, \pm 1\}^{\mathbb{N}}$; as with the lexicographical ordering, σ is also defined on the set $\{0, \pm 1\}^*$ of finite words by sending $\mathbf{w} \in \{0, \pm 1\}^*$ to $\mathbf{w}0^\infty$. We remark that for each $T \in \{S_\alpha, T_\alpha, B\}$, the left shift of the T-expansion of x equals the T-expansion of $T(x)$.

Definition 2.2.13. A word $\mathbf{d} \in \mathcal{B}$ satisfies *Property M* if, for each $j \geq 0$, both $\sigma^j(\mathbf{d}) \preceq \mathbf{d}$ and $\sigma^j(\overline{\varphi(\mathbf{d})}) \preceq \mathbf{d}$. Denote by $\mathcal{M} \subset \mathcal{B}$ the collection of all words $\mathbf{d}$ satisfying Property M. We call 10 and 1001 the *exceptional* words in $\mathcal{M}$ and denote by $\mathcal{M}_U \coloneqq \mathcal{M}\backslash\{10, 1001\}$ the collection of *unexceptional* words in $\mathcal{M}$.

Example 2.2.14. Let $\mathbf{d} \in \mathcal{B}$ be as in Example 2.2.12. Then

$$\varphi(\mathbf{d}) = \overline{0\mathbf{w}_0\mathbf{w}_2\mathbf{w}_1\mathbf{w}_2 0} = \overline{00001001010},$$

and since both $\sigma^j(\mathbf{d}) \preceq \mathbf{d}$ and $\sigma^j(\overline{\varphi(\mathbf{d})}) \preceq \mathbf{d}$ for all $j \geq 0$, we have $\mathbf{d} \in \mathcal{M}$.

We shall see that Property M classifies matching words of the maps S_α. To show that $\mathcal{M}$ contains all matching words we need the following observation, which is not novel, but for which a proof is included for completeness:

Lemma 2.2.15. *Fix $\alpha \in [1, \beta]$ and $x, y \in [-1, 1]$. Then $x < y$ if and only if $(s_{\alpha,j}(x))_{j\geq 1} \prec (s_{\alpha,j}(y))_{j\geq 1}$. Similarly, for $x, y \in [0, 1]$, $x < y$ if and only if $(b_j(x))_{j\geq 1} \prec (b_j(y))_{j\geq 1}$.*

Proof. Suppose $x, y \in [-1, 1]$ with $x < y$, and let

$$n \coloneqq \min_{j\geq 1}\{s_{\alpha,j}(x) \neq s_{\alpha,j}(y)\}.$$

We first claim for each $0 \leq j < n$ that $S_\alpha^j(x) < S_\alpha^j(y)$. This is true by assumption for $j = 0$. If $n = 1$, we're finished. Assume $n > 1$ and that the claim holds for some $j = k - 1$ with $0 \leq k - 1 < n - 1$. Since $s_{\alpha,k}(x) = s_{\alpha,k}(y)$, we have that S_α restricts to a linear function with positive slope on an interval containing $S_\alpha^{k-1}(x)$ and $S_\alpha^{k-1}(y)$. But $S_\alpha^{k-1}(x) < S_\alpha^{k-1}(y)$ by assumption, so also $S_\alpha^k(x) < S_\alpha^k(y)$ and the claim holds. Since $s_{\alpha,n}(x) \neq s_{\alpha,n}(y)$ and $S_\alpha^{n-1}(x) < S_\alpha^{n-1}(y)$, it must be true that $s_{\alpha,n}(x) < s_{\alpha,n}(y)$ and hence $(s_{\alpha,j}(x))_{j\geq 1} \prec (s_{\alpha,j}(y))_{j\geq 1}$.

Now suppose $x \geq y$. If equality holds, then by uniqueness of S_α-expansions, $(s_{\alpha,j}(x))_{j\geq 1} = (s_{\alpha,j}(y))_{j\geq 1}$. If the inequality is strict, the argument above applies with x and y interchanged.

The proof of the second statement is identical, *mutatis mutandis*. □

Proposition 2.2.16. *Suppose for some $\alpha \in [1, \beta]$ that S_α has matching with index $m \coloneqq m(\alpha)$, and let $\mathbf{d} \coloneqq d_1 \cdots d_m$ denote the corresponding matching word. Then $\mathbf{d} \in \mathcal{M}$, and $\mathbf{e} \coloneqq \varphi(\mathbf{d})$ agrees with the first m digits $e_1 \cdots e_m$ of the S_α-expansion of $1 - \alpha$.*

Proof. From the cases of §2.2.1, the result holds for $\alpha \notin (1, 1+1/\beta^3)$; in particular, $\alpha \in (1 + 1/\beta^2, \beta]$ and $\alpha \in (1 + 1/\beta^3, 1 + 1/\beta^2)$ correspond to the exceptional words $\mathbf{d} = 10$ and $\mathbf{d} = 1001$, respectively, in

$\mathcal{M}$, and $\varphi(10) = \overline{01},\ \varphi(1001) = \overline{0010}$. Now assume $\alpha \in (1, 1+1/\beta^3)$. Note that $d_1 = 1,\ e_1 = 0$, and

$$S_\alpha(1) - S_\alpha(1-\alpha) = (\beta - \alpha) - \beta(1-\alpha) = \alpha/\beta.$$

Recall from Equation (2.5) and the discussion preceding it that

$$\left(\frac{d_{m-2}d_{m-1}d_m}{e_{m-2}e_{m-1}e_m}\right) \in \left\{\left(\frac{001}{010}\right), \left(\frac{010}{001}\right)\right\} = \left\{\left(\frac{\mathbf{w}_0 1}{\mathbf{w}_2 0}\right), \left(\frac{\mathbf{w}_2 0}{\mathbf{w}_0 1}\right)\right\},$$

and $S_\alpha^{m-3}(1) - S_\alpha^{m-3}(1-\alpha) = \alpha/\beta$. The remaining digits

$$\left(\frac{d_2 d_3 \cdots d_{m-3}}{e_2 e_3 \cdots e_{m-3}}\right)$$

are thus determined by edge labels of cycles in the graph of Figure 2.2 beginning and ending at vertex α/β. There are three possible cycles, whose edge labels give

$$\left(\frac{d_j d_{j+1}}{e_j e_{j+1}}\right) = \left(\frac{01}{00}\right) = \left(\frac{\mathbf{w}_2}{\mathbf{w}_0}\right), \quad \left(\frac{d_j d_{j+1}}{e_j e_{j+1}}\right) = \left(\frac{00}{01}\right) = \left(\frac{\mathbf{w}_0}{\mathbf{w}_2}\right),$$

and

$$\left(\frac{d_j d_{j+1} d_{j+2}}{e_j e_{j+1} e_{j+2}}\right) = \left(\frac{001}{001}\right) = \left(\frac{\mathbf{w}_1}{\mathbf{w}_1}\right).$$

It follows that

$$\mathbf{d} = 1\mathbf{w}_{i_1}\mathbf{w}_{i_2}\cdots\mathbf{w}_{i_n}(1 - i_n/2)$$

and

$$e_1 \cdots e_m = \overline{0\mathbf{w}_{2-i_1}\mathbf{w}_{2-i_2}\cdots\mathbf{w}_{2-i_n}(i_n/2)}$$

for some $i_1, \ldots, i_n \in \{0, 1, 2\},\ n \geq 1$ and $i_n \neq 1$. Moreover, note from case (v) of §2.2.1 that $d_1 d_2 d_3 d_4 = 1010$, so $i_1 = 2$. Thus $\mathbf{d} \in \mathcal{B}$ and $\mathbf{e} = e_1 \cdots e_m = \varphi(\mathbf{d})$. From Lemma 2.2.15, the facts that $S_\alpha^j(1),\ S_\alpha^j(1-\alpha) \in [-1, 1]$ for each $j \geq 0$ imply that $\sigma^j(\mathbf{d}), \sigma^j(\overline{\mathbf{e}}) \preceq \mathbf{d}$ for each $j \geq 0$: if $\sigma^j(\mathbf{d}) \succ \mathbf{d}$, then $d_{j+1}d_{j+2}\cdots \succ d_1 d_2 \cdots$ implies $S_\alpha^j(1) > 1$, and similarly (using symmetry of S_α) if $\sigma^j(\overline{\mathbf{e}}) \succ \mathbf{d}$ then $S_\alpha^j(\alpha - 1) > 1$. Thus $\mathbf{d} \in \mathcal{M}$. □

The previous result states that every matching word belongs to $\mathcal{M}$. Before proving the converse (Propositions 2.2.19 and 2.2.21), we define and investigate properties of the *valuation function* $v : \mathcal{S} \to \mathbb{R}$ given by the (absolutely) convergent series

$$v((w_j)_{j\geq 1}) \coloneqq \sum_{j\geq 1} w_j/\beta^j,$$

where $\mathcal{S} \subset \mathbb{Z}^{\mathbb{N}}$ consists of all sequences $(w_j)_{j\geq 1}$ whose entries are bounded above and below. The valuation function is also defined on the set $\mathcal{S}^* \subset \mathcal{S}$ of finite words by considering the corresponding finite sum and setting $v(\varepsilon) = 0$ for the empty word ε. It is not difficult to check for finite words $\mathbf{w}, \mathbf{w}' \in \{0,1\}^* \cup \{0,-1\}^*$ with no consecutive nonzero digits that $\mathbf{w} \prec \mathbf{w}'$ if and only if $v(\mathbf{w}) < v(\mathbf{w}')$.

Lemma 2.2.17. *If $\mathbf{w} \coloneqq w_1 w_2 \cdots w_k \in \{0,1,2\}^*$ is ε (in which case we set $k = 0$) or is any combination of blocks of 01's and 002's, then*

$$v(\mathbf{w}) = 1/\beta - 1/\beta^{k+1}.$$

Proof. The case that $\mathbf{w} = \varepsilon$ is trivial, so suppose $\mathbf{w} \neq \varepsilon$. One easily verifies that

$$v(01002) = v(00201) \qquad \text{and} \qquad v((01)^3) = v((002)^2).$$

Moreover, for each $1 \leq j \leq k$,

$$v(\mathbf{w}) = v(w_1 \cdots w_j) + (1/\beta^j) v(w_{j+1} \cdots w_k).$$

Thus, by moving all blocks 002 to the left and replacing each $(01)^3$ by $(002)^2$, we find

$$v(\mathbf{w}) = \begin{cases} v((002)^{k/3}), & k \equiv 0 \pmod 3, \\ v((002)^{(k-4)/3}(01)^2), & k \equiv 1 \pmod 3, \\ v((002)^{(k-2)/3}01), & k \equiv 2 \pmod 3. \end{cases}$$

Notice that for any $j \geq 1$,

$$\begin{aligned} v((002)^j) &= 2\sum_{i=1}^{j}(1/\beta^3)^i = 2 \cdot \frac{1/\beta^3 - 1/\beta^{3j+3}}{1 - 1/\beta^3} \\ &= 2 \cdot \frac{1 - 1/\beta^{3j}}{\beta^3 - 1} = 2 \cdot \frac{1 - 1/\beta^{3j}}{2\beta} \\ &= 1/\beta - 1/\beta^{3j+1}. \end{aligned}$$

If $k \equiv 0 \pmod 3$, setting $j = k/3$ gives the result. If $k \equiv 1 \pmod 3$, we compute

$$\begin{aligned} v(\mathbf{w}) &= v((002)^{(k-4)/3}(01)^2) \\ &= v((002)^{(k-4)/3}) + (1/\beta^{k-4})v((01)^2) \\ &= 1/\beta - 1/\beta^{k-3} + (1/\beta^{k-4})(1/\beta^2 + 1/\beta^4) \\ &= 1/\beta - 1/\beta^{k-3} + 1/\beta^{k-2} + 1/\beta^k \\ &= 1/\beta - 1/\beta^{k+1}. \end{aligned}$$

Similarly, if $k \equiv 2 \pmod 3$,

$$\begin{aligned} v(\mathbf{w}) &= v((002)^{(k-2)/3}01) = v((002)^{(k-2)/3}) + (1/\beta^{k-2})v(01) \\ &= 1/\beta - 1/\beta^{k-1} + 1/\beta^k = 1/\beta - 1/\beta^{k+1}. \end{aligned}$$ □

For equal-length words $\mathbf{x}, \mathbf{y} \in \{0, \pm 1\}^*$, define $\mathbf{x} + \mathbf{y}, \mathbf{x} - \mathbf{y} \in \{0, \pm 1, \pm 2\}^*$ where addition and subtraction, respectively, are performed entry-wise. Note that

$$\mathbf{w}_2 - \overline{\mathbf{w}_0} = 01, \qquad \mathbf{w}_0 - \overline{\mathbf{w}_2} = 01, \qquad \text{and} \qquad \mathbf{w}_1 - \overline{\mathbf{w}_1} = 002.$$

Suppose $\mathbf{d}$ satisfies Property M with $m := \text{len}(\mathbf{d})$ the length of the word $\mathbf{d}$. Since $\mathbf{d}$ is in admissible block form, the definition of $\mathbf{e} := \varphi(\mathbf{d})$ implies that $\mathbf{d} - \mathbf{e} = 1\mathbf{w}1$ for some word $\mathbf{w}$ consisting solely of blocks of 01's and 002's or $\mathbf{w} = \varepsilon$. Using Lemma 2.2.17, we compute

$$v(\mathbf{d} - \mathbf{e}) = v(1\mathbf{w}1) = 1/\beta + (1/\beta)(1/\beta - 1/\beta^{m-1}) + 1/\beta^m = 1.$$

This proves the following:

Proposition 2.2.18. *If* $\mathbf{d} \in \mathcal{M}$ *and* $\mathbf{e} \coloneqq \varphi(\mathbf{d})$*, then*

$$v(\mathbf{d}) - v(\mathbf{e}) = v(\mathbf{d} - \mathbf{e}) = 1.$$

For $\mathbf{d} = 10$, set $I_{\mathbf{d}} = (\alpha_{\mathbf{d}}^-, \alpha_{\mathbf{d}}^+] \coloneqq (1 + 1/\beta^2, \beta]$, and for all other $\mathbf{d} = d_1 \cdots d_m \in \mathcal{M}$, define

$$I_{\mathbf{d}} = (\alpha_{\mathbf{d}}^-, \alpha_{\mathbf{d}}^+) \coloneqq \left(\frac{\beta^m + \beta^{d_m}}{\beta^m v(\mathbf{d}) + \beta^{d_m}}, \frac{\beta^m - \beta^{1-d_m}}{\beta^m v(\mathbf{d}) - \beta^{1-d_m}} \right). \tag{2.8}$$

Proposition 2.2.19. *For each* $\mathbf{d} \in \mathcal{M}$*,* $I_{\mathbf{d}}$ *is a subinterval of* $(1, \beta]$ *containing* $1/v(\mathbf{d})$*.*

Proof. The result is true for $\mathbf{d} = 10$ by definition of I_{10} and the fact that $1/v(10) = \beta$. Now suppose that $\mathbf{d} \neq 10$. We first show that $I_{\mathbf{d}}$ contains $1/v(\mathbf{d})$, i.e., that

$$\frac{\beta^m + \beta^{d_m}}{\beta^m v(\mathbf{d}) + \beta^{d_m}} < \frac{1}{v(\mathbf{d})} < \frac{\beta^m - \beta^{1-d_m}}{\beta^m v(\mathbf{d}) - \beta^{1-d_m}}.$$

The denominator of the expression on the left is positive, and—since $d_1 = 1$ implies $v(\mathbf{d}) \geq 1/\beta$—so is the denominator of the expression in the center. Moreover, $m > 2$ implies $\beta^m v(\mathbf{d}) \geq \beta^{m-1} \geq \beta^2 > \beta^{1-d_m}$, so the denominator of the right-hand side is positive. Multiplying through by denominators, the inequalities thus hold if both

$$\beta^m v(\mathbf{d}) + \beta^{d_m} v(\mathbf{d}) < \beta^m v(\mathbf{d}) + \beta^{d_m}$$

and

$$\beta^m v(\mathbf{d}) - \beta^{1-d_m} < \beta^m v(\mathbf{d}) - \beta^{1-d_m} v(\mathbf{d}).$$

Since $\mathbf{d}$ has no consecutive 1's, one finds that $v(\mathbf{d}) < v((10)^\infty) = 1$ (see also Lemma 1 of [159]), so both inequalities are satisfied.

Next we show that $I_{\mathbf{d}} \subset (1, \beta]$. The left endpoint of $I_{\mathbf{d}}$ is greater than 1 again since $v(\mathbf{d}) < 1$. It remains to show that

$$\frac{\beta^m - \beta^{1-d_m}}{\beta^m v(\mathbf{d}) - \beta^{1-d_m}} \leq \beta.$$

Recall that $d_1 = 1$, and if $d_m = 0$, then $d_{m-1} = 1$; thus $v(\mathbf{d}) \geq 1/\beta + \beta^{1-d_m}/\beta^m$, and

$$\beta^{m+1}v(\mathbf{d}) - \beta^{2-d_m} \geq \beta^{m+1}(1/\beta + \beta^{1-d_m}/\beta^m) - \beta^{2-d_m} > \beta^m - \beta^{1-d_m}.$$

Dividing both sides by $\beta^m v(\mathbf{d}) - \beta^{1-d_m} > 0$ gives the desired inequality. □

For each $\mathbf{u} \in \{0,1\}^*$, let $\Delta(\mathbf{u})$ denote the cylinder of points $x \in [0,1)$ for which the β-expansion of x begins with $\mathbf{u}$. One finds for each $\mathbf{u} = u_1 \cdots u_n$ with $u_j u_{j+1} = 0,\ 1 \leq j < n$, that

$$\Delta(\mathbf{u}) = \begin{cases} [v(\mathbf{u}), v(\mathbf{u}) + 1/\beta^n), & u_n = 0, \\ [v(\mathbf{u}), v(\mathbf{u}) + 1/\beta^{n+1}), & u_n = 1. \end{cases} \tag{2.9}$$

The following lemma is needed in Proposition 2.2.21 below.

Lemma 2.2.20. *Let $\mathbf{d} \in \mathcal{M}_U$. Then $B^j(1/\alpha_{\mathbf{d}}^-) \leq 1/\alpha_{\mathbf{d}}^-$ and $B^j(1 - 1/\alpha_{\mathbf{d}}^+) \leq 1/\alpha_{\mathbf{d}}^+$ for all $j > 0$.*

Proof. This is a corollary of two technical results (Lemmas 2.4.1 and 2.4.2), whose statements and proofs are provided in the appendix. □

The next result—together with Proposition 2.2.19—states that every word $\mathbf{d} \in \mathcal{M}$ is in fact a matching word, thus completing our classification of matching words as the set $\mathcal{M}$. Moreover, it states that the interval $I_{\mathbf{d}}$ is contained in a matching interval corresponding to the matching word $\mathbf{d}$.

Proposition 2.2.21. *For any $\mathbf{d} \in \mathcal{M}$ and $\alpha \in I_{\mathbf{d}}$, the S_α-expansions of 1 and $1 - \alpha$ begin with $\mathbf{d}$ and $\varphi(\mathbf{d})$, respectively. Moreover, S_α has matching with matching index $m(\alpha) = \mathit{len}(\mathbf{d})$.*

Proof. The result is shown for exceptional words $\mathbf{d} \in \{10, 1001\}$ in §2.2.1, so assume $\mathbf{d} \in \mathcal{M}_U$. Suppose the first statement holds. That S_α has matching with index $m(\alpha) = \mathrm{len}(\mathbf{d})$ is implied by the final

three digits of $\mathbf{d}$ and $\varphi(\mathbf{d})$ (see the discussion surrounding Equation (2.5)), so we need only prove the first statement. Let $\alpha \in I_{\mathbf{d}}$, and write $\mathbf{d} = d_1 \cdots d_m$ and $\mathbf{e} \coloneqq \varphi(\mathbf{d}) = e_1 \cdots e_m$. We must show that

$$d_{\alpha,1} \cdots d_{\alpha,m} = d_1 \cdots d_m \qquad \text{and} \qquad e_{\alpha,1} \cdots e_{\alpha,m} = e_1 \cdots e_m.$$

Assume that

$$\begin{pmatrix} d_{m-2} d_{m-1} d_m \\ e_{m-2} e_{m-1} e_m \end{pmatrix} = \begin{pmatrix} 001 \\ \overline{010} \end{pmatrix},$$

and set $\alpha_0 \coloneqq 1/v(\mathbf{d})$ (the case that $d_m = 0$ is similar, up to changes of indices). By Proposition 2.2.19, $\alpha^- < \alpha_0 < \alpha^+$, where, for ease of notation, $\alpha^\pm \coloneqq \alpha^\pm_{\mathbf{d}}$. Recall the definition of $\ell_\alpha(x)$ from (2.6). We claim that it suffices to show the following:

(i) if $\alpha \in (\alpha^-, \alpha_0)$, then $\ell_\alpha(1) > m-1,\ \ell_\alpha(1-\alpha) = m-2$,

$$b_1(1/\alpha) \cdots b_m(1/\alpha) = d_1 \cdots d_m$$

and

$$b_1(1-1/\alpha) \cdots b_{m-2}(1-1/\alpha) = \overline{e_1 \cdots e_{m-2}};$$

(ii) if $\alpha \in (\alpha_0, \alpha^+)$, then $\ell_\alpha(1) = m-1,\ \ell_\alpha(1-\alpha) > m-2$,

$$b_1(1/\alpha) \cdots b_{m-1}(1/\alpha) = d_1 \cdots d_{m-1}$$

and

$$b_1(1-1/\alpha) \cdots b_{m-1}(1-1/\alpha) = \overline{e_1 \cdots e_{m-1}};$$

and

(iii) if $\alpha = \alpha_0$, then $\ell_\alpha(1) = m-1,\ \ell_\alpha(1-\alpha) = m-2$,

$$b_1(1/\alpha) \cdots b_{m-1}(1/\alpha) = d_1 \cdots d_{m-1},$$

$$b_1(1-1/\alpha) \cdots b_{m-2}(1-1/\alpha) = \overline{e_1 \cdots e_{m-2}}$$

and

$$B^{m-1}(1/\alpha) = B^{m-2}(1-1/\alpha) = 1/\beta.$$

Indeed, suppose (i) holds. Lemma 2.2.6 implies

$$d_{\alpha,1}\cdots d_{\alpha,m} = d_1\cdots d_m \qquad \text{and} \qquad e_{\alpha,1}\cdots e_{\alpha,m-2} = e_1\cdots e_{m-2}.$$

Since $\ell_\alpha(1-\alpha) = m-2$, Corollary 2.2.5 gives $S_\alpha^{m-2}(1-\alpha) \in [-\alpha/\beta, -1/\beta)$, so $e_{\alpha,m-1} = -1$ and $e_{\alpha,m} = 0$. In case (ii), Lemma 2.2.6 again gives

$$d_{\alpha,1}\cdots d_{\alpha,m-1} = d_1\cdots d_{m-1} \quad \text{and} \quad e_{\alpha,1}\cdots e_{\alpha,m-1} = e_1\cdots e_{m-1}.$$

Moreover, by Corollary 2.2.5, $\ell_\alpha(1) = m-1$ implies $S_\alpha^{m-1}(1) \in (1/\beta, \alpha/\beta]$ and hence $d_{\alpha,m} = 1$. Since $e_{\alpha,m-1} = e_{m-1} = -1$, it follows that $e_{\alpha,m} = 0 = e_m$. In (iii), Lemma 2.2.6 implies

$$d_{\alpha,1}\cdots d_{\alpha,m-1} = d_1\cdots d_{m-1} \quad \text{and} \quad e_{\alpha,1}\cdots e_{\alpha,m-2} = e_1\cdots e_{m-2}.$$

Moreover, Lemma 2.2.6 gives $S_\alpha^{m-1}(1) = -S_\alpha^{m-2}(1-\alpha) = \alpha/\beta$, so $d_{\alpha,m} = \overline{e_{\alpha,m-1}} = 1$ and $e_{\alpha,m} = 0$.

By Corollary 2.2.7, (i), (ii) and (iii) are implied by showing:

(a) $1/\overline{I_{\mathbf{d}}} \subsetneq \Delta(d_1\cdots d_{m-1})$ and $1 - 1/\overline{I_{\mathbf{d}}} \subsetneq \Delta(\overline{e_1\cdots e_{m-2}})$, where $1/\overline{I_{\mathbf{d}}}$ denotes the set of reciprocals of points in the closure of $I_{\mathbf{d}}$;

(b) $B^j(1/\alpha) \notin (1/\beta\alpha, 1/\beta]$ for each $0 \le j < m-1$, and $B^j(1-1/\alpha) \notin (1/\beta\alpha, 1/\beta]$ for each $0 \le j < m-2$;

(c) if $\alpha \in (\alpha^-, \alpha_0)$, then $B^{m-1}(1/\alpha) > 1/\beta$ and $B^{m-2}(1-1/\alpha) \in (1/\beta\alpha, 1/\beta]$;

(d) if $\alpha \in (\alpha_0, \alpha^+)$, then $B^{m-1}(1/\alpha) \in (1/\beta\alpha, 1/\beta]$ and $B^{m-2}(1-1/\alpha) > 1/\beta$; and

(e) if $\alpha = \alpha_0$, then $B^{m-1}(1/\alpha) = B^{m-2}(1-1/\alpha) = 1/\beta$.

We prove each of (a), (b), (c), (d) and (e):

(a) The first inclusion is implied by showing

$$v(d_1 \cdots d_{m-1}) < 1/\alpha^+ < 1/\alpha^- < v(d_1 \cdots d_{m-1}) + 1/\beta^{m-1}. \tag{2.10}$$

Note that $v(d_1 \cdots d_{m-1}) < 1/\alpha^+$ if and only if

$$v(\mathbf{d}) - 1/\beta^m < \frac{\beta^m v(\mathbf{d}) - 1}{\beta^m - 1}.$$

Multiplying both sides by $\beta^m - 1$, cancelling and rearranging terms, this is equivalent to $v(\mathbf{d}) > 1/\beta^m$. This latter inequality holds since $v(\mathbf{d}) \geq v(d_1) = 1/\beta$ and $m > 1$. Next, $1/\alpha^- < v(d_1 \cdots d_{m-1}) + 1/\beta^{m-1}$ if and only if

$$\frac{\beta^m v(\mathbf{d}) + \beta}{\beta^m + \beta} < v(\mathbf{d}) - 1/\beta^m + 1/\beta^{m-1}.$$

Using the fact that $1/\beta^{m-1} = 1/\beta^m + 1/\beta^{m+1}$ and multiplying both sides by $\beta^m + \beta$, this is equivalent to

$$\beta^m v(\mathbf{d}) + \beta < (\beta^m + \beta)(v(\mathbf{d}) + 1/\beta^{m+1}),$$

or

$$\beta^m v(\mathbf{d}) + \beta < \beta^m v(\mathbf{d}) + 1/\beta + \beta v(\mathbf{d}) + 1/\beta^m.$$

Simplifying, this is equivalent to $1 < \beta v(\mathbf{d}) + 1/\beta^m$, which again holds since $v(\mathbf{d}) \geq 1/\beta$. Thus $1/\overline{I_{\mathbf{d}}} \subsetneq \Delta(d_1 \cdots d_{m-1})$.

The second inclusion is implied by showing

$$\begin{aligned} v(\overline{e_1 \cdots e_{m-2}}) &< 1 - 1/\alpha^- < 1 - 1/\alpha^+ \\ &< v(\overline{e_1 \cdots e_{m-2}}) + 1/\beta^{m-2}. \end{aligned}$$

Now $v(\overline{e_1 \cdots e_{m-2}}) < 1 - 1/\alpha^-$ if and only if $1/\alpha^- < 1 - (v(\overline{\mathbf{e}}) - 1/\beta^{m-1})$. By Proposition 2.2.18, the fact that $v(\overline{\mathbf{e}}) = -v(\mathbf{e})$ and (2.10),

$$\begin{aligned} 1 - (v(\overline{\mathbf{e}}) - 1/\beta^{m-1}) &= v(\mathbf{d}) + 1/\beta^{m-1} \\ &> v(d_1 \cdots d_{m-1}) + 1/\beta^{m-1} > 1/\alpha^-. \end{aligned}$$

Lastly, $1 - 1/\alpha^+ < v(\overline{e_1 \cdots e_{m-2}}) + 1/\beta^{m-2}$ if and only if $1 - 1/\alpha^+ < v(\overline{\mathbf{e}}) - 1/\beta^{m-1} + 1/\beta^{m-2}$, or $v(\mathbf{d}) < 1/\alpha^+ + 1/\beta^m$. From (2.10), we find

$$v(\mathbf{d}) - 1/\beta^m = v(d_1 \cdots d_{m-1}) < 1/\alpha^+.$$

Thus $1 - 1/\overline{I_{\mathbf{d}}} \subsetneq \Delta(\overline{e_1 \cdots e_{m-2}})$.

(b) We will prove that $B^j(1/\alpha) \notin (1/\beta\alpha, 1/\beta]$ for each $0 \leq j < m-1$; the proof that $B^j(1-1/\alpha) \notin (1/\beta\alpha, 1/\beta]$ for each $0 \leq j < m-2$ is similar. Fix $0 \leq j < m-1$. Since $1/\alpha > 1/\beta$, we may assume $j > 0$. By part (a), $b_{j+1}(x) = d_{j+1}$ for all $x \in 1/\overline{I_{\mathbf{d}}}$, and by Equation (2.7),

$$B^j(x) = \beta^j(x - v(d_1 \cdots d_j)). \tag{2.11}$$

Since $1/\alpha \in 1/\overline{I_{\mathbf{d}}}$, it suffices to show that $B^j(1/\alpha) > 1/\beta$ if $d_{j+1} = 1$ and $B^j(1/\alpha) \leq 1/\beta\alpha$ if $d_{j+1} = 0$.

If $d_{j+1} = 1$, then $1/\alpha^+ \in 1/\overline{I_{\mathbf{d}}}$ implies $B^j(1/\alpha^+) \geq 1/\beta$, and since $1/\alpha > 1/\alpha^+$, Equation (2.11) implies $B^j(1/\alpha) > B^j(1/\alpha^+) \geq 1/\beta$ as desired.

Now suppose $d_{j+1} = 0$. We claim that it suffices to show that $B^j(1/\alpha^-) \leq 1/\beta\alpha^-$. Using (2.11), we find that for any $x \in 1/\overline{I_{\mathbf{d}}}$, $B^j(x) \leq x/\beta$ if and only if $x \leq \beta^j v(d_1 \cdots d_j)/(\beta^j - 1/\beta)$. Since $1/\alpha^- \in 1/\overline{I_{\mathbf{d}}}$, showing $B^j(1/\alpha^-) \leq 1/\beta\alpha^-$ will imply $1/\alpha < 1/\alpha^- \leq \beta^j v(d_1 \cdots d_j)/(\beta^j - 1/\beta)$ and hence $B^j(1/\alpha) \leq 1/\beta\alpha$ as claimed. Now suppose for the sake of contradiction that $B^j(1/\alpha^-) > 1/\beta\alpha^-$. Since $b_{j+1}(1/\alpha^-) = d_{j+1} = 0$, the definition of B gives $B^{j+1}(1/\alpha^-) > 1/\alpha^-$, contradicting Lemma 2.2.20, and the result follows.

(c) Suppose $\alpha \in (\alpha^-, \alpha_0)$. From Equation (2.7) and part (a), we have for each $x \in 1/\overline{I_{\mathbf{d}}}$ that

$$\begin{aligned} B^{m-1}(x) &= \beta^{m-1}(x - v(d_1 \cdots d_{m-1})) \\ &= \beta^{m-1}(x - (v(\mathbf{d}) - 1/\beta^m)). \end{aligned} \tag{2.12}$$

Since $1/\alpha > 1/\alpha_0 = v(\mathbf{d})$, we have $B^{m-1}(1/\alpha) > 1/\beta$. Also from Equation (2.7), part (a) and Proposition 2.2.18, for each $x \in 1/\overline{I_{\mathbf{d}}}$,

$$\begin{aligned} B^{m-2}(1-x) &= \beta^{m-2}(1 - x - v(\overline{e_1 \cdots e_{m-2}})) \\ &= \beta^{m-2}(1 - x + v(\mathbf{e}) + 1/\beta^{m-1}) \\ &= \beta^{m-2}(-x + v(\mathbf{d}) + 1/\beta^{m-1}) \\ &= -\beta^{m-2}x + \beta^{m-2}v(\mathbf{d}) + 1/\beta. \end{aligned} \tag{2.13}$$

Hence

$$B^{m-2}(1 - 1/\alpha) < B^{m-2}(1 - 1/\alpha_0) = 1/\beta,$$

and $B^{m-2}(1 - 1/\alpha) > 1/\beta\alpha$ if and only if

$$\frac{\beta^{m-2}v(\mathbf{d}) + 1/\beta}{\beta^{m-2} + 1/\beta} > 1/\alpha.$$

But the left hand side equals $1/\alpha^-$, so the inequality holds.

(d) Suppose $\alpha \in (\alpha_0, \alpha^+)$. From Equation (2.12), $1/\alpha < 1/\alpha_0 = v(\mathbf{d})$ implies $B^{m-1}(1/\alpha) < 1/\beta$. Moreover, $B^{m-1}(1/\alpha) > 1/\beta\alpha$ if and only if

$$1/\alpha > \frac{\beta^{m-1}v(\mathbf{d}) - 1/\beta}{\beta^{m-1} - 1/\beta}.$$

The right-hand side equals $1/\alpha^+$, and $\alpha < \alpha^+$ by assumption. We also find from Equation (2.13) that

$$B^{m-2}(1 - 1/\alpha) = \beta^{m-2}(v(\mathbf{d}) - 1/\alpha) + 1/\beta > 1/\beta$$

since $1/\alpha < 1/\alpha_0 = v(\mathbf{d})$.

(e) This again follows from Equations (2.12) and (2.13), setting $x = 1/\alpha_0 = v(\mathbf{d})$. □

The following proposition states that the interval $I_{\mathbf{d}}$ contains the matching intervals corresponding to the matching word $\mathbf{d}$; together with Proposition 2.2.21, this characterises matching intervals as the collection $\{I_{\mathbf{d}}\}_{\mathbf{d}\in\mathcal{M}}$.

Proposition 2.2.22. *If S_α has matching with $m(\alpha) = m$, then $\alpha \in I_{\mathbf{d}}$, where $\mathbf{d} = d_1 \cdots d_m$ is the beginning of the S_α-expansion of 1.*

Proof. By Proposition 2.2.16, $\mathbf{d} \in \mathcal{M}$, so $I_{\mathbf{d}}$ is defined. The result holds for $m \leq 2$ by the cases in §2.2.1, so assume $m > 2$ and let $\mathbf{e} = e_1 \cdots e_m$ denote the beginning of the S_α-expansion of $1 - \alpha$. Recall from Equation (2.5) that

$$\begin{pmatrix} d_{m-2}d_{m-1}d_m \\ e_{m-2}e_{m-1}e_m \end{pmatrix} \in \left\{ \begin{pmatrix} 010 \\ \overline{001} \end{pmatrix}, \begin{pmatrix} 001 \\ \overline{010} \end{pmatrix} \right\}.$$

Assume $d_m = 0$ (the other case is similar). By Lemma 2.2.4, $\ell_\alpha \in \{m-1, m-2\}$. Suppose that $\ell_\alpha = \ell_\alpha(1)$. By Corollary 2.2.5, either $S_\alpha^{m-2}(1)$ or $S_\alpha^{m-1}(1)$ belongs to $(1/\beta, \alpha/\beta]$. Since $d_{m-1} = 1$ and $d_m = 0$, we must have $S_\alpha^{m-2}(1) \in (1/\beta, \alpha/\beta]$. A similar analysis in the case that $\ell_\alpha = \ell_\alpha(1-\alpha)$ gives $S_\alpha^{m-1}(1-\alpha) \in [-\alpha/\beta, -1/\beta)$. That is, either

$$\text{(i) } S_\alpha^{m-2}(1) \in (1/\beta, \alpha/\beta] \quad \text{or} \quad \text{(ii) } S_\alpha^{m-1}(1-\alpha) \in [-\alpha/\beta, -1/\beta).$$

It suffices to show that both (i) and (ii) imply

$$\alpha \in I_{\mathbf{d}} = \left(\frac{\beta^m + 1}{\beta^m v(\mathbf{d}) + 1}, \frac{\beta^m - \beta}{\beta^m v(\mathbf{d}) - \beta} \right).$$

(i) Equation (2.3) gives

$$S_\alpha^{m-2}(1) = \beta^{m-2}(1 - \alpha v(d_1 \cdots d_{m-2})) \in (1/\beta, \alpha/\beta].$$

Note that $v(d_1 \cdots d_{m-2}) = v(\mathbf{d}) - 1/\beta^{m-1}$, so

$$1 - \alpha(v(\mathbf{d}) - 1/\beta^{m-1}) \in (1/\beta^{m-1}, \alpha/\beta^{m-1}].$$

Now

$$1 - \alpha(v(\mathbf{d}) - 1/\beta^{m-1}) > 1/\beta^{m-1}$$

and the fact that $v(\mathbf{d}) \geq 1/\beta > 1/\beta^{m-1}$ imply

$$\alpha < \frac{1 - 1/\beta^{m-1}}{v(\mathbf{d}) - 1/\beta^{m-1}} = \frac{\beta^m - \beta}{\beta^m v(\mathbf{d}) - \beta}.$$

Moreover,

$$1 - \alpha(v(\mathbf{d}) - 1/\beta^{m-1}) \leq \alpha/\beta^{m-1}$$

gives $1 \leq \alpha v(\mathbf{d})$. Thus we have

$$\alpha \in \left[\frac{1}{v(\mathbf{d})}, \frac{\beta^m - \beta}{\beta^m v(\mathbf{d}) - \beta}\right),$$

which—by Proposition 2.2.19—proves the claim.

(ii) Again from Equation (2.3),

$$S_\alpha^{m-1}(1-\alpha) = \beta^{m-1}(1-\alpha(1+v(e_1 \cdots e_{m-1}))) \in [-\alpha/\beta, -1/\beta).$$

The assumption that $e_m = -1$ together with Proposition 2.2.18 give

$$1 + v(e_1 \cdots e_{m-1}) = 1 + v(\mathbf{e}) + 1/\beta^m = v(\mathbf{d}) + 1/\beta^m,$$

so

$$1 - \alpha(v(\mathbf{d}) + 1/\beta^m) \in [-\alpha/\beta^m, -1/\beta^m).$$

Now

$$1 - \alpha(v(\mathbf{d}) + 1/\beta^m) \geq -\alpha/\beta^m$$

implies $1 \geq \alpha v(\mathbf{d})$. Furthermore,

$$1 - \alpha(v(\mathbf{d}) + 1/\beta^m) < -1/\beta^m$$

gives

$$\alpha > \frac{1 + 1/\beta^m}{v(\mathbf{d}) + 1/\beta^m} = \frac{\beta^m + 1}{\beta^m v(\mathbf{d}) + 1}.$$

Hence

$$\alpha \in \left(\frac{\beta^m + 1}{\beta^m v(\mathbf{d}) + 1}, \frac{1}{v(\mathbf{d})} \right],$$

which—again by Proposition 2.2.19—proves the claim. □

The implications of Propositions 2.2.16, 2.2.19, 2.2.21 and 2.2.22 are summarised in the following:

Corollary 2.2.23. *The sets $\mathcal{M}$ and $\{I_{\mathbf{d}}\}_{\mathbf{d}\in\mathcal{M}}$ classify the matching words and matching intervals, respectively, of the maps S_α.*

Remark 2.2.24. The results of this subsection also imply that $\varphi(\mathcal{M})$ classifies the first $m(\alpha) < \infty$ digits of the S_α-expansions of $1-\alpha$ for matching parameters $\alpha \in [1, \beta]$. Moreover, the intervals $I_{\mathbf{d}}$ in $\{I_{\mathbf{d}}\}_{\mathbf{d}\in\mathcal{M}} = \{I_{\varphi^{-1}(\mathbf{e})}\}_{\mathbf{e}\in\varphi(\mathcal{M})}$ classify the maximal subintervals of matching parameters α for which these first $m(\alpha)$ digits coincide (and equal $\mathbf{e} = \varphi(\mathbf{d})$).

Remark 2.2.25. While not needed for our purposes, we briefly mention that the sets $\mathcal{M}$ (or $\varphi(\mathcal{M})$) and $\{I_{\mathbf{d}}\}_{\mathbf{d}\in\mathcal{M}}$ also give rise to classifications of the T_α-expansions of 1 (resp. $\beta(1-\alpha)$) before matching and the maximal intervals of parameters α on which these expansions coincide. In particular, if $\mathbf{d} \in \mathcal{M}$ (resp. $\mathbf{e} \coloneqq \varphi(\mathbf{d}) \in \varphi(\mathcal{M})$), then the corresponding T_α-word $\mathbf{d}'$ (resp. $\mathbf{e}'$) 'forgets' each non-terminal 0 which immediately follows a 1 (resp. -1, and $\mathbf{e}'$ also forgets the initial 0 of $\mathbf{e}$). The matching intervals $I_{\mathbf{d}}$ are unchanged. For instance, $\mathbf{d} = 10100001$ and $\mathbf{e} = \varphi(\mathbf{d}) = \overline{00001010}$ give rise to the words $\mathbf{d}' = 110001$ and $\mathbf{e}' = \overline{000110}$ for T_α, and each of these words corresponds to the matching interval $I_{\mathbf{d}} = \left(\frac{\beta^8+\beta}{\beta^7+\beta^5+\beta^2}, \frac{\beta^8-1}{\beta^7+\beta^5} \right)$.

2.2.3 Cascades of matching intervals

Here, it is shown that each *unexceptional matching interval* $I_{\mathbf{d}}$, $\mathbf{d} \in \mathcal{M}_U$, generates a whole 'cascade' of unexceptional matching intervals with adjacent endpoints. Define $\psi : \mathcal{M}_U \to \{0,1\}^*$, where for $\mathbf{d} =$

$d_1 \cdots d_m \in \mathcal{M}_U$ and $\mathbf{e} \coloneqq \varphi(\mathbf{d}) = e_1 \cdots e_m$,

$$\psi(\mathbf{d}) \coloneqq \begin{cases} \mathbf{d}\overline{\mathbf{e}}, & d_m = 0, \\ \mathbf{d}\overline{e_2 \cdots e_m}, & d_m = 1. \end{cases}$$

Recall the definition of the matching interval $I_{\mathbf{d}} = (\alpha_{\mathbf{d}}^-, \alpha_{\mathbf{d}}^+)$ from (2.8).

Proposition 2.2.26. *Property M is preserved under the map ψ, i.e., $\psi(\mathcal{M}_U) \subset \mathcal{M}_U$. Moreover, $\alpha_{\mathbf{d}}^- = \alpha_{\psi(\mathbf{d})}^+$ for each $\mathbf{d} \in \mathcal{M}_U$.*

Proof. Let $\mathbf{d} = d_1 \cdots d_m \in \mathcal{M}_U$, and assume $d_m = 0$ (the other case is similar). We first show $\alpha_{\mathbf{d}}^- = \alpha_{\psi(\mathbf{d})}^+$, assuming $\psi(\mathcal{M}_U) \subset \mathcal{M}_U$. Using the fact that the final digit of $\psi(\mathbf{d}) = \mathbf{d}\overline{\mathbf{e}}$ is 1 together with Proposition 2.2.18, we compute

$$\begin{aligned} \alpha_{\psi(\mathbf{d})}^+ &= \frac{\beta^{2m} - 1}{\beta^{2m} v(\mathbf{d}\overline{\mathbf{e}}) - 1} = \frac{(\beta^m + 1)(\beta^m - 1)}{\beta^{2m}(v(\mathbf{d}) - (1/\beta^m)v(\mathbf{e})) - 1} \\ &= \frac{(\beta^m + 1)(\beta^m - 1)}{\beta^{2m} v(\mathbf{d}) - \beta^m (v(\mathbf{d}) - 1) - 1} = \frac{(\beta^m + 1)(\beta^m - 1)}{(\beta^m v(\mathbf{d}) + 1)(\beta^m - 1)} \\ &= \frac{\beta^m + 1}{\beta^m v(\mathbf{d}) + 1} = \alpha_{\mathbf{d}}^- \end{aligned}$$

as desired. Now we prove that $\mathbf{d}' \coloneqq \psi(\mathbf{d}) \in \mathcal{M}_U$. Clearly $\mathbf{d}' \notin \{10, 1001\}$, so we need only show $\mathbf{d}' \in \mathcal{M}$. Write

$$\mathbf{d} = 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n} 0$$

with $i_n = 2$ and

$$\mathbf{e} = \varphi(\mathbf{d}) = \overline{0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n} 1}.$$

Then

$$\begin{aligned} \mathbf{d}' = \mathbf{d}\overline{\mathbf{e}} &= 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n} 00 \mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n} 1 \\ &= 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n} \mathbf{w}_0 \mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n} 1, \end{aligned}$$

so $\mathbf{d}' \in \mathcal{B}$ is in admissible block form. To prove $\mathbf{d}' \in \mathcal{M}$, it remains to show for each $j \geq 0$ that (i) $\sigma^j(\mathbf{d}') \preceq \mathbf{d}'$ and (ii) $\sigma^j(\overline{\varphi(\mathbf{d}')}) \preceq \mathbf{d}'$. (Recall that $\mathbf{d} \in \mathcal{M}$ implies the analogous inequalities hold for $\mathbf{d}$.)

(i) If $j \geq m$, then

$$\sigma^j(\mathbf{d}') = \sigma^j(\mathbf{d}\overline{\mathbf{e}}) = \sigma^{j-m}(\overline{\mathbf{e}}) \preceq \mathbf{d} \preceq \mathbf{d}'.$$

Assume $j < m$, and suppose for the sake of contradiction that $\sigma^j(\mathbf{d}') \succ \mathbf{d}'$. Since $\mathbf{d}'$ begins with 1, so does $\sigma^j(\mathbf{d}')$. Thus either

$$\sigma^j(\mathbf{d}') = 1\mathbf{w}_{i_\ell} \cdots \mathbf{w}_{i_n}\mathbf{w}_0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}1$$

for some $1 < \ell \leq n$, or

$$\sigma^j(\mathbf{d}') = 1\mathbf{w}_0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}1.$$

Since $\mathbf{w}_0 \prec \mathbf{w}_2 = \mathbf{w}_{i_1}$, the second case is impossible and we must have

$$1\mathbf{w}_{i_\ell} \cdots \mathbf{w}_{i_n}\mathbf{w}_0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}1 \succ 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n}\mathbf{w}_0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}1$$

for some ℓ. Since $\sigma^j(\mathbf{d}) \preceq \mathbf{d}$, it follows that

$$1\mathbf{w}_{i_\ell} \cdots \mathbf{w}_{i_n} = 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_{n-\ell+1}}$$

and thus

$$\mathbf{w}_0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}1 \succ \mathbf{w}_{i_{n-\ell+2}} \cdots \mathbf{w}_{i_n}\mathbf{w}_0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}1.$$

Then either there is some $1 \leq p \leq \ell - 3$ for which

$$(0, 2 - i_1, \ldots, 2 - i_{p-1}) = (i_{n-\ell+2}, i_{n-\ell+3}, \ldots, i_{n+p-\ell+1})$$

and $2 - i_p > i_{n+p-\ell+2}$, or

$$(0, 2 - i_1, \ldots, 2 - i_{\ell-2}) = (i_{n-\ell+2}, i_{n-\ell+3}, \ldots, i_n).$$

In the first case,

$$(2 - i_{n-\ell+2}, 2 - i_{n-\ell+3}, \ldots, 2 - i_{n+p-\ell+1}) = (2, i_1, \ldots, i_{p-1})$$

and $2 - i_{n+p-\ell+2} > i_p$. Thus there exists some $k \geq 0$ for which

$$\begin{aligned}\sigma^k(\overline{\mathbf{e}}) &= 1\mathbf{w}_{2-i_{n-\ell+3}} \cdots \mathbf{w}_{2-i_{n+p-\ell+1}} \mathbf{w}_{2-i_{n+p-\ell+2}} \cdots \mathbf{w}_{2-i_n} 1 \\ &\succ 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_{p-1}} \mathbf{w}_{i_p} \cdots \mathbf{w}_{i_n} 0 = \mathbf{d},\end{aligned}$$

contradicting the fact that $\mathbf{d} \in \mathcal{M}$. In the second case,

$$(2 - i_{n-\ell+2}, 2 - i_{n-\ell+3}, \ldots, 2 - i_n) = (2, i_1, \ldots, i_{\ell-2}).$$

Since $i_n = 2$ implies $i_{\ell-2} = 0$, there is again some $k \geq 0$ for which

$$\begin{aligned}\sigma^k(\overline{\mathbf{e}}) &= 1\mathbf{w}_{2-i_{n-\ell+3}} \cdots \mathbf{w}_{2-i_{n-1}} \mathbf{w}_{2-i_n} 1 \\ &= 1\mathbf{w}_{2-i_{n-\ell+3}} \cdots \mathbf{w}_{2-i_{n-1}} \mathbf{w}_1 \\ &\succ 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_{\ell-3}} \mathbf{w}_{i_{\ell-2}} \cdots \mathbf{w}_{i_n} 0 = \mathbf{d},\end{aligned}$$

contradicting $\mathbf{d} \in \mathcal{M}$.

(ii) Set $\mathbf{e}' := \varphi(\mathbf{d}')$, and recall that $d_m = 0$ implies $e_m = -1$, where $\mathbf{e} = \varphi(\mathbf{d}) = e_1 \cdots e_m$. Then

$$\mathbf{e}' = \overline{0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n} \mathbf{w}_2 \mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n} 0} = e_1 \cdots e_{m-1} 0 \overline{\mathbf{d}}.$$

If $j < m - 1$, then

$$\sigma^j(\overline{\mathbf{e}'}) = \overline{e_{j+1} \cdots e_{m-1} 0 \mathbf{d}} \prec \overline{e_{j+1} \cdots e_m} = \sigma^j(\overline{\mathbf{e}}) \preceq \mathbf{d} \preceq \mathbf{d}'.$$

If $j = m - 1$, then

$$\sigma^j(\overline{\mathbf{e}'}) = 0\mathbf{d} \prec \mathbf{d}',$$

and if $j \geq m$, then

$$\sigma^j(\overline{\mathbf{e}'}) = \sigma^{j-m}(\mathbf{d}) \preceq \mathbf{d} \preceq \mathbf{d}'.$$

This concludes the proof that $\mathbf{d}' = \psi(\mathbf{d}) \in \mathcal{M}$ and thus $\psi(\mathcal{M}_U) \subset \mathcal{M}_U$. □

2.3 Invariant measures and frequencies of digits

As noted above, our main interest in matching arises from results of [112] which provide explicit expressions for the densities of absolutely continuous invariant measures. These densities depend on the orbits of the left and right limits at critical points and are in general infinite sums of (finite) step functions; however, the infinite sum becomes finite when either matching or a Markov partition occurs. These observations are used in this section to obtain explicit invariant probability measures ν_α and μ_α for the maps S_α and T_α, respectively, and asymptotic relative frequencies of digits occurring in their respective generic expansions. These measures and frequencies are used in the proofs of Theorems 2.1.1 and 2.1.2.

For $\alpha \in (1, \beta]$, we shall find that the invariant probability measures ν_α and μ_α for S_α and T_α are unique, and hence ergodic, and we can apply Birkhoff's ergodic theorem to compute the asymptotic relative frequencies of digits. The case of $\alpha = 1$ must be handled separately, as the maps S_1 and T_1 each have precisely two ergodic, absolutely continuous invariant probability measures. For each of these maps, the averages ν_1 and μ_1 of the two ergodic measures is invariant, though not ergodic. Nevertheless, the asymptotic relative frequencies of digits occurring in generic S_1- and T_1-expansions may still be described in terms of these non-ergodic measures, as we now illustrate.

Recall that $B(x) \coloneqq \beta x \pmod 1$. It is well known that

$$h(x) \coloneqq \begin{cases} \frac{5+3\sqrt{5}}{10}, & x \in [0, 1/\beta), \\ \frac{5+\sqrt{5}}{10}, & x \in [1/\beta, 1], \end{cases}$$

is the density of a unique, ergodic, B-invariant probability measure which is equivalent to Lebesgue measure λ ([165]). By Birkhoff's

ergodic theorem, the frequency of 0 in λ–a.e. β-expansion is

$$\int_{[0,1/\beta)} h d\lambda = \frac{5+\sqrt{5}}{10}.$$

When $\alpha = 1$, the map $S_\alpha = S_1$ restricts on $[0,1]\backslash\{1/\beta\}$ to B and on $[-1,0]\backslash\{-1/\beta\}$ to $-B(-x)$. Since S_1 is invariant on $\pm[0,1]$, we find that the frequency of 0 in λ–a.e. S_1-expansion is also $(5+\sqrt{5})/10$. Define $f_1 : [-1,1] \to [-1,1]$ by $f_1(x) \coloneqq h(|x|)/2$, and recall the definitions of the subintervals $J_i \subset [-1,1],\ i \in \{-1,0,1\}$ from §2.1. Note, then, that the measure ν_1 defined on Lebesgue-measurable $A \subset [-1,1]$ by $\nu_1(A) \coloneqq \int_A f_1 d\lambda$ satisfies $\nu_1(J_0) \coloneqq (5+\sqrt{5})/10$.

A similar analysis (with Lebesgue measure) reveals that the frequency of 0 in λ–a.e. T_1-expansion is $1/\beta$. Setting $\mu_1 \coloneqq \lambda/2$ as normalised Lebesgue measure gives $\mu_1(J_0) = 1/\beta$. In what follows we consider $\alpha \neq 1$.

2.3.1 Invariant measures

Let $\alpha \in (1,\beta]$. Following a procedure completely analogous to that in §2.1 of [45], results of [112] imply that the collection of absolutely continuous S_α-invariant measures forms a real, one-dimensional linear space and thus there is a unique—and hence ergodic—absolutely continuous invariant probability measure ν_α. Moreover, its corresponding probability density is given explicitly by

$$f_\alpha(x) \coloneqq \frac{1}{C}\sum_{t\geq 0}\frac{1}{\beta^{t+1}}\big(1_{[-1,S_\alpha^t(\alpha-1))}(x) - 1_{[-1,S_\alpha^t(-1))}(x) + 1_{[-1,S_\alpha^t(1))}(x) - 1_{[-1,S_\alpha^t(1-\alpha))}(x)\big),$$

where $C \in \mathbb{R}$ is some normalising constant. Symmetry of S_α together with Proposition 2.2.1 allow us to rewrite $f_\alpha(x)$ as

$$f_\alpha(x) = \frac{1}{C}\sum_{t\geq 0}\frac{1}{\beta^{t+1}}\left(1_{[S_\alpha^t(-1),S_\alpha^t(\alpha-1))}(x) + 1_{[S_\alpha^t(1-\alpha),S_\alpha^t(1))}(x)\right). \tag{2.14}$$

Note that f_α is bounded away from 0 on $[-1,1)$, so ν_α is in fact equivalent to Lebesgue measure λ. Also observe that when matching (or a Markov partition) occurs, the summation becomes a finite sum and $f_\alpha(x)$ is a (finite) step function (see Figure 2.3).

The measure ν_α can now be used to obtain a unique, absolutely continuous T_α-invariant probability measure $\mu_\alpha = \int g_\alpha d\lambda$. For each $\alpha \in (1,\beta]$, define a probability measure

$$\mu_\alpha(A) \coloneqq \frac{\nu_\alpha\left(S_\alpha^{-1}(A) \cap J_0\right)}{\nu_\alpha(J_0)} \tag{2.15}$$

on $[-1,1]$, where $A \subset [-1,1]$ is Lebesgue-measurable. Note that $S_\alpha^{-1}(A) \cap J_0 = \frac{1}{\beta}A$, so μ_α may be written $\mu_\alpha(A) = \nu_\alpha(\frac{1}{\beta}A)/\nu_\alpha(J_0)$.

Theorem 2.3.1. *The measure μ_α is the unique—hence ergodic—invariant probability measure for T_α which is absolutely continuous with respect to Lebesgue measure. Moreover, μ_α is equivalent to Lebesgue measure.*

Proof. Since T_α is an expanding, piecewise C^2 monotone map, results of [124] imply the existence of an invariant probability measure ρ_α for T_α which is absolutely continuous with respect to Lebesgue measure. Let $J_{\pm 1} \coloneqq J_{-1} \cup J_1$. As T_α is a jump transformation for S_α, the measure ρ_α induces an S_α-invariant measure defined by

$$\tilde{\rho}_\alpha(A) \coloneqq \rho_\alpha(A) + \rho_\alpha\left(S_\alpha^{-1}(A) \cap J_{\pm 1}\right) \tag{2.16}$$

(see, e.g., Proposition 11.4.1 of [46]). Note that for any $A \subset J_{\pm 1}$ we have $S_\alpha^{-1}(A) \subset J_0$, so (2.16) gives $\tilde{\rho}_\alpha(A) = \rho_\alpha(A)$. Then for any measurable $A \subset [-1,1]$,

$$\tilde{\rho}_\alpha\left(S_\alpha^{-1}(A) \cap J_{\pm 1}\right) = \rho_\alpha\left(S_\alpha^{-1}(A) \cap J_{\pm 1}\right),$$

and (2.16) gives

$$\rho_\alpha(A) = \tilde{\rho}_\alpha(A) - \tilde{\rho}_\alpha\left(S_\alpha^{-1}(A) \cap J_{\pm 1}\right).$$

Since $\tilde{\rho}_\alpha$ is S_α-invariant, the previous line may be rewritten

$$\rho_\alpha(A) = \tilde{\rho}_\alpha(S_\alpha^{-1}(A)) - \tilde{\rho}_\alpha\left(S_\alpha^{-1}(A) \cap J_{\pm 1}\right) = \tilde{\rho}_\alpha(S_\alpha^{-1}(A) \cap J_0).$$

Recall that ν_α is the unique invariant, absolutely continuous probability measure for S_α, so $\tilde{\rho}_\alpha = c\nu_\alpha$ for some $c > 0$. Thus

$$\rho_\alpha(A) = c\nu_\alpha\left(S_\alpha^{-1}(A) \cap J_0\right),$$

and setting $A = [-1, 1]$ gives $c = 1/\nu_\alpha(J_0)$. Hence $\rho_\alpha = \mu_\alpha$.

That μ_α is equivalent to Lebesgue measure λ follows immediately from the fact that ν_α is equivalent to λ and the observation above that $\mu_\alpha(A) = \nu_\alpha(\frac{1}{\beta}A)/\nu_\alpha(J_0)$. □

We are now ready to prove Theorem 2.1.1:

Proof of Theorem 2.1.1. Theorem 2.3.1 asserts the existence of a unique, absolutely continuous T_α-invariant probability measure μ_α which is in fact equivalent to Lebesgue measure. By Propositions 2.2.8 and 2.2.10 and Corollary 2.2.23, the union of the countable collection of disjoint, open matching intervals $\{I_{\mathbf{d}}\}_{\mathbf{d}\in\mathcal{M}}$ has full Lebesgue measure in $[1, \beta]$. It remains to show that for fixed $\mathbf{d} \in \mathcal{M}$, the density g_α of each μ_α, $\alpha \in I_{\mathbf{d}}$, is a step function with at most the same, finite number of jumps. Using a change of variables, one finds that

$$\begin{aligned}\mu_\alpha(A) =& \frac{\nu_\alpha(\frac{1}{\beta}A)}{\nu_\alpha(J_0)} = \frac{1}{\nu_\alpha(J_0)}\int_{\frac{1}{\beta}A} f_\alpha(x)d\lambda(x)\\ =& \frac{1}{\beta\nu_\alpha(J_0)}\int_A f_\alpha(x/\beta)d\lambda(x),\end{aligned}$$

so

$$g_\alpha(x) = \frac{f_\alpha(x/\beta)}{\beta\nu_\alpha(J_0)}.$$

Since, by (2.14), f_α is a linear combination of at most $2m(\alpha)$ indicator functions and $m(\alpha)$ is constant on $I_{\mathbf{d}}$, the result follows. □

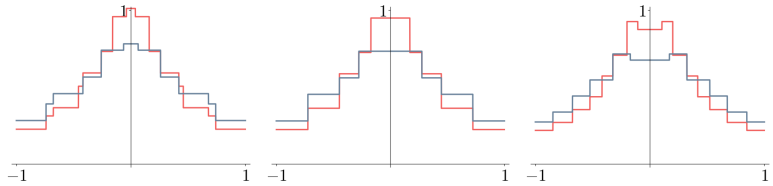


Figure 2.3: The invariant densities f_α for S_α (red) and g_α for T_α (blue) with $\alpha = 1.16$ (left), $\alpha = 1/v(1010) \approx 1.17082\ldots$ (center) and $\alpha = 1.2$ (right).

Remark 2.3.2. The number of jumps of the invariant densities f_α and g_α for S_α and T_α, respectively, are non-constant on matching intervals $I_{\mathbf{d}}$. Figure 2.3 shows these densities for three values of α in the matching interval $I_{\mathbf{d}} \approx (1.14589\ldots, 1.23606\ldots)$ with $\mathbf{d} = 1010$. Note that the number of jumps is fewer for $\alpha = 1/v(\mathbf{d})$. One can show that this phenomenon generalises to all matching intervals; in fact, for each $\mathbf{d} \in \mathcal{M}$, the number of jumps of f_α and g_α, respectively, are constant for all but finitely many $\alpha \in I_{\mathbf{d}}$, and the number of jumps decreases for $\alpha = 1/v(\mathbf{d}) \in I_{\mathbf{d}}$.

2.3.2 Frequencies of digits

We are now in a position to determine the frequencies of digits in generic S_α- and T_α-expansions. Define $\mathfrak{f}_S, \mathfrak{f}_T : [1, \beta] \to [0, 1]$ by

$$\mathfrak{f}_S(\alpha) \coloneqq \nu_\alpha(J_0) \qquad \text{and} \qquad \mathfrak{f}_T(\alpha) \coloneqq \mu_\alpha(J_0).$$

For $\alpha \neq 1$, Birkhoff's ergodic theorem—together with the equivalence of the ergodic measures ν_α and μ_α with Lebesgue measure λ—implies that the asymptotic frequencies

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} 1_{J_0}(S_\alpha^i(x)) \qquad \text{and} \qquad \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} 1_{J_0}(T_\alpha^i(x))$$

of the digit 0 in Lebesgue–a.e. S_α- and T_α-expansion are given by $\mathfrak{f}_S(\alpha)$ and $\mathfrak{f}_T(\alpha)$, respectively. Indeed, with the discussion and notation given at the beginning of §2.3, $\mathfrak{f}_S(1)$ and $\mathfrak{f}_T(1)$ also give the generic asymptotic frequencies of the digit 0. Note, too, that the frequencies of the digits ± 1 are readily obtained from the frequency of 0.

As in the proof of Theorem 2.3.1, set $J_{\pm 1} \coloneqq J_{-1} \cup J_1$. Using (2.15) and the S_α-invariance of ν_α, one has for any measurable $A \subset [-1, 1]$,

$$\begin{aligned}\mu_\alpha(A) =& \frac{\nu_\alpha(S_\alpha^{-1}(A)) - \nu_\alpha(S_\alpha^{-1}(A) \cap J_{\pm 1})}{\nu_\alpha(J_0)} \\ =& \frac{\nu_\alpha(A) - \nu_\alpha(S_\alpha^{-1}(A) \cap J_{\pm 1})}{\nu_\alpha(J_0)}.\end{aligned}$$

Setting $A = J_0$ and using the fact that $S_\alpha^{-1}(J_0) \cap J_{\pm 1} = J_{\pm 1}$, we find

$$\mu_\alpha(J_0) = \frac{\nu_\alpha(J_0) - \nu_\alpha(J_{\pm 1})}{\nu_\alpha(J_0)} = \frac{\nu_\alpha(J_0) - (1 - \nu_\alpha(J_0))}{\nu_\alpha(J_0)},$$

or

$$\mathfrak{f}_T(\alpha) = 2 - \frac{1}{\mathfrak{f}_S(\alpha)}. \tag{2.17}$$

Proposition 2.3.3. *The frequency functions $\mathfrak{f}_S$ and $\mathfrak{f}_T$ are continuous.*

Proof. Arguments completely analogous to those in §4 of [45] give that $\mathfrak{f}_S$ is continuous. Continuity of $\mathfrak{f}_T$ is immediate from 2.17. □

The remainder of this subsection is devoted to finding—for matching parameters α—an explicit expression for $\mathfrak{f}_S(\alpha)$ in terms of α and its corresponding matching word $\mathbf{d}$ (see Figure 2.4). Density of matching parameters in $[1, \beta]$, continuity of $\mathfrak{f}_S$ and Equation (2.17) then allow us to determine $\mathfrak{f}_S(\alpha)$ and $\mathfrak{f}_T(\alpha)$ for any $\alpha \in [1, \beta]$ as limits of these explicit expressions. These expressions are then used in

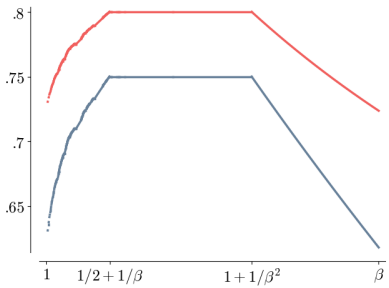


Figure 2.4: The frequency functions $\mathfrak{f}_S(\alpha)$ (red) and $\mathfrak{f}_T(\alpha)$ (blue) plotted on all matching intervals $I_{\mathbf{d}}$ with $\mathrm{len}(\mathbf{d}) \leq 20$. The visible plateaux correspond to the interval $[1/2 + 1/\beta, 1 + 1/\beta^2]$.

§2.3.3 to determine the maximal frequency of the digit 0 occurring in generic S_α- and T_α-expansions, and it is shown that these maximal values are attained for α in the interval $[1/2 + 1/\beta, 1 + 1/\beta^2]$.

Assume that $\alpha \in I_{\mathbf{d}}$, $\mathbf{d} \in \mathcal{M}$, with matching index $m := m(\alpha) < \infty$, and recall the density f_α from Equation (2.14). We first find an expression for the normalising constant C. By symmetry of S_α,

$$\begin{aligned} 1 = \nu_\alpha([-1,1]) &= \int_{-1}^{1} f_\alpha(x) d\lambda(x) \\ &= \frac{2}{C} \sum_{t=0}^{m-1} \int_{-1}^{1} \frac{1}{\beta^{t+1}} 1_{[S_\alpha^t(1-\alpha), S_\alpha^t(1))}(x) d\lambda(x) \\ &= \frac{2}{C} \sum_{t=0}^{m-1} \frac{1}{\beta^{t+1}} \left(S_\alpha^t(1) - S_\alpha^t(1-\alpha) \right). \end{aligned}$$

Assume $\alpha < 1 + 1/\beta^2$ and write

$$\mathbf{d} = d_1 \cdots d_m = 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n}(1 - i_n/2).$$

For each $i \in 0, 1, 2$, let $\ell(i) \in \{2, 3\}$ denote the length of the block $\mathbf{w}_i$—explicitly, $\ell(0) = \ell(2) = 2$ and $\ell(1) = 3$—and let $p = p_{\mathbf{d}}$:

$\{1,\dots,n\} \to \{1,\dots,m-3\}$ be defined by $p(k) \coloneqq 1 + \sum_{j=1}^{k-1} \ell(i_j)$ so that $\sigma^{p(k)}(\mathbf{d}) = \mathbf{w}_{i_k} \cdots \mathbf{w}_{i_n}(1 - i_n/2)$. Recall from Figure 2.2 that $S_\alpha^0(1) - S_\alpha^0(1-\alpha) = \alpha$, $S_\alpha^{m-1}(1) - S_\alpha^{m-1}(1-\alpha) = \alpha/\beta$, and that the remaining differences $S_\alpha^t(1) - S_\alpha^t(1-\alpha)$ are determined by cycles of length two or three beginning at vertex α/β. In particular, if $i_k \in \{0,2\}$, then $S_\alpha^{p(k)}(1) - S_\alpha^{p(k)}(1-\alpha) = \alpha/\beta$ and $S_\alpha^{p(k)+1}(1) - S_\alpha^{p(k)+1}(1-\alpha) = \alpha$ give a cycle of length two, while if $i_k = 1$, $S_\alpha^{p(k)}(1) - S_\alpha^{p(k)}(1-\alpha) = \alpha/\beta$, $S_\alpha^{p(k)+1}(1) - S_\alpha^{p(k)+1}(1-\alpha) = \alpha$ and $S_\alpha^{p(k)+2}(1) - S_\alpha^{p(k)+2}(1-\alpha) = \beta\alpha$ give a cycle of length three. We find for each $k \in \{1,\dots,n\}$ that

$$\sum_{t=p(k)}^{p(k)+\ell(i_k)-1} \frac{1}{\beta^{t+1}} \left(S_\alpha^t(1) - S_\alpha^t(1-\alpha)\right) = \frac{\ell(i_k)}{\beta^{p(k)+2}}\alpha,$$

and thus

$$\begin{aligned} 1 &= \frac{2}{C} \sum_{t=0}^{m-1} \frac{1}{\beta^{t+1}} \left(S_\alpha^t(1) - S_\alpha^t(1-\alpha)\right) \\ &= \frac{2}{C} \left(\frac{\alpha}{\beta} + \sum_{k=1}^{n} \sum_{t=p(k)}^{p(k)+\ell(i_k)-1} \frac{1}{\beta^{t+1}} \left(S_\alpha^t(1) - S_\alpha^t(1-\alpha)\right) + \frac{\alpha}{\beta^{m+1}} \right) \\ &= \frac{2\alpha}{C} \left(\frac{1}{\beta} + \sum_{k=1}^{n} \frac{\ell(i_k)}{\beta^{p(k)+2}} + \frac{1}{\beta^{m+1}} \right). \end{aligned} \tag{2.18}$$

Note that (2.18) also holds for $\alpha > 1 + 1/\beta^2$ (i.e., $\mathbf{d} = 10$), interpreting the summation over k set to zero. Define a substitution $\xi : \{\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2\} \to \{02, 030\}$ by $\xi(\mathbf{w}_0) = \xi(\mathbf{w}_2) \coloneqq 02$ and $\xi(\mathbf{w}_1) \coloneqq 030$, and let $\Xi : \mathcal{M} \to \{0,1,2,3\}^*$ be given by $\Xi(\mathbf{d}) \coloneqq 101$ if $\mathbf{d} = 10$, and

$$\Xi(\mathbf{d}) \coloneqq 1\xi(\mathbf{w}_{i_1}) \cdots \xi(\mathbf{w}_{i_n})01$$

if $\mathbf{d} = 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n}(1 - i_n/2) \in \mathcal{M}\backslash\{10\}$. The left- and right-most sides of (2.18) may be written more succinctly as $1 = \frac{2\alpha}{C} v(\Xi(\mathbf{d}))$, and thus $C = 2\alpha v(\Xi(\mathbf{d}))$.

Having found C, we are now in a position to determine $\mathfrak{f}_S(\alpha)$. Again by symmetry of S_α,

$$\begin{aligned}
\mathfrak{f}_S(\alpha) &= \nu_\alpha(J_0) \\
&= 1 - \nu_\alpha(J_{-1}) - \nu_\alpha(J_1) \\
&= 1 - \int_{-1}^{-1/\beta} f_\alpha(x)d\lambda(x) - \int_{1/\beta}^{1} f_\alpha(x)d\lambda(x) \\
&= 1 - \frac{2}{C}\sum_{t=0}^{m-1}\left(\int_{-1}^{-1/\beta}\frac{1}{\beta^{t+1}}1_{[S_\alpha^t(1-\alpha),S_\alpha^t(1))}(x)d\lambda(x)\right. \\
&\qquad\qquad \left. + \int_{1/\beta}^{1}\frac{1}{\beta^{t+1}}1_{[S_\alpha^t(1-\alpha),S_\alpha^t(1))}(x)d\lambda(x)\right).
\end{aligned}$$

Write $\mathbf{e} \coloneqq \varphi(\mathbf{d}) = e_1 \cdots e_m$. Proposition 2.2.1 and the fact that the diameters of J_1 and J_{-1} are $1/\beta^2 < \alpha/\beta$ imply that for $t < m$, $S_\alpha^t(1) \notin J_{-1}$ and $S_\alpha^t(1-\alpha) \notin J_1$. Thus the previous line may be rewritten as

$$\begin{aligned}
\mathfrak{f}_S(\alpha) &= 1 - \frac{2}{C}\left(\sum_{\substack{0\le t\le m-1\\ e_{t+1}=-1}}\frac{1}{\beta^{t+1}}(-1/\beta - S_\alpha^t(1-\alpha))\right. \\
&\qquad\qquad \left. + \sum_{\substack{0\le t\le m-1\\ d_{t+1}=1}}\frac{1}{\beta^{t+1}}(S_\alpha^t(1) - 1/\beta)\right) \\
&= 1 - \frac{2}{C}\left(\sum_{\substack{0\le t\le m-1\\ d_{t+1}=1}}\frac{1}{\beta^{t+1}}S_\alpha^t(1)\right. \\
&\qquad\qquad \left. - \sum_{\substack{0\le t\le m-1\\ e_{t+1}=-1}}\frac{1}{\beta^{t+1}}S_\alpha^t(1-\alpha) - 1/\beta\right),
\end{aligned}$$

where we have used Proposition 2.2.18 together with the facts that

$$\sum_{\substack{0\le t\le m-1\\ e_{t+1}=-1}} 1/\beta^{t+1} = -v(\mathbf{e}) \qquad \text{and} \qquad \sum_{\substack{0\le t\le m-1\\ d_{t+1}=1}} 1/\beta^{t+1} = v(\mathbf{d}).$$

Let $\mathbf{d}_1^0 = \mathbf{e}_1^0 = \varepsilon$ be the empty word, and for $1 \le t \le m-1$ set $\mathbf{d}_1^t \coloneqq d_1 \cdots d_t$ and $\mathbf{e}_1^t \coloneqq e_1 \cdots e_t$. For each $0 \le t \le m-1$, Equation (2.3) gives $S_\alpha^t(1) = \beta^t(1-\alpha v(\mathbf{d}_1^t))$ and $S_\alpha^t(1-\alpha) = \beta^t(1-\alpha-\alpha v(\mathbf{e}_1^t))$. Setting

$$\mathfrak{n}(\mathbf{d}) \coloneqq \#\{1 \le j \le m \mid d_j = 1\} - \#\{1 \le j \le m \mid e_j = -1\}, \quad (2.19)$$

the frequency function may be written as

$$\begin{aligned}
\mathfrak{f}_S(\alpha) &= 1 - \frac{2}{C}\left(\sum_{\substack{0\le t\le m-1\\ d_{t+1}=1}} \frac{1}{\beta^{t+1}}\beta^t(1-\alpha v(\mathbf{d}_1^t))\right.\\
&\qquad\left. - \sum_{\substack{0\le t\le m-1\\ e_{t+1}=-1}} \frac{1}{\beta^{t+1}}\beta^t(1-\alpha-\alpha v(\mathbf{e}_1^t)) - 1/\beta\right)\\
&= 1 - \frac{2}{\beta C}\left(\sum_{\substack{0\le t\le m-1\\ d_{t+1}=1}} (1-\alpha v(\mathbf{d}_1^t))\right.\\
&\qquad\left. - \sum_{\substack{0\le t\le m-1\\ e_{t+1}=-1}} (1-\alpha-\alpha v(\mathbf{e}_1^t)) - 1\right)
\end{aligned}$$

$$= 1 - \frac{2}{\beta C}\left(\mathfrak{n}(\mathbf{d}) - \alpha\left(\sum_{\substack{0\le t\le m-1\\ d_{t+1}=1}} v(\mathbf{d}_1^t) - \sum_{\substack{0\le t\le m-1\\ e_{t+1}=-1}} (1+v(\mathbf{e}_1^t))\right) - 1\right).$$

Letting

$$K_{\mathbf{d}} \coloneqq \sum_{\substack{0\le t\le m-1\\ d_{t+1}=1}} v(\mathbf{d}_1^t) - \sum_{\substack{0\le t\le m-1\\ e_{t+1}=-1}} (1+v(\mathbf{e}_1^t))$$

and recalling that $C = 2\alpha v(\Xi(\mathbf{d}))$, we find

$$\mathfrak{f}_S(\alpha) = 1 - \frac{1}{\beta v(\Xi(\mathbf{d}))}\left(\frac{\mathfrak{n}(\mathbf{d})-1}{\alpha} - K_{\mathbf{d}}\right). \tag{2.20}$$

Example 2.3.4. Let $\mathbf{d} = 1001$. Then $\mathbf{e} = \overline{0010}$, so $\mathfrak{n}(\mathbf{d}) = 1$. Moreover,

$$v(\Xi(\mathbf{d})) = v(10201) = \frac{1}{\beta} + \frac{2}{\beta^3} + \frac{1}{\beta^5}$$

and

$$K_{\mathbf{d}} = v(\varepsilon) + v(100) - (1 + v(\overline{00})) = -\frac{1}{\beta^2}.$$

Thus for all $\alpha \in I_{1001}$,

$$\mathfrak{f}_S(\alpha) = 1 - \frac{1}{\beta^3(1/\beta + 2/\beta^3 + 1/\beta^5)} = 4/5.$$

A similar calculation with $\mathbf{d} = 1010$ reveals that $\mathfrak{f}_S(\alpha) = 4/5$ also for all $\alpha \in I_{1010}$.

Note that by (2.20), the frequency function $\mathfrak{f}_S$ is monotone on each matching interval $I_{\mathbf{d}}$. Since $v(\Xi(\mathbf{d})) > 0$, we have:

Corollary 2.3.5. *For each* $\mathbf{d} \in \mathcal{M}$*, the frequency function* $\mathfrak{f}_S$ *restricted to* $I_{\mathbf{d}}$ *is strictly increasing with* α *for* $\mathfrak{n}(\mathbf{d}) > 1$*, strictly decreasing for* $\mathfrak{n}(\mathbf{d}) < 1$ *and constant for* $\mathfrak{n}(\mathbf{d}) = 1$*. By (2.17), the same monotonicity conditions hold for* $\mathfrak{f}_T$*.*

Remark 2.3.6. Each of the three monotonicity cases of Corollary 2.3.5 occurs on infinitely many matching intervals. Indeed, for any integer $n > 1$, one has that $\mathbf{d} \coloneqq 1(01)^n0 \in \mathcal{M}$ with $\mathbf{e} \coloneqq \varphi(\mathbf{d}) = \overline{0(00)^n1}$, and $\mathfrak{n}(\mathbf{d}) = n > 1$ implies $\mathfrak{f}_S$ is strictly increasing on $I_{\mathbf{d}}$. On the other hand, for any $n > 2$, one finds

$$\mathbf{d} \coloneqq 10101(00001)^n010 \in \mathcal{M}$$

with

$$\mathbf{e} \coloneqq \varphi(\mathbf{d}) = \overline{00000(01001)^n001},$$

and $\mathfrak{n}(\mathbf{d}) = (n+4) - (2n+1) = 3 - n < 1$ implies $\mathfrak{f}_S$ is strictly decreasing on $I_{\mathbf{d}}$. That there are infinitely many $\mathbf{d} \in \mathcal{M}$ for which $\mathfrak{n}(\mathbf{d}) = 1$ follows from Lemma 2.3.7 below.

Before turning toward the maximal frequency of the digit 0, we give an alternate expression for $K_{\mathbf{d}}$ which will be helpful below. Note that the first summation in the definition of $K_{\mathbf{d}}$ may be rewritten as the sum of all $v(\mathbf{d}_1^t)$, $1 \le t \le m$, for which $d_t{=}1$, excluding the greatest such index t. The second sum may be similarly rewritten (though an extra term 1 appears from the first non-zero summand of the original sum). Now suppose $\mathbf{d} \neq 10$. Recalling that $\mathbf{d} = 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n}(1 - i_n/2)$, $\mathbf{e} = \overline{0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}(i_n/2)}$ and $\{d_{m-2}d_{m-1}d_m, \overline{e_{m-2}e_{m-1}e_m}\} = \{001, 010\}$, we have

$$K_{\mathbf{d}} = \sum_{\substack{1 \le t \le m-3 \\ d_t = 1}} v(\mathbf{d}_1^t) - \left(1 + \sum_{\substack{1 \le t \le m-3 \\ e_t = -1}} (1 + v(\mathbf{e}_1^t)) \right)$$

$$
\begin{aligned}
=&v(1)+\sum_{\substack{1\le k\le n-1\\ i_k\in\{1,2\}}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})\\
&-\left(1+\sum_{\substack{1\le k\le n-1\\ 2-i_k\in\{1,2\}}}(1-v(0\mathbf{w}_{2-i_1}\cdots\mathbf{w}_{2-i_k}))\right).
\end{aligned}
$$

Recall that $p(k+1)$, $1\le k\le n-1$, gives the power for which $\sigma^{p(k+1)}(\mathbf{d})=\mathbf{w}_{i_{k+1}}\cdots\mathbf{w}_{i_n}(1-i_n/2)$; in particular, $p(k+1)$ equals the length of $1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k}$. By Lemma 2.2.17,

$$
\begin{aligned}
v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})+v(0\mathbf{w}_{2-i_1}\cdots\mathbf{w}_{2-i_k}) =&\frac{1}{\beta}+\frac{1}{\beta}\left(\frac{1}{\beta}-\frac{1}{\beta^{p(k+1)}}\right)\\
=&1-1/\beta^{p(k+1)+1}.
\end{aligned}
$$

Then

$$
\begin{aligned}
K_{\mathbf{d}}=&\frac{1}{\beta}+\sum_{\substack{1\le k\le n-1\\ i_k\in\{1,2\}}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})\\
&-\left(1+\sum_{\substack{1\le k\le n-1\\ 2-i_k\in\{1,2\}}}\left(v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})+1/\beta^{p(k+1)+1}\right)\right)\\
=&-\frac{1}{\beta^2}+\sum_{\substack{1\le k\le n-1\\ i_k=2}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})-\sum_{\substack{1\le k\le n-1\\ i_k=0}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})\\
&-\sum_{\substack{1\le k\le n-1\\ 2-i_k\in\{1,2\}}} 1/\beta^{p(k+1)+1}.
\end{aligned}
$$

The latter summation equals

$$\begin{aligned}\sum_{\substack{1\le k\le n-1\\ 2-i_k\in\{1,2\}}} 1/\beta^{p(k+1)+1} &= \frac{1}{\beta}v(0\mathbf{w}_{2-i_1}\cdots\mathbf{w}_{2-i_{n-1}})\\ &= \frac{1}{\beta}\left(v(\overline{\mathbf{e}})-\frac{1}{\beta^{m-3}}v(\mathbf{w}_{2-i_n}i_n/2)\right)\\ &= \frac{1}{\beta}\left(1-v(\mathbf{d})-\frac{1}{\beta^{m-3}}v(\mathbf{w}_{2-i_n}i_n/2)\right)\\ &= \frac{1}{\beta}\left(1-v(d_1\cdots d_{m-3})-\frac{1}{\beta^{m-3}}v(011)\right)\\ &= \frac{1}{\beta}-\frac{1}{\beta}v(d_1\cdots d_{m-3})-\frac{1}{\beta^{m-1}},\end{aligned}$$

and thus for $\mathbf{d}\in\mathcal{M}\backslash\{10\}$,

$$\begin{aligned}K_{\mathbf{d}} = -1+\sum_{\substack{1\le k\le n-1\\ i_k=2}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})-\sum_{\substack{1\le k\le n-1\\ i_k=0}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})\\ +\frac{1}{\beta}v(d_1\cdots d_{m-3})+\frac{1}{\beta^{m-1}}.\end{aligned}\quad(2.21)$$

2.3.3 Maximal frequency of zero

Here, we prove that the frequency functions $\mathfrak{f}_S$ and $\mathfrak{f}_T$ attain their maximums on the maximal interval $[1/2+1/\beta, 1+1/\beta^2]$, but we first need some preliminary results. Recall from Corollary 2.3.5 that the frequency function $\mathfrak{f}_S$ is constant on matching intervals $I_{\mathbf{d}}$ with $\mathfrak{n}(\mathbf{d})=1$. The first of our preliminary results states that $\mathfrak{f}_S$ (and hence $\mathfrak{f}_T$) is constant on 'cascade' intervals:

Lemma 2.3.7. *For each $\mathbf{d}\in\mathcal{M}_U$, we have $\mathfrak{n}(\psi(\mathbf{d}))=1$. In particular, for each $\mathbf{d}\in\mathcal{M}_U$, the frequency function $\mathfrak{f}_S$ is constant on $[\lim_{n\to\infty}\alpha^-_{\psi^n(\mathbf{d})},\alpha^-_{\mathbf{d}}]$.*

Proof. It suffices to prove the first statement; the second follows immediately from this, Proposition 2.2.26 and continuity of $\mathfrak{f}_S$. Write

$$\mathbf{d} = d_1 \cdots d_m = 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n}(1 - i_n/2)$$

and

$$\mathbf{e} \coloneqq \varphi(\mathbf{d}) = e_1 \cdots e_m = \overline{0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}(i_n/2)}.$$

Observe that

$$\begin{aligned}
\mathbf{d}' \coloneqq \psi(\mathbf{d}) &= \begin{cases} \mathbf{d}\overline{\mathbf{e}}, & d_m = 0, \\ \mathbf{d}\overline{e_2 \cdots e_m}, & d_m = 1, \end{cases} \\
&= \begin{cases} 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n} 00 \mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}(i_n/2), & d_m = 0, \\ 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_{n-1}} 001 \mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}(i_n/2), & d_m = 1, \end{cases} \\
&= \begin{cases} 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n} \mathbf{w}_0 \mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}(i_n/2), & d_m = 0, \\ 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_{n-1}} \mathbf{w}_1 \mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n}(i_n/2), & d_m = 1, \end{cases}
\end{aligned}$$

so

$$\begin{aligned}
\mathbf{e}' &\coloneqq \varphi(\mathbf{d}') \\
&= \begin{cases} \overline{0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_n} \mathbf{w}_2 \mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n}(1 - i_n/2)}, & d_m = 0, \\ \overline{0\mathbf{w}_{2-i_1} \cdots \mathbf{w}_{2-i_{n-1}} \mathbf{w}_1 \mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n}(1 - i_n/2)}, & d_m = 1, \end{cases} \\
&= \begin{cases} e_1 \cdots e_{m-1} 0\overline{\mathbf{d}}, & d_m = 0, \\ e_1 \cdots e_{m-2} 0\overline{\mathbf{d}}, & d_m = 1. \end{cases}
\end{aligned}$$

Recall that if $d_m = 0$, then $\overline{e_m} = 1$. In this case $\mathbf{d}'$ has exactly one more digit 1 than does $\overline{\mathbf{e}'}$. If $d_m = 1$, then $\overline{e_{m-1}e_m} = 10$. Since $e_1 = 0$, we see that in this case, too, $\mathbf{d}'$ has exactly one more digit 1 than does $\overline{\mathbf{e}'}$. Thus in both cases $\mathfrak{n}(\mathbf{d}') = 1$. □

We make note here of some computations which will be useful

below. Let $c, \ell \in \mathbb{Z}$ with $\ell \geq 0$:

$$v((0c)^\ell) = c\sum_{j=1}^{\ell} 1/\beta^{2j} = \frac{c}{\beta^2} \cdot \frac{1-1/\beta^{2\ell}}{1-1/\beta^2} = \frac{c}{\beta}(1-1/\beta^{2\ell}), \quad (2.22)$$

$$v((00c)^\ell) = c\sum_{j=1}^{\ell} 1/\beta^{3j} = \frac{c}{\beta^3} \cdot \frac{1-1/\beta^{3\ell}}{1-1/\beta^3} = \frac{c}{2\beta}(1-1/\beta^{3\ell}), \quad (2.23)$$

$$v((0c0)^\ell) = \beta v((00c)^\ell) = \frac{c}{2}(1-1/\beta^{3\ell}), \quad (2.24)$$

$$v((000c)^\ell) = c\sum_{j=1}^{\ell} 1/\beta^{4j} = \frac{c}{\beta^4}\frac{1-1/\beta^{4\ell}}{1-1/\beta^4} = \frac{c}{\beta(\beta^2+1)}(1-1/\beta^{4\ell}), \quad (2.25)$$

and

$$v((0c00)^\ell) = \beta^2 v((000c)^\ell) = \frac{c\beta}{\beta^2+1}(1-1/\beta^{4\ell}). \quad (2.26)$$

The second of our preliminary results analyses the frequency function $\mathfrak{f}_S$ on matching intervals $I_{\mathbf{d}}$ for which $\mathfrak{n}(\mathbf{d}) = 1$, i.e., for which $\mathfrak{f}_S$ is constant. While the proof is technical, we include it in the main text due to the lemma's importance in Theorem 2.3.9—and hence Theorem 2.1.2—below.

Lemma 2.3.8. *If $\alpha \in I_{\mathbf{d}}$ for some $\mathbf{d} \in \mathcal{M}$ with $\mathfrak{n}(\mathbf{d}) = 1$, then $\mathfrak{f}_S(\alpha) \leq 4/5$. Moreover, equality holds if and only if $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$.*

Proof. Suppose $\mathbf{d} \in \mathcal{M}$ with $\mathfrak{n}(\mathbf{d}) = 1$ and $\alpha \in I_{\mathbf{d}}$. The proof consists of three main steps:

(i) we rewrite the desired inequality $\mathfrak{f}_S(\alpha) \leq 4/5$ in terms of blocks of $\mathbf{w}_{2i}$'s and $\mathbf{w}_1$'s occurring in $\mathbf{d}$ (Equation (2.31) below);

(ii) we prove the inequality obtained in Step (i)—and characterise when equality holds—by bounding both sides of (2.31) in terms

of the number of $\mathbf{w}_{2i}$'s occurring in $\mathbf{d}$ (Equation (2.35) below); and

(iii) we prove that our characterisation of equality in Step (ii) is equivalent to $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$.

Step (i)

By Equation (2.20), $\mathfrak{f}_S(\alpha) = 1 + K_{\mathbf{d}}/\beta v(\Xi(\mathbf{d}))$. Rearranging terms, it suffices to show that

$$\beta v(\Xi(\mathbf{d})) + 5K_{\mathbf{d}} \leq 0 \tag{2.27}$$

with equality if and only if $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$. Note that $\mathfrak{n}(10) = 0$, so we may assume $\mathbf{d} \succ 10$. That $\mathfrak{f}_S(\alpha) = 4/5$ for all $\alpha \in I_{1010} \cup I_{1001}$ was shown in Example 2.3.4. Thus we may assume that $\mathbf{d} \succ 1010$. Write

$$\begin{aligned}\mathbf{d} = d_1 \cdots d_m =& 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_n}(1 - i_n/2) \\ =& 1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_t\mathbf{Y}_t\mathbf{w}_{i_n}(1 - i_n/2),\end{aligned}$$

where each $\mathbf{X}_s$ and $\mathbf{Y}_s$, $1 \leq s \leq t$, consists solely of $\mathbf{w}_{2i}$'s and $\mathbf{w}_1$'s, respectively, and each $\mathbf{X}_s$, $\mathbf{Y}_s \neq \varepsilon$ except possibly $\mathbf{Y}_t$. Let $\ell_{2s-1} \coloneqq \frac{1}{2}\mathrm{len}(\mathbf{X}_s)$ and $\ell_{2s} \coloneqq \frac{1}{3}\mathrm{len}(\mathbf{Y}_s)$ denote the number of blocks $\mathbf{w}_i$ in $\mathbf{X}_s$ and $\mathbf{Y}_s$, respectively, and set $\ell_j \coloneqq 0$ for $j > 2t$. Analogous to the function $p = p_{\mathbf{d}}$ defined in §2.3.2, set $p_1 \coloneqq 1$ and for each $s \geq 1$, let $p_{2s} \coloneqq p_{2s-1} + 2\ell_{2s-1}$ and $p_{2s+1} \coloneqq p_{2s} + 3\ell_{2s}$; note, then, that

$$\sigma^{p_{2s-1}}(\mathbf{d}) = \mathbf{X}_s\mathbf{Y}_s \cdots \mathbf{X}_t\mathbf{Y}_t\mathbf{w}_{i_n}(1 - i_n/2)$$

and

$$\sigma^{p_{2s}}(\mathbf{d}) = \mathbf{Y}_s\mathbf{X}_{s+1} \cdots \mathbf{X}_t\mathbf{Y}_t\mathbf{w}_{i_n}(1 - i_n/2).$$

Let $k_{2s-1}, k_{2s} \in \{1, \ldots, n\}$ be the indices for which

$$\sigma^{p_{2s-1}}(\mathbf{d}) = \mathbf{w}_{i_{k_{2s-1}}} \cdots \mathbf{w}_{i_{n-1}}\mathbf{w}_{i_n}(1 - i_n/2)$$

and

$$\sigma^{p_{2s}}(\mathbf{d}) = \mathbf{w}_{i_{k_{2s}}} \cdots \mathbf{w}_{i_{n-1}}\mathbf{w}_{i_n}(1 - i_n/2).$$

We express each of $v(\Xi(\mathbf{d}))$ and $K_{\mathbf{d}}$ from Equation (2.27) in terms of the numbers ℓ_j and p_j. Using (2.22) and (2.24), we compute

$$\begin{aligned} v(\Xi(\mathbf{d})) &= v(1(02)^{\ell_1}(030)^{\ell_2}\cdots(02)^{\ell_{2t-1}}(030)^{\ell_{2t}}0201) \\ &= \frac{1}{\beta} + \sum_{s=1}^{t}\left(\frac{1}{\beta^{p_{2s-1}}}v((02)^{\ell_{2s-1}}) + \frac{1}{\beta^{p_{2s}}}v((030)^{\ell_{2s}})\right) \\ &\qquad + \frac{1}{\beta^{m-3}}v(0201) \\ &= \frac{1}{\beta} + \sum_{s=1}^{t}\left(\frac{2}{\beta^{p_{2s-1}+1}}(1-1/\beta^{2\ell_{2s-1}}) + \frac{3}{2\beta^{p_{2s}}}(1-1/\beta^{3\ell_{2s}})\right) \\ &\qquad + \frac{1}{\beta^{m-3}}(2/\beta^2+1/\beta^4). \end{aligned}$$

Moreover, (2.23) gives

$$\begin{aligned} v(d_1\cdots d_{m-3}) &= \frac{1}{\beta} + \sum_{s=1}^{t}\left(\frac{1}{\beta^{p_{2s-1}}}v(\mathbf{X}_s) + \frac{1}{\beta^{p_{2s}}}v(\mathbf{Y}_s)\right) \\ &= \frac{1}{\beta} + \sum_{s=1}^{t}\left(\frac{1}{\beta^{p_{2s-1}}}v(\mathbf{X}_s) + \frac{1}{\beta^{p_{2s}}}v((001)^{\ell_{2s}})\right) \\ &= \frac{1}{\beta} + \sum_{s=1}^{t}\left(\frac{1}{\beta^{p_{2s-1}}}v(\mathbf{X}_s) + \frac{1}{2\beta^{p_{2s}+1}}(1-1/\beta^{3\ell_{2s}})\right), \end{aligned}$$

so Equation (2.21) becomes

$$\begin{aligned} K_{\mathbf{d}} = &-\frac{1}{\beta} + \sum_{\substack{1\le k\le n-1\\ i_k=2}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k}) - \sum_{\substack{1\le k\le n-1\\ i_k=0}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k}) \\ &+ \sum_{s=1}^{t}\left(\frac{1}{\beta^{p_{2s-1}+1}}v(\mathbf{X}_s) + \frac{1}{2\beta^{p_{2s}+2}}(1-1/\beta^{3\ell_{2s}})\right) + \frac{1}{\beta^{m-1}}. \end{aligned}$$

Then the left-hand side of (2.27) equals

$$
\begin{aligned}
&1+\sum_{s=1}^{t}\left(\frac{2}{\beta^{p_{2s-1}}}(1-1/\beta^{2\ell_{2s-1}})+\frac{3\beta}{2\beta^{p_{2s}}}(1-1/\beta^{3\ell_{2s}})\right)\\
&\qquad\qquad+\frac{1}{\beta^{m-3}}(2/\beta+1/\beta^3)\\
&-\frac{5}{\beta}+5\left(\sum_{\substack{1\le k\le n-1\\ i_k=2}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})-\sum_{\substack{1\le k\le n-1\\ i_k=0}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})\right)\\
&\quad+5\sum_{s=1}^{t}\left(\frac{1}{\beta^{p_{2s-1}+1}}v(\mathbf{X}_s)+\frac{1}{2\beta^{p_{2s}+2}}(1-1/\beta^{3\ell_{2s}})\right)+\frac{5}{\beta^{m-1}}\\
&=1-\frac{5}{\beta}+\sum_{s=1}^{t}\frac{1}{\beta^{p_{2s}}}\left(3\beta/2+5/2\beta^2\right)(1-1/\beta^{3\ell_{2s}})\\
&\qquad\qquad+\frac{1}{\beta^{m-3}}(2/\beta+5/\beta^2+1/\beta^3)\\
&\quad+\sum_{s=1}^{t}\left(\frac{2}{\beta^{p_{2s-1}}}(1-1/\beta^{2\ell_{2s-1}})+\frac{5}{\beta^{p_{2s-1}+1}}v(\mathbf{X}_s)\right)\\
&\quad+5\left(\sum_{\substack{1\le k\le n-1\\ i_k=2}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})-\sum_{\substack{1\le k\le n-1\\ i_k=0}} v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})\right).
\end{aligned}
$$

One easily verifies that both $3\beta/2+5/2\beta^2$ and $2/\beta+5/\beta^2+1/\beta^3$

equal $c := 5 - \beta$. We claim that it suffices to show that

$$\sum_{s=1}^{t} \left(\frac{2}{\beta^{p_{2s-1}}}(1 - 1/\beta^{2\ell_{2s-1}}) + \frac{5}{\beta^{p_{2s-1}+1}} v(\mathbf{X}_s) \right) + 5 \left(\sum_{\substack{1 \le k \le n-1 \\ i_k = 2}} v(1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_k}) - \sum_{\substack{1 \le k \le n-1 \\ i_k = 0}} v(1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_k}) \right) \le \sum_{s=1}^{t} \frac{c}{\beta^{p_{2s-1}}}(1 - 1/\beta^{2\ell_{2s-1}}), \quad (2.28)$$

with equality if and only if $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$. Indeed, suppose the claim holds. Then the computation above becomes

$$\begin{aligned} \beta v(\Xi(\mathbf{d})) + 5K_{\mathbf{d}} &\le 1 - \frac{5}{\beta} + c \sum_{s=1}^{t} \left(\frac{1}{\beta^{p_{2s-1}}}(1 - 1/\beta^{2\ell_{2s-1}}) \right. \\ &\qquad \left. + \frac{1}{\beta^{p_{2s}}}(1 - 1/\beta^{3\ell_{2s}}) \right) + \frac{c}{\beta^{m-3}} \\ &= 1 - \frac{5}{\beta} + c \sum_{s=1}^{t} (1/\beta^{p_{2s-1}} - 1/\beta^{p_{2s}} + 1/\beta^{p_{2s}} \\ &\qquad - 1/\beta^{p_{2s+1}}) + \frac{c}{\beta^{m-3}} \\ &= 1 - \frac{5}{\beta} + c(1/\beta - 1/\beta^{m-3}) + \frac{c}{\beta^{m-3}} \\ &= 1 - \frac{5}{\beta} + \frac{c}{\beta} = 0 \end{aligned}$$

with equality if and only if $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$, as desired; see (2.27).

It remains to show the claim of (2.28). The constant c defined above may be rewritten as $c = 2 + 5/(\beta^2 + 1)$. Subtracting $\sum_{s=1}^{t} (2/\beta^{p_{2s-1}})(1 - 1/\beta^{2\ell_{2s-1}})$ from both sides, dividing by 5 and noting that $i_k \in \{0, 2\}$ only when $k_{2s-1} \le k < k_{2s}$, $1 \le s \le t$,

Equation (2.28) becomes

$$\sum_{s=1}^{t}\left(\frac{1}{\beta^{p_{2s-1}+1}}v(\mathbf{X}_s)+\sum_{\substack{k_{2s-1}\le k<k_{2s}\\ i_k=2}}v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})\right.$$
$$\left.-\sum_{\substack{k_{2s-1}\le k<k_{2s}\\ i_k=0}}v(1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k})\right) \tag{2.29}$$
$$\le\frac{1}{\beta^2+1}\sum_{s=1}^{t}\frac{1}{\beta^{p_{2s-1}}}(1-1/\beta^{2\ell_{2s-1}}).$$

Fix $1\le s\le t$, and recall that $\mathbf{X}_s$ consists solely of $\mathbf{w}_{2i}$'s. Hence we may write

$$\mathbf{X}_s=\mathbf{w}_2^{n_{s,1}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,2}}\mathbf{w}_0^{n_{s,3}}(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4}}\cdots \tag{2.30}$$
$$\mathbf{w}_2^{n_{s,4r_s-3}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4r_s-2}}\mathbf{w}_0^{n_{s,4r_s-1}}(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4r_s}},$$

where the powers $n_{s,\ell}\ge 0$ are chosen so that $\sum_{\ell=1}^{4r_s}n_{s,\ell}$ is minimal and no three consecutive $n_{s,\ell}$ are zero except possibly the first or final three $n_{s,\ell}$. Set $p_{s,1}\coloneqq p_{2s-1}$ and for each $1\le j\le r_s$,

$$p_{s,4j-2}\coloneqq p_{s,4j-3}+2n_{s,4j-3},\qquad p_{s,4j-1}\coloneqq p_{s,4j-2}+4n_{s,4j-2},$$
$$p_{s,4j}\coloneqq p_{s,4j-1}+2n_{s,4j-1},\qquad p_{s,4j+1}\coloneqq p_{s,4j}+4n_{s,4j}.$$

Note that with these definitions, $p_{s,4r_s+1}=p_{2s}$. Equations (2.22)–(2.26) give that the first term in the summand on the left of (2.29) is

$$\frac{1}{\beta^{p_{2s-1}+1}}v(\mathbf{X}_s)=\frac{1}{\beta}\sum_{j=1}^{r_s}\left(\frac{1}{\beta^{p_{s,4j-3}}}v(\mathbf{w}_2^{n_{s,4j-3}})\right.$$
$$+\frac{1}{\beta^{p_{s,4j-2}}}v((\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}})$$
$$\left.+\frac{1}{\beta^{p_{s,4j-1}}}v(\mathbf{w}_0^{n_{s,4j-1}})+\frac{1}{\beta^{p_{s,4j}}}v((\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j}})\right)$$

$$
\begin{aligned}
&= \sum_{j=1}^{r_s} \bigg(\frac{1}{\beta^{p_{s,4j-3}}} \frac{1}{\beta^2} (1 - 1/\beta^{2n_{s,4j-3}}) \\
&\quad + \frac{1}{\beta^{p_{s,4j-2}}} \frac{1}{\beta^2 + 1} (1 - 1/\beta^{4n_{s,4j-2}}) \\
&\quad + \frac{1}{\beta^{p_{s,4j}}} \frac{1}{\beta^2(\beta^2 + 1)} (1 - 1/\beta^{4n_{s,4j}}) \bigg),
\end{aligned}
$$

and that the second and third terms in the summand on the left of (2.29) are

$$
\begin{aligned}
&\sum_{\substack{k_{2s-1} \le k < k_{2s} \\ i_k = 2}} v(1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_k}) - \sum_{\substack{k_{2s-1} \le k < k_{2s} \\ i_k = 0}} v(1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_k}) \\
&= \sum_{j=1}^{r_s} \bigg(\sum_{\ell=1}^{n_{s,4j-3}} v(1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}} \cdots (\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-4}} \mathbf{w}_2^{\ell}) \\
&\qquad - \sum_{\ell=1}^{n_{s,4j-1}} v(1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}} \cdots (\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}} \mathbf{w}_0^{\ell}) \\
&\qquad\quad + v(1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}} \cdots (\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j}}) \\
&\qquad - v(1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}} \cdots (\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}} \mathbf{w}_0^{n_{s,4j-1}} \mathbf{w}_0) \bigg) \\
&= \sum_{j=1}^{r_s} \bigg(\sum_{\ell=1}^{n_{s,4j-3}} v(1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}} \cdots (\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-4}} \mathbf{w}_2^{\ell}) \\
&\qquad - n_{s,4j-1} v(1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}} \cdots (\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}) \\
&\qquad\quad + \frac{1}{\beta^{p_{s,4j}}} \frac{1}{\beta(\beta^2 + 1)} (1 - 1/\beta^{4n_{s,4j}}) \bigg).
\end{aligned}
$$

Thus the left-hand side of (2.29) equals

$$\sum_{s=1}^{t}\sum_{j=1}^{r_s}\Bigg(\frac{1}{\beta^{p_{s,4j-3}}}\frac{1}{\beta^2}(1-1/\beta^{2n_{s,4j-3}}) + \frac{1}{\beta^{p_{s,4j-2}}}\frac{1}{\beta^2+1}(1-1/\beta^{4n_{s,4j-2}}) + \frac{1}{\beta^{p_{s,4j}}}\frac{1}{\beta^2+1}(1-1/\beta^{4n_{s,4j}}) + \sum_{\ell=1}^{n_{s,4j-3}} v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-4}}\mathbf{w}_2^{\ell}) - n_{s,4j-1}v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}})\Bigg).$$

Moreover, using the definition of $p_{s,4j-i}$, we find that each summand on the right-hand side of (2.29) may be expanded

$$\begin{aligned}\frac{1}{\beta^{p_{2s-1}}}(1-1/\beta^{2\ell_{2s-1}}) &= 1/\beta^{p_{2s-1}} - 1/\beta^{p_{2s}}\\ &= 1/\beta^{p_{s,1}} - 1/\beta^{p_{s,4r_s+1}}\\ &= \sum_{j=1}^{r_s}\Bigg(\frac{1}{\beta^{p_{s,4j-3}}}(1-1/\beta^{2n_{s,4j-3}})\\ &\quad + \frac{1}{\beta^{p_{s,4j-2}}}(1-1/\beta^{4n_{s,4j-2}})\\ &\quad + \frac{1}{\beta^{p_{s,4j-1}}}(1-1/\beta^{2n_{s,4j-1}})\\ &\quad + \frac{1}{\beta^{p_{s,4j}}}(1-1/\beta^{4n_{s,4j}})\Bigg).\end{aligned}$$

Subtracting $\sum_{s=1}^{t}\sum_{j=1}^{r_s}\frac{1}{\beta^{p_{s,4j-i}}}\frac{1}{\beta^2+1}(1-1/\beta^{4n_{s,4j-i}})$, $i=0,2$, from

both sides, (2.29) becomes

$$\begin{aligned}
&\sum_{s=1}^{t}\sum_{j=1}^{r_s}\Bigg(\frac{1}{\beta^{p_{s,4j-3}}}\frac{1}{\beta^2}(1-1/\beta^{2n_{s,4j-3}})\\
&+\sum_{\ell=1}^{n_{s,4j-3}} v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-4}}\mathbf{w}_2^{\ell})\\
&-n_{s,4j-1}v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}})\Bigg)\\
&\le\sum_{s=1}^{t}\sum_{j=1}^{r_s}\Bigg(\frac{1}{\beta^{p_{s,4j-3}}}\frac{1}{\beta^2+1}(1-1/\beta^{2n_{s,4j-3}})\\
&\qquad+\frac{1}{\beta^{p_{s,4j-1}}}\frac{1}{\beta^2+1}(1-1/\beta^{2n_{s,4j-1}})\Bigg).
\end{aligned}$$

Rearranging and using the fact that $1/\beta^2-1/(\beta^2+1)=1/(\beta^2(\beta^2+1))$, the previous inequality is equivalent to

$$\begin{aligned}
&\sum_{s=1}^{t}\sum_{j=1}^{r_s}\Bigg(\sum_{\ell=1}^{n_{s,4j-3}} v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-4}}\mathbf{w}_2^{\ell})\\
&\qquad+\frac{1}{\beta^{p_{s,4j-3}}}\frac{1}{\beta^2(\beta^2+1)}(1-1/\beta^{2n_{s,4j-3}})\Bigg)\\
&\le\sum_{s=1}^{t}\sum_{j=1}^{r_s}\Bigg(n_{s,4j-1}v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}})\\
&\qquad+\frac{1}{\beta^{p_{s,4j-1}}}\frac{1}{\beta^2+1}(1-1/\beta^{2n_{s,4j-1}})\Bigg).
\end{aligned}\tag{2.31}$$

It thus suffices to show (2.31) with equality if and only if $\mathbf{d}\prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$.

Step (ii)

Consider the summand (with respect to the summation over j) on the left-hand side of the previous inequality. We will show that this is less than or equal to $n_{s,4j-3}v(\mathbf{d})$, with equality if and only if $n_{s,4j-3}=0$. If $n_{s,4j-3}=0$, both the summand and $n_{s,4j-3}v(\mathbf{d})$ are zero; assume $n_{s,4j-3}>0$. We must show

$$\sum_{\ell=1}^{n_{s,4j-3}} (v(\mathbf{d}) - v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-4}}\mathbf{w}_2^{\ell})) > \frac{1}{\beta^{p_{s,4j-3}}}\frac{1}{\beta^2(\beta^2+1)}(1-1/\beta^{2n_{s,4j-3}}). \quad (2.32)$$

The left-hand side of the previous line equals

$$\sum_{\ell=1}^{n_{s,4j-3}} \left(\frac{1}{\beta^{p_{s,4j-3}+2\ell}} v(\mathbf{w}_2^{n_{s,4j-3}-\ell}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}\cdots\mathbf{w}_{i_n}(1-i_n/2))\right).$$

Note that $(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}\cdots\mathbf{w}_{i_n}(1-i_n/2) \succ (\mathbf{w}_0\mathbf{w}_2)^{\infty}$: if not, then the former word begins with $(\mathbf{w}_0\mathbf{w}_2)^{n'}\mathbf{w}_0\mathbf{w}_i$ for some $n'\geq 0$ and $i\in\{0,1\}$. But then

$$\begin{aligned}&\mathbf{w}_2^{n_{s,4j-3}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}\cdots\mathbf{w}_{i_n}(1-i_n/2)\\ &=\mathbf{w}_2^{n_{s,4j-3}}(\mathbf{w}_0\mathbf{w}_2)^{n'}\mathbf{w}_0\mathbf{w}_i\cdots\mathbf{w}_{i_n}(1-i_n/2)\\ &=\mathbf{w}_2^{n_{s,4j-3}-1}(\mathbf{w}_2\mathbf{w}_0)^{n'+1}\mathbf{w}_i\cdots\mathbf{w}_{i_n}(1-i_n/2),\end{aligned}$$

contradicting the minimality of the sum of powers $\sum_{\ell=1}^{4r_s} n_{s,\ell}$. Thus the left-hand side of (2.32) is strictly greater than

$$\begin{aligned}&\sum_{\ell=1}^{n_{s,4j-3}} \left(\frac{1}{\beta^{p_{s,4j-3}+2\ell}} v(\mathbf{w}_2^{n_{s,4j-3}-\ell}) + \frac{1}{\beta^{p_{s,4j-2}}} v((\mathbf{w}_0\mathbf{w}_2)^{\infty})\right)\\ =&\frac{1}{\beta^{p_{s,4j-3}}}\sum_{\ell=1}^{n_{s,4j-3}}\left(\frac{1}{\beta^{2\ell+1}}(1-1/\beta^{2n_{s,4j-3}-2\ell}) + \frac{1}{\beta^2+1}\frac{1}{\beta^{2n_{s,4j-3}+1}}\right)\end{aligned}$$

$$= \frac{1}{\beta^{p_{s,4j-3}}} \left(\frac{1}{\beta^2}(1 - 1/\beta^{2n_{s,4j-3}}) - \frac{n_{s,4j-3}}{\beta^{2n_{s,4j-3}+1}} + \frac{1}{\beta^2+1} \frac{n_{s,4j-3}}{\beta^{2n_{s,4j-3}+1}} \right)$$
$$= \frac{1}{\beta^{p_{s,4j-3}}} \left(\frac{1}{\beta^2}(1 - 1/\beta^{2n_{s,4j-3}}) - \frac{\beta^2}{\beta^2+1} \frac{n_{s,4j-3}}{\beta^{2n_{s,4j-3}+1}} \right).$$

It suffices to show that the right-hand side of the previous line is greater than or equal to the right-hand side of (2.32). Multiplying both quantities by $\beta^{p_{s,4j-3}+2}(\beta^2+1)$, this is equivalent to showing

$$(\beta^2+1)(1 - 1/\beta^{2n_{s,4j-3}}) - \beta^3 \frac{n_{s,4j-3}}{\beta^{2n_{s,4j-3}}} \geq 1 - \frac{1}{\beta^{2n_{s,4j-3}}},$$

which simplifies to

$$1 - \frac{1}{\beta^{2n_{s,4j-3}}} \geq \beta \frac{n_{s,4j-3}}{\beta^{2n_{s,4j-3}}}.$$

The left- and right-hand sides of the previous line increase and decrease, respectively, as functions of integers $n_{s,4j-3} > 0$. Since the inequality holds for $n_{s,4j-3} = 1$, we conclude that (2.32) holds. Thus the summand on the left-hand side of (2.31) is less than or equal to $n_{s,4j-3}v(\mathbf{d})$, with equality if and only if $n_{s,4j-3} = 0$.

Next, consider the summand on the right-hand side of (2.31). We shall show that this is greater than or equal to $n_{s,4j-1}v(\mathbf{d})$ with equality if and only if $n_{s,4j-1} = 0$. Again if $n_{s,4j-1} = 0$, both the summand and $n_{s,4j-1}v(\mathbf{d})$ equal zero, so assume $n_{s,4j-1} > 0$. The desired inequality is equivalent to

$$n_{s,4j-1}(v(\mathbf{d}) - v(1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}} \cdots (\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}))$$
$$< \frac{1}{\beta^{p_{s,4j-1}}} \frac{1}{\beta^2+1}(1 - 1/\beta^{2n_{s,4j-1}}). \quad (2.33)$$

The left-hand side of the previous line equals

$$\frac{n_{s,4j-1}}{\beta^{p_{s,4j-1}}} v(\mathbf{w}_0^{n_{s,4j-1}}(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j}} \cdots \mathbf{w}_{i_n}(1 - i_n/2))$$
$$= \frac{n_{s,4j-1}}{\beta^{p_{s,4j}}} v((\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j}} \cdots \mathbf{w}_{i_n}(1 - i_n/2)).$$

For similar reasons as above, one finds that $(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j}} \cdots \mathbf{w}_{i_n}(1 - i_n/2) \prec (\mathbf{w}_2\mathbf{w}_0)^\infty$. It follows that the left-hand side of (2.33) is strictly less than

$$\frac{n_{s,4j-1}}{\beta^{p_{s,4j}}} v((\mathbf{w}_2\mathbf{w}_0)^\infty) = \frac{n_{s,4j-1}}{\beta^{p_{s,4j}}} \frac{\beta}{\beta^2+1}.$$

Multiplying both sides of (2.33) by $\beta^{p_{s,4j}}(\beta^2+1)$ and recalling that $p_{s,4j} - p_{s,4j-1} = 2n_{s,4j-1}$, it thus suffices to show that

$$\beta n_{s,4j-1} \leq \beta^{2n_{s,4j-1}} - 1,$$

which clearly holds for each $n_{s,4j-1} \geq 1$. This proves that the summand on the right-hand side of (2.31) is greater than or equal to $n_{s,4j-1}v(\mathbf{d})$ with equality if and only if $n_{s,4j-1} = 0$.

Note that (2.19) may be rewritten as

$$\begin{aligned}\mathfrak{n}(\mathbf{d}) &= (1 + \#\{1 \leq k \leq n-1 \mid i_k \in \{1,2\}\} + 1) \\ &\quad - (\#\{1 \leq k \leq n-1 \mid 2 - i_k \in \{1,2\}\} + 1) \\ &= 1 + \#\{1 \leq k \leq n-1 \mid i_k = 2\} - \#\{1 \leq k \leq n-1 \mid i_k = 0\}.\end{aligned}$$

Since $\mathfrak{n}(\mathbf{d}) = 1$ by assumption, we have

$$\#\{1 \leq k \leq n-1 \mid i_k = 2\} = \#\{1 \leq k \leq n-1 \mid i_k = 0\}.$$

Recalling that $\mathbf{d} = 1\mathbf{X}_1\mathbf{Y}_1 \cdots \mathbf{X}_t\mathbf{Y}_t\mathbf{w}_{i_n}(1 - i_n/2)$, (2.30) and the fact that each $\mathbf{Y}_s$ consists solely of w_1's, we find

$$\#\{1 \leq k \leq n-1 \mid i_k = 2\} = \sum_{s=1}^{t} \sum_{j=1}^{r_s} (n_{s,4j-3} + n_{s,4j-2} + n_{s,4j})$$

and

$$\#\{1 \leq k \leq n-1 \mid i_k = 0\} = \sum_{s=1}^{t} \sum_{j=1}^{r_s} (n_{s,4j-2} + n_{s,4j-1} + n_{s,4j}),$$

so

$$\sum_{s=1}^{t}\sum_{j=1}^{r_s} n_{s,4j-3} = \sum_{s=1}^{t}\sum_{j=1}^{r_s} n_{s,4j-1}. \tag{2.34}$$

Using this and our prior observations regarding the left- and right-hand sides of (2.31), we have

$$\begin{aligned}
&\sum_{s=1}^{t}\sum_{j=1}^{r_s}\Bigg(\sum_{\ell=1}^{n_{s,4j-3}} v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-4}}\mathbf{w}_2^{\ell}) \\
&\qquad\qquad + \frac{1}{\beta^{p_{s,4j-3}}}\frac{1}{\beta^2(\beta^2+1)}(1-1/\beta^{2n_{s,4j-3}})\Bigg) \\
&\le v(\mathbf{d})\sum_{s=1}^{t}\sum_{j=1}^{r_s} n_{s,4j-3} = v(\mathbf{d})\sum_{s=1}^{t}\sum_{j=1}^{r_s} n_{s,4j-1} \\
&\le \sum_{s=1}^{t}\sum_{j=1}^{r_s}\Bigg(n_{s,4j-1}v(1\mathbf{X}_1\mathbf{Y}_1\cdots\mathbf{X}_{s-1}\mathbf{Y}_{s-1}\mathbf{w}_2^{n_{s,1}}\cdots(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}) \\
&\qquad\qquad + \frac{1}{\beta^{p_{s,4j-1}}}\frac{1}{\beta^2+1}(1-1/\beta^{2n_{s,4j-1}})\Bigg)
\end{aligned} \tag{2.35}$$

with equality throughout if and only if each $n_{s,4j-1} = n_{s,4j-3} = 0$. Thus the inequality in (2.31) holds, and it remains to show that $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$ if and only if each $n_{s,4j-1} = n_{s,4j-3} = 0$.

Step (iii)

Suppose $\mathbf{d} \succeq 1(\mathbf{w}_2\mathbf{w}_0)^\infty$. Then $\mathbf{d}$ begins with $1(\mathbf{w}_2\mathbf{w}_0)^{n'}\mathbf{w}_2\mathbf{w}_i$ for some $n' \ge 0$ and $i \in \{1,2\}$. This implies that either $n_{1,1}$ or $n_{1,5}$ is positive. For the converse, suppose that some $n_{s,4j-3}$ or $n_{s,4j-1}$ is positive. By (2.34), we can choose some $n_{s,4j-3} > 0$ with (s,j) (lexicographically) minimal. Note that

$$\sigma^{p_{s,4j-3}-1}(\mathbf{d}) = d_i\mathbf{w}_2^{n_{s,4j-3}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}\cdots\mathbf{w}_{i_n}(1-i_n/2)$$

with $d_i \in \{0,1\}$. Suppose $d_i = 0$ (the case that $d_i = 1$ is similar). If $s = 1$, then since $d_1 = 1$ we must have $j > 1$. Similarly, if $s > 1$,

then since $\mathbf{Y}_{s-1}$ ends with a digit 1 we must have $j > 1$. Hence, in either case, $j > 1$ and

$$\sigma^{p_{s,4j-7}}(\mathbf{d}) = \mathbf{w}_2^{n_{s,4j-7}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-6}}\mathbf{w}_0^{n_{s,4j-5}}(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-4}} \mathbf{w}_2^{n_{s,4j-3}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}\cdots\mathbf{w}_{i_n}(1-i_n/2).$$

Since $d_i = 0$, we must have $n_{s,4j-4} = 0$, and by minimality of j, also $n_{s,4j-7} = 0$. Moreover, $n_{s,4j-5} > 0$ contradicts the minimality of $\sum_{\ell=1}^{4r_s} n_{s,\ell}$, so $n_{s,4j-5} = 0$. Since no three consecutive $n_{s,\ell}$'s can be zero (except possibly the first and final three $n_{s,\ell}$), it follows that $n_{s,4j-6} > 0$. Thus

$$\sigma^{p_{s,4j-6}-1}(\mathbf{d}) = d_{i'}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-6}}\mathbf{w}_2^{n_{s,4j-3}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}\cdots\mathbf{w}_{i_n}(1-i_n/2) \tag{2.36}$$

for some $d_{i'} \in \{0,1\}$. Suppose $d_{i'} = 0$. Then for similar reasons as above, $j > 2$ and

$$\sigma^{p_{s,4j-11}}(\mathbf{d}) = \mathbf{w}_2^{n_{s,4j-11}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-10}}\mathbf{w}_0^{n_{s,4j-9}}(\mathbf{w}_0\mathbf{w}_2)^{n_{s,4j-8}} (\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-6}}\mathbf{w}_2^{n_{s,4j-3}}(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-2}}\cdots\mathbf{w}_{i_n}(1-i_n/2).$$

Since $d_{i'} = 0$, we have $n_{s,4j-8} = 0$. If $n_{s,4j-9} > 0$, the fact that $n_{s,4j-3} > 0$ contradicts the minimality of $\sum_{\ell=1}^{4r_s} n_{s,\ell}$. But $n_{s,4j-9} = 0$ is also a contradiction since this implies three consecutive $n_{s,\ell}$'s are zero. Thus $d_{i'} = 1$. Now suppose $\sigma^{p_{s,4j-6}-1}(\mathbf{d}) \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$. From (2.36), we find that $n_{s,4j-3} = 1$ and

$$\sigma^{p_{s,4j-6}-1}(\mathbf{d}) = 1(\mathbf{w}_2\mathbf{w}_0)^{n_{s,4j-6}}\mathbf{w}_2(\mathbf{w}_0\mathbf{w}_2)^{n'}\mathbf{w}_0\mathbf{w}_{i''}\cdots\mathbf{w}_{i_n}(1-i_n/2)$$

for some $n' \geq 0$ and $i'' \in \{0,1\}$. In any case, this contradicts the minimality of $\sum_{\ell=1}^{4r_s} n_{s,\ell}$. Thus (using the fact that $\mathbf{d} \in \mathcal{M}$),

$$\mathbf{d} \succeq \sigma^{p_{s,4j-6}-1}(\mathbf{d}) \succeq 1(\mathbf{w}_2\mathbf{w}_0)^\infty,$$

and we conclude that $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$ if and only if each $n_{s,4j-1} = n_{s,4j-3} = 0$. □

Note that for each $n \geq 1$, the word $\mathbf{d}^n := 1(\mathbf{w}_2\mathbf{w}_0)^n001 \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$ satisfies Property M. Furthermore, $v(\mathbf{d}^n)$ approaches $v(1(\mathbf{w}_2\mathbf{w}_0)^\infty) = 2\beta/(\beta+2)$ from below, and thus $1/v(\mathbf{d}^n)$ approaches $(\beta+2)/(2\beta) = 1/2 + 1/\beta$ from above. If $\mathbf{d} \in \mathcal{M}$ satisfies $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$, then there is some $n \geq 1$ for which $\mathbf{d} \prec \mathbf{d}^n \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$ and $1/2 + 1/\beta < 1/v(\mathbf{d}^n) < 1/v(\mathbf{d})$. Since $I_{\mathbf{d}^n} \cap I_{\mathbf{d}} = \varnothing$ and $I_{\mathbf{d}^n}$ and $I_{\mathbf{d}}$ contain $1/v(\mathbf{d}^n)$ and $1/v(\mathbf{d})$, respectively (see Proposition 2.2.19), it follows that $I_{\mathbf{d}} \subset (1/2+1/\beta, \beta]$. Similarly reasoning shows that if $\mathbf{d} \succ 1(\mathbf{w}_2\mathbf{w}_0)^\infty$, then $I_{\mathbf{d}} \subset (1, 1/2 + 1/\beta)$, and in fact $1/2 + 1/\beta$ is a non-matching parameter.

With these observations and the previous lemmas, we are now ready to prove the main result of this section:

Theorem 2.3.9. *The frequency functions* $\mathfrak{f}_S, \mathfrak{f}_T : [1, \beta] \to [0, 1]$ *attain their maximums* $\mathfrak{f}_S(\alpha) = 4/5$ *and* $\mathfrak{f}_T(\alpha) = 3/4$ *on the maximal interval* $[1/2 + 1/\beta, 1 + 1/\beta^2]$.

Proof. By (2.17), it suffices to show the statement for $\mathfrak{f}_S$. Recall from Example 2.3.4 that $\mathfrak{f}_S$ equals $4/5$ on $I_{1010} \cup I_{1001} = (1 + 1/\beta^4, 1 + 1/\beta^2)\backslash\{1 + 1/\beta^3\}$. Moreover, $\mathfrak{f}_S$ is decreasing on $I_{10} = (1 + 1/\beta^2, \beta]$ since $\mathfrak{n}(10) = 0$. By continuity of $\mathfrak{f}_S$, the statement has been proven for $\alpha \in [1 + 1/\beta^4, \beta]$.

We now show that $\mathfrak{f}_S(\alpha) \leq 4/5$ for $\alpha \in [1, 1 + 1/\beta^4)$, with equality if $\alpha \geq 1/2 + 1/\beta$. Since $\mathfrak{f}_S$ is continuous and is monotone on each matching interval $I_{\mathbf{d}}$, and since the set of matching parameters $\cup_{\mathbf{d}\in\mathcal{M}} I_{\mathbf{d}}$ is dense, it suffices to show the desired statements for the endpoints $\alpha_{\mathbf{d}}^{\pm}$ of matching intervals in $[1, 1 + 1/\beta^4)$. We claim that each endpoint $\alpha_{\mathbf{d}}^+, \alpha_{\mathbf{d}}^- \in [1, 1+1/\beta^4)$ is the limit (from above) of some sequence of endpoints of cascade intervals. If $\mathbf{d} \in \psi(\mathcal{M}_U)$, then $I_{\mathbf{d}}$ is itself a cascade interval and we take the constant sequence. Suppose $\mathbf{d} \in \mathcal{M}_U\backslash\psi(\mathcal{M}_U)$. Since each lower endpoint $\alpha_{\mathbf{d}}^-$ equals the upper endpoint $\alpha_{\psi(\mathbf{d})}^+$ of $I_{\psi(\mathbf{d})}$ by Proposition 2.2.26, we can again take the constant sequence. Now consider $\alpha_{\mathbf{d}}^+$. Let $\varepsilon > 0$, and choose some matching parameter $\alpha' \in I_{\mathbf{d}'}$ satisfying $\alpha_{\mathbf{d}}^+ < \alpha' < \alpha_{\mathbf{d}}^+ + \varepsilon$. Since matching intervals are disjoint, Proposition 2.2.26 implies that the

cascade interval $I_{\psi(\mathbf{d}')}$ lies strictly between $\alpha_{\mathbf{d}}^+$ and α', and thus its endpoints are within a distance of ε of $\alpha_{\mathbf{d}}^+$. It follows $\alpha_{\mathbf{d}}^+$ is the limit (from above) of a sequence of endpoints of cascade intervals, and the claim holds. Again by continuity of $\mathfrak{f}_S$, it now suffices to show the desired statements for endpoints of cascade intervals. These follow directly from Lemmas 2.3.7 and 2.3.8 and the observation above that if $I_{\mathbf{d}} \subset (1/2 + 1/\beta, \beta]$, then $\mathbf{d} \prec 1(\mathbf{w}_2\mathbf{w}_0)^\infty$.

Maximality of the interval $[1/2 + 1/\beta, 1 + 1/\beta^2]$ follows from the fact that $\mathfrak{f}_S$ is strictly decreasing on $(1+1/\beta^2, \beta]$, density of matching parameters in $[1, \beta]$ and Lemmas 2.3.7 and 2.3.8. □

Theorem 2.1.2 is now a collection of previous results:

Proof of Theorem 2.1.2. This is a direct consequence of Proposition 2.3.3, Theorem 2.3.9 and Corollary 2.3.5. □

2.4 Appendix: Proofs of technical lemmas

We include here two technical results, which together with Lemma 2.2.15 prove Lemma 2.2.20. Recall that $\Delta(\mathbf{u})$ denotes the cylinder set of points $x \in [0, 1)$ for which the β-expansion of x begins with $\mathbf{u}$.

Lemma 2.4.1. *Let* $\mathbf{d} = d_1 \cdots d_m \in \mathcal{M}_U$ *and* $\mathbf{e} \coloneqq \varphi(\mathbf{d}) = e_1 \cdots e_m$. *The* β*-expansions of* $1/\alpha_{\mathbf{d}}^-$, $1/\alpha_{\mathbf{d}}^+$ *and* $1 - 1/\alpha_{\mathbf{d}}^+$ *are given by*

$$(b_j(1/\alpha_{\mathbf{d}}^-))_{j\geq 1} = \begin{cases} (\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)^\infty, & d_m = 1, \\ (\mathbf{d}\overline{e_1 \cdots e_{m-1}}0)^\infty, & d_m = 0, \end{cases}$$

$$(b_j(1/\alpha_{\mathbf{d}}^+))_{j\geq 1} = \begin{cases} (d_1 \cdots d_{m-1}0)^\infty, & d_m = 1, \\ (d_1 \cdots d_{m-2}0)^\infty, & d_m = 0, \end{cases} \qquad \textit{and}$$

$$(b_j(1 - 1/\alpha_{\mathbf{d}}^+))_{j\geq 1} = \begin{cases} \overline{\mathbf{e}}^\infty, & d_m = 1, \\ 0(\overline{e_2 \cdots e_m})^\infty, & d_m = 0. \end{cases}$$

Proof. We consider only the β-expansion of $1/\alpha_{\mathbf{d}}^-$ for $d_m = 1$; the proofs of the other expansions are similar. It suffices to show that $1/\alpha_{\mathbf{d}}^- \in \Delta(\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)$ and $B^{2m-2}(1/\alpha_{\mathbf{d}}^-) = 1/\alpha_{\mathbf{d}}^-$. First, note that

$$\begin{aligned} v(\mathbf{d}\overline{e_2 \cdots e_{m-2}}0) &= v(\mathbf{d}) - (1/\beta^m)v(e_2 \cdots e_{m-2}) \\ &= v(\mathbf{d}) - (1/\beta^{m-1})v(e_1 \cdots e_{m-2}) \\ &= v(\mathbf{d}) - (1/\beta^{m-1})(v(\mathbf{e}) + 1/\beta^{m-1}) \\ &= v(\mathbf{d}) - (1/\beta^{m-1})(v(\mathbf{d}) - 1 + 1/\beta^{m-1}) \\ &= (1 - 1/\beta^{m-1})(v(\mathbf{d}) + 1/\beta^{m-1}). \end{aligned}$$

Using this and Equation (2.9), $1/\alpha_{\mathbf{d}}^- \in \Delta(\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)$ if and only if

$$(1-1/\beta^{m-1})(v(\mathbf{d})+1/\beta^{m-1}) \le 1/\alpha_{\mathbf{d}}^- < (1-1/\beta^{m-1})v(\mathbf{d})+1/\beta^{m-1}.$$

Since $d_m = 1$, the first inequality holds if and only if

$$(1 - 1/\beta^{m-1})(v(\mathbf{d}) + 1/\beta^{m-1}) \le \frac{\beta^m v(\mathbf{d}) + \beta}{\beta^m + \beta},$$

or

$$(\beta^m + \beta)(1 - 1/\beta^{m-1})(v(\mathbf{d}) + 1/\beta^{m-1}) \le \beta^m v(\mathbf{d}) + \beta.$$

Factoring β^m from the first and multiplying it through the third term, the left-hand side is equal to

$$(1+1/\beta^{m-1})(1-1/\beta^{m-1})(\beta^m v(\mathbf{d})+\beta) = (1-1/\beta^{2m-2})(\beta^m v(\mathbf{d})+\beta),$$

which is less than $\beta^m v(\mathbf{d}) + \beta$. The second inequality is true if and only if

$$\frac{\beta^m v(\mathbf{d}) + \beta}{\beta^m + \beta} < (1 - 1/\beta^{m-1})v(\mathbf{d}) + 1/\beta^{m-1}.$$

Multiplying both sides by $\beta^m + \beta$, this is equivalent to

$$\beta^m v(\mathbf{d}) + \beta < (\beta^m - 1/\beta^{m-2})v(\mathbf{d}) + \beta + 1/\beta^{m-2},$$

or $(1/\beta^{m-2})v(\mathbf{d}) < 1/\beta^{m-2}$. This holds since $v(\mathbf{d}) < v((10)^\infty) = 1$. Thus $1/\alpha_{\mathbf{d}}^- \in \Delta(\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)$. With this and Equation (2.7),

$$\begin{aligned}
&B^{2m-2}(1/\alpha_{\mathbf{d}}^-) \\
&= \beta^{2m-2}(1/\alpha_{\mathbf{d}}^- - v(\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)) \\
&= \beta^{2m-2}\left(\frac{\beta^m v(\mathbf{d}) + \beta}{\beta^m + \beta} - (1 - 1/\beta^{m-1})(v(\mathbf{d}) + 1/\beta^{m-1})\right) \\
&= \beta^{2m-2}\left(\frac{\beta^m v(\mathbf{d}) + \beta - (\beta^m - 1/\beta^{m-2})(v(\mathbf{d}) + 1/\beta^{m-1})}{\beta^m + \beta}\right) \\
&= \frac{\beta^m v(\mathbf{d}) + \beta}{\beta^m + \beta} = 1/\alpha_{\mathbf{d}}^-.
\end{aligned}$$

□

Lemma 2.4.2. *Let* $\mathbf{d} = d_1 \cdots d_m \in \mathcal{M}_U$ *and* $\mathbf{e} \coloneqq \varphi(\mathbf{d}) = e_1 \cdots e_m$. *If* $d_m = 1$, *then for each* $j \geq 0$,

$$\sigma^j((\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)^\infty) \preceq (\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)^\infty$$

and

$$\sigma^j(\overline{\mathbf{e}}^\infty) \preceq (d_1 \cdots d_{m-1}0)^\infty.$$

If $d_m = 0$, *then for each* $j \geq 0$,

$$\sigma^j((\mathbf{d}\overline{e_1 \cdots e_{m-1}}0)^\infty) \preceq (\mathbf{d}\overline{e_1 \cdots e_{m-1}}0)^\infty$$

and

$$\sigma^j(0(\overline{e_2 \cdots e_m})^\infty) \preceq (d_1 \cdots d_{m-2}0)^\infty.$$

Proof. In each statement $j = 0$ holds trivially, so assume $j > 0$. We prove the statements for $d_m = 1$; the other proofs are similar. Write

$$\mathbf{d} = 1\mathbf{w}_{i_1}\mathbf{w}_{i_2} \cdots \mathbf{w}_{i_n}(1 - i_n/2) \tag{2.37}$$

and

$$\mathbf{e} = \overline{0\mathbf{w}_{2-i_1}\mathbf{w}_{2-i_2} \cdots \mathbf{w}_{2-i_n}(i_n/2)} \tag{2.38}$$

with each $i_k \in \{0,1,2\}$ and $i_n = 0$. Due to periodicity, it suffices to show the first inequality for $j < 2m-2$. Note that $d_m = 1$ implies $\overline{e_{m-1}} = 1$. If $j \geq m$, then

$$\begin{aligned}\sigma^j((\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)^\infty) =& (\overline{e_{j-m+2} \cdots e_{m-2}}0\mathbf{d}\overline{e_2 \cdots e_{j-m+1}})^\infty \\ \prec& \overline{e_{j-m+2} \cdots e_{m-1}e_m} \\ \preceq& \mathbf{d} \prec (\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)^\infty,\end{aligned}$$

where the penultimate inequality follows from Property M. Now suppose $j < m$. It suffices to show that

$$d_{j+1} \cdots d_m \overline{e_2 \cdots e_{m-2}}0 d_1 \cdots d_j \preceq \mathbf{d}\overline{e_2 \cdots e_{m-2}}0.$$

If $j = m-1$, the inequality follows from the fact that $\mathbf{d} \in \mathcal{M}_U$ implies $d_2 d_3 = 01$ and $\overline{e_2 e_3} = 00$. Now suppose $j < m-1$. Since $\sigma^j(\mathbf{d}) \preceq \mathbf{d}$, we have $d_{j+1} \cdots d_m \preceq d_1 \cdots d_{m-j}$. If this inequality is strict, we are finished. Suppose equality holds. Then we wish to show

$$\overline{e_2 \cdots e_{m-2}}0 d_1 \cdots d_j \preceq d_{m-j+1} \cdots d_m \overline{e_2 \cdots e_{m-2}}0.$$

Since $\overline{e_{m-1}} = 1$, it suffices to show

$$\overline{e_2 \cdots e_{m-1}} \preceq d_{m-j+1} \cdots d_m \overline{e_2 \cdots e_{m-j-1}}. \tag{2.39}$$

By assumption, $d_{j+1} \cdots d_m = d_1 \cdots d_{m-j}$, so $d_{j+1} = d_1 = 1 = d_m = d_{m-j}$. Now $d_2 = 0$ implies $j \neq m-2$, and similarly $d_{m-2} = 0$ implies $j \neq m-3$. Hence $j < m-3$, and $d_{j+1} = d_{m-j} = 1$ imply that d_{j+2} and d_{m-j+1} are the beginnings of some blocks $\mathbf{w}_{i_p}$ and $\mathbf{w}_{i_\ell}$, respectively. (Similarly, $\overline{e_{j+2}}$ and $\overline{e_{m-j+1}}$ are the beginnings of $\mathbf{w}_{2-i_p}$ and $\mathbf{w}_{2-i_\ell}$, respectively.) Then $d_1 \cdots d_{m-j} = d_{j+1} \cdots d_m$ may be written as

$$1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_{\ell-1}} = 1\mathbf{w}_{i_p} \cdots \mathbf{w}_{i_{n-1}}001.$$

In particular, $i_{\ell-1} = 1$, and $\mathbf{w}_{2-i_{\ell-1}} = \mathbf{w}_1$ implies

$$\overline{e_{m-j-2}e_{m-j-1}e_{m-j}} = 001.$$

Omitting the final two zeros from the right-hand side of (2.39), the desired inequality may be written in terms of blocks:

$$\mathbf{w}_{2-i_1}\cdots\mathbf{w}_{2-i_n} \preceq \mathbf{w}_{i_\ell}\cdots\mathbf{w}_{i_{n-1}}\mathbf{w}_1\mathbf{w}_{2-i_1}\cdots\mathbf{w}_{2-i_{\ell-2}}.$$

Suppose for the sake of contradiction that this inequality does not hold, and let $1 \le k \le n$ be minimal such that $\mathbf{w}_{2-i_k}$ differs from the kth block on the right-hand side. Then

$$\mathbf{w}_{2-i_k} \succ \begin{cases} \mathbf{w}_{i_{\ell+k-1}}, & k < n-\ell+1, \\ \mathbf{w}_1, & k = n-\ell+1, \\ \mathbf{w}_{2-i_{k-(n-\ell)-1}}, & k > n-\ell+1, \end{cases}$$

and we consider these three cases separately:

(i) If $k < n-\ell+1$, then

$$(2-i_1,\dots,2-i_{k-1}) = (i_\ell,\dots,i_{\ell+k-2})$$

and $2-i_k > i_{\ell+k-1}$ imply

$$(2-i_\ell,\dots,2-i_{\ell+k-2}) = (i_1,\dots,i_{k-1})$$

and $2-i_{\ell+k-1} > i_k$. This gives

$$1\mathbf{w}_{2-i_\ell}\cdots\mathbf{w}_{2-i_{\ell+k-1}} \succ 1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_k}.$$

Recall that $\overline{e_{m-j+1}}$ is the beginning of the block $\mathbf{w}_{2-i_\ell}$, so the previous line together with $\overline{e_{m-j}} = 1$ imply $\sigma^{m-j-1}(\overline{\mathbf{e}}) \succ \mathbf{d}$, a contradiction.

(ii) If $k = n-\ell+1$, then

$$(2-i_1,\dots,2-i_{n-\ell}) = (i_\ell,\dots,i_{n-1})$$

and $2-i_{n-\ell+1} > 1$ imply

$$(2-i_\ell,\dots,2-i_{n-1}) = (i_1,\dots,i_{n-\ell})$$

and $i_{n-\ell+1} = 0$. Since $2 - i_n = 2$, this implies

$$1\mathbf{w}_{2-i_\ell} \cdots \mathbf{w}_{2-i_n} \succ 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_{n-\ell+1}}.$$

As in case (i), this gives the contradiction that $\sigma^{m-j-1}(\overline{\mathbf{e}}) \succ \mathbf{d}$.

(iii) If $k > n - \ell + 1$, then

$$(2 - i_1, \ldots, 2 - i_{n-\ell}) = (i_\ell, \ldots, i_{n-1})$$

and $2 - i_{n-\ell+1} = 1$ implies

$$(2 - i_\ell, \ldots, 2 - i_{n-1}) = (i_1 \ldots, i_{n-\ell})$$

and $i_{n-\ell+1} = 1$. Again since $2 - i_n = 2$,

$$1\mathbf{w}_{2-i_\ell} \cdots \mathbf{w}_{2-i_n} \succ 1\mathbf{w}_{i_1} \cdots \mathbf{w}_{i_{n-\ell+1}},$$

and the contradiction of cases (i) and (ii) arises.

This proves for each $j > 0$ that

$$\sigma^j((\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)^\infty) \preceq (\mathbf{d}\overline{e_2 \cdots e_{m-2}}0)^\infty.$$

It remains to show that

$$\sigma^j(\overline{\mathbf{e}}^\infty) \preceq (d_1 \cdots d_{m-1}0)^\infty,$$

or, equivalently,

$$\overline{e_{j+1} \cdots e_m e_1 \cdots e_j} \preceq d_1 \cdots d_{m-1}0$$

for $j < m$. If $j = m - 1$, then since $d_m = 1$ implies $\overline{e_m} = 0$, the inequality holds. Now let $j < m - 1$, and suppose for the sake of contradiction that the desired inequality does not hold. If there is some $k \leq m - j$ for which

$$\overline{e_{j+1} \cdots e_{j+k}} \succ d_1 \cdots d_k,$$

then $\sigma^j(\overline{\mathbf{e}}) \succ \mathbf{d}$, a contradiction. Thus either $\overline{e_{j+1}\cdots e_m e_1 \cdots e_{j-1}} = d_1 \cdots d_{m-1}$ and $\overline{e_j} > 0$, or there is some minimal $1 \le k < j$ for which

$$\overline{e_{j+1}\cdots e_m e_1 \cdots e_k} \succ d_1 \cdots d_{m-j+k}.$$

In the first case, $\overline{e_{j+1}} = d_1 = 1$ implies $\overline{e_j} = 0$, so this is impossible. In the second case, using the fact that $\overline{e_{j+1}} = d_1 = 1$, (2.37), (2.38) and the fact that that $d_m = 1$ implies $\overline{e_{m-2}e_{m-1}e_m e_1} = 0100 = \mathbf{w}_2\mathbf{w}_0$, the previous line may be written in block form

$$1\mathbf{w}_{2-i_\ell}\cdots\mathbf{w}_{2-i_{n-1}}\mathbf{w}_2\mathbf{w}_0\mathbf{w}_{2-i_1}\cdots\mathbf{w}_{2-i_p} \succ 1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_q}$$

for some $\ell, p, q \in \{1, \ldots, n\}$. In particular,

$$(0, 2-i_1, \ldots, 2-i_{p-1}) = (i_{q-p}, i_{q-p+1}, \ldots, i_{q-1})$$

and $2 - i_p > i_q$ imply

$$(2-i_{q-p}, 2-i_{q-p+1}, \ldots, 2-i_{q-1}) = (2, i_1, \ldots, i_{p-1})$$

and $2 - i_q > i_p$. Since $\mathbf{w}_{2-i_{q-p}} = 01$, there is some $s \ge 0$ such that

$$\begin{aligned}\sigma^s(\overline{\mathbf{e}}) =& 1\mathbf{w}_{2-i_{q-p+1}}\cdots\mathbf{w}_{2-i_{q-1}}\mathbf{w}_{2-i_q}\cdots\mathbf{w}_{2-i_n}0\\ \succ& 1\mathbf{w}_{i_1}\cdots\mathbf{w}_{i_{p-1}}\mathbf{w}_{i_p}\cdots\mathbf{w}_{i_n}1 = \mathbf{d},\end{aligned}$$

contrary to the assumption that $\mathbf{d} \in \mathcal{M}$. □

Chapter 3

Constructing lattice surfaces with prescribed Veech groups: an algorithm

3.1 Introduction

The study of translation surfaces may be approached from numerous different angles—complex analysis, differential geometry, algebraic topology and number theory, for instance, each offer unique insights into these objects. These different perspectives lead to a wealth of questions regarding translation surfaces and their dynamics. One such avenue of inquiry is the well-known action of $\mathrm{SL}_2\mathbb{R}$ on the stratum $\mathcal{H}_1(d_1, \ldots, d_\kappa)$ of unit-area translation surfaces with singularities of orders $d_1, \ldots, d_\kappa$. Stabilizers (i.e., Veech groups) and orbits of translation surfaces under this action have proven to be of great interest. Analogues of Ratner's theorems con-

cerning unipotent flows on homogeneous spaces (see, e.g., [149]) are proven in the celebrated works of Eskin, Mirzakhani and Mohammadi; in particular, all $\mathrm{SL}_2\mathbb{R}$-orbit closures are affine invariant 'suborbifolds' of $\mathcal{H}_1(d_1, \ldots, d_\kappa)$ ([67, 68]). Despite having been well-studied since as early as the 1980's, some immediate questions regarding Veech groups have been non-trivial to address. General methods of computing Veech groups were unknown until recently (see [27, 28, 63, 64, 151, 189] and the special cases of [73, 171]), and while some universal properties of Veech groups are known—for instance, they are necessarily discrete and non-cocompact ([187])—a complete classification of which subgroups of $\mathrm{SL}_2\mathbb{R}$ are realised as Veech groups remains an open problem ([85]).

The first non-arithmetic examples were discovered by Veech, who proved that the odd index Hecke triangle groups and an index two subgroup of the even index Hecke triangle groups are realised as Veech groups ([187]). More general triangle groups are studied in [25, 83], where the authors show that there are Veech groups commensurable to all triangle groups $\Delta(m, n, \infty)$, $2 \leq m < n < \infty$. On the other hand, triangle groups whose orientation preserving subgroups are never contained in a Veech group are also given in [83]. Infinitely generated Veech groups are shown to exist in [86] and [140], as are non-trivial Veech groups with no parabolic elements in [84]. While this brief survey is far from an exhaustive overview of the literature, it serves to highlight that the question of realisability of Veech groups has been a problem of interest spanning decades.

Of particular interest within the space of all translation surfaces are so-called *lattice surfaces* (or *Veech surfaces*), which are those whose Veech groups are lattices (i.e., have finite covolume) in $\mathrm{SL}_2\mathbb{R}$. These surfaces admit especially nice dynamics ([187]), but the list of known families of lattice surfaces is relatively short ([59]). Lattice surfaces are also interesting as they are precisely those surfaces whose $\mathrm{SL}_2\mathbb{R}$-orbits are closed with respect to the analytic topology on the stratum ([177, 188]), and the projection of a closed orbit to the moduli space of Riemann surfaces is an algebraic curve—called a *Teichmüller curve*—isometrically immersed with respect to the

Teichmüller metric (see, say, [193]).

We present an explicit algorithm which constructs all translation surfaces with a given lattice Veech group in any given stratum:

Algorithm 3.1.1. *Input: Non-negative integers* $d_1 \leq \cdots \leq d_\kappa$ *whose sum is even and a finite set of generators of a lattice* $\Gamma \leq \mathrm{SL}_2\mathbb{R}$.

Output: All translation surfaces $(X, \omega) \in \mathcal{H}_1(d_1, \ldots, d_\kappa)$ *with Veech group* $\mathrm{SL}(X, \omega) = \Gamma$.

An implicit algorithm is given in [178] which enumerates all affine equivalence classes of lattice surfaces in terms of a parameter measuring the infimum of areas of triangles within the surface. Algorithm 3.1.1 similarly returns lattice surfaces, but it fundamentally differs both in its goal of finding surfaces with specific, prescribed Veech groups and in its method of doing so.

In developing Algorithm 3.1.1, we obtain a new proof of a finiteness result of Smillie and Weiss (Corollary 1.7, [179]; see also [146]):

Theorem 3.1.2 (Smillie–Weiss, 2010 [179])**.** *There are at most finitely many unit-area lattice surfaces with a given Veech group in any given stratum.*

Our methods essentially reverse the algorithm for computing Veech groups presented in [63, 64]. There the authors associate to each stratum a canonical (infinite area, disconnected in general) flat surface $\mathcal{O}$, within which any closed, connected translation surface (X, ω) of the stratum may be naturally represented. Data regarding the translation surface and its saddle connections are recorded via pairs of points in $\mathcal{O}$ termed *orientation-paired marked segments*, and the authors show that (X, ω) may be recovered from a certain finite subset of these marked segments—the *marked Voronoi staples* (§3.4 of [64]). Furthermore, a classification of the Veech group of (X, ω) in terms of its orientation-paired marked segments and affine automorphisms of $\mathcal{O}$ is given in Proposition 17 of [64].

Rather than beginning with a lattice surface (X, ω) and using its corresponding orientation-paired marked segments to compute

elements of the Veech group $\Gamma = \mathrm{SL}(X, \omega)$, Algorithm 3.1.1 begins with a lattice $\Gamma \leq \mathrm{SL}_2\mathbb{R}$, simulates normalised subsets of orientation-paired marked segments in $\mathcal{O}$ using Γ, and constructs candidate translation surfaces with Veech group Γ from finite unions of appropriately scaled versions of these 'simulations.' Theorem 3.1.2 follows from the facts that

(i) a lattice Γ determines only finitely many such simulations (Lemma 3.4.11),

(ii) the orientation-paired marked segments of any unit-area translation surface with Veech group Γ are realised as a finite union of scaled simulations (Theorem 3.4.14), and

(iii) for any finite collection of simulations there are only finitely many scalars for which the union of scaled simulations could possibly coincide with the orientation-paired marked segments of a unit-area translation surface.

For this latter point, we introduce the technical but important notions of *permissible triples* and *permissible scalars* of particular subsets of $\mathcal{O}$, prove finiteness results regarding them (Proposition 3.3.14 and Lemma 3.3.16), and apply these results to orientation-paired marked segments and marked Voronoi staples (Proposition 3.4.17). Our novel proof differs from that of Smillie and Weiss, who use finiteness results of Markov partitions corresponding to hyperbolic affine automorphisms of surfaces ([179]).

This work should be viewed in light of the aforementioned open question regarding the realisation of subgroups of $\mathrm{SL}_2\mathbb{R}$ as Veech groups: a full implementation[1] of Algorithm 3.1.1 could lead to the discovery of new lattice surfaces and experimental conjectures as to which groups are realised as Veech groups. Moreover, further

[1] A partial implementation has been written by the author using [181] to confirm some low genus examples with small generating sets, though an efficient version of the algorithm which handles more interesting examples awaits implementation.

work to determine a halting criterion for Algorithm 3.1.1 would give a general, explicit procedure to determine whether or not a given lattice is a Veech group in any particular stratum (see comments at the end of §3.5). At present, the methods of this chapter may be used in special cases to obtain obstructions for lattices being realised as Veech groups. In particular, we prove:

Theorem 3.1.3. *The square torus is the only translation surface in* $\cup_{g>0}\mathcal{H}(2g-2)$ *with Veech group* $\mathrm{SL}(X,\omega) = \mathrm{SL}_2\mathbb{Z}$.

This chapter is organised as follows. Section 3.2 gives a brief background on translation surfaces (§3.2.1) and their Veech groups (§3.2.2). Section 3.3 concerns the canonical surface $\mathcal{O}$ studied in [63] and [64]: definitions and immediate results regarding $\mathcal{O}$ are given in §3.3.1, permissible triples and permissible scalars are introduced and studied in §3.3.2, and in §3.3.3 we recall definitions and results regarding marked segments and Voronoi staples of a translation surface. Section 3.4 further develops the machinery for Algorithm 3.1.1 and uses these ideas to give a new proof of Theorem 3.1.2: a characterisation of lattice groups in terms of directions announced by linear transformations is given in §3.4.1; subsections 3.4.2 and 3.4.3 explore a group action on marked segments and how a lattice group may be used to simulate orbits of this action; subsection 3.4.4 relates marked Voronoi staples to permissible triples; and a new proof of Theorem 3.1.2 is given in §3.4.5. Algorithm 3.1.1 is presented in §3.5 and Theorem 3.1.3 is proven in §3.6.

3.2 Background

3.2.1 Translation surfaces

We begin with basic definitions and notation; see, say, the surveys [193] and [195] for more details.

Translation surfaces: various perspectives

A *translation surface* (X, ω) is a Riemann surface X together with a non-zero, holomorphic one-form ω. Equivalently, a translation surface is defined as a real surface X such that off of a finite set Σ, the surface $X \backslash \Sigma$ is equipped with a translation atlas, and the resulting flat structure extends to all of X to give conical singularities at points of Σ whose angles are positive integral multiples of 2π. Any translation surface may be *polygonally represented* as a collection of polygons in the plane together with an identification of edges in pairs, such that identified edges are parallel, equal length and of opposite orientation. A translation surface is *closed* if it is compact and has no boundary.

In these three perspectives—which we move fluidly between—the zeros of ω, elements of Σ of cone angles $2\pi(d+1),\ d > 0$, and, after identifying edges, the vertices of angles $2\pi(d+1),\ d > 0$, of a polygonally represented translation surface each correspond to the same set of *non-removable singularities* of the surface. Throughout, the *singular set* Σ denotes these singularities together with possibly finitely many *removable singularities* (or *marked points*), i.e., points of cone angle 2π. We refer to any element of Σ as a singularity (of cone angle $2\pi(d+1),\ d \geq 0$), bearing in mind that it may in fact be removable. Points in the complement $X \backslash \Sigma$ are called *regular*.

Metric, Lebesgue measure, saddle connections and holonomy vectors

A translation surface (X, ω) comes equipped with a metric defined as follows: given $p, q \in X$, the distance from p to q is given by

$$d(p, q) \coloneqq \inf_{\gamma} \left| \int_{\gamma} \omega \right|,$$

where the infimum is over all piecewise-smooth curves $\gamma \subset X$ from p to q. The infimum is realised by a straight line segment (with

respect to the flat structure) on (X, ω) or by a union of such segments meeting at singularities. Lebesgue measure in the plane pulls back via coordinate maps of the translation atlas of (X, ω) to give a measure on $X \backslash \Sigma$; we extend this measure to all of X by declaring Σ to be a null set.

A *separatrix* is a straight line segment on (X, ω) emanating from a singularity and having no singularities in its interior. A *saddle connection* is a separatrix (of positive length) which also ends at a singularity. The *holonomy vector* of a saddle connection s is defined as $\mathrm{hol}(s) \coloneqq \int_s \omega$. The collection of all holonomy vectors of (X, ω) forms a subset of the plane with no limit points and whose directions are dense in S^1 (Proposition 3.1, [191]).

Voronoi decomposition

Every translation surface (X, ω) has a (unique) *Voronoi decomposition* subordinate to its singular set Σ, which is described as follows (see also [27, 137]). For any $x \in X$, let $d(x, \Sigma)$ denote the minimum of the set $\{d(x, \sigma)\}_{\sigma \in \Sigma}$. To each $\sigma \in \Sigma$ there is associated an open, connected 2-cell, $\mathcal{C}_\sigma$, comprised of the points $x \in X$ for which $d(x, \Sigma)$ is realised by a unique length-minimising path ending at σ. The boundary of $\mathcal{C}_\sigma$ is the union of 1-cells of points x for which $d(x, \Sigma)$ is realised by precisely two length-minimising paths, at least one of which ends at σ, and of 0-cells of points x for which $d(x, \Sigma)$ is realised by three or more length-minimising paths, at least one of which ends at σ.

3.2.2 Veech groups

Affine diffeomorphisms and Veech groups

An *affine diffeomorphism* $f : (X_1, \omega_1) \to (X_2, \omega_2)$ between (not necessarily closed nor connected) translation surfaces is a diffeomorphism from X_1 to X_2 sending the singular set Σ_1 of X_1 into that of X_2, and which—on the complement of Σ_1—is locally an affine

map of the plane of constant linear part[2] (i.e., a map of the form $v \mapsto Av + b$ for some global $A \in \mathrm{GL}_2\mathbb{R}$ and local $b \in \mathbb{R}^2$). An *affine automorphism* is an affine diffeomorphism from a translation surface (X, ω) to itself. The set of all affine automorphisms of (X, ω), denoted $\mathrm{Aff}(X, \omega)$, forms a group under composition, as does the subset $\mathrm{Aff}^+(X, \omega) \subset \mathrm{Aff}(X, \omega)$ of orientation-preserving elements. The map $\mathrm{der} : \mathrm{Aff}(X, \omega) \to \mathrm{GL}_2\mathbb{R}$ sending an affine automorphism to its linear part in local coordinates gives a group homomorphism. The image of $\mathrm{Aff}^+(X, \omega)$ under this map, denoted $\mathrm{SL}(X, \omega)$, is called the *Veech group* of (X, ω), and its kernel is the group of translations denoted by $\mathrm{Trans}(X, \omega)$. Veech shows in [187] that for closed, connected (X, ω), the Veech group $\mathrm{SL}(X, \omega)$ is a discrete subgroup of $\mathrm{SL}_2\mathbb{R}$—and thus its image in $\mathrm{PSL}_2\mathbb{R}$ is a Fuchsian group—and, moreover, $\mathrm{SL}(X, \omega)$ is always non-cocompact.

Strata, a $\mathrm{GL}_2\mathbb{R}$-action and lattices

We call (X_1, ω_1) and (X_2, ω_2) *translation equivalent* if there is an affine diffeomorphism between them with trivial linear part. For non-negative integers $d_1 \leq \cdots \leq d_\kappa$, let $\mathcal{H}(d_1, \ldots, d_\kappa)$ denote the *stratum* of all closed, connected translation surfaces—up to translation equivalence—with singularities of cone angles $2\pi(d_1 + 1) \leq \cdots \leq 2\pi(d_\kappa + 1)$ and no other singularities. (Recall that we allow for 'removable singularities,' so some d_i may be zero, and with our definitions, $\mathrm{SL}(X, \omega)$ depends on these marked points in Σ. In particular, Algorithm 3.1.1 applies not only to standard strata where each $d_i > 0$, but it also allows for the construction of lattice surfaces (X, ω) with a specified number of marked points which remain invariant under $\mathrm{Aff}^+(X, \omega)$.) Abusing notation, we denote an element of a stratum by any of its translation-equivalent representatives. By the Riemann–Roch Theorem, every translation surface $(X, \omega) \in \mathcal{H}(d_1, \ldots, d_\kappa)$ has the same genus $g > 0$, and the sum of

[2]The translation structures guarantee that the linear part of a locally affine map is constant on components of (X_1, ω_1); this latter statement in the definition is thus intended for disconnected translation surfaces.

the d_i equals $2g-2$.

There is a natural action of $\mathrm{GL}_2\mathbb{R}$ on each stratum, given by post-composing the coordinate charts of $(X,\omega) \in \mathcal{H}(d_1,\ldots,d_\kappa)$ with the usual action of a matrix in the plane via a linear transformation (one can verify that this action is well-defined with respect to translation-equivalence). Any translation surface may be normalised by the action of a diagonal matrix so as to have unit-area; we denote the collection of unit-area translation surfaces in $\mathcal{H}(d_1,\ldots,d_\kappa)$ by $\mathcal{H}_1(d_1,\ldots,d_\kappa)$. There is a corresponding action of $\mathrm{SL}_2\mathbb{R}$ on $\mathcal{H}_1(d_1,\ldots,d_\kappa)$, and the stabilizer of (X,ω) under this $\mathrm{SL}_2\mathbb{R}$-action is isomorphic to the Veech group $\mathrm{SL}(X,\omega)$ as defined above. We call (X,ω) a *lattice surface* if $\mathrm{SL}(X,\omega)$ is a lattice group, i.e., $\mathrm{SL}(X,\omega)$ has finite covolume in $\mathrm{SL}_2\mathbb{R}$.

Remark 3.2.1. Recall that a Veech group is always non-cocompact, and thus a lattice Veech group is non-uniform (i.e., it is a non-cocompact discrete group of finite covolume). Other necessary properties of lattice Veech groups are known; for example, the trace field of $\mathrm{SL}(X,\omega)$ is a totally real number field of degree at most g, where g is the genus of (X,ω) ([81, 84, 107]). Throughout, when speaking of a lattice $\Gamma \leq \mathrm{SL}_2\mathbb{R}$ the reader may wish to impose such known restrictions, though they are not strictly necessary for our statements. By a lattice we simply mean a discrete subgroup of $\mathrm{SL}_2\mathbb{R}$ of finite covolume.

3.3 A canonical surface for each stratum

3.3.1 Canonical surface

Here, we recall the canonical (infinite area and disconnected in general) flat surface associated to each stratum $\mathcal{H}(d_1,\ldots,d_\kappa)$ studied in [63, 64] and explore some immediate results. For each $1 \leq i \leq \kappa$,

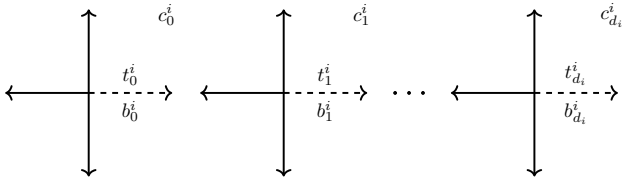


Figure 3.1: The surface $\mathcal{O}_i$, where the bottom of the slit, b_j^i, is identified with the top of the slit, t_{j+1}^i, for each $j \in \mathbb{Z}_{d_i+1}$.

let $\mathcal{O}_i \coloneqq (\mathbb{C}, z^{d_i}dz)$, and set

$$\mathcal{O} = \mathcal{O}(d_1, \ldots, d_\kappa) \coloneqq \bigsqcup_{i=1}^{\kappa} \mathcal{O}_i.$$

Each component $\mathcal{O}_i \subset \mathcal{O}$ is an infinite translation surface with a sole singularity of cone angle $2\pi(d_i + 1)$ at the origin, which we denote by $0 \in \mathcal{O}_i$ (the context will be clear as to which component an origin 0 belongs). Intuitively, we may think of $\mathcal{O}_i$ as $d_i + 1$ copies of the plane, denoted $c_0^i, \ldots, c_{d_i}^i$, each of which is slit along the non-negative real axis and glued so that the bottom edge, b_j^i, of the slit of c_j^i is identified with the top edge, t_{j+1}^i, of the slit of c_{j+1}^i for each $j \in \mathbb{Z}_{d_i+1}$; see Figure 3.1. After making these identifications, we (arbitrarily) consider $t_{j+1}^i \sim b_j^i$ as part of c_{j+1}^i and not of c_j^i. In particular, $c_j^i \cap c_k^i = \{0\}$ for distinct $j, k \in \mathbb{Z}_{d_i+1}$.

Define $\boldsymbol{\pi}_i : \mathcal{O}_i \to \mathbb{C}$ by $\boldsymbol{\pi}_i(p) \coloneqq \int_\gamma z^{d_i}dz$, where γ is any piecewise-smooth curve from 0 to p (note that the integral is independent of path due to Cauchy's integral theorem), and extend to a map $\boldsymbol{\pi} : \mathcal{O} \to \mathbb{C}$ by setting $\boldsymbol{\pi}|_{\mathcal{O}_i} \coloneqq \boldsymbol{\pi}_i$ for each $i \in \{1, \ldots, \kappa\}$. Observe that $\boldsymbol{\pi}_i(\cdot)$ is a degree $d_i + 1$ branched covering with a single branch point $0 \in \mathbb{C}$, and the extended map $\boldsymbol{\pi}(\cdot)$ is a degree $\sum_{i=1}^{\kappa}(d_i + 1) = 2g - 2 + \kappa$ branched covering with sole branch point $0 \in \mathbb{C}$. Throughout, we denote by z_x and z_y the real and imaginary parts, respectively of $z \in \mathbb{C}$; hence $\boldsymbol{\pi}(p) = \boldsymbol{\pi}(p)_x + i\boldsymbol{\pi}(p)_y$. Note that

on sufficiently small neighborhoods in $\mathcal{O}$ (and away from singularities), the map $\boldsymbol{\pi}(\cdot)$ restricts to give a coordinate chart for the translation atlas associated to $\mathcal{O}$. Moreover, each $\mathcal{O}_i$ is naturally equipped with generalised polar coordinates: every point $p \in c_j^i \subset \mathcal{O}_i$ is determined by a magnitude $|p| := d(0,p) = |\boldsymbol{\pi}(p)|$ and, for $p \neq 0$, a principal argument value $\arg(p) := \arg(\boldsymbol{\pi}(p)) + 2\pi j \in [2\pi j, 2\pi(j+1))$, where $|z|$ and $\arg(z) \in [0, 2\pi)$ are the Euclidean norm and principal value of the argument function, respectively, of $z \in \mathbb{C}$.

For each $i \in \{1, \ldots, \kappa\}$, define an affine automorphism $\rho_i : \mathcal{O}_i \to \mathcal{O}_i$ which acts as a counterclockwise rotation of angle 2π about the origin; note that ρ_i belongs to $\mathrm{Trans}(\mathcal{O}_i)$ since, in local coordinates given by $\boldsymbol{\pi}(\cdot)$, ρ_i has trivial linear part. We extend ρ_i to $\mathcal{O}$ by acting as the identity on all other components and observe that $\rho_i \in \mathrm{Trans}(\mathcal{O})$. For each integer $d \geq 0$, let $n(d) = \#\{1 \leq i \leq \kappa \mid d_i = d\}$. Having fixed d, let $i_1, \ldots, i_{n(d)}$ be the distinct indices of the $d_1, \ldots, d_\kappa$ which equal d, and let $S_{n(d)}$ denote the symmetric group on the set $\{i_1, \ldots, i_{n(d)}\}$. Extend $S_{n(d)}$ to act as the identity on all other indices, and for each $\alpha \in S_{n(d)}$, let $f_\alpha : \mathcal{O} \to \mathcal{O}$ permute the indices of $\mathcal{O}_1, \ldots, \mathcal{O}_\kappa$ by α, respecting polar coordinates. We again have that $f_\alpha \in \mathrm{Trans}(\mathcal{O})$ for each $\alpha \in S_{n(d)}$ with $n(d) \geq 1$.

Lemma 3.3.1. *There is a one-to-one correspondence between the group Trans($\mathcal{O}$) and the Cartesian product*

$$\left(\prod_{\substack{d \geq 0, \\ n(d) \geq 1}} S_{n(d)} \right) \times \left(\prod_{i=1}^{\kappa} C_{d_i+1} \right),$$

where C_{d_i+1} is the cyclic group of order $d_i + 1$.

Sketch. Lemma 2 of [64] shows that $\mathrm{Trans}(\mathcal{O})$ is generated by an action of

$$\prod_{\substack{d \geq 0, \\ n(d) \geq 1}} S_{n(d)}$$

together with an action of $\prod_{i=1}^{\kappa} C_{d_i+1}$ on $\mathcal{O}$, as naturally determined from the definitions of f_α and ρ_i above. That is, any element $\tau \in \mathrm{Trans}(\mathcal{O})$ may be written as a finite composition of various f_α's and ρ_i's. We may rearrange the order of this composition (possibly altering the indices of some ρ_i's) to collect all f_α's with α belonging to the same $S_{n(d)}$ and all ρ_i's corresponding to the same $\mathcal{O}_i$. That τ may be uniquely written in such a way is verified by considering the indices of images of 2π-sectors c_j^i. On the other hand, taking compositions of any collection of the maps f_α and ρ_i discussed above, we see that each element of the Cartesian product corresponds to an element of $\mathrm{Trans}(\mathcal{O})$. □

Lemma 3.3.2. *Let $f \in Aff(\mathcal{O})$ with $der(f) = A$. Then*

$$\boldsymbol{\pi} \circ f = A \cdot \boldsymbol{\pi},$$

where the notation on the right denotes the usual action of a matrix A on the plane as a linear transformation.

Proof. Since $\boldsymbol{\pi}(\cdot)$ restricts on sufficiently small neighborhoods to give a (bijective) coordinate map for the translation atlas of $\mathcal{O}$, we have—by definition of f—that for each $v \in \mathbb{R}^2$ in the image of such a neighborhood,

$$\boldsymbol{\pi} \circ f \circ \boldsymbol{\pi}^{-1}(v) = Av + b$$

for some $b \in \mathbb{R}^2$. This composition agrees on the images of the intersections of such neighborhoods; in particular, b depends only on the component $\mathcal{O}_i$ to which the neighborhood belongs. Choose some neighborhood as above which contains $0 \in \mathcal{O}_i$ on its boundary. The above composition (and each of the maps comprising it) extends continuously to the closure of its domain. Since f sends singularities to singularities, setting $v = 0$ gives $b = 0$. As $\mathcal{O}_i$ was arbitrary, the result follows from the previous observations. □

In Lemma 3 of [64], it is shown that $\mathrm{der}(\mathrm{Aff}(\mathcal{O})) = \mathrm{GL}_2\mathbb{R}$. To each $A \in \mathrm{GL}_2^+\mathbb{R}$ we associate a canonical $f_A \in \mathrm{Aff}^+(\mathcal{O})$ with $\mathrm{der}(f_A) = A$ as follows. Fix some $g_A \in \mathrm{Aff}^+(\mathcal{O})$ with $\mathrm{der}(g_A) = A$,

and choose $\tau \in \mathrm{Trans}(\mathcal{O})$ so that the composition $f_A \coloneqq \tau \circ g_A$ satisfies for each $1 \leq i \leq \kappa$ both (i) $f_A(\mathcal{O}_i) = \mathcal{O}_i$ and (ii) for each point $p \in \mathcal{O}_i$, the angle measured counterclockwise from p to $f_A(p)$ is non-negative and less than 2π. Existence of τ follows from Lemma 3.3.1. Notice that $\mathrm{der}(f_A) = A$ since $\mathrm{der}(\cdot)$ is a group homomorphism, $\tau \in \mathrm{Trans}(\mathcal{O}) = \ker(\mathrm{der})$ and $\mathrm{der}(g_A) = A$. Furthermore, conditions (i) and (ii) guarantee that f_A is unique, regardless of the initial choice of g_A (and subsequent choice of τ).

We also note that any $g_A \in \mathrm{Aff}^+(\mathcal{O})$ with $\mathrm{der}(g_A) = A \in \mathrm{GL}_2^+\mathbb{R}$ may be written uniquely as $g_A = \tau' \circ f_A$ for some $\tau' \in \mathrm{Trans}(O)$ (namely $\tau' = \tau^{-1}$ for τ as above), with f_A the canonical affine automorphism associated to A.

Definition 3.3.3. With notation as above, let

$$\mathrm{Aff}_{\mathrm{C}}^+(\mathcal{O}) \coloneqq \{f_A \in \mathrm{Aff}^+(\mathcal{O}) \mid A \in \mathrm{GL}_2^+\mathbb{R}\}$$

be the collection of canonical affine automorphisms of $\mathcal{O}$. For any $r \in \mathbb{R}_+$ and $p \in \mathcal{O}$, let

$$rp \coloneqq f_{D(r)}(p),$$

where $f_{D(r)} \in \mathrm{Aff}_{\mathrm{C}}^+(\mathcal{O})$ with $D(r) = \left(\begin{smallmatrix} r & 0 \\ 0 & r \end{smallmatrix}\right)$.

In particular, if $p \in \mathcal{O}_i$ with polar coordinates $(|p|, \theta)$, then $rp \in \mathcal{O}_i$ with polar coordinates $(r|p|, \theta)$. Note then that $\boldsymbol{\pi}(rp) = r\boldsymbol{\pi}(p)$.

While $\mathrm{Trans}(\mathcal{O})$ is generally non-abelian, we have the following commutativity result:

Lemma 3.3.4. *The subgroup $Trans(\mathcal{O}) \leq Aff(\mathcal{O})$ belongs to the centraliser of $Aff_C^+(\mathcal{O})$, i.e.,*

$$\tau \circ f_A = f_A \circ \tau$$

for each $\tau \in Trans(\mathcal{O})$ and $f_A \in Aff_C^+(\mathcal{O})$.

Proof. Let $\tau \in \mathrm{Trans}(\mathcal{O})$, $f_A \in \mathrm{Aff}_{\mathrm{C}}^+(\mathcal{O})$ and $p \in \mathcal{O}$. We must show

$$\tau \circ f_A(p) = f_A \circ \tau(p).$$

Since $\mathrm{der}(\tau \circ f_A) = \mathrm{der}(f_A \circ \tau) = A$, both sides of the previous line are sent under $\boldsymbol{\pi}(\cdot)$ to the same point $z \in \mathbb{C}$ (Lemma 3.3.2). Suppose $p \in c_k^i$ and $\tau(\mathcal{O}_i) = \mathcal{O}_j$ with $\tau(c_k^i) = c_\ell^j$. Let r_k^i denote the set of points in $\mathcal{O}_i$ with argument $2\pi k$ (i.e., r_k^i is the ray along the positive real axis in c_k^i). Note that $\boldsymbol{\pi}(\cdot)$ sends each of $\tau \circ f_A(c_k^i)$ and $f_A \circ \tau(c_k^i)$ bijectively onto $\mathbb{C}$, and these sets are completely determined by the images $\tau \circ f_A(r_k^i)$ and $f_A \circ \tau(r_k^i)$ in $\mathcal{O}_j$: the former are the sets of points within angle 2π counterclockwise of the respective images of the ray r_k^i. Since there is only one point in each 2π-sector which maps to $z \in \mathbb{C}$ under $\boldsymbol{\pi}(\cdot)$, it suffices to show that $\tau \circ f_A(r_k^i) = f_A \circ \tau(r_k^i)$. By definition of f_A and r_k^i, the image $f_A(r_k^i)$ belongs to c_k^i, and thus $\tau \circ f_A(r_k^i)$ belongs to c_ℓ^j. But also $\tau(r_k^i)$ belongs to c_ℓ^j by assumption, and again by definition of f_A, the image $f_A \circ \tau(r_k^i)$ belongs to c_ℓ^j as well. Since the images $\tau \circ f_A(r_k^i)$ and $f_A \circ \tau(r_k^i)$ are rays emanating from 0 in c_ℓ^j and pointing in the same direction, we have $\tau \circ f_A(r_k^i) = f_A \circ \tau(r_k^i)$ and the result follows. □

3.3.2 Permissible triples

Here, we introduce terminology and present results regarding particular subsets of the canonical surface $\mathcal{O} = \mathcal{O}(d_1, \ldots, d_\kappa)$. While the material of this subsection becomes technical, the underlying notions are rooted in elementary Euclidean geometry. As we shall see in §3.4.4 and §3.4.5, the results proven here will be crucial for our proof of Theorem 3.1.2. We begin with definitions and notation.

Definition 3.3.5. The *half-space* determined by $p \in \mathcal{O}_i$ is

$$H(p) \coloneqq \{q \in \mathcal{O}_i \mid d(0, q) \leq d(p, q)\}.$$

Note that $H(p)$ is convex in the sense that the length-minimising path between any two points in $H(p)$ is also contained in $H(p)$. Unless otherwise noted, a (closed or open) *θ-sector* of $\mathcal{O}_i$ is a (closed or open) sector of infinite radius, centered at $0 \in \mathcal{O}_i$, and of angle θ. For the following definition and subsequent discussion, see Figure 3.2.

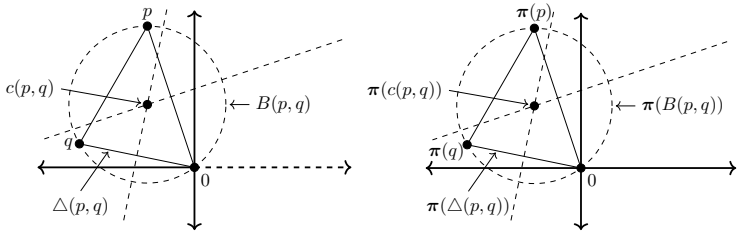


Figure 3.2: Left: The triangle $\triangle(p,q)$, circumcenter $c(p,q)$, and ball $B(p,q)$ determined by p and q in $\mathcal{O}_i$. The circumcircle $C(p,q)$ is the boundary of $B(p,q)$, and the straight dashed lines through $c(p,q)$ are the boundaries of the half-spaces $H(p)$ and $H(q)$. Right: The image in $\mathbb{C}$ under $\boldsymbol{\pi}(\cdot)$.

Definition 3.3.6. Let $p, q \in \mathcal{O}_i \backslash \{0\}$ be two points with distinct arguments in the same open π-sector of $\mathcal{O}_i$. The *triangle* determined by p and q, denoted $\triangle(p,q)$, is the union of straight line segments from $0 \in \mathcal{O}_i$ to p, from p to q, and from q to 0. Let

$$c(p,q) \coloneqq \partial H(p) \cap \partial H(q)$$

denote the *circumcenter* determined by p and q,

$$C(p,q) \coloneqq \{z \in \mathcal{O}_i \mid d(z, c(p,q)) = |c(p,q)|\}$$

denote the *circumcircle* determined by p and q, and

$$B(p,q) \coloneqq \{z \in \mathcal{O}_i \mid d(z, c(p,q)) < |c(p,q)|\}$$

denote the *ball* determined by p and q.

Locally, and away from a non-removable singularity, the geometry on $\mathcal{O}_i$ is Euclidean, so the circumcenter, circumcircle and ball of Definition 3.3.6 may be viewed as isometric copies of their Euclidean namesakes in the plane. To be more precise, Lemma 4 of [64] implies

that for any two points $p_1, p_2 \in \mathcal{O}_i$ belonging to the same closed π-sector, the distance $d(p_1, p_2)$ between p_1 and p_2 in $\mathcal{O}_i$ equals the distance between their respective images $\boldsymbol{\pi}(p_1)$ and $\boldsymbol{\pi}(p_2)$ in the plane. Hence on any such sector, $\boldsymbol{\pi}(\cdot)$ is an isometry. Using Euclidean geometry, we find that $\boldsymbol{\pi}(c(p,q))$ is the center of the Euclidean circle $\boldsymbol{\pi}(C(p,q))$ of radius $|c(p,q)|$, and $\boldsymbol{\pi}(B(p,q))$ is the open ball whose boundary is $\boldsymbol{\pi}(C(p,q))$. Moreover, $\boldsymbol{\pi}(C(p,q))$ is the circumcircle of the Euclidean triangle $\boldsymbol{\pi}(\triangle(p,q))$ with vertices $0, \boldsymbol{\pi}(p), \boldsymbol{\pi}(q)$, and $0, \boldsymbol{\pi}(p), \boldsymbol{\pi}(q) \in \boldsymbol{\pi}(C(p,q))$ implies $0, p, q \in C(p,q) = \partial B(p,q)$.

Remark 3.3.7. We use the same notation and terminology from Definition 3.3.6 for the analogous objects determined by two points in the Euclidean plane $\mathbb{C}$.

Definition 3.3.8. Define the set of *oppositely projected pairs* of points in $\mathcal{O}$ by

$$\mathbb{P}(\mathcal{O}) \coloneqq \{\{p, p^-\} \subset \mathcal{O} \mid \boldsymbol{\pi}(p^-) = -\boldsymbol{\pi}(p) \neq 0\},$$

and for any subset $P \subset \mathbb{P}(\mathcal{O})$ of oppositely projected pairs, define the *forgotten* version of P by

$$P_F \coloneqq \bigcup_{\{p,p^-\} \in P} \{p, p^-\};$$

that is, the pairing inherent to P is 'forgotten' in the set $P_F \subset \mathcal{O}$. For $r \in \mathbb{R}_+$ and $P \subset \mathbb{P}(\mathcal{O})$, let $rP \coloneqq \{\{rp, rp^-\} \mid \{p, p^-\} \in P\}$ and $rP_F \coloneqq (rP)_F$. Call a subset $P \subset \mathbb{P}(\mathcal{O})$ *limit-point free* if its forgotten version P_F has no limit points in $\mathcal{O}$.

Recall that $\boldsymbol{\pi}(\cdot)$ is a degree $2g - 2 + \kappa$ branched covering with a single branch point $0 \in \mathbb{C}$. Thus for any $p \in \mathcal{O}$ with $\boldsymbol{\pi}(p) \neq 0$, there are precisely $2g - 2 + \kappa$ points $p^- \in \mathcal{O}$ in the fiber above $-\boldsymbol{\pi}(p)$. That is,

$$\left|\{p^- \mid \{p, p^-\} \in \mathbb{P}(\mathcal{O})\}\right| = 2g - 2 + \kappa.$$

The subsets $P \subset \mathbb{P}(\mathcal{O})$ of interest to us do not have this multiplicity of elements:

Definition 3.3.9. Call a nonempty subset $P \subset \mathbb{P}(\mathcal{O})$ of oppositely projected pairs *distinctive* if for any $p \in P_F$,

$$\left|\{p^- \mid \{p, p^-\} \in P\}\right| = 1.$$

Now let $(p, q) \in \mathcal{O}^2$ be an ordered pair of two regular points with different arguments which belong to the same open π-sector of the same component of $\mathcal{O}$ (see the left-hand side of Figure 3.3), and consider the difference $\boldsymbol{\pi}(q) - \boldsymbol{\pi}(p)$ of their images under $\boldsymbol{\pi}(\cdot)$ in $\mathbb{C}$. As above, there are exactly $2g - 2 + \kappa$ points $u \in \mathcal{O}$ satisfying $\boldsymbol{\pi}(u) = \boldsymbol{\pi}(q) - \boldsymbol{\pi}(p)$. Each of the $2g-2+\kappa$ oppositely projected pairs $\{p, p^-\} \in \mathbb{P}(\mathcal{O})$ naturally announces a unique such point u, namely the u lying in the same open π-sector of the same component of $\mathcal{O}$ as p^-. A distinctive set P determines a unique point p^- and hence a unique point u. We fix special notation for this point u in the following:

Definition 3.3.10. Let $P, Q \subset \mathbb{P}(\mathcal{O})$ be distinctive subsets of oppositely projected pairs and $(\{p, p^-\}, \{q, q^-\}) \in P \times Q$. If p and q belong to the same open π-sector of the same component of $\mathcal{O}$ and satisfy $\arg(p) \neq \arg(q)$, then we denote by $[p, q]$ the unique element of $\mathcal{O}$ for which

(i) $\boldsymbol{\pi}([p, q]) = \boldsymbol{\pi}(q) - \boldsymbol{\pi}(p)$, and

(ii) $[p, q]$ belongs to the same open π-sector of the same component of $\mathcal{O}$ as the point p^-.

See Figure 3.3.

Remark 3.3.11. The reader should note that the point $[p, q]$ depends on the distinctive set P, as distinctiveness uniquely determines the point p^- in Definition 3.3.10. However, as P shall be clear by context, we omit it from the notation $[p, q]$.

Notice from Definition 3.3.10 that $[p, q]$ is defined if and only if $[q, p]$ is defined. The following technical result is needed for Proposition 3.3.15 below and is due to the fact that for distinctive sets,

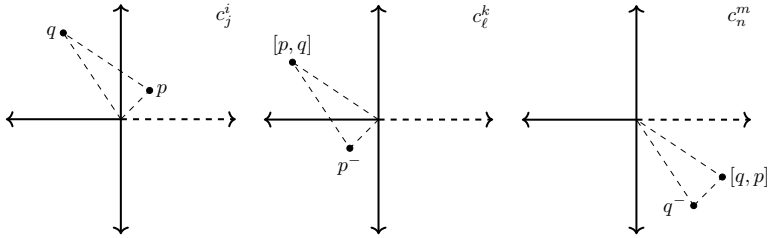


Figure 3.3: The points $[p,q]$ and $[q,p]$, as determined by $\{p,p^-\}$ and $\{q,q^-\}$.

the points p and q naturally determine three isometric triangles in $\mathcal{O}$. See again Figure 3.3.

Lemma 3.3.12. *Let $P,Q,U \subset \mathbb{P}(\mathcal{O})$ be distinctive subsets of oppositely projected pairs. For any $(\{p,p^-\},\{q,q^-\}) \in P \times Q$ for which $[p,q] \in \mathcal{O}$ is defined, we have that $\{[p,q],[q,p]\} \in \mathbb{P}(\mathcal{O})$ is an oppositely projected pair. Moreover, if $\{[p,q],[q,p]\} \in U$ then $[p^-,[p,q]]$ and $[[p,q],p^-]$ are defined and equal q and q^-, respectively.*

Proof. Definition 3.3.10 gives that

$$\boldsymbol{\pi}([p,q]) = \boldsymbol{\pi}(q) - \boldsymbol{\pi}(p) = -\boldsymbol{\pi}([q,p]).$$

Since p and q belong to the same π-sector with $\arg(p) \neq \arg(q)$, we have $\boldsymbol{\pi}(q) - \boldsymbol{\pi}(p) \neq 0$, so $\{[p,q],[q,p]\} \in \mathbb{P}(\mathcal{O})$.

Condition (ii) of Definition 3.3.10 guarantees that p^- and $[p,q]$ belong to the same open π-sector of the same component. Since p and q belong to the same open π-sector with different arguments, condition (i) guarantees that $\arg(p^-) \neq \arg([p,q])$, so $[p^-,[p,q]]$ and $[[p,q],p^-]$ are defined. Next we show $[p^-,[p,q]] = q$. Condition (ii) of Definition 3.3.10 (applied to $[p^-,[p,q]]$) gives that p and $[p^-,[p,q]]$ belong to the same open π-sector of the same component of $\mathcal{O}$; by assumption, the same is true of p and q, so we find that q and $[p^-,[p,q]]$ belong to the same open 2π-sector. As $\boldsymbol{\pi}(\cdot)$ is injective on

open 2π-sectors, it suffices to show that these latter two points are sent under this map to the same point in the plane. We compute

$$\boldsymbol{\pi}([p^-,[p,q]]) = \boldsymbol{\pi}([p,q]) - \boldsymbol{\pi}(p^-) = \boldsymbol{\pi}(q) - \boldsymbol{\pi}(p) + \boldsymbol{\pi}(p) = \boldsymbol{\pi}(q)$$

as desired. The proof that $[[p,q],p^-]$ equals q^- is similar. □

The following definition and subsequent results are needed in §3.4.4 and §3.4.5 below.

Definition 3.3.13. Let P, Q and U be distinctive, limit-point free subsets of $\mathbb{P}(\mathcal{O})$ and

$$(\{p,p^-\},\{q,q^-\},\{u,u^-\}) \in P \times Q \times U$$

a triple of oppositely projected pairs. We call $(p,q,u) \in P_F \times Q_F \times U_F$ a *permissible triple* if there exist *permissible scalars* $(r,s,t) \in \mathbb{R}^3_+$ such that, for $(\{rp,rp^-\},\{sq,sq^-\},\{tu,tu^-\}) \in rP \times sQ \times tU$,

(i) $[rp,sq]$ and $[sq,rp]$ are defined and equal tu and tu^-, respectively, and

(ii) $\big(B(rp,sq) \cup B(tu,rp^-) \cup B(sq^-,tu^-)\big) \cap (rP_F \cup sQ_F \cup tU_F) = \varnothing$;

see Figure 3.4. For a collection $\mathbf{P} = \{P_i\}_{i\in I}$ of distinctive, limit-point free subsets of $\mathbb{P}(\mathcal{O})$, we let $\mathcal{P}(\mathbf{P})$ denote the *set of all permissible triples* arising from $\mathbf{P}$; that is, $\mathcal{P}(\mathbf{P})$ is the set

$$\left\{ (p_i,p_j,p_k) \in (P_i)_F \times (P_j)_F \times (P_k)_F \;\middle|\; \begin{array}{l} i,j,k \in I \text{ and} \\ (p_i,p_j,p_k) \text{ is a} \\ \text{permissible triple} \end{array} \right\}.$$

One immediate, but important, observation regarding permissible triples and permissible scalars is the following:

Proposition 3.3.14. *Let (p,q,u) be a permissible triple with permissible scalars $(r,s,t) \in \mathbb{R}^3_+$. The scalars s and t are uniquely determined by $p,q,u \in \mathcal{O}$ and $r \in \mathbb{R}_+$. In particular, the set of all permissible scalars for (p,q,u) is precisely the ray $\{(ar,as,at) \mid a \in \mathbb{R}_+\}$.*

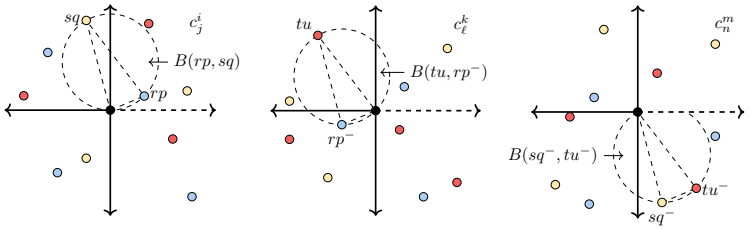


Figure 3.4: A permissible triple (p, q, u) with permissible scalars (r, s, t). Points of rP_F, sQ_F and tU_F are in blue, tan, and red, respectively. Condition (i) of Definition 3.3.13 is met, as $[rp, sq] = tu$ and $[sq, rp] = tu^-$. Condition (ii) is met since none of the open balls $B(rp, sq)$, $B(tu, rp^-)$ or $B(sq^-, tu^-)$ contain any points of rP_F, sQ_F or tU_F.

Proof. By Definitions 3.3.10 and 3.3.13, we have

$$t\boldsymbol{\pi}(u) = \boldsymbol{\pi}([rp, sq]) = s\boldsymbol{\pi}(q) - r\boldsymbol{\pi}(p). \tag{3.1}$$

Rearranging, we find that s and t are solutions to

$$\begin{pmatrix} \boldsymbol{\pi}(q)_x & -\boldsymbol{\pi}(u)_x \\ \boldsymbol{\pi}(q)_y & -\boldsymbol{\pi}(u)_y \end{pmatrix} \begin{pmatrix} s \\ t \end{pmatrix} = \begin{pmatrix} r\boldsymbol{\pi}(p)_x \\ r\boldsymbol{\pi}(p)_y \end{pmatrix}.$$

The fact that rp and sq belong to the same open π-sector with different arguments implies that $\boldsymbol{\pi}(p)$ and $\boldsymbol{\pi}(q)$ are $\mathbb{R}$-linearly independent. This, together with Equation (3.1), implies that also $\boldsymbol{\pi}(u)$ and $\boldsymbol{\pi}(q)$ are $\mathbb{R}$-linearly independent. Hence the matrix on the left side of the previous equation is invertible, so s and t are uniquely determined.

For the second statement, let $P_1, P_2 \subset \mathbb{P}(O)$ be distinctive and $a \in \mathbb{R}_+$. From Definition 3.3.10, one finds for $(\{p_1, p_1^-\}, \{p_2, p_2^-\}) \in P_1 \times P_2$ and $(\{ap_1, ap_1^-\}, \{ap_2, ap_2^-\}) \in aP_1 \times aP_2$ that if $[p_1, p_2]$ is defined then $[ap_1, ap_2]$ is defined and equals $a[p_1, p_2]$. It follows then from Definition 3.3.13 that if (r, s, t) are permissible scalars for

(p, q, u), then so are (ar, as, at) for any $a \in \mathbb{R}_+$. On the other hand, if (r_1, s_1, t_1) and (r_2, s_2, t_2) are both permissible scalars for (p, q, u), then so are $(1, s_1/r_1, t_1/r_1)$ and $(1, s_2/r_2, t_2/r_2)$. The first statement of this proposition implies that (r_1, s_1, t_1) and (r_2, s_2, t_2) belong to a common ray. □

Note that the order of entries of a permissible triple (p, q, u) is important in Definition 3.3.13; for instance, the first two entries p and q must necessarily belong to the same open π-sector of the same component of $\mathcal{O}$ for condition (i) to hold. Nevertheless, this ordering does admit some flexibility.

Proposition 3.3.15. *With notation as in Definition 3.3.13, the following statements are equivalent:*

(a) (p, q, u) *is a permissible triple with permissible scalars* (r, s, t),

(b) (q^-, u^-, p) *is a permissible triple with permissible scalars* (s, t, r), *and*

(c) (u, p^-, q^-) *is a permissible triple with permissible scalars* (t, r, s).

Proof. We need only show (a) implies (b): the same argument will give (b) implies (c) and (c) implies (a). Let (p, q, u) be a permissible triple with permissible scalars (r, s, t). We claim that (q^-, u^-, p) is a permissible triple with permissible scalars (s, t, r).

(i) We must show that $[sq^-, tu^-]$ and $[tu^-, sq^-]$ are defined and equal rp and rp^-, respectively. We have by assumption that $[sq, rp]$ is defined and equals tu^-. By Lemma 3.3.12, both $[sq^-, [sq, rp]] = [sq^-, tu^-]$ and $[[sq, rp], sq^-] = [tu^-, sq^-]$ are defined. By the same lemma, these equal rp and rp^-, respectively.

(ii) We have that

$$\big(B(sq^-, tu^-) \cup B(rp, sq) \cup B(tu, rp^-)\big) \cap (sQ_F \cup tU_F \cup rP_F)$$

is empty by assumption. □

One might suspect from Definition 3.3.13 that it is difficult for some $(p, q, u) \in P_F \times Q_F \times U_F$ to be a permissible triple. Indeed, this suspicion is confirmed by the following:

Lemma 3.3.16. *Let $\mathbf{P} = \{P_i\}_{i=1}^n$ be a finite collection of distinctive, limit-point free subsets of $\mathbb{P}(\mathcal{O})$. Then the set $\mathcal{P}(\mathbf{P})$ of all permissible triples arising from $\mathbf{P}$ is finite.*

Proof. It suffices to show that for any distinctive, limit-point free $P, Q, U \subset \mathbb{P}(\mathcal{O})$, the set of permissible triples $(p, q, u) \in P_F \times Q_F \times U_F$ is finite. Suppose on the contrary that $\{(p_k, q_k, u_k)\}_{k\in\mathbb{N}} \subset P_F \times Q_F \times U_F$ is an infinite set of permissible triples with corresponding permissible scalars $\{(r_k, s_k, t_k)\}_{k\in\mathbb{N}} \subset \mathbb{R}^3_+$. At least one of the sets $\{p_k\}_{k\in\mathbb{N}}, \{q_k\}_{k\in\mathbb{N}}$ or $\{u_k\}_{k\in\mathbb{N}}$ is infinite; by Proposition 3.3.15, we may assume that either $\{p_k\}_{k\in\mathbb{R}}$ or $\{p_k^-\}_{k\in\mathbb{N}}$ is infinite. Since P is distinctive, these sets have the same cardinality and hence are both infinite. As the set of possible directions of points in each $\mathcal{O}_i$ is compact and there are only finitely many components $\mathcal{O}_i$ of $\mathcal{O}$, we may also assume that each p_k (resp. p_k^-) belongs to some small sector of a fixed component of $\mathcal{O}$ and that $\lim_k \arg(\boldsymbol{\pi}(p_k)) = \theta_P$ (resp. $\lim_k \arg(\boldsymbol{\pi}(p_k^-)) = \theta_P^-$) exists. Rotating each of P, Q and U by $-\theta_P$, assume for simplicity that $\theta_P = 0$ (and hence $\theta_P^- = \pi$).

Rescaling each (r_k, s_k, t_k) as necessary (see Proposition 3.3.14), we further assume that $|r_k p_k| = 1$ for all k. Under these assumptions, we find that $\{r_k \boldsymbol{\pi}(p_k)\}_{k\in\mathbb{N}} \subset S^1$ with $r_k \boldsymbol{\pi}(p_k) \to 1 \in \mathbb{C}$. Also note that since P is limit-point free, $|p_k| \to \infty$ and hence $r_k = 1/|p_k| \to 0$.

Condition (ii) of Definition 3.3.13 (together with the fact that $\boldsymbol{\pi}(\cdot)$ is an isometry on π-sectors) guarantees that in the plane, we have for each $k \in \mathbb{N}$ both

$$B(r_k \boldsymbol{\pi}(p_k), s_k \boldsymbol{\pi}(q_k)) \cap \{r_k \boldsymbol{\pi}(p_\ell)\}_{\ell\in\mathbb{N}} = \varnothing \tag{3.2}$$

and

$$B(t_k \boldsymbol{\pi}(u_k), r_k \boldsymbol{\pi}(p_k^-)) \cap \{r_k \boldsymbol{\pi}(p_\ell^-)\}_{\ell\in\mathbb{N}} = \varnothing.$$

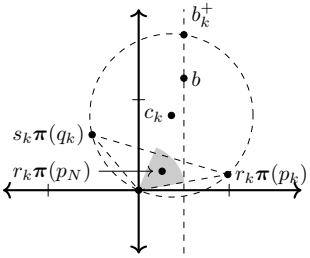


Figure 3.5: Illustration of the proof of Lemma 3.3.16. The vertical dashed line represents $x = 1/2$, and the shaded region is the subset S_k of the ball $B_k = B(r_k\boldsymbol{\pi}(p_k), s_k\boldsymbol{\pi}(q_k))$.

This latter intersection may be rewritten

$$B(t_k\boldsymbol{\pi}(u_k), -r_k\boldsymbol{\pi}(p_k)) \cap \{-r_k\boldsymbol{\pi}(p_\ell)\}_{\ell\in\mathbb{N}} = \varnothing. \tag{3.3}$$

For each k, let

$$c_k \coloneqq c(r_k\boldsymbol{\pi}(p_k), s_k\boldsymbol{\pi}(q_k)) \quad \text{and} \quad B_k \coloneqq B(r_k\boldsymbol{\pi}(p_k), s_k\boldsymbol{\pi}(q_k))$$

be the circumcenter and ball determined by $r_k\boldsymbol{\pi}(p_k)$ and $s_k\boldsymbol{\pi}(q_k)$. The circumcenter c_k is the intersection of the perpendicular bisectors of the straight line segments from the origin to $r_k\boldsymbol{\pi}(p_k)$ and from the origin to $s_k\boldsymbol{\pi}(q_k)$. Since $\arg(\boldsymbol{\pi}(p_k)) \to 0$, for large k the former perpendicular bisector does not intersect the negative real axis, so for all large k we have $\arg(c_k) \in [0,\pi)\cup(\pi, 2\pi)$. Passing to a subsequence if necessary, assume that $\arg(c_k)$ belongs to one of these intervals for all k, say $\arg(c_k) \in [0,\pi)$ (the other case is similar).

We consider two cases (see Figure 3.5):

(i) Suppose there is a subsequence for which $\boldsymbol{\pi}(p_k)_y \geq 0$ for all k. Since $\arg(\boldsymbol{\pi}(p_k)) \to 0$, we may pass to a subsequence to assume $\arg(\boldsymbol{\pi}(p_{k+1})) \leq \arg(\boldsymbol{\pi}(p_k))$ for all k. For large enough k, the x-coordinate of $r_k\boldsymbol{\pi}(p_k) \in S^1$ is greater than $1/2$. Since both 0

and $r_k\boldsymbol{\pi}(p_k)$ belong to $C_k \coloneqq \partial B_k$, we find that the vertical line $x = 1/2$ intersects C_k at two distinct points; let b_k^+ denote the point of intersection on the upper-semicircle of C_k. Note that $b_y \coloneqq \inf_k\{(b_k^+)_y\} > 0$; otherwise $\arg(c_k) \in (\pi, 2\pi)$ contrary to our assumption. Set $b \coloneqq (1/2, b_y)$ and note that for large enough k, we have $\arg(\boldsymbol{\pi}(p_k)) < \arg(b)$. Let

$$S_k = \{p \in \mathbb{C} \mid |p| < 1/2,\ \arg(p) \in (\arg(\boldsymbol{\pi}(p_k)), \arg(b))\},$$

and note that S_k is contained in the triangle with vertices 0, $r_k\boldsymbol{\pi}(p_k)$ and b_k^+, which is contained in B_k; hence $S_k \subset B_k$ for each k. Choose some $N \in \mathbb{N}$ large enough that $\arg(\boldsymbol{\pi}(p_N)) \in [0, \arg(b))$. For each $k > N$, we have $\arg(\boldsymbol{\pi}(p_k)) \leq \arg(\boldsymbol{\pi}(p_N))$ by assumption, so $\arg(\boldsymbol{\pi}(p_N)) \in [\arg(\boldsymbol{\pi}(p_k)), \arg(b))$. Taking k large enough, we also have $|r_k\boldsymbol{\pi}(p_N)| = |\boldsymbol{\pi}(p_N)|/|\boldsymbol{\pi}(p_k)| < 1/2$. Hence $r_k\boldsymbol{\pi}(p_N) \in S_k \subset B_k = B(r_k\boldsymbol{\pi}(p_k), s_k\boldsymbol{\pi}(q_k))$, contradicting Equation (3.2).

(ii) Suppose there is no subsequence for which $\boldsymbol{\pi}(p_k)_y \geq 0$ for all k. Then there is some subsequence for which $\boldsymbol{\pi}(p_k)_y < 0$—and hence $-\boldsymbol{\pi}(p_k)_y > 0$—for all k. Notice that

$$c(t_k\boldsymbol{\pi}(u_k), -r_k\boldsymbol{\pi}(p_k)) = c_k - r_k\boldsymbol{\pi}(p_k),$$

so the argument of this circumcenter belongs to $(0, \pi)$. An analogous proof (reflecting about the y-axis) to that of case (i) implies that for some fixed N and large enough k, $-r_k\boldsymbol{\pi}(p_N) \in B(t_k\boldsymbol{\pi}(u_k), -r_k\boldsymbol{\pi}(p_k))$, which contradicts Equation (3.3).

We conclude that the set of permissible triples must be finite. □

3.3.3 Marked segments and Voronoi staples

In this subsection we fix a stratum $\mathcal{H}(d_1, \ldots, d_\kappa)$, its corresponding canonical surface $\mathcal{O} = \mathcal{O}(d_1, \ldots, d_\kappa)$ and a translation surface $(X, \omega) \in \mathcal{H}(d_1, \ldots, d_\kappa)$. Label the singularities of (X, ω) as $\sigma_1, \ldots, \sigma_\kappa$

so that σ_i has cone angle $2\pi(d_i+1)$. In a sufficiently small neighborhood of σ_i define generalised polar coordinates in a fashion analogous to that on $\mathcal{O}_i$ (see §3.3.1 and the proof of Lemma 5 of [64]).

Let s be a separatrix emanating from σ_i of length $|s| \geq 0$. If $|s| > 0$ and s has corresponding angle $\theta \in [0, 2\pi(d_i+1))$, denote by $\hat{s}$ the point of $\mathcal{O}_i$ with polar coordinates $(|s|, \theta)$. If $|s| = 0$, set $\hat{s} \coloneqq 0 \in \mathcal{O}_i$. If s is in fact a saddle connection, we call $\hat{s}$ the *marked segment* determined by s; note that in this case, $\mathrm{hol}(s) = \boldsymbol{\pi}(\hat{s}) \in \mathbb{C}$. Denote by s' the identical, but oppositely-oriented saddle connection to s. Let $\mathcal{M}(X,\omega)$ be the set of all pairs $\{s, s'\}$ of oppositely-oriented saddle connections of (X,ω) and

$$\mathcal{M}_F(X,\omega) \coloneqq \bigcup_{\{s,s'\} \in \mathcal{M}(X,\omega)} \{s, s'\}$$

the set of all oriented saddle connections on (X,ω). Let $\widehat{\mathcal{M}}(X,\omega)$ denote the set of all *orientation-paired marked segments* $\{\hat{s}, \hat{s}'\}$ determined by oppositely oriented saddle connections $\{s, s'\} \in \mathcal{M}(X,\omega)$.

Proposition 3.3.17. *The set $\widehat{\mathcal{M}}(X,\omega)$ is a limit-point free subset of the set $\mathbb{P}(\mathcal{O})$ of oppositely projected pairs. Moreover, if $\hat{s}$ and $\hat{t}$ are marked segments with identical arguments belonging to the same component of $\mathcal{O}$, then in fact $\{\hat{s}, \hat{s}'\} = \{\hat{t}, \hat{t}'\}$. In particular, $\widehat{\mathcal{M}}(X,\omega)$ is distinctive.*

Proof. Suppose $\{\hat{s}, \hat{s}'\} \in \widehat{\mathcal{M}}(X,\omega)$. Since the saddle connections s and s' are identical but oppositely-oriented, it is clear that $\boldsymbol{\pi}(\hat{s}') = -\boldsymbol{\pi}(\hat{s})$, so $\{\hat{s}, \hat{s}'\} \in \mathbb{P}(\mathcal{O})$ and $\widehat{\mathcal{M}}(X,\omega) \subset \mathbb{P}(\mathcal{O})$. The image of $(\widehat{\mathcal{M}}(X,\omega))_F$ under $\boldsymbol{\pi}(\cdot)$ equals the image of $\mathcal{M}_F(X,\omega)$ under $\mathrm{hol}(\cdot)$. Since the set of holonomy vectors has no limit points (see §3.2.1) and $\boldsymbol{\pi}(\cdot)$ is a homeomorphism on sufficiently small neighborhoods of regular points, we have that $(\widehat{\mathcal{M}}(X,\omega))_F$ has no limit points and hence $\widehat{\mathcal{M}}(X,\omega)$ is limit-point free.

Now suppose $\hat{s}$ and $\hat{t}$ are marked segments with identical arguments in the same component $\mathcal{O}_i \subset \mathcal{O}$. Then the underlying saddle

connections s and t both emanate in the same direction from $\sigma_i \in \Sigma$. If their lengths differ, say $|\hat{s}| < |\hat{t}|$, then $|s| < |t|$. This implies that t has a singularity in its interior, which is impossible. If $|\hat{s}| = |\hat{t}|$, then in fact $s = t$, and so $\{\hat{s}, \hat{s}'\} = \{\hat{t}, \hat{t}'\}$. It follows that $\widehat{\mathcal{M}}(X, \omega)$ is distinctive. □

Let $\widehat{\mathcal{M}}_F(X, \omega) \coloneqq (\widehat{\mathcal{M}}(X, \omega))_F$ denote the set of all marked segments determined by saddle connections on (X, ω). The *star domain* for $\sigma_i \in \Sigma$ is the subset of $\mathcal{O}_i$ given by

$$\operatorname{star}_i(X, \omega) \coloneqq \{\hat{s} \mid s \text{ is a separatrix on } (X, \omega) \text{ emanating from } \sigma_i\}.$$

Note that $\operatorname{star}_i(X, \omega)$ consists of the union of closed rays emanating from $0 \in \mathcal{O}_i$ which stop only when meeting a marked segment (and thus almost every such ray is infinite). The *star domain* for (X, ω) is

$$\operatorname{star}(X, \omega) = \bigsqcup_{i=1}^{\kappa} \operatorname{star}_i(X, \omega) \subset \mathcal{O}.$$

Define a map $\eta : \operatorname{star}(X, \omega) \to (X, \omega)$, where for each point $p \in \operatorname{star}_i(X, \omega)$, if s is the separatrix for which $p = \hat{s}$, then $\eta(p)$ is the endpoint of s on (X, ω). In other words, if $p \in \operatorname{star}(X, \omega)$ has polar coordinates $(|p|, \theta)$, then $\eta(p)$ is the endpoint of the separatrix of length $|p|$ emanating from $\sigma_i \in \Sigma$ with angle θ.

For each x in the Voronoi 2-cell $\mathcal{C}_i \coloneqq \mathcal{C}_{\sigma_i}$, let s_x be the unique length-minimising separatrix from σ_i to x. We claim that η is injective—and thus invertible—on the set

$$\{\hat{s}_x \mid x \in \mathcal{C}_i\} \subset \operatorname{star}_i(X, \omega).$$

Indeed, if $\eta(\hat{s}_x) = \eta(\hat{s}_y)$ for $x, y \in \mathcal{C}_i$, then $x = y$ and $s_x = s_y$ by the aforementioned uniqueness of these separatrices. Hence $\hat{s}_x = \hat{s}_y$. Let $\iota_i : \mathcal{C}_i \to \{\hat{s}_x \mid x \in \mathcal{C}_i\}$ denote the corresponding inverse, namely $x \mapsto \hat{s}_x$ for each $x \in \mathcal{C}_i$, and define $\iota : \bigsqcup_{i=1}^{\kappa} \mathcal{C}_i \to \mathcal{O}$ by setting $\iota|_{\mathcal{C}_i} \coloneqq \iota_i$ for each i; see Figure 3.6.

The translation surface (X, ω) is isometric to the quotient space of $\bigsqcup_{i=1}^{\kappa} \overline{\mathcal{C}_i}$ under the equivalence relation which identifies shared

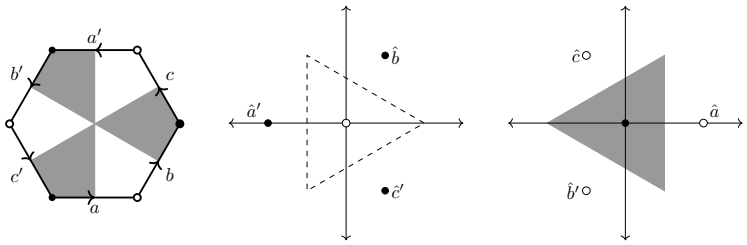


Figure 3.6: Left: Voronoi decomposition of a translation surface subordinate to two removable singularities, σ_1 (white) and σ_2 (black). The 2-cell $\mathcal{C}_1 \coloneqq \mathcal{C}_{\sigma_1}$ (resp. $\mathcal{C}_2 \coloneqq \mathcal{C}_{\sigma_2}$) is the open region in white (resp. gray). Center: The image $\iota(\mathcal{C}_1)$ in $\mathcal{O}_1$, along with three marked segments. Right: The image $\iota(\mathcal{C}_2)$ in $\mathcal{O}_2$, along with three marked segments.

edges of Voronoi 2-cells. Proposition 7 of [64] shows that in a similar fashion, (X, ω) may be recovered from the closure of the image under ι of its Voronoi 2-cells by identifying appropriate edges of the various $\overline{\iota(\mathcal{C}_i)}$. We provide a brief overview of the method by which these edge identifications are made. Recall that the half-space $H(p)$ determined by $p \in \mathcal{O}_i$ is convex, and thus so is any intersection of such half-spaces (see Definition 3.3.5).

Definition 3.3.18. For S a subset of $\mathcal{O}$ with no limit points, the *convex body* of $\mathcal{O}_i$ subordinate to S is defined by

$$\Omega_i(S) \coloneqq \bigcap_{p \in S \cap \mathcal{O}_i} H(p).$$

The set of *essential points* of $\Omega_i(S)$ is the (unique) minimal subset $\mathcal{E}_i(S) \subset S$ for which

$$\Omega_i(S) = \bigcap_{p \in \mathcal{E}_i(S)} H(p).$$

When $S = \widehat{\mathcal{M}}_F(X,\omega)$ is the set of all marked segments of (X,ω), we use the suppressed notation $\Omega_i \coloneqq \Omega_i(\widehat{\mathcal{M}}_F(X,\omega))$ and $\mathcal{E}_i \coloneqq \mathcal{E}_i(\widehat{\mathcal{M}}_F(X,\omega))$. Call distinct points $p,q \in S$ *adjacent* within S if $p,q \neq 0$ belong to the same component $\mathcal{O}_i$, and either of the two open sectors centered at 0 between p and q contains no points of S.

Proposition 14 of [64] shows that the convex body Ω_i is precisely the set $\overline{\iota(\mathcal{C}_i)}$. Furthermore, Proposition 14 and Definition 15 of [64] show that there is a subset $\widehat{\mathcal{S}}(X,\omega)$ of $\widehat{\mathcal{M}}(X,\omega)$ for which the union of essential points $\mathcal{E} \coloneqq \sqcup_{i=1}^{\kappa}\mathcal{E}_i$ equals $\widehat{\mathcal{S}}_F(X,\omega) \coloneqq (\widehat{\mathcal{S}}(X,\omega))_F$, i.e., the essential points come equipped with a natural pairing. Elements $\{\hat{s},\hat{s}'\}$ of $\widehat{\mathcal{S}}(X,\omega)$ are called *marked Voronoi staples* and their underlying pairs of saddle connections $\{s,s'\}$ in $\mathcal{M}(X,\omega)$ are called *Voronoi staples*. It follows from the definition of $\mathcal{E}_i$ that each edge on the boundary of $\Omega_i = \overline{\iota(\mathcal{C}_i)}$ belongs to the boundary of a half-space $H(\hat{s})$ for some $\hat{s} \in \mathcal{E}_i \subset \widehat{\mathcal{S}}_F(X,\omega)$; conversely, the boundary of the half-space determined by each $\hat{s} \in \widehat{\mathcal{S}}_F(X,\omega)$ contains an edge on the boundary of some $\Omega_i = \overline{\iota(\mathcal{C}_i)}$. Propositions 7 and 14 of [64] show that the edges of the convex bodies of $\sqcup_{i=1}^{\kappa}\Omega_i$ corresponding to $\hat{s}$ and its orientation-paired $\hat{s}'$ are equal length, and that (X,ω) is isometric to the quotient space of $\sqcup_{i=1}^{\kappa}\Omega_i$ under the equivalence relation given by identifying these edges via translation.

Of import for our purposes is the observation that a translation surface is uniquely determined by its marked Voronoi staples $\widehat{\mathcal{S}}(X,\omega)$, and that $\widehat{\mathcal{S}}_F(X,\omega)$ is precisely the union of essential points $\mathcal{E} = \sqcup_{i=1}^{\kappa}\mathcal{E}_i$.

Remark 3.3.19. Under this identification of edges, the vertices of the various Ω_i are regarded as regular points on the resulting translation surface (they correspond to the Voronoi 0-cells of (X,ω)). The origins $0 \in \Omega_i \subset \mathcal{O}_i$ become singularities of the resulting translation surface; we require this also if $\mathcal{O}_i = (\mathbb{C}, dz)$, in which case the singularity is a marked point.

Example 3.3.20. In Figure 3.6, the marked Voronoi staples are $\{\hat{a},\hat{a}'\}$, $\{\hat{b},\hat{b}'\}$ and $\{\hat{c},\hat{c}'\}$. The hexagonal translation surface on

the left is recovered by identifying the edges of the convex bodies corresponding to these respective pairs.

We conclude this subsection a brief observation:

Proposition 3.3.21. *Any two adjacent elements $\hat{s}$ and $\hat{t}$ of $\widehat{\mathcal{S}}_F(X,\omega)$ belong to the same open π-sector.*

Proof. If not, then $\partial H(\hat{s}) \cap \partial H(\hat{t}) = \varnothing$, and the corresponding convex body has infinite area. This would imply that the closed translation surface (X,ω) has infinite area, which is a contradiction. □

3.4 Fanning groups, simulations and finiteness of lattices in strata

3.4.1 Fanning groups and lattices

In this subsection we consider a set of directions $\Theta_\Gamma \subset S^1$ announced by a discrete subset $\Gamma \subset \mathrm{SL}_2\mathbb{R}$ and give an equivalent definition of lattice groups in terms of this set of directions (Proposition 3.4.4). Abusing notation, we write Γ for both a discrete subset of $\mathrm{SL}_2\mathbb{R}$ whose elements act as linear transformations on $\mathbb{R}^2$ and for its image as a discrete subset of $\mathrm{PSL}_2\mathbb{R}$ whose elements act as Möbius transformations on the closure of the upper-half plane $\mathbb{H}$. Similarly, we denote elements of Γ in both of these settings by the same notation $A \in \Gamma$. When there is risk of confusion, we shall explicitly mention the setting in which to consider Γ.

Definition 3.4.1. Let Γ be a discrete subset of $\mathrm{SL}_2\mathbb{R}$ and Δ the open unit disk. Set

$$\Theta_\Gamma \coloneqq S^1 \setminus \bigcup_{A\in\Gamma} A \cdot \Delta,$$

where $A \cdot \Delta$ denotes the usual action of a matrix on $\mathbb{R}^2$ as a linear transformation. A group Γ for which Θ_Γ is finite is called a *fanning*

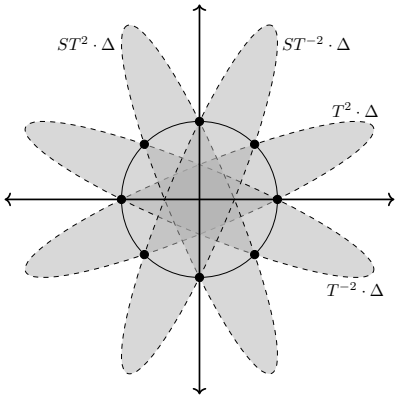


Figure 3.7: The set of eighth roots of unity contains Θ_Γ with $\Gamma = \langle S, T^2 \rangle$ and S and T defined as in Example 3.4.2.

group. We call (X, ω) a *fanning surface* if its Veech group $\mathrm{SL}(X, \omega)$ is fanning.

As we shall see in Corollary 3.4.8 below, Θ_Γ contains the directions of certain short marked segments of any translation surface with Veech group Γ. These directions will be instrumental in our construction of such surfaces.

Example 3.4.2. Let $S \coloneqq \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $T \coloneqq \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, and set $\Gamma \coloneqq \langle S, T^2 \rangle$. Figure 3.7 shows that Θ_Γ is contained in the set of eighth roots of unity and thus Γ a fanning group. One verifies that in fact Θ_Γ equals this set.

We introduce notation to help us relate lattices to fanning groups. For $A \in \mathrm{PSL}_2\mathbb{R}$, set

$$H(A) \coloneqq \{z \in \mathbb{H} \mid d_{\mathbb{H}}(i, z) \leq d_{\mathbb{H}}(Ai, z)\},$$

where $d_{\mathbb{H}}$ denotes the hyperbolic metric on $\mathbb{H}$, and for a discrete

subset $\Gamma \subset \mathrm{PSL}_2\mathbb{R}$, define

$$D(\Gamma) \coloneqq \bigcap_{A \in \Gamma} H(A)$$

(compare with Definitions 3.3.5 and 3.3.18). If Γ is a Fuchsian group which trivially stabilizes $i \in \mathbb{H}$—or, equivalently, $\Gamma \cap \mathrm{PSO}_2\mathbb{R} = \{\mathrm{Id}\}$—then $D(\Gamma)$ is the Dirichlet region centered at i for Γ, and in this case, $D(\Gamma)$ is a convex fundamental polygon for Γ; see, say, [11, 102]. For any subset $S \subset \mathbb{H}$, let $\overline{S}$ and ∂S denote the closure and boundary, respectively, of S in $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. Define

$$S(A) \coloneqq \{x/y \mid (x, y)^T \in S^1 \backslash A \cdot \Delta\}$$

and

$$S_\Gamma \coloneqq \bigcap_{A \in \Gamma} S(A) = \{x/y \mid (x, y)^T \in \Theta_\Gamma\},$$

so that $S(A)$ and S_Γ belong to $\partial\mathbb{H} = \mathbb{R} \cup \{\infty\}$. The following lemma states that the reciprocals of the slopes of directions in $\Theta_\Gamma \subset S^1$ coincide with the points of $\overline{D(\Gamma)}$ in $\partial\mathbb{H}$.

Lemma 3.4.3. *For any discrete* $\Gamma \subset \mathrm{SL}_2\mathbb{R}$,

$$S_\Gamma = \overline{D(\Gamma)} \cap \partial\mathbb{H}.$$

Proof. We first show that for any $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2\mathbb{R}$,

$$S(A) = \overline{H(A)} \cap \partial\mathbb{H}.$$

If $A \in \mathrm{SO}_2\mathbb{R}$, then both sets are all of $\partial\mathbb{H}$, so suppose $A \notin \mathrm{SO}_2\mathbb{R}$. Assume that A satisfies $Ai = i + 2t_0$ for some $t_0 \in \mathbb{R}_+$. Then $H(A)$ consists of all points $\tau \in \mathbb{H}$ with $\mathrm{Re}(\tau) \leq t_0$, and

$$\overline{H(A)} \cap \partial\mathbb{H} = \{t \in \mathbb{R} \mid t \leq t_0\} \cup \{\infty\}.$$

From $(ai + b)/(ci + d) = i + 2t_0$ we find that

$$a = 2ct_0 + d \qquad \text{and} \qquad b = 2dt_0 - c,$$

and, moreover,

$$1 = \det(A) = ad - bc = c^2 + d^2.$$

Now $S(A)$ is the complement of the set $S^C(A) \coloneqq \{x/y \mid (x,y)^T \in S^1 \cap A \cdot \Delta\}$, and $(x,y)^T \in S^1 \cap A \cdot \Delta$ if and only if $x^2 + y^2 = 1$ and $|A^{-1}(x,y)^T| < 1$. Using the relations above, one computes

$$\begin{aligned}\left|A^{-1}(x,y)^T\right|^2 =& (c^2+d^2)x^2 - 2(ac+bd)xy + (a^2+b^2)xy \\ =& x^2 - 4t_0xy + (4t_0^2+1)y^2,\end{aligned}$$

and—using the fact that $x^2+y^2 = 1$—this is less than 1 if and only if $4t_0^2y^2 < 4t_0xy$. For this to be satisfied, we must have $y \neq 0$; dividing both sides by $4t_0y^2 > 0$, this is equivalent to $t_0 < x/y < \infty$. Hence $S^C(A) = \{t \in \mathbb{R} \mid t > t_0\}$, and

$$S(A) = \{t \in \mathbb{R} \mid t \leq t_0\} \cup \{\infty\} = \overline{H(A)} \cap \partial\mathbb{H}.$$

Now consider general $A \in \Gamma\backslash \mathrm{SO}_2\mathbb{R}$, and fix $B \in \mathrm{SO}_2\mathbb{R}$ such that $BAi = i + 2t_0$ for some $t_0 \in \mathbb{R}_+$. From the argument above, we have

$$S(BA) = \overline{H(BA)} \cap \partial\mathbb{H}.$$

We claim that $S(BA) = BS(A)$ and $H(BA) = BH(A)$. By definition, $t \in S(BA)$ if and only if there is some $(x,y)^T \in S^1\backslash BA \cdot \Delta$ with $x/y = t$. The latter inclusion is equivalent to $B^{-1}(x,y)^T \in B^{-1}(S^1\backslash BA \cdot \Delta) = S^1\backslash A \cdot \Delta$ since $B^{-1} \in \mathrm{SO}_2\mathbb{R}$. Writing $B^{-1} = \left(\begin{smallmatrix} e & f \\ g & h \end{smallmatrix}\right)$, we find $B^{-1}(x,y)^T = (ex+fy, gx+hy)^T$. But $B^{-1}t = B^{-1}(x/y) = (ex+fy)/(gx+hy)$ is the reciprocal of the slope of $B^{-1}(x,y)^T \in S^1\backslash A \cdot \Delta$, so $t \in S(BA)$ if and only if $B^{-1}t \in S(A)$, as desired. For the second claim, note that $\tau \in H(BA)$ if and only if $d_{\mathbb{H}}(i,\tau) \leq d_{\mathbb{H}}(BAi,\tau)$. Since B^{-1} is an isometry fixing i, this inequality is equivalent to $d_{\mathbb{H}}(i, B^{-1}\tau) \leq d_{\mathbb{H}}(Ai, B^{-1}\tau)$, which is true if and only if $B^{-1}\tau \in H(A)$. We thus have $BS(A) = \overline{BH(A)} \cap \partial\mathbb{H}$. Applying B^{-1}—which acts as a homeomorphism on $\partial\mathbb{H}$—to both sides, this implies $S(A) = \overline{H(A)} \cap \partial\mathbb{H}$, as claimed.

Next we show that

$$\overline{D(\Gamma)} = \bigcap_{A\in\Gamma} \overline{H(A)}.$$

The forward inclusion holds as the closure of an intersection of sets is always contained in the intersection of the closures of the sets. For the reverse inclusion, suppose $\tau \in \bigcap_{A\in\Gamma} \overline{H(A)}\backslash\{i\}$, and let γ be a geodesic segment in $\mathbb{H}$ without endpoints, for which $\overline{\gamma}$ has endpoints i and τ. Since $\tau \in \overline{H(A)}$ for each $A \in \Gamma$ and each $H(A)$ is convex with i in its interior, we have $\gamma \subset \bigcap_{A\in\Gamma} H(A) = D(\Gamma)$. Hence $\tau \in \overline{\gamma} \subset \overline{D(\Gamma)}$, so the claim holds.

It follows that

$$\begin{aligned} S_\Gamma &= \bigcap_{A\in\Gamma} S(A) = \bigcap_{A\in\Gamma} \left(\overline{H(A)} \cap \partial\mathbb{H}\right) \\ &= \left(\bigcap_{A\in\Gamma} \overline{H(A)}\right) \cap \partial\mathbb{H} = \overline{D(\Gamma)} \cap \partial\mathbb{H}. \end{aligned}$$ □

With the aid of Lemma 3.4.3, lattices may be characterised as the finitely generated discrete groups Γ for which the set of directions Θ_Γ is finite. Furthermore, for a lattice Γ, the set of directions Θ_Γ may be computed in finite time:

Proposition 3.4.4. *A discrete subgroup $\Gamma \subset \mathrm{SL}_2\mathbb{R}$ is a lattice if and only if it is a finitely generated fanning group. In this case there is some finite subset $\Gamma_n \subset \Gamma$ for which $\Theta_\Gamma = \Theta_{\Gamma_n}$.*

Proof. Let Λ denote the image in $\mathrm{PSL}_2\mathbb{R}$ of $\Gamma \cap \mathrm{SO}_2\mathbb{R}$. As Λ contains only elliptic elements, it is a finite cyclic group of order, say, n (Corollary 2.4.2 of [102]). Let $A_0 \in \Lambda$ be a generator of this group. Recall that if $n = 1$, then $D(\Gamma)$ is the Dirichlet region centered at i for Γ and is thus a convex fundamental polygon. If $n > 1$, then $D(\Gamma)$ is no longer a fundamental domain: one finds that $A_0 D(\Gamma) = D(\Gamma)$, and the interior of $D(\Gamma)$ contains n points from each orbit $\Gamma\tau$, $\tau \in \mathbb{H}$. Let γ_0 denote a geodesic segment beginning at i and ending on $\partial D(\Gamma)$.

For each $0 \leq j < n$, the geodesic segment $\gamma_j := A_0^j \gamma_0$ also begins and ends at i and $\partial D(\Gamma)$. Let $\mathcal{F}_0 \subset D(\Gamma)$ denote the convex polygon bounded by γ_0, γ_1 and $\partial D(\Gamma)$. A slight generalisation of the proof of Theorem 3.2.2 of [102] shows that $\mathcal{F}_0$ is a convex fundamental polygon for Γ. In particular,

$$\mu_{\mathbb{H}}(D(\Gamma)) = n\mu_{\mathbb{H}}(\mathcal{F}_0) = n\mu_{\mathbb{H}}(\Gamma\backslash\mathbb{H}),$$

where $\mu_{\mathbb{H}}$ denotes hyperbolic area. Hence Γ is a lattice if and only if $\mu_{\mathbb{H}}(D(\Gamma))$ is finite.

Suppose Γ is a lattice. Then Γ is geometrically finite and hence finitely generated. If Γ is not a fanning group, then Θ_Γ—and, consequently, S_Γ—is infinite. By Lemma 3.4.3, $\overline{D(\Gamma)} \cap \partial\mathbb{H}$ is infinite, and by Gauss-Bonnet, $D(\Gamma)$ has infinite hyperbolic area. This is a contradiction, so Γ is a finitely generated fanning group.

Now assume Γ is a finitely generated fanning group. Again by Lemma 3.4.3, this implies that $\overline{D(\Gamma)} \cap \partial\mathbb{H}$ is finite, so $\mathcal{F}_0 \subset D(\Gamma)$ has no free sides and at most finitely many cusps. Since Γ is finitely generated, it is geometrically finite and thus every convex fundamental polygon for Γ has finitely many sides (Theorem 10.1.2 of [11]). By Gauss-Bonnet, Γ is a lattice, and the first statement is proven.

For the second statement, suppose Γ is a lattice. From the previous paragraph, we know that $\mathcal{F}_0$ has finitely many sides, and thus the same is true of $D(\Gamma)$. But then there is some finite subset $\Gamma_n \subset \Gamma$ for which $D(\Gamma_n) = D(\Gamma)$, and by Lemma 3.4.3, $S_\Gamma = S_{\Gamma_n}$, which implies $\Theta_\Gamma = \Theta_{\Gamma_n}$. □

3.4.2 The group $\mathbf{Aff}^+_{\mathcal{O}}(X,\omega)$ and its action on marked segments

Let $(X,\omega) \in \mathcal{H}(d_1,\ldots,d_\kappa)$ and $\mathcal{O} = \mathcal{O}(d_1,\ldots,d_\kappa)$. A central result of [63] (Theorem 18) and [64] (Proposition 17) is that a matrix $A \in \mathrm{SL}_2\mathbb{R}$ belongs to $\mathrm{SL}(X,\omega)$ if and only if there is some $f \in \mathrm{Aff}^+(\mathcal{O})$ with $\mathrm{der}(f) = A$ satisfying $f(\widehat{\mathcal{S}}(X,\omega)) \subset \widehat{\mathcal{M}}(X,\omega)$. Theorem 17 of [63] shows that the same statement holds with the latter subset

inclusion replaced by the equality $f(\widehat{\mathcal{M}}(X,\omega)) = \widehat{\mathcal{M}}(X,\omega)$. Note, in particular, that these statements require the affine automorphism f to respect the orientation-pairing of marked segments—it is not enough that the set of marked segments $\widehat{\mathcal{M}}_F(X,\omega)$ is invariant under f to conclude that $\mathrm{der}(f) \in \mathrm{SL}(X,\omega)$ (see Example 21 of [64]). We introduce the following notation for the collection of such affine automorphisms of $\mathcal{O}$:

Definition 3.4.5. Let

$$\mathrm{Aff}^+_{\mathcal{O}}(X,\omega) \coloneqq \{f \in \mathrm{Aff}^+(\mathcal{O}) \mid f(\widehat{\mathcal{M}}(X,\omega)) = \widehat{\mathcal{M}}(X,\omega)\}.$$

Note that $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ is a subgroup of $\mathrm{Aff}^+(\mathcal{O})$. From the comments above we see that the image of $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ under $\mathrm{der}(\cdot)$ is precisely the Veech group $\Gamma = \mathrm{SL}(X,\omega)$ (though $\mathrm{der}(\cdot)$ restricted to $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ need not be injective, so $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ and $\mathrm{SL}(X,\omega)$ are in general non-isomorphic). The group $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ acts on $\widehat{\mathcal{M}}(X,\omega)$ via

$$\begin{aligned}\mathrm{Aff}^+_{\mathcal{O}}(X,\omega) \times \widehat{\mathcal{M}}(X,\omega) &\to \widehat{\mathcal{M}}(X,\omega)\\ (f, \{\hat{s}, \hat{s}'\}) &\mapsto \{f(\hat{s}), f(\hat{s}')\}.\end{aligned}$$

Note in particular that $f(\hat{s}') = f(\hat{s})'$. For any $\{\hat{s}, \hat{s}'\} \in \widehat{\mathcal{M}}(X,\omega)$, let $[\{\hat{s}, \hat{s}'\}]$ denote the orbit of $\{\hat{s}, \hat{s}'\}$ under this action.

Recall from §3.3.3 that (X,ω) may be reconstructed from its marked Voronoi staples $\widehat{\mathcal{S}}(X,\omega) \subset \widehat{\mathcal{M}}(X,\omega)$. The goal of Algorithm 3.1.1 is to construct—using only $\Gamma = \mathrm{SL}(X,\omega)$ and $d_1 \leq \cdots \leq d_\kappa$—increasing subsets of $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbits $[\{\hat{s}, \hat{s}'\}]$ whose union eventually contains $\widehat{\mathcal{S}}(X,\omega)$. To this end, the two main obstacles we face are

(i) determining $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$, and

(ii) determining a representative $\{\hat{s}, \hat{s}'\}$ of each of the $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbits $[\{\hat{s}, \hat{s}'\}]$ intersecting $\widehat{\mathcal{S}}(X,\omega)$

using only the data Γ and $d_1 \leq \cdots \leq d_\kappa$. Recall the collection $\mathrm{Aff}^+_{\mathrm{C}}(\mathcal{O})$ of canonical affine automorphisms of $\mathcal{O}$ from Definition 3.3.3. The first challenge is addressed by the following:

Lemma 3.4.6. *Let G be a set of generators of $\Gamma \leq \mathrm{SL}_2\mathbb{R}$. Then for any $(X,\omega) \in \mathcal{H}(d_1,\ldots,d_\kappa)$ with $\mathrm{SL}(X,\omega) = \Gamma$, the group $Aff^+_{\mathcal{O}}(X,\omega)$ is generated by a subset of*

$$G_{\mathcal{O}} \coloneqq \{\tau \circ f_A \mid \tau \in \mathit{Trans}(\mathcal{O}),\ f_A \in \mathit{Aff}^+_C(\mathcal{O}),\ A \in G\}.$$

In particular, if G is finite then $Aff^+_{\mathcal{O}}(X,\omega)$ is generated by a subset of the finite set $G_{\mathcal{O}}$.

Proof. For each $A \in G$, fix $g_A \in \mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ with $\mathrm{der}(g_A) = A$. Recall that g_A may be written uniquely as $g_A = \tau_A \circ f_A$, where $\tau_A \in \mathrm{Trans}(\mathcal{O})$ and $f_A \in \mathrm{Aff}^+_{\mathrm{C}}(\mathcal{O})$, so the set of such g_A is contained in $G_{\mathcal{O}}$. Let Λ be the subgroup of $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ generated by these g_A. Since $\mathrm{Trans}(\mathcal{O})$ is closed under composition, it suffices to show that $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega) = \cup\, \tau\Lambda$, where the union is over some collection of $\tau \in \mathrm{Trans}(\mathcal{O})$.

Now let $f \in \mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ be arbitrary and $A \coloneqq \mathrm{der}(f) \in \mathrm{SL}(X,\omega)$. The matrix A may be written as a product $A = A_1^{\delta_1} \cdots A_m^{\delta_m}$, $\delta_j \in \{\pm 1\}$, of elements A_j in the generating set G and their inverses. Set $g \coloneqq g_{A_1}^{\delta_1} \ldots g_{A_m}^{\delta_m} \in \Lambda$, with each g_{A_j} a chosen generator of Λ as above. Then $\mathrm{der}(f \circ g^{-1}) = \mathrm{Id}$, so $\tau \coloneqq f \circ g^{-1} \in \mathrm{Trans}(\mathcal{O})$ and $f \in \tau\Lambda$ as desired.

The final statement of the lemma follows immediately from finiteness of both $\mathrm{Trans}(\mathcal{O})$ (Lemma 3.3.1) and G. □

For challenge (ii) mentioned above, we wish to determine the generalised polar coordinates of representatives of $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbits. In the proof of the following lemma—which is also crucial for Algorithm 3.1.1—it is shown that the directions of the shortest representatives $\{\hat{s},\hat{s}'\}$ of each orbit $[\{\hat{s},\hat{s}'\}]$ are announced by the set $\Theta_{\mathrm{SL}(X,\omega)}$. The more delicate procedure of determining the lengths of such representatives is addressed in §3.4.5.

Lemma 3.4.7. *If (X,ω) is a fanning surface, then the orbit space*

$$\widehat{\mathcal{M}}(X,\omega)/Aff^+_{\mathcal{O}}(X,\omega) = \left\{ [\{\hat{s},\hat{s}'\}] \;\middle|\; \{\hat{s},\hat{s}'\} \in \widehat{\mathcal{M}}(X,\omega) \right\}$$

is finite.

Proof. Let $[\{\hat{s},\hat{s}'\}] \in \widehat{\mathcal{M}}(X,\omega)/\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$. The image under $\boldsymbol{\pi}(\cdot)$ of $[\{\hat{s},\hat{s}'\}]$ consists of pairs $\{\pm v\}$ of holonomy vectors in $\mathbb{C}$, among which there is some pair of minimal length. The preimage under $\boldsymbol{\pi}(\cdot)$ of this pair is a finite set in $\mathcal{O}$ containing a pair of orientation-paired marked segments $\{\hat{s},\hat{s}'\}$ of minimal length in $[\{\hat{s},\hat{s}'\}]$. Rescaling (X,ω) if necessary, we may assume $|\hat{s}| = |\hat{s}'| = 1$. We claim that

$$\{\boldsymbol{\pi}(\hat{s}), \boldsymbol{\pi}(\hat{s}')\} \subset \Theta_{\mathrm{SL}(X,\omega)}.$$

Note for each $A \in \mathrm{SL}(X,\omega)$ that $\boldsymbol{\pi}(\hat{s}) \notin S^1 \cap A\cdot\Delta$: otherwise $A^{-1}\cdot \boldsymbol{\pi}(\hat{s}) \in \Delta$, but the comments following Definition 3.4.5 together with Lemma 3.3.2 imply $A^{-1}\cdot\boldsymbol{\pi}(\hat{s}) = \boldsymbol{\pi}\circ f(\hat{s})$ for some $f \in \mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ with $\mathrm{der}(f) = A^{-1}$. Since $1 > |\boldsymbol{\pi}\circ f(\hat{s})| = |f(\hat{s})|$, this contradicts the assumption that $\{\hat{s},\hat{s}'\}$ is a pair of minimal length in $[\{\hat{s},\hat{s}'\}]$. The same is true for $\boldsymbol{\pi}(\hat{s}')$, so we find that

$$\{\boldsymbol{\pi}(\hat{s}), \boldsymbol{\pi}(\hat{s}')\} \subset \bigcap_{A\in\mathrm{SL}(X,\omega)} (S^1 \setminus A\cdot\Delta) = S^1 \setminus \bigcup_{A\in\mathrm{SL}(X,\omega)} A\cdot\Delta = \Theta_{\mathrm{SL}(X,\omega)}$$

as claimed.

For each $v \in S^1$, let r_v denote the infinite open ray emanating from $0 \in \mathbb{C}$ in the direction of v. Note that the preimage $\boldsymbol{\pi}^{-1}(r_v) \subset \mathcal{O}$ consists of $\sum_{i=1}^{\kappa}(d_i+1) = 2g-2+\kappa$ infinite open rays emanating from the origins of the various components of $\mathcal{O}$, and each of these rays contains at most one marked segment of (X,ω) (Proposition 3.3.17). From the claim above, the minimal-length representatives of each $[\{\hat{s},\hat{s}'\}] \in \widehat{\mathcal{M}}(X,\omega)/\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ belong to one of the $2g-2+\kappa$ open rays of $\boldsymbol{\pi}^{-1}(r_v)$ for some $v \in \Theta_{\mathrm{SL}(X,\omega)}$. Hence

$$\left|\widehat{\mathcal{M}}(X,\omega)/\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)\right| \leq (2g-2+\kappa)\cdot|\Theta_{\mathrm{SL}(X,\omega)}|,$$

where $|\cdot|$ denotes cardinality. Since (X,ω) is a fanning surface, the right-hand side is finite. □

The proof of Lemma 3.4.7 also gives the following:

Corollary 3.4.8. *Let (X,ω) be a fanning surface and $[\{\hat{s},\hat{s}'\}] \in \widehat{\mathcal{M}}(X,\omega)/Aff^+_{\mathcal{O}}(X,\omega)$. If $\{\hat{s},\hat{s}'\}$ are orientation-paired marked segments of minimal length in $[\{\hat{s},\hat{s}'\}]$, then the directions $\boldsymbol{\pi}(\hat{s})/|\boldsymbol{\pi}(\hat{s})|$ and $\boldsymbol{\pi}(\hat{s}')/|\boldsymbol{\pi}(\hat{s}')|$ belong to the finite set $\Theta_{\mathrm{SL}(X,\omega)}$.*

3.4.3 Simulating normalised $\mathbf{Aff}^+_{\mathcal{O}}(X,\omega)$-orbits

The set of orientation-paired marked segments $\widehat{\mathcal{M}}(X,\omega)$ (and the set of $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbits into which it partitions) depends intrinsically on (X,ω) and its geometry. The results of the previous subsection suggest that in the case of a fanning surface, much of this information is encoded in the Veech group $\mathrm{SL}(X,\omega)$. This subsection further explores these ideas by introducing *simulations* of $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbits which are constructed via generators of a fanning group. We again fix a stratum $\mathcal{H}(d_1,\ldots,d_\kappa)$.

Definition 3.4.9. Let $\Gamma \leq \mathrm{SL}_2\mathbb{R}$ be a fanning group generated by $G \subset \Gamma$ and set

$$\mathfrak{S}_G \coloneqq \{(H,\{p,p^-\}) \in 2^{G_{\mathcal{O}}}\backslash\{\varnothing\} \times \mathbb{P}(\mathcal{O}) \mid \boldsymbol{\pi}(p),\boldsymbol{\pi}(p^-) \in \Theta_\Gamma\},$$

where $G_{\mathcal{O}}$ is defined as in Lemma 3.4.6 and $2^{G_{\mathcal{O}}}$ is the power set of $G_{\mathcal{O}}$. For each $\mathfrak{s} \coloneqq (H,\{p,p^-\}) \in \mathfrak{S}_G$, define the *stage* 0 *simulation* determined by $\mathfrak{s}$ by $\mathrm{sim}^0(\mathfrak{s}) \coloneqq \{\{p,p^-\}\} \subset \mathbb{P}(\mathcal{O})$, and for $n \geq 1$ define the *stage* n *simulation* determined by $\mathfrak{s}$ recursively as

$$\mathrm{sim}^n(\mathfrak{s}) \coloneqq \bigcup_{f\in H\cup\{\mathrm{Id}\}} f^{\pm 1}(\mathrm{sim}^{n-1}(\mathfrak{s})) \subset \mathbb{P}(\mathcal{O}).$$

The *simulation* determined by $\mathfrak{s}$ is the union $\mathrm{sim}(\mathfrak{s}) \coloneqq \bigcup_{n\geq 0} \mathrm{sim}^n(\mathfrak{s})$ of all stage n simulations determined by $\mathfrak{s}$. Denote by

$$\mathrm{Sims}_G \coloneqq \{\mathrm{sim}(\mathfrak{s}) \mid \mathrm{sim}(\mathfrak{s}) \text{ is distinctive, } \mathfrak{s} \in \mathfrak{S}_G\}$$

and

$$\mathrm{Sims}_G^n \coloneqq \{\mathrm{sim}^n(\mathfrak{s}) \mid \mathrm{sim}^n(\mathfrak{s}) \text{ is distinctive, } \mathfrak{s} \in \mathfrak{S}_G\}$$

the *set of all distinctive simulations* and *set of all stage n distinctive simulations*, respectively, determined by G. For any $r \in \mathbb{R}_+$, the *r-scaled (stage n)* simulation is $r\mathrm{sim}(\mathfrak{s})$ ($r\mathrm{sim}^n(\mathfrak{s})$). Finally, set $\mathrm{sim}_F(\mathfrak{s}) \coloneqq (\mathrm{sim}(\mathfrak{s}))_F \subset \mathcal{O}$ and $\mathrm{sim}_F^n(\mathfrak{s}) \coloneqq (\mathrm{sim}^n(\mathfrak{s}))_F \subset \mathcal{O}$ (recall Definition 3.3.8).

Note that by construction, the stage n simulation $\mathrm{sim}^n(\mathfrak{s})$ is the set of images in $\mathcal{O}$ of the oppositely projected pair $\{p, p^-\}$ under all compositions of at most n elements of H and their inverses, and the simulation $\mathrm{sim}(\mathfrak{s})$ is simply the $\langle H\rangle$-orbit of $\{p, p^-\}$. Furthermore, the distinctive simulations in Sims_G and Sims_G^n are limit-point free since Γ is discrete.

Recall S, T and Γ from Example 3.4.2.

Example 3.4.10. Let $G \coloneqq \{S, T^2\}$, $\mathcal{O} = \mathcal{O}(2)$, and set

$$\mathfrak{s}_0 \coloneqq (H, \{p_0, p_0^-\}) \in \mathfrak{S}_G,$$

where $H = \{\rho_1^2 \circ f_S, f_{T^2}\}$, and $p_0, p_0^- \in \mathcal{O}_1 = \mathcal{O}$ have unit length (this is required since $\boldsymbol{\pi}(p_0), \boldsymbol{\pi}(p_0^-) \in \Theta_\Gamma \subset S^1$) and arguments $\arg(p_0) = 0$ and $\arg(p_0^-) = 5\pi$. The map $\rho_1^2 \circ f_S$ acts as a counterclockwise rotation of $\mathcal{O}$ by an angle of $\pi/2 + 4\pi = 9\pi/2$, and f_{T^2} is a bijective horizontal shear on each c_j^1. A subset of the simulation $\mathrm{sim}(\mathfrak{s}_0)$ is shown in blue in Figure 3.8. In red and tan are subsets of $\mathrm{sim}(\mathfrak{s}_1)$ and $\mathrm{sim}(\mathfrak{s}_2)$, respectively, for

$$\mathfrak{s}_1 \coloneqq (H, \{p_1, p_1^-\}) \in \mathfrak{S}_G \qquad \text{and} \qquad \mathfrak{s}_2 \coloneqq (H, \{p_2, p_2^-\}) \in \mathfrak{S}_G,$$

where $\arg(p_1) = 4\pi$, $\arg(p_1^-) = \pi$, $\arg(p_2) = \pi/4$ and $\arg(p_2^-) = 13\pi/4$.

Observe that Definition 3.4.9, Proposition 3.4.4 and Lemma 3.4.6 immediately imply:

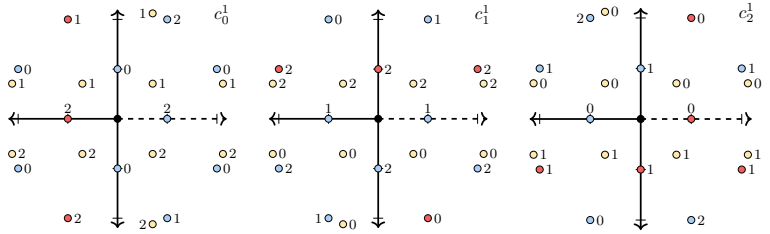


Figure 3.8: Three (subsets) of the simulations $\mathrm{sim}(\mathfrak{s}_0)$, $\mathrm{sim}(\mathfrak{s}_1)$, $\mathrm{sim}(\mathfrak{s}_2) \in \mathrm{Sims}_G$ from Example 3.4.10 are shown in blue, red and tan, respectively. A label j next to a point p indicates to which c_j^1 the point p^- belongs.

Lemma 3.4.11. *If Γ is a lattice, then there exists a set of generators G of Γ for which the set of simulations Sims_G is finite.*

Remark 3.4.12. Recall from Proposition 3.4.4 that for a lattice Γ, the set of directions Θ_Γ may be computed in finite time. Indeed, letting Γ_n be the subset of Γ consisting of all words of length no greater than n in the generators G and their inverses, there is some $n \in \mathbb{N}$ for which $\Theta_{\Gamma_n} = \Theta_\Gamma$. In general, $\Theta_{\Gamma_n} \supset \Theta_\Gamma$, and it may be difficult to determine when equality holds. Thus in practice, the set Θ_Γ in the definition of $\mathfrak{S}_G$ (Definition 3.4.9) is replaced by some finite Θ_{Γ_n} containing Θ_Γ. Such a replacement does not affect the statements of the results in the remainder of the chapter, though it could increase the size of the—still finite—set of simulations Sims_G.

Definition 3.4.13. Let $r \in \mathbb{R}_+$ be the minimal length of orientation-paired marked segments in $[\{\hat{s}, \hat{s}'\}] \in \widehat{\mathcal{M}}(X,\omega)/\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$. For any marked segment $\hat{s}$ belonging to a pair in the orbit $[\{\hat{s}, \hat{s}'\}]$, the *normalised marked segment* corresponding to $\hat{s}$ is $n(\hat{s}) \coloneqq (1/r)\hat{s}$. Set $n(\hat{s})' \coloneqq n(\hat{s}')$, and define the *normalised $\mathit{Aff}^+_{\mathcal{O}}(X,\omega)$-orbit* corresponding to $[\{\hat{s}, \hat{s}'\}]$ as

$$n([\{\hat{s}, \hat{s}'\}]) \coloneqq \big\{\{n(\hat{s}), n(\hat{s})'\} \;\big|\; \{\hat{s}, \hat{s}'\} \in [\{\hat{s}, \hat{s}'\}]\big\}.$$

Thus the normalised $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbit corresponding to $[\{\hat{s},\hat{s}'\}]$ takes all points of the orbit $[\{\hat{s},\hat{s}'\}]$ and rescales them (by a constant scalar $1/r$) so that the minimal length of a pair in the orbit is one.

Theorem 3.4.14. *If (X,ω) is a fanning surface, then $\widehat{\mathcal{M}}(X,\omega)$ is a finite union of scaled simulations. In particular, if $\Gamma \leq \mathrm{SL}_2\mathbb{R}$ is a fanning group generated by $G \subset \Gamma$, then*

$$\left\{ n([\{\hat{s},\hat{s}'\}]) \;\middle|\; \begin{array}{l} [\{\hat{s},\hat{s}'\}] \in \widehat{\mathcal{M}}(X,\omega)/\mathit{Aff}^+_{\mathcal{O}}(X,\omega) \\ \text{for some } (X,\omega) \text{ with } \mathrm{SL}(X,\omega) = \Gamma \end{array} \right\} \subset \mathit{Sims}_G.$$

Proof. We will prove the second statement; the first follows from this together with Lemma 3.4.7. Let (X,ω) be a translation surface with $\mathrm{SL}(X,\omega) = \Gamma$, and let G be a (finite or infinite) set of generators for Γ. By Lemma 3.4.6, there is some subset $H \subset G_{\mathcal{O}}$ which generates $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$. Now let $[\{\hat{s},\hat{s}'\}] \in \widehat{\mathcal{M}}(X,\omega)/\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ with $\{\hat{s},\hat{s}'\}$ a pair of minimal length in $[\{\hat{s},\hat{s}'\}]$. Set $p \coloneqq n(\hat{s})$ and $p^- \coloneqq n(\hat{s})'$. By Corollary 3.4.8, $\boldsymbol{\pi}(p) = \boldsymbol{\pi}(\hat{s})/|\boldsymbol{\pi}(\hat{s})|$ and $\boldsymbol{\pi}(p^-) = \boldsymbol{\pi}(\hat{s}')/|\boldsymbol{\pi}(\hat{s}')|$ belong to the finite set Θ_Γ, so $\mathfrak{s} \coloneqq (H,\{p,p^-\}) \in \mathfrak{S}_G$. Furthermore, $\mathrm{sim}(\mathfrak{s})$ is the $\langle H\rangle = \mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbit of $\{p,p^-\} = \{n(\hat{s}), n(\hat{s})'\}$, and this is precisely the set $n([\{\hat{s},\hat{s}'\}])$. By Proposition 3.3.17, $[\{\hat{s},\hat{s}'\}]$—and thus also $n([\{\hat{s},\hat{s}'\}])$—is distinctive; hence $n([\{\hat{s},\hat{s}'\}]) \in \mathrm{Sims}_G$. □

Theorem 3.4.14 implies that for any fanning group Γ, the orientation-paired marked segments $\widehat{\mathcal{M}}(X,\omega)$ of any (X,ω) with Veech group Γ are given by a finite collection of simulations, each of which is appropriately scaled. Recall that the marked Voronoi staples $\widehat{\mathcal{S}}(X,\omega)$—from which (X,ω) may be reconstructed—are determined by $\widehat{\mathcal{M}}(X,\omega)$ (§3.3.3). In §3.4.5 , we show that for any finite collection of simulations, there are at most finitely many scalars for which the union of scaled simulations agrees with the marked segments of a unit-area translation surface. Together with Lemma 3.4.11, this will prove Theorem 3.1.2.

3.4.4 Marked Voronoi staples determine permissible triples

Let $(X,\omega) \in \mathcal{H}(d_1,\dots,d_\kappa)$ and $\mathcal{O} = \mathcal{O}(d_1,\dots,d_\kappa)$. In this subsection we apply the results of §3.3.2 to orientation-paired marked segments and their $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbits. In particular, we shall see that adjacent elements of $\widehat{\mathcal{S}}_F(X,\omega)$ naturally determine permissible triples (Proposition 3.4.17). To this end, we begin with the following:

Lemma 3.4.15. *For any two adjacent $\hat{s},\hat{t} \in \widehat{\mathcal{S}}_F(X,\omega)$, we have*

$$B(\hat{s},\hat{t}) \cap \widehat{\mathcal{M}}_F(X,\omega) = \varnothing.$$

Moreover, $[\hat{s},\hat{t}] \in \widehat{\mathcal{M}}_F(X,\omega)$ with $[\hat{s},\hat{t}]' = [\hat{t},\hat{s}]$.

Proof. Let i be the index for which $\hat{s},\hat{t} \in \mathcal{O}_i$, and recall from Definition 3.3.6 that the circumcenter $c \coloneqq c(\hat{s},\hat{t})$ determined by $\hat{s},\hat{t} \in \mathcal{O}_i$ is the intersection of $\partial H(\hat{s})$ and $\partial H(\hat{t})$. Since $\hat{s},\hat{t} \in \widehat{\mathcal{S}}_F(X,\omega)$ are essential points of Ω_i, there are two (adjacent) edges of Ω_i contained in $\partial H(\hat{s})$ and $\partial H(\hat{t})$, respectively. The intersection of these edges is precisely the circumcenter c, so $c \in \Omega_i$. Suppose for the sake of contradiction that there is some marked segment $\hat{u} \in B(\hat{s},\hat{t})$. Then

$$d(\hat{u},c) < |c| = d(0,c),$$

and hence $c \notin H(\hat{u})$. This contradicts the fact that $c \in \Omega_i$, so we have

$$B(\hat{s},\hat{t}) \cap \widehat{\mathcal{M}}_F(X,\omega) = \varnothing.$$

This result—together with the convexity of $B(\hat{s},\hat{t})$ and the fact that $0 \in \partial B(\hat{s},\hat{t})$—gives that $B(\hat{s},\hat{t}) \subset \mathrm{star}_i(X,\omega)$, so η is defined on all of $B(\hat{s},\hat{t})$. Let γ be the straight line segment in $\mathcal{O}_i$ from $\hat{s}$ to $\hat{t}$, and let $\sigma_j,\sigma_k \in \Sigma$ be the singularities at which the oriented saddle connections s and t end, respectively. Then, recycling notation, $u \coloneqq \eta(\gamma)$ is a saddle connection from σ_j to σ_k, and $\hat{u} \in \widehat{\mathcal{M}}_F(X,\omega)$. We claim that $\hat{u} = [\hat{s},\hat{t}]$.

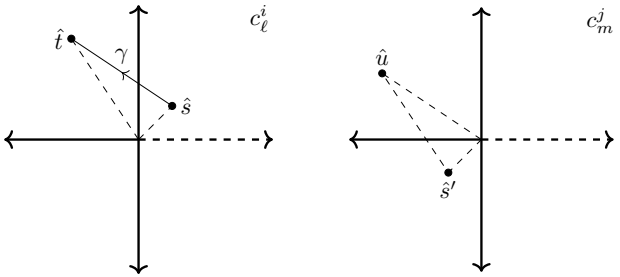


Figure 3.9: Illustration of the proof of Lemma 3.4.15, showing that $\hat{u} = [\hat{s}, \hat{t}]$ is a marked segment for adjacent $\hat{s}, \hat{t} \in \widehat{\mathcal{S}}_F(X, \omega)$.

By Proposition 3.3.21, $\hat{s}$ and $\hat{t}$ belong to the same open π-sector of $\mathcal{O}_i$. We also have by Proposition 3.3.17 that $\arg(\hat{s}) \neq \arg(\hat{t})$, so $[\hat{s}, \hat{t}]$ is in fact defined. It is clear from the definition of $\hat{u}$ that $\hat{s}'$ and $\hat{u}$ belong to the same open π-sector of the same component $\mathcal{O}_j$. Furthermore, $\boldsymbol{\pi}(\hat{u}) = \boldsymbol{\pi}(\hat{t}) - \boldsymbol{\pi}(\hat{s})$; see Figure 3.9. By Definition 3.3.10, we have $\hat{u} = [\hat{s}, \hat{t}]$. Reversing the orientation of γ in the argument above gives that $\hat{u}' = [\hat{t}, \hat{s}]$. □

Recall that permissible triples are defined in terms of three distinctive, limit-point free subsets of $\mathbb{P}(\mathcal{O})$ (Definition 3.3.13). We introduce notation for a set of—a priori non-permissible—triples determined by adjacent $\hat{s}, \hat{t} \in \widehat{\mathcal{S}}_F(X, \omega)$ and subsequently prove that these in fact belong to the set $\mathcal{P}(\mathrm{Sim}_G)$ of all permissible triples arising from Sim_G, where $\mathrm{SL}(X, \omega) = \langle G \rangle$.

Definition 3.4.16. For $(X, \omega) \in \mathcal{H}(d_1, \ldots, d_\kappa)$, let $\mathcal{P}(X, \omega)$ denote the set

$$\left\{ (n(\hat{s}), n(\hat{t}), n([\hat{s}, \hat{t}])) \;\middle|\; \hat{s},\ \hat{t} \text{ are adjacent elements of } \widehat{\mathcal{S}}_F(X, \omega) \right\}.$$

Proposition 3.4.17. *For any fanning group* $\Gamma \leq \mathrm{SL}_2\mathbb{R}$ *generated by* $G \subset \Gamma$ *and any* $(X, \omega) \in \mathcal{H}(d_1, \ldots, d_\kappa)$ *with Veech group* $\mathrm{SL}(X, \omega) =$

Γ, *we have* $\mathcal{P}(X,\omega) \subset \mathcal{P}(Sims_G)$. *Moreover, if* G *is finite (i.e.,* Γ *is a lattice), then there exists some* $N \in \mathbb{N}$ *such that for any* $(X,\omega) \in \mathcal{H}(d_1,\ldots,d_\kappa)$ *with Veech group* $\mathrm{SL}(X,\omega) = \Gamma$, *we have* $\mathcal{P}(X,\omega) \subset \mathcal{P}(Sims_G) \subset \mathcal{P}(Sims_G^N)$.

Remark 3.4.18. At present, it is unknown whether or not the number N in Proposition 3.4.17 can be determined explicitly; see also the discussion at the end of §3.5 below.

Proof of Proposition 3.4.17. Let $(n(\hat{s}), n(\hat{t}), n([\hat{s},\hat{t}])) \in \mathcal{P}(X,\omega)$ for some (X,ω) with Veech group Γ. By Theorem 3.4.14, there are $\mathfrak{s}_{\hat{s}}, \mathfrak{s}_{\hat{t}}, \mathfrak{s}_{[\hat{s},\hat{t}]} \in \mathfrak{S}_G$ for which

$$n([\{\hat{s},\hat{s}'\}]) = \mathrm{sim}(\mathfrak{s}_{\hat{s}}), \qquad n([\{\hat{t},\hat{t}'\}]) = \mathrm{sim}(\mathfrak{s}_{\hat{t}}),$$

and

$$n([\{[\hat{s},\hat{t}],[\hat{s},\hat{t}]'\}]) = \mathrm{sim}(\mathfrak{s}_{[\hat{s},\hat{t}]}).$$

Hence

$$(n(\hat{s}), n(\hat{t}), n([\hat{s},\hat{t}])) \in \mathrm{sim}_F(\mathfrak{s}_{\hat{s}}) \times \mathrm{sim}_F(\mathfrak{s}_{\hat{t}}) \times \mathrm{sim}_F(\mathfrak{s}_{[\hat{s},\hat{t}]}),$$

and we must show that $(n(\hat{s}), n(\hat{t}), n([\hat{s},\hat{t}]))$ is a permissible triple. The corresponding permissible scalars we shall use are the minimal lengths $(r_{\hat{s}}, r_{\hat{t}}, r_{[\hat{s},\hat{t}]}) \in \mathbb{R}^3_+$ of pairs in the respective $\mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$-orbits $[\{\hat{s},\hat{s}'\}]$, $[\{\hat{t},\hat{t}'\}]$ and $[\{[\hat{s},\hat{t}],[\hat{s},\hat{t}]'\}]$. In particular, $r_{\hat{s}} n(\hat{s}) = \hat{s}$, $r_{\hat{t}} n(\hat{t}) = \hat{t}$ and $r_{[\hat{s},\hat{t}]} n([\hat{s},\hat{t}]) = [\hat{s},\hat{t}]$. Moreover, $r_{\hat{s}}\mathrm{sim}(\mathfrak{s}_{\hat{s}}) = [\{\hat{s},\hat{s}'\}]$, $r_{\hat{t}}\mathrm{sim}(\mathfrak{s}_{\hat{t}}) = [\{\hat{t},\hat{t}'\}]$ and $r_{[\hat{s},\hat{t}]}\mathrm{sim}(\mathfrak{s}_{[\hat{s},\hat{t}]}) = [\{[\hat{s},\hat{t}],[\hat{s},\hat{t}]'\}]$. We now verify the two criteria of Definition 3.3.13 on the triple $(n(\hat{s}), n(\hat{t}), n([\hat{s},\hat{t}]))$ and the scalars $(r_{\hat{s}}, r_{\hat{t}}, r_{[\hat{s},\hat{t}]})$.

(i) From the observations above, we have $[r_{\hat{s}} n(\hat{s}), r_{\hat{t}} n(\hat{t})] = [\hat{s},\hat{t}]$ and $[r_{\hat{t}} n(\hat{t}), r_{\hat{s}} n(\hat{s})] = [\hat{t},\hat{s}]$. Since $\hat{s}$ and $\hat{t}$ are adjacent elements of $\widehat{\mathcal{S}}_F(X,\omega)$, Lemma 3.4.15 guarantees these are defined. Moreover,

$$[r_{\hat{s}} n(\hat{s}), r_{\hat{t}} n(\hat{t})] = [\hat{s},\hat{t}] = r_{[\hat{s},\hat{t}]} n([\hat{s},\hat{t}])$$

and

$$[r_{\hat{t}}n(\hat{t}), r_{\hat{s}}n(\hat{s})] = [\hat{t}, \hat{s}] = [\hat{s}, \hat{t}]' = (r_{[\hat{s},\hat{t}]}n([\hat{s}, \hat{t}]))' = r_{[\hat{s},\hat{t}]}n([\hat{s}, \hat{t}])'.$$

(ii) We must show that the union of

$$B(r_{\hat{s}}n(\hat{s}), r_{\hat{t}}n(\hat{t})), \qquad B(r_{[\hat{s},\hat{t}]}n([\hat{s}, \hat{t}]), r_{\hat{s}}n(\hat{s})'),$$

and

$$B(r_{\hat{t}}n(\hat{t})', r_{[\hat{s},\hat{t}]}n([\hat{s}, \hat{t}])')$$

does not intersect

$$r_{\hat{s}}\mathrm{sim}_F(\mathfrak{s}_{\hat{s}}) \cup r_{\hat{t}}\mathrm{sim}_F(\mathfrak{s}_{\hat{t}}) \cup r_{[\hat{s},\hat{t}]}\mathrm{sim}_F(\mathfrak{s}_{[\hat{s},\hat{t}]}),$$

or, equivalently,

$$\begin{aligned}\big(B(\hat{s}, \hat{t}) \cup B([\hat{s}, \hat{t}], \hat{s}') \cup B(\hat{t}', [\hat{s}, \hat{t}]')\big) \\ \cap \big([\{\hat{s}, \hat{s}'\}]_F \cup [\{\hat{t}, \hat{t}'\}]_F \cup [\{[\hat{s}, \hat{t}], [\hat{s}, \hat{t}]'\}]_F\big) = \varnothing.\end{aligned}$$

Since each of the $\mathrm{Aff}^+_{\mathcal{O}}(X, \omega)$-orbits are contained in $\widehat{\mathcal{M}}(X, \omega)$, it suffices to show

$$\big(B(\hat{s}, \hat{t}) \cup B([\hat{s}, \hat{t}], \hat{s}') \cup B(\hat{t}', [\hat{s}, \hat{t}]')\big) \cap \widehat{\mathcal{M}}_F(X, \omega) = \varnothing.$$

Lemma 3.4.15 gives that

$$B(\hat{s}, \hat{t}) \cap \widehat{\mathcal{M}}_F(X, \omega) = \varnothing.$$

Now suppose for the sake of contradiction that

$$B([\hat{s}, \hat{t}], \hat{s}') \cap \widehat{\mathcal{M}}_F(X, \omega) \neq \varnothing,$$

and let j be the index for which $\hat{s}' \in \mathcal{O}_j$. Since $\widehat{\mathcal{M}}_F(X, \omega)$ has no limit points, there are at most finitely many marked segments belonging to this intersection, and by Proposition 3.3.17, none of these marked segments have the same argument

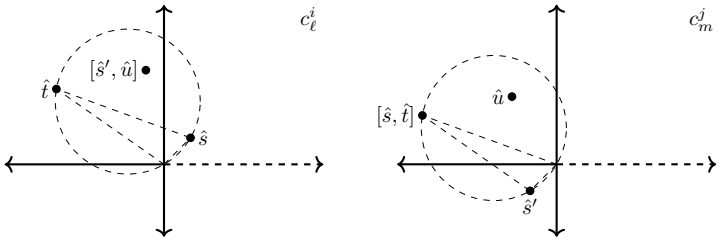


Figure 3.10: Illustration of the proof of Proposition 3.4.17.

as $\hat{s}'$. Choose some $\hat{u}$ in the intersection with argument nearest that of $\hat{s}'$. Then the straight-line segment from $\hat{s}'$ to $\hat{u}$ belongs to $\mathrm{star}_j(X, \omega)$, and a similar argument to that in the proof of Lemma 3.4.15 gives that $[\hat{s}', \hat{u}] \in \widehat{\mathcal{M}}(X, \omega)$. By definition of $[\hat{s}', \hat{u}]$, this marked segment belongs to the same open π-sector as $\hat{s}$ and hence to $B(\hat{s}, \hat{t})$, contradicting Lemma 3.4.15 (see Figure 3.10). A similar argument gives

$$B(\hat{t}', [\hat{s}, \hat{t}]') \cap \widehat{\mathcal{M}}_F(X, \omega) = \varnothing.$$

Thus $(n(\hat{s}), n(\hat{t}), n([\hat{s}, \hat{t}]))$ is a permissible triple, and $\mathcal{P}(X, \omega) \subset \mathcal{P}(\mathrm{Sims}_G)$.

Now suppose that G is finite. Then the set Sims_G of all simulations is finite, and by Lemma 3.3.16, the set $\mathcal{P}(\mathrm{Sims}_G)$ is finite. Hence for each $(p, q, u) \in \mathcal{P}(\mathrm{Sims}_G)$ there is some n for which $(p, q, u) \in \mathcal{P}(\mathrm{Sims}_G^n)$. Taking N to be the maximum of these, we have $\mathcal{P}(X, \omega) \subset \mathcal{P}(\mathrm{Sims}_G) \subset \mathcal{P}(\mathrm{Sims}_G^N)$. □

3.4.5 Finiteness of lattice surfaces with given Veech groups

Throughout this subsection, let G be a set of generators of a fanning group Γ. As noted at the end of §3.4.3, to prove Theorem 3.1.2 it suffices to show that for any finite collection of simulations determined

by G, there are at most finitely many scalars for which the union of scaled simulations coincides with the marked segments of a unit-area translation surface. Propositions 3.3.14, 3.3.15 and 3.4.17 suggest that for adjacent elements $\hat{s}, \hat{t} \in \widehat{\mathcal{S}}_F(X, \omega)$ of a translation surface (X, ω) with Veech group Γ, the desired scalars for the simulations corresponding to $\hat{s}, \hat{t}$ and $[\hat{s}, \hat{t}]$ are uniquely determined by one another. Hence if the elements of $\widehat{\mathcal{S}}_F(X, \omega)$ are all—loosely speaking—'transitively adjacent' to one another (or their orientation-paired inverses), then an arbitrary choice of one scalar uniquely determines all other scalars.

We make these vague notions precise by introducing a *permissible triples graph* corresponding to a set P of permissible triples whose vertices are the distinctive, limit-point free sets corresponding to the triples and whose edges are defined between the vertices corresponding to entries of the same permissible triple in P. The 'transitive adjacency' alluded to above refers to the connectedness of such a graph.

Definition 3.4.19. Let $\mathbf{Q} = \{Q_i\}_{i \in I}$ be a collection of distinctive, limit-point free subsets of $\mathbb{P}(\mathcal{O})$. For a subset $P \subset \mathcal{P}(\mathbf{Q})$ of permissible triples, define the *permissible triples graph* determined by P to be the graph $\mathfrak{G}_P \coloneqq (\mathfrak{V}_P, \mathfrak{E}_P)$ with vertex set

$$\mathfrak{V}_P \coloneqq \left\{ Q_i, Q_j, Q_k \;\middle|\; \begin{array}{l} \text{there exists some} \\ (q_i, q_j, q_k) \in P \cap ((Q_i)_F \times (Q_j)_F \times (Q_k)_F) \end{array} \right\}$$

consisting of the sets in $\mathbf{Q}$ corresponding to the permissible triples in P, and edge set

$$\mathfrak{E}_P \coloneqq \left\{ \{Q_\ell, Q_m\} \;\middle|\; \begin{array}{l} \ell, m \in \{i, j, k\} \text{ for some} \\ (q_i, q_j, q_k) \in P \cap ((Q_i)_F \times (Q_j)_F \times (Q_k)_F) \end{array} \right\},$$

where an edge connects vertices Q_ℓ and Q_m if there is some permissible triple in P to which both Q_ℓ and Q_m correspond. We call $(r_i)_{i \in I} \in \mathbb{R}_+^{|I|}$ *consistent scalars* for the graph $\mathfrak{G}_P$ if (r_i, r_j, r_k)

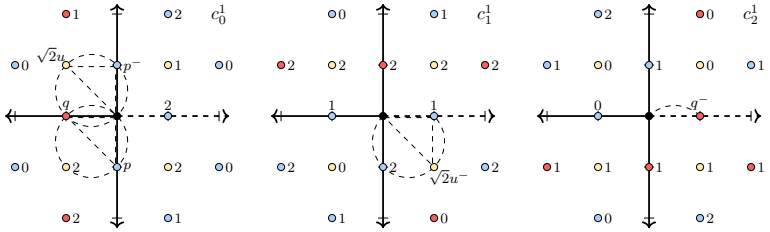


Figure 3.11: Subsets of the scaled simulations $\mathrm{sim}(\mathfrak{s}_0)$, $\mathrm{sim}(\mathfrak{s}_1)$ and $\sqrt{2}\mathrm{sim}(\mathfrak{s}_2)$, with $\mathfrak{s}_0$, $\mathfrak{s}_1$ $\mathfrak{s}_2 \in \mathfrak{S}_G$ as defined in Example 3.4.10, and balls showing that $(p, q, u) \in \mathcal{P}(\mathrm{Sims}_G)$ is a permissible triple with permissible scalars $(1, 1, \sqrt{2})$.

are permissible scalars for each permissible triple $(q_i, q_j, q_k) \in P \cap ((Q_i)_F \times (Q_j)_F \times (Q_k)_F)$. We call the graph $\mathfrak{G}_P$ *consistent* if it admits consistent scalars.

Example 3.4.20. Consider again Example 3.4.10 and Figure 3.8; let $p \in \mathrm{sim}(\mathfrak{s}_0)$ be the point with argument $3\pi/2$, $q \in \mathrm{sim}(\mathfrak{s}_1)$ the point with argument π and $u \in \mathrm{sim}(\mathfrak{s}_2)$ the point with argument $3\pi/4$. Figure 3.11 shows that $(p, q, u) \in \mathcal{P}(\mathrm{Sims}_G)$ is a permissible triple with permissible scalars $(1, 1, \sqrt{2})$. Let $P \coloneqq \{(p, q, u)\}$. Then $\mathfrak{G}_P$ is the graph with vertex set $\mathfrak{V}_P = \{\mathrm{sim}(\mathfrak{s}_0), \mathrm{sim}(\mathfrak{s}_1), \mathrm{sim}(\mathfrak{s}_2)\}$ and edge set

$$\mathfrak{E}_P = \{\{\mathrm{sim}(\mathfrak{s}_0), \mathrm{sim}(\mathfrak{s}_1)\}, \{\mathrm{sim}(\mathfrak{s}_1), \mathrm{sim}(\mathfrak{s}_2)\}, \{\mathrm{sim}(\mathfrak{s}_2), \mathrm{sim}(\mathfrak{s}_0)\}\},$$

and $(1, 1, \sqrt{2})$ are consistent scalars for $\mathfrak{G}_P$.

Proposition 3.4.21. *Let (X, ω) be a fanning surface with Veech group $\mathrm{SL}(X, \omega) = \Gamma$, and let $P \coloneqq \mathcal{P}(X, \omega) \subset \mathcal{P}(Sims_G)$. Then the permissible triples graph $\mathfrak{G}_P$ is connected and consistent, and if $(r_{\mathfrak{s}})_{sim(\mathfrak{s}) \in \mathfrak{V}_P}$ are consistent scalars for $\mathfrak{G}_P$, then any other consistent scalars are of the form $(ar_{\mathfrak{s}})_{sim(\mathfrak{s}) \in \mathfrak{V}_P}$ with $a \in \mathbb{R}_+$.*

Proof. Theorem 3.4.14 implies that the vertex set $\mathfrak{V}_P$ is finite, say $\mathfrak{V}_P = \{\mathrm{sim}(\mathfrak{s}_i)\}_{i=1}^n$. The same theorem also gives that

$$\bigcup_{i=1}^{n} r_i \mathrm{sim}(\mathfrak{s}_i) \subset \widehat{\mathcal{M}}(X, \omega)$$

for some $(r_i)_{i=1}^n \in \mathbb{R}_+^n$. In fact, each r_i is the minimal length of a pair in the $\mathrm{Aff}_{\mathcal{O}}^+(X, \omega)$-orbit for which $r_i \mathrm{sim}(\mathfrak{s}_i) = [\{\hat{s}, \hat{s}'\}]$. The proof of Proposition 3.4.17 shows that the scalars $(r_i)_{i=1}^n \in \mathbb{R}_+^n$ are consistent, so the graph $\mathfrak{G}_P$ is consistent.

Now suppose that $(r_i')_{i=1}^n \in \mathbb{R}_+^n$ are also consistent scalars for $\mathfrak{G}_P$, and recall the definition of the edge set $\mathfrak{E}_P$. Induction with Propositions 3.3.14 and 3.3.15 implies that any of the scalars r_i' corresponding to a simulation of a connected component of $\mathfrak{G}_P$ uniquely determines the remaining scalars r_j' of the simulations of the component. In particular, if $r_i' = ar_i$ for some $a \in \mathbb{R}_+$, then $r_j' = ar_j$ for each of these scalars. Hence it suffices to show that $\mathfrak{G}_P$ is connected.

Consider the subgraph (V, E) of $(\mathfrak{V}_P, \mathfrak{E}_P)$ with vertex set V consisting of all simulations in $\mathfrak{V}_P$ which contain a pair of normalised marked Voronoi staples and edge set E consisting of edges $\{\mathrm{sim}(\mathfrak{s}_i), \mathrm{sim}(\mathfrak{s}_j)\}$, where $\mathrm{sim}(\mathfrak{s}_i)$ and $\mathrm{sim}(\mathfrak{s}_j)$ contain $\{n(\hat{s}), n(\hat{s})'\}$ and $\{n(\hat{t}), n(\hat{t})'\}$, respectively, for adjacent $\hat{s}, \hat{t} \in \widehat{\mathcal{S}}_F(X, \omega)$. Recall that (X, ω) is isometric to the quotient space of $\bigsqcup_{i=1}^{\kappa} \Omega_i$ under the equivalence relation given by identifying the two edges of convex bodies which correspond to a pair of marked Voronoi staples (§3.3.3 above, and also §3 of [64]). If (V, E) were disconnected, then after this identification of edges of convex bodies the resulting surface—and hence (X, ω)—would be disconnected. This is a contradiction, so the subgraph (V, E) is connected. By definition of $P = \mathcal{P}(X, \omega)$ and $\mathfrak{V}_P$, any vertex $\mathrm{sim}(\mathfrak{s}_i) \in \mathfrak{V}_P \backslash V$ must contain $\{n([\hat{s}, \hat{t}]), n([\hat{s}, \hat{t}])'\}$ for some adjacent $\hat{s}, \hat{t} \in \widehat{\mathcal{S}}_F(X, \omega)$. By definition of $\mathfrak{E}_P$, there are edges of $\mathfrak{G}_P$ from $\mathrm{sim}(\mathfrak{s}_i)$ to the simulations containing $\{n(\hat{s}), n(\hat{s})'\}$ and $\{n(\hat{t}), n(\hat{t})'\}$, both of which belong to $V \subset \mathfrak{V}_P$. Thus $\mathfrak{G}_P$ is connected, and the result follows. □

We are now ready to prove Theorem 3.1.2:

Proof of Theorem 3.1.2. Let $\Gamma \leq \mathrm{SL}_2\mathbb{R}$ be a lattice. By Proposition 3.4.4, Γ is a finitely generated fanning group. Let G be a finite set of generators of Γ and fix $\mathcal{H}_1(d_1, \ldots, d_\kappa)$. We must show that the set of all $(X, \omega) \in \mathcal{H}_1(d_1, \ldots, d_\kappa)$ for which $\mathrm{SL}(X, \omega) = \Gamma$ is finite. By Proposition 3.4.17, each of these translation surfaces belongs to the set

$$\{(X, \omega) \in \mathcal{H}_1(d_1, \ldots, d_\kappa) \mid \mathcal{P}(X, \omega) \subset \mathcal{P}(\mathrm{Sims}_G)\},$$

so it suffices to show that this latter set is finite. By Lemmas 3.3.16 and 3.4.11, the set $\mathcal{P}(\mathrm{Sims}_G)$ of all permissible triples is finite, so it suffices to show for any $P \subset \mathcal{P}(\mathrm{Sims}_G)$ that the set

$$\{(X, \omega) \in \mathcal{H}_1(d_1, \ldots, d_\kappa) \mid \mathcal{P}(X, \omega) = P\}$$

is finite. Let $\mathfrak{V}_P = \{\mathrm{sim}(\mathfrak{s}_i)\}_{i=1}^n$ be the vertex set of the permissible triples graph $\mathfrak{G}_P$. As in the proof of Proposition 3.4.21, for each (X, ω) with $\mathcal{P}(X, \omega) = P$, there are consistent scalars $(r_i)_{i=1}^n \in \mathbb{R}_+^n$ for which

$$\bigcup_{i=1}^n r_i \mathrm{sim}(\mathfrak{s}_i) \subset \widehat{\mathcal{M}}(X, \omega).$$

By definition of $\mathfrak{G}_P$ and its vertex set $\mathfrak{V}_P$, the set of marked Voronoi staples, $\widehat{\mathcal{S}}(X, \omega)$—from which (X, ω) may be recovered—is contained in this union of scaled simulations. Again from Proposition 3.4.21, if $(Y, \eta) \in \mathcal{H}_1(d_1, \ldots, d_\kappa)$ also satisfies $\mathcal{P}(Y, \eta) = P$, then its corresponding consistent scalars are of the form $(ar_i)_{i=1}^n$ for some $a \in \mathbb{R}_+^n$. From §3.3.3, we have

$$\bigsqcup_{j=1}^\kappa \Omega_j(\widehat{\mathcal{S}}_F(X, \omega)) = \bigsqcup_{j=1}^\kappa \Omega_j \left(\bigcup_{i=1}^n r_i \mathrm{sim}_F(\mathfrak{s}_i) \right)$$

and

$$\bigsqcup_{j=1}^{\kappa} \Omega_j(\widehat{\mathcal{S}}_F(Y,\eta)) = \bigsqcup_{j=1}^{\kappa} \Omega_j \left(\bigcup_{i=1}^{n} a r_i \mathrm{sim}_F(\mathfrak{s}_i) \right)$$
$$= a \left(\bigsqcup_{j=1}^{\kappa} \Omega_j \left(\bigcup_{i=1}^{n} r_i \mathrm{sim}_F(\mathfrak{s}_i) \right) \right).$$

Since (X,ω) and (Y,η) are isometric to the quotient spaces of these respective sets under an equivalence relation identifying edges, and since both translation surfaces have area one, we conclude that $a = 1$ and thus $(X,\omega) = (Y,\eta)$. □

3.5 The algorithm

The constructive results of the previous sections suggest an algorithm (Algorithm 3.1.1) which returns all translation surfaces with a given lattice Veech group Γ in any given stratum $\mathcal{H}_1(d_1,\ldots,d_\kappa)$. We highlight the main steps of this algorithm here, providing necessary references and further commentary in italics. Recall that G is a given finite set of generators of the lattice $\Gamma \leq \mathrm{SL}_2\mathbb{R}$.

(1) Compute a finite set $\Theta_{\Gamma_m} \subset S^1$ containing Θ_Γ, and let

$$\mathfrak{S}_G \coloneqq \{(H,\{p,p^-\}) \in 2^{G_\mathcal{O}} \setminus \{\varnothing\} \times \mathbb{P}(\mathcal{O}) \mid \boldsymbol{\pi}(p), \boldsymbol{\pi}(p^-) \in \Theta_{\Gamma_m}\}.$$

(*See Definition 3.4.9 and the remark following Lemma 3.4.11.*)

(2) For each $n \in \mathbb{N}$:

   (a) Compute the set $\mathrm{Sims}_G^n = \{\mathrm{sim}^n(\mathfrak{s}) \mid \mathfrak{s} \in \mathfrak{S}_G\}$ of all stage n distinctive simulations determined by G.
   (*See Defintion 3.4.9.*)

   (b) Compute the set $\mathcal{P}(\mathrm{Sims}_G^n)$ of all permissible triples arising from Sims_G^n along with their respective rays of permissible scalars.

(*See Definition 3.3.13 and Proposition 3.3.14. Recall from Definition 3.4.16 and Proposition 3.4.17 that there is some* $N \in \mathbb{N}$ *such that for any* $(Y,\eta) \in \mathcal{H}_1(d_1,\dots,d_\kappa)$ *with* $\mathrm{SL}(Y,\eta) = \Gamma$*, the set* $\mathcal{P}(Y,\eta)$ *determined by the marked Voronoi staples* $\widehat{\mathcal{S}}(Y,\eta)$ *is contained in the finite set of permissible triples* $\mathcal{P}(\mathit{Sims}_G^N)$.)

(c) For each $P \subset \mathcal{P}(\mathrm{Sims}_G^n)\backslash\{\varnothing\}$:
(*For* $n = N$ *and any* (Y,η) *as above, this loop will set* $P = \mathcal{P}(Y,\eta)$ *for some* P.)

(i) Construct the permissible triples graph $\mathfrak{G}_P = (\mathfrak{V}_P, \mathfrak{E}_P)$. If $\mathfrak{G}_P$ is connected and consistent with consistent scalars $(r_{\mathfrak{s}})_{\mathrm{sim}^n(\mathfrak{s})\in\mathfrak{V}_P^n}$, then set

$$S := \bigcup_{\mathrm{sim}^n(\mathfrak{s})\in\mathfrak{V}_P} r_{\mathfrak{s}}\mathrm{sim}^n(\mathfrak{s}),$$

construct the convex bodies $\Omega_i(S_F)$ and determine the corresponding essential points $\mathcal{E}_i(S_F)$ for each $i \in \{1,\dots,\kappa\}$.
(*See Definition 3.4.19 and §3.3.3.*)

(ii) If each convex body $\Omega_i(S_F)$ is compact, the essential points $\sqcup_{i=1}^{\kappa}\mathcal{E}_i(S_F)$ come in pairs $\{p,p^-\}$ determined by the simulations $\mathrm{sim}^n(\mathfrak{s}) \in \mathfrak{V}_P$, and the edges of the convex bodies determined by p and p^- are equal length, then construct the translation surface (X,ω) by identifying these edges via translation.
(*See §3.3.3. When* $P = \mathcal{P}(Y,\eta)$*, the proof of Proposition 3.4.21 shows that for some* $a \in \mathbb{R}_+$*, the set* aS *contains the set of marked Voronoi staples* $\widehat{\mathcal{S}}(Y,\eta)$*, and thus the union* $\sqcup_{i=1}^{\kappa}a\mathcal{E}_i(S_F)$ *of scaled essential points is precisely* $\widehat{\mathcal{S}}_F(Y,\eta)$*. Hence* (X,ω) *is a scaled version of* (Y,η).)

(iii) Rescale (X,ω) if necessary so that it has area one, and verify that $\mathrm{SL}(X,\omega) = \Gamma$.

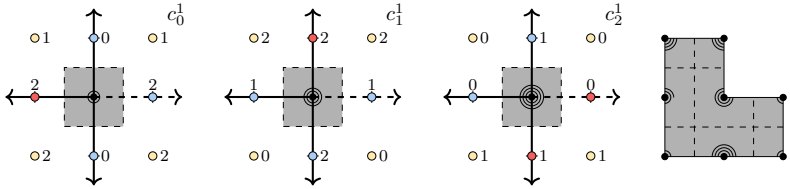


Figure 3.12: The convex body and resulting translation surface (X, ω) (with opposite sides identified) using the scaled simulations $\mathrm{sim}(\mathfrak{s}_0)$, $\mathrm{sim}(\mathfrak{s}_1)$ and $\sqrt{2}\mathrm{sim}(\mathfrak{s}_2)$ from Examples 3.4.10 and 3.4.20. The arcs about the singularity of $\mathcal{O}$ and of (X, ω) indicate arguments of generalised polar coordinates.

(*Computing $SL(X, \omega)$ is done using one of the algorithms mentioned in §3.1.*)

Example 3.5.1. Examples 3.4.10 and 3.4.20 illustrate (parts of) Steps 1, 2.a, 2.b and 2.c.i of the algorithm. Figure 3.12 illustrates the construction of convex bodies and corresponding essential points in Step 2.c.i as well as the resulting translation surface (X, ω) constructed in Step 2.c.ii. Notice that the essential points come only from $\mathrm{sim}(\mathfrak{s}_0)$ and $\mathrm{sim}(\mathfrak{s}_1)$; however, $\mathrm{sim}(\mathfrak{s}_2)$ was necessary to determine the permissible scalars $(1, 1, \sqrt{2})$ (Example 3.4.20). Rescaling to unit-area, one verifies that indeed $\mathrm{SL}(X, \omega) = \langle S, T^2 \rangle$.

A number of improvements to Algorithm 3.1.1 can be made for computational efficiency. We mention three in particular:

- It is optimal to choose a minimal set of generators G for Γ, as the size of $\mathfrak{S}_G$—and hence of Sims_G—depends on $|G|$.
- Recall from Proposition 3.3.17 that if $\hat{s}, \hat{t} \in \widehat{\mathcal{M}}_F(X, \omega)$ are two marked segments with identical arguments in the same component $\mathcal{O}_i \subset \mathcal{O}$, then in fact $\{\hat{s}, \hat{s}'\} = \{\hat{t}, \hat{t}'\}$. This requirement restricts the permissible triples graphs that one needs to consider in step 2.c.

- In Step 2.b, one need only consider permissible triples (p, q, u) in $\mathrm{sim}_F^n(\mathfrak{r}) \times \mathrm{sim}_F^n(\mathfrak{s}) \times \mathrm{sim}_F^n(\mathfrak{t})$ where $\mathfrak{r}$, $\mathfrak{s}$ and $\mathfrak{t}$ are defined using the same subset $H \subset G_{\mathcal{O}}$. A similar restriction applies to the subsets $P \subset \mathcal{P}(\mathrm{Sims}_G^n)$ considered in Step 2.c.

While Algorithm 3.1.1 does produce the finite set of lattice surfaces asserted by Theorem 3.1.2 in finite time, it does not give a halting criterion to determine when the set of returned translation surfaces is exhaustive. An explicit test to determine the $N \in \mathbb{N}$ for which $\mathcal{P}(\mathrm{Sims}_G) \subset \mathcal{P}(\mathrm{Sims}_G^N)$ would give such a halting criterion (see Proposition 3.4.17). In particular, Algorithm 3.1.1 together with this halting criterion would give a general procedure to determine whether or not any given lattice group is realised as a Veech group in any given stratum. However, even without this criterion, the ideas of this chapter may still be used in special cases to show that certain lattice groups are not realised in certain strata; see §3.6.

As the set of all strata is countable, Algorithm 3.1.1—if allowed to run indefinitely—is easily adapted to produce all unit-area translation surfaces with a given lattice Veech group in *any* stratum. A full implementation could also provide experimental evidence for lattices which are never realised as Veech groups, though—even with the aforementioned stratum-wise halting criterion—a criterion to definitively assert that a lattice is not a Veech group in any stratum seems to the author to be elusive.

We also mention that as the set of generators for any fanning group is at most countable, Algorithm 3.1.1 may be adapted to construct surfaces (X, ω) with given fanning Veech groups in given strata. However, for countably infinite generating sets G, the set $\mathfrak{S}_G$ is uncountable, so a construction of such surfaces may not be exhaustive.

3.6 The modular group in minimal strata

Here, we give an example showing how the ideas of the previous sections may be used—in certain cases—to give obstructions for lat-

tices being realised as Veech groups in strata.[3] In particular, we show that the square torus is the only translation surface with Veech group $\mathrm{SL}_2\mathbb{Z}$ in the collection of all minimal strata $\mathcal{H}(2g-2),\ g>0$. Throughout this section, fix $g>0$ and $(X,\omega)\in\mathcal{H}(2g-2)$. Let $\sigma:=\sigma_1$ be the sole singularity of (X,ω), $\Omega:=\Omega_1$ the convex body subordinate to $\widehat{\mathcal{M}}_F(X,\omega)$ and $\mathcal{O}=\mathcal{O}(2g-2)$ the canonical surface associated to $\mathcal{H}(2g-2)$. We begin with two lemmas.

Lemma 3.6.1. *Let $\widehat{\mathcal{S}}(X,\omega)=\{\{\hat{s}_i,\hat{s}_i'\}\}_{i=1}^n$ be the marked Voronoi staples of (X,ω). If the counterclockwise angle measured from $\hat{s}_i$ to $\hat{s}_i'$ equals that measured from $\hat{s}_i'$ to $\hat{s}_i$ for each i, then $n=2g$ for n even and $n=2g+1$ for n odd.*

Proof. Since $\widehat{\mathcal{S}}_F(X,\omega)$ is the set of all essential points of Ω, the convex body Ω has $2n$ edges. Decompose Ω into $2n$ closed triangular regions with disjoint interiors, where each triangle has one vertex at the origin $0\in\mathcal{O}$, and remaining two vertices at the endpoints of an edge of Ω. The sum of the interior angles of all triangles in this decomposition is $2n\pi$. The origin $0\in\mathcal{O}$ is a singularity of cone angle $2(2g-1)\pi$, so the sum of the interior angles of Ω is

$$2n\pi-2(2g-1)\pi=2(n-2g+1)\pi.$$

Recall that (X,ω) is isometric to the quotient space of Ω under the equivalence relation given by identifying edges determined by the marked Voronoi staples $\widehat{\mathcal{S}}(X,\omega)$, and the vertices of Ω under this identification correspond to Voronoi 0-cells of (X,ω) (see §3.3.3). In particular, the sum of interior angles of Ω which are identified as a single point must equal 2π. Our assumptions on the counterclockwise angles between $\hat{s}_i$ and $\hat{s}_i'$ and between $\hat{s}_i'$ and $\hat{s}_i$ imply that for n even, the vertices of Ω become a single point under the identification, while for n odd, the vertices of Ω become two distinct points under the identification (the argument here is analogous to

[3] Due to time constraints, we have only considered the group $\mathrm{SL}_2\mathbb{Z}$ in minimal strata. However, the author suspects that similar methods can be applied more generally to obtain further restrictions for different groups and strata.

that which shows the translation surface constructed by identifying opposite edges of a regular $2m$-gon has one singularity when m is even and two singularities when m is odd). In particular, for n even,

$$2(n-2g+1)\pi = 2\pi$$

implies $n = 2g$, while for n odd,

$$2(n-2g+1)\pi = 4\pi$$

implies $n = 2g+1$. □

Lemma 3.6.2. *Suppose that the minimal length of a saddle connection on (X,ω) is one. If the set*

$$P \coloneqq \{p \in \mathcal{O} \mid |p| = 1,\ arg(\boldsymbol{\pi}(p)) \in \{0, \pi/2, \pi, 3\pi/2\}\}$$

belongs to the set of marked segments $\widehat{\mathcal{M}}_F(X,\omega)$, then in fact $P = \widehat{\mathcal{S}}_F(X,\omega)$.

Proof. It suffices to show that P is the set of essential points of the convex body Ω, i.e., that (i) $\Omega = \cap_{p\in P} H(p)$ and (ii) for any $q \in P$, $\Omega \subsetneq \cap_{p\in P\setminus\{q\}} H(p)$. Condition (ii) is immediate from the fact that Ω is compact while $\cap_{p\in P\setminus\{q\}} H(p)$ is non-compact for any $q \in P$. Now

$$\Omega = \bigcap_{\hat{s}\in\widehat{\mathcal{M}}_F(X,\omega)} H(\hat{s}) = \left(\bigcap_{p\in P} H(p)\right) \cap \left(\bigcap_{\hat{s}\in\widehat{\mathcal{M}}_F(X,\omega)\setminus P} H(\hat{s})\right),$$

so it suffices to show for each marked segment $\hat{s} \in \widehat{\mathcal{M}}_F(X,\omega)\setminus P$, that $\cap_{p\in P} H(p) \subset H(\hat{s})$.

We claim that for each $p \in P$, the ball centered at p of radius one contains no marked segments other than p, i.e.,

$$B_1(p) \cap \widehat{\mathcal{M}}_F(X,\omega)\setminus\{p\} = \varnothing.$$

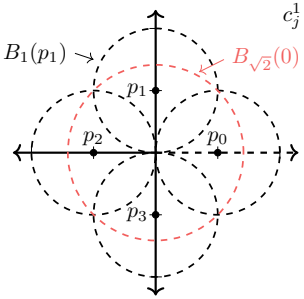


Figure 3.13: The ball $B_{\sqrt{2}}(0)$ is contained in the union of balls $B_1(p)$, $p \in P$. Note that this figure shows only a portion of the ball $B_{\sqrt{2}}(0)$, as this ball is centered at the singularity $0 \in \mathcal{O}$. Similarly, only the upper half of $B_1(p_0)$ is shown.

Suppose on the contrary that the intersection is non-empty. As in the proof of Proposition 3.4.17, we can choose some $\hat{u}$ in the intersection with argument nearest that of p and find that $[p, \hat{u}]$ is a marked segment of length strictly less than one. Its corresponding saddle connection also has length strictly less than one, contrary to our assumptions, so the claim holds.

Next, we claim that the ball centered at the origin of radius $\sqrt{2}$ is contained in the union of balls of radius one centered at the various $p \in P$,

$$B_{\sqrt{2}}(0) \subset \bigcup_{p \in P} B_1(p).$$

This is clear from elementary Euclidean geometry, restricting to each 2π-sector c_j^1; see Figure 3.13. In particular, the previous two claims imply that

$$B_{\sqrt{2}}(0) \cap \left(\widehat{\mathcal{M}}_F(X, \omega) \backslash P \right) = \varnothing. \tag{3.4}$$

Notice that the maximum distance from the origin to any point in $\cap_{p \in P} H(p)$ is $\sqrt{2}/2$ (this distance being realised at the vertices

of this intersection). Let $\hat{s} \in \widehat{\mathcal{M}}_F(X,\omega)\backslash P$ and let q belong to the complement of $H(\hat{s})$. Certainly $|q| > |\hat{s}|/2$. Equation (3.4) implies $|\hat{s}| \geq \sqrt{2}$, so $|q| > \sqrt{2}/2$. Hence $q \notin \cap_{p\in P} H(p)$, so the complement of $H(\hat{s})$ does not intersect $\cap_{p\in P} H(p)$. Thus $\cap_{p\in P} H(p) \subset H(\hat{s})$ as desired. □

With these two lemmas, we now prove the main result of this section.

Proof of Theorem 3.1.3. Let $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, fix $\tau_S, \tau_T \in \mathrm{Trans}(\mathcal{O})$, and set $g_S = \tau_S \circ f_S$ and $g_T = \tau_T \circ f_T$ where $f_S, f_T \in \mathrm{Aff}^+_{\mathbb{C}}(\mathcal{O})$ (see Definition 3.3.3). We first show that all elements of

$$P \coloneqq \{p \in \mathcal{O} \mid |p| = 1,\ \arg(\boldsymbol{\pi}(p)) \in \{0, \pi/2, \pi, 3\pi/2\}\}$$

belong to the same $\langle g_S, g_T\rangle$-orbit. Note that $\mathrm{Trans}(\mathcal{O}) = \langle \rho_1 \rangle$; let $n \in \mathbb{Z}_{2g-1}$ be such that $\tau_T = \rho_1^n$. Set

$$P' \coloneqq \{p \in \mathcal{O} \mid |p| = \sqrt{2},\ \arg(\boldsymbol{\pi}(p)) \in \{\pi/4, 3\pi/4, 5\pi/4, 7\pi/4\}\}.$$

Beginning with $p_0 \in P \cap c_0^1$, $\arg(p_0) = 0$ and sweeping counterclockwise, label the elements of $P \cup P'$ as $p_0, p_1, \ldots, p_{16g-9}$; see Figure 3.14. For each $k \in \mathbb{Z}_{16g-8}$, we have

$$f_S(p_k) = p_{k+2},$$

and for $k \not\equiv 1 \pmod 4$,

$$f_T(p_k) = \begin{cases} p_k & \text{if } k \equiv 0 \pmod 4, \\ p_{k-1} & \text{if } k \equiv 2, 3 \pmod 4. \end{cases}$$

Moreover, for each $k \in \mathbb{Z}_{16g-8}$,

$$\tau_T(p_k) = p_{k+8n}.$$

Set $f \coloneqq g_T^{-1} \circ g_S^{-1} \circ g_T^{-1} \circ g_S$ and $h \coloneqq g_S^{-1} \circ g_T^{-1} \circ g_S \circ g_T^{-1}$. Using the relations above—together with Lemma 3.3.4 and the fact that

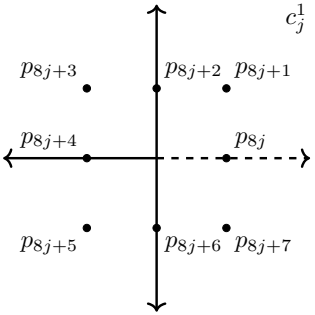


Figure 3.14: Labelling of the points $p_0, \ldots, p_{16g-9}$ of $P \cup P'$ for $j \in \mathbb{Z}_{2g-1}$.

$\mathrm{Trans}(\mathcal{O}) = \langle \rho_1 \rangle$ is abelian—we find for each $k \in \mathbb{Z}_{16g-8}$ that

$$\begin{aligned} f(p_{4k}) &= g_T^{-1} \circ g_S^{-1} \circ g_T^{-1} \circ g_S(p_{4k}) \\ &= (f_T^{-1} \circ \tau_T^{-1}) \circ (f_S^{-1} \circ \tau_S^{-1}) \circ (f_T^{-1} \circ \tau_T^{-1}) \circ (\tau_S \circ f_S)(p_{4k}) \\ &= \tau_T^{-2} \circ f_T^{-1} \circ f_S^{-1} \circ f_T^{-1} \circ f_S(p_{4k}) \\ &= \tau_T^{-2} \circ f_T^{-1} \circ f_S^{-1} \circ f_T^{-1}(p_{4k+2}) \\ &= \tau_T^{-2} \circ f_T^{-1} \circ f_S^{-1}(p_{4k+3}) \\ &= \tau_T^{-2} \circ f_T^{-1}(p_{4k+1}) \\ &= \tau_T^{-2}(p_{4k+2}) \\ &= p_{4k-16n+2} \end{aligned}$$

and

$$\begin{aligned} h(p_{4k+2}) &= g_S^{-1} \circ g_T^{-1} \circ g_S \circ g_T^{-1}(p_{4k+2}) \\ &= (f_S^{-1} \circ \tau_S^{-1}) \circ (f_T^{-1} \circ \tau_T^{-1}) \circ (\tau_S \circ f_S) \circ (f_T^{-1} \circ \tau_T^{-1})(p_{4k+2}) \\ &= \tau_T^{-2} \circ f_S^{-1} \circ f_T^{-1} \circ f_S \circ f_T^{-1}(p_{4k+2}) \\ &= \tau_T^{-2} \circ f_S^{-1} \circ f_T^{-1} \circ f_S(p_{4k+3}) \end{aligned}$$

$$\begin{aligned}
&= \tau_T^{-2} \circ f_S^{-1} \circ f_T^{-1}(p_{4k+5}) \\
&= \tau_T^{-2} \circ f_S^{-1}(p_{4k+6}) \\
&= \tau_T^{-2}(p_{4k+4}) \\
&= p_{4k-16n+4}.
\end{aligned}$$

Then

$$\begin{aligned}
g_T^8 \circ h \circ f \circ h \circ f(p_0) &= g_T^8 \circ h \circ f \circ h(p_{-16n+2}) \\
&= g_T^8 \circ h \circ f(p_{-32n+4}) \\
&= g_T^8 \circ h(p_{-48n+6}) \\
&= g_T^8(p_{-64n+8}) \\
&= \tau_T^8 \circ f_T^8(p_{-64n+8}) \\
&= \tau_T^8(p_{-64n+8}) \\
&= p_8.
\end{aligned}$$

Iterating this latter composition shows that each p_{8k} belongs to the same $\langle g_S, g_T \rangle$-orbit. Injectivity of g_S implies that the union of the images under g_S, g_S^2 and g_S^3 of these p_{8k} is all of P, as claimed.

Now suppose $(X, \omega) \in \mathcal{H}(2g-2)$ with $\mathrm{SL}(X, \omega) = \mathrm{SL}_2\mathbb{Z} = \langle S, T \rangle$ for some $g > 0$. Normalise (X, ω) if necessary so that its shortest saddle connection has unit length. One computes

$$S^1 \backslash (T \cdot \Delta \cup T^{-1} \cdot \Delta \cup ST \cdot \Delta \cup ST^{-1} \cdot \Delta) = \{(\pm 1, 0), (0, \pm 1)\}.$$

This set contains $\Theta_{\mathrm{SL}_2\mathbb{Z}} = \Theta_{\mathrm{SL}(X,\omega)}$, so Corollary 3.4.8 implies that the shortest saddle connection of (X, ω) is horizontal or vertical. Since the orthogonal matrix S belongs to $\mathrm{SL}(X, \omega)$ and any affine automorphism of (X, ω) sends saddle connections to saddle connections, (X, ω) has both horizontal and vertical saddle connections of minimal length one.

Recycling notation, let $f \in \mathrm{Aff}^+(\mathcal{O})$ and $\{p, p^-\} \in \mathbb{P}(\mathcal{O})$. Lemma 3.3.2 together with the fact that f is a self-homeomorphism of $\mathcal{O} = \mathcal{O}(2g-2)$ implies that the counterclockwise angle measured from p

to p^- equals that measured from $f(p)$ to $f(p^-)$. In particular, this is true of any $f \in \mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ and horizontal, unit-length $\{\hat{s},\hat{s}'\} \in \widehat{\mathcal{M}}(X,\omega)$. Since $S,T \in \mathrm{SL}(X,\omega)$, there exist $g_S, g_T \in \mathrm{Aff}^+_{\mathcal{O}}(X,\omega)$ with $\mathrm{der}(g_S) = S$, $\mathrm{der}(g_T) = T$ (see §3.4.2). The argument above shows that the set P belongs to the $\langle g_S, g_T\rangle$-orbit of $\hat{s}$. Since this orbit belongs to $\widehat{\mathcal{M}}_F(X,\omega)$, Lemma 3.6.2 implies that $P = \widehat{\mathcal{S}}_F(X,\omega)$. Moreover, $\hat{s}'$ belongs to this orbit, so the observation above implies that the counterclockwise angle measured from $\hat{s}$ to $\hat{s}'$ equals that measured from $\hat{s}'$ to $\hat{s}$, and the same is true of any pair of marked Voronoi staples in $\widehat{\mathcal{S}}(X,\omega)$. Since

$$|\widehat{\mathcal{S}}(X,\omega)| = |\widehat{\mathcal{S}}_F(X,\omega)|/2 = |P|/2 = 2(2g-1)$$

is even, Lemma 3.6.1 implies that $2(2g-1) = 2g$, which is only true for $g = 1$. When $g = 1$, the convex body Ω is the unit-square centered at $0 \in \mathcal{O} = (\mathbb{C}, dz)$, and the reconstruction of (X,ω) from Ω gives the square torus. □

Discussion

We conclude with a summary of the major results from each chapter and directions for future work.

Chapter 1 – Inducing contractions and a unifying metrical theory for continued fractions

In Chapter 1 we couple two acceleration techniques—*induced transformations* and *contractions*—in the setting of Ito's natural extension of the Farey tent map. The resulting *contracted Farey expansions* yield a large collection of new (e.g., the *superoptimal continued fractions* of §1.9) and well-studied continued fraction algorithms. In particular, contracted Farey expansions include regular continued fractions (RCFs), Kraaikamp's large class of S-expansions and Nakada's α-continued fractions for all $0 < \alpha \leq 1$ (§1.7).

Moreover, induced transformations of Ito's natural extension are shown to provide a unifying theory for metrical results on RCF-convergents and mediant convergents (§1.8). In particular, we realise and extend classical results of Legendre, Lévy, Bosma–Jager–Wiedijk and their analogues for RCF-convergents and mediants in a single, cohesive framework.

The results of this chapter lead to a number of new directions and questions for future work, including:

(i) Employ similar methods (induced transformations governing

contractions) to other continued fraction algorithms producing expansions with non-integer digits, e.g., (α-) Rosen continued fractions and Hurwitz's complex continued fractions.

(ii) Investigate the connection between our induced systems $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ and geodesic flow on the modular surface, as established for the Gauss map ([174]), the Farey tent map ([82]) and the α-continued fraction maps ([4]).

(iii) Classify isomorphic induced systems $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$. (Since the natural extensions of Nakada's α-continued fractions arise as such systems, this would answer an open problem posed in [120].)

(iv) Use skew products of Ito's natural extension and its induced systems—analogous to the skew products introduced in [48] (see also [99])—to study subsequences of RCF-convergents and mediants with numerators and denominators belonging to certain arithmetic progressions. (Suggested by V. Berthé at the workshop, *Continued fractions: multifractal and dynamical aspects*, CIRM, September 2023.)

(v) Study normality for contracted Farey expansions. (Suggested by T. A. Schmidt in Delft, July 2024.) In particular, is the set of normal numbers with respect to $(R, \mathcal{B}, \bar{\mu}_R, \mathcal{F}_R)$ independent of the choice of a $\bar{\mu}$-continuity, proper inducible set $R \subset \Omega$? (See [116] and [51], where the analogous question is answered affirmatively for S-expansions and α-continued fractions.)

Chapter 2 – Ergodic properties of a parameterised family of symmetric golden maps

In this chapter we introduce a new, one-parameter family of piecewise-linear interval maps $\{T_\alpha\}_{\alpha\in[1,\beta]}$, $\beta = (\sqrt{5}+1)/2$, called *skewed symmetric golden maps*. Each T_α generates a unique expansion of every number $x \in [-1, 1]$ whose digits depend on the T_α-orbit of x.

Theorem 2.1.1 asserts the existence of a unique, absolutely continuous invariant probability measure μ_α for each T_α, $\alpha > 1$. On a set of full-measure—the union of *matching intervals*—the density of μ_α is an explicitly determined step function. Using these measures, we study the asymptotic relative frequency of the digit 0 in Lebesgue–a.e. T_α-expansion, show that this frequency varies continuously as a function of α, and show that the frequency function is monotone on each matching interval (Theorem 2.1.2).

These results are extracted from analogous results proven for the simpler family $\{S_\alpha\}_{\alpha\in[1,\beta]}$ of *symmetric golden maps* of constant slope β, for which the skewed symmetric golden maps are jump transformations. The central tool used to determine the density of the absolutely continuous invariant probability measure for S_α is the property of *matching*, where at each critical point of S_α, the orbits of the left- and right-limits agree after some finite number of steps. These explicit, invariant measures are used together with Birkhoff's ergodic theorem to determine the asymptotic relative frequency of 0 in a.e. S_α-expansion.

This chapter opens several directions for further study:

(i) For each $m \in (1,2]$ and $\alpha \in [1,m]$, define $S_{\alpha,m} : [-1,1] \to [-1,1]$ by

$$S_{\alpha,m}(x) = \begin{cases} mx + \alpha, & x \in [-1, -1/m), \\ mx, & x \in [-1/m, 1/m], \\ mx - \alpha, & x \in (1/m, 1]. \end{cases}$$

Note that when $m = \beta$, the one-parameter family $\{S_{\alpha,\beta}\}_{\alpha\in[1,\beta]}$ coincides with the symmetric golden maps, while for $m = 2$, $\{S_{\alpha,2}\}_{\alpha\in[1,2]}$ gives the symmetric doubling maps from [45]. For which other slopes m can matching be used to obtain results analogous to Theorems 2.1.1 and 2.1.2?

(ii) Is there an explicit, orientation-preserving homeomorphism between $[0,1]$ and $[1,\beta]$ which sends the matching intervals of

Nakada's α-continued fraction maps to the matching intervals of the (skewed) symmetric golden maps? (Such a homeomorphism exists between the parameter spaces of Nakada's α-continued fraction maps and the symmetric doubling maps; see §3.2 of [45].)

(iii) Determine the Hausdorff dimension of the set of non-matching parameters for the (skewed) symmetric golden maps.

Chapter 3 – Constructing lattice surfaces with prescribed Veech groups: an algorithm

Chapter 3 presents a novel algorithm to construct lattice translation surfaces with given Veech groups (Algorithm 3.1.1). In developing this algorithm, we give a new proof of a result of Smillie and Weiss ([179]), which states that there are at most finitely many unit-area lattice surfaces with a given Veech group in any given stratum (Theorem 3.1.2). Moreover, we prove that the square torus is the only translation surface in any minimal stratum whose Veech group is the modular group (Theorem 3.1.3).

As with Chapters 1 and 2, several directions for future work present themselves here:

(i) Determine a halting criterion for Algorithm 3.1.1.

(ii) Fully implement Algorithm 3.1.1 in, e.g., SageMath ([181]).

(iii) Can Algorithm 3.1.1 be extended to non-lattices? For instance, to groups $\Gamma \leq \mathrm{SL}_2\mathbb{R}$ containing two transverse parabolic elements (i.e., to construct *prelattice*, or *bouillabaisse*, surfaces)?

(iv) Can methods similar to those in §3.6 be used to classify all translation surfaces with Veech group $\mathrm{SL}(X, \omega) = \mathrm{SL}_2\mathbb{Z}$? Can obstructions/classifications of other groups $\Gamma \leq \mathrm{SL}_2\mathbb{R}$ realisable as Veech groups be found with these methods?

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Nederlandse samenvatting

Dynamica van getaluitbreidingen en translatieoppervlakken

Dit proefschrift bestaat uit drie hoofdstukken. De eerste twee gaan over de dynamica van verschillende typen getaluitbreidingen, en het derde gaat over de dynamica van een $\mathrm{SL}_2\mathbb{R}$-actie op families van translatieoppervlakken.

Hoofdstuk 1 – Het induceren van contracties en een verenigende metrische theorie voor kettingbreuken

Het eerste hoofdstuk introduceert een nieuwe, grote klasse van kettingbreuk algoritmen die zogenaamde *gecontracteerde Farey uitbreidingen* produceren en bestudeert hun metrische eigenschappen. Deze algoritmen worden gedefinieerd door twee versnellingstechnieken te koppelen—*geïnduceerde transformaties* en *contractie*—in de setting van Shunji Ito's 1989 ([93]) natuurlijke uitbreiding van de Farey tent afbeelding. Een geïnduceerde transformatie is een versnelling van een dynamisch systeem via een first-return afbeelding naar een gegeven subregio van het domein; contractie is een oude, schijnbaar over het hoofd geziene rekenkundige procedure die kettingbreuken

versnelt door voorgeschreven convergenten te verwijderen; en de Farey tent afbeelding produceert 'langzame' semi-regelmatige kettingbreuk uitbreidingen waarvan de convergenten bestaan uit alle regelmatige kettingbreuk convergenten en zogenaamde *mediante convergenten*. Kort gezegd gebruiken onze algoritmen geïnduceerde transformaties van Ito's natuurlijke uitbreiding om contracties van Farey uitbreidingen te besturen.

Onze constructie is analoog aan de manier waarop Kraaikamp S-expansies definieert, die gebruik maken van geïnduceerde transformaties van de natuurlijke uitbreiding van de Gauss afbeelding om *singularisaties*—speciale gevallen van contracties—van regelmatige kettingbreuken te besturen. In [115] laat Kraaikamp zien dat verschillende kettingbreuk algoritmen worden gerealiseerd als S-expansies, inclusief Nakada's α-kettingbreuken voor $1/2 \leq \alpha \leq 1$. De α-kettingbreuken zijn echter gedefinieerd voor alle $0 \leq \alpha \leq 1$. Deze gedeeltelijk overlappende families worden beide gevonden in onze verenigde setting: dat wil zeggen, zowel S-expansies als α-kettingbreuken voor alle $0 < \alpha \leq 1$ worden verkregen als specifieke voorbeelden van gecontracteerde Farey expansies (§1.7.2 en §1.7.3). In het bijzonder geven we een nieuwe beschrijving van planaire natuurlijke uitbreidingen van de α-kettingbreuken als expliciete geïnduceerde transformaties van Ito's natuurlijke uitbreiding (cf. [120]).

Naast het definiëren van nieuwe algoritmen voor kettingbreuken (bijvoorbeeld de *superoptimale kettingbreuken* van §1.9), ontwikkelen we ook een verenigende metrische theorie voor deelrijen van Farey convergenten (d.w.z. regelmatige kettingbreuk convergenten en medianten). In §1.8 bewijzen we een schat aan oude en nieuwe resultaten over deze deelrijen en hun Diophantische eigenschappen. In het bijzonder breiden we een klassiek resultaat van Lévy ([127]) over de groeisnelheid van de noemers van regelmatige kettingbreuk convergeten uit naar deelrijen van Farey convergenten; de Doeblin–Lenstra conjectuur—bewezen door Bosma, Jager en Wiedijk ([21])—over de distributie van zogenaamde *approximatie coëfficiënten*; het dynamische systeem van Jager ([95, 96]) dat wordt gebruikt om

opeenvolgende approximatie coëfficiënten te bestuderen; en verschillende analogen en verfijningen van een klassieke stelling van Legendre ([125]).

Hoofdstuk 1 is gebaseerd op drie artikelen. De theorie van gecontracteerde Farey expansies komt voor in [52], terwijl de metrische theorie van §1.8 grotendeels is overgenomen uit [53]. De superoptimale kettingbreuken die zijn geïntroduceerd in §1.9 zijn gebaseerd op [169].

Hoofdstuk 2 – Ergodische eigenschappen van een geparametriseerde familie van symmetrische gouden afbeeldingen

In Hoofdstuk 2 introduceren we een nieuwe, één-parameter familie van stuksgewijs lineaire interval afbeeldingen en bestuderen we eigenschappen van de getaluitbreidingen die ze produceren. Onze *scheve symmetrische gouden afbeeldingen* hebben hellingen bepaald door de gouden snede, $(\sqrt{5}+1)/2$. We tonen aan dat al onze afbeeldingen (op één na) een unieke—en dus ergodische—absoluut continue invariante waarschijnlijkheidsmaat hebben, en we verkrijgen expliciete dichtheden voor deze maten op een aftelbare verzameling van disjuncte, open subintervallen van de parameterruimte waarvan de vereniging een volledige Lebesgue maat heeft. Analoge resultaten worden eerst bewezen voor een eenvoudigere familie van *symmetrische gouden afbeeldingen* met een constante helling gelijk aan de gouden snede. We gebruiken het feit dat de oorspronkelijke scheve afbeeldingen onstaan als sprongtransformaties van de symmetrische gouden afbeeldingen om uitspraken over de laatstgenoemde familie over te dragen naar uitspraken over de eerstgenoemde.

Het belangrijkste hulpmiddel dat wordt gebruikt om de dichtheden van de invariante maten van de symmetrische gouden afbeeldingen te bepalen is *matching*—een fenomeen dat de laatste tijd veel aandacht heeft gekregen (zie bijvoorbeeld [16, 31, 32, 37, 38, 39, 42, 44, 45, 47, 120, 156]). Een interval afbeelding heeft matching als op elk kritiek punt de banen van de limieten van links en van

rechts overeenkomen na een eindig aantal stappen. De resultaten van Kopf ([112]) impliceren dat wanneer matching plaatsvindt, de dichtheid een stapfunctie is die alleen afhankelijk is van punten in de banen vóór matching. Met deze expliciete dichtheden in handen gebruiken we de ergodische stelling van Birkhoff om de relatieve frequenties te berekenen van getallen die typisch voorkomen in getaluitbreidingen geproduceerd door de symmetrische—en dus de scheef symmetrische—gouden afbeeldingen, en we bestuderen hoe deze frequenties continu variëren met de parameters.

Met uitzondering van kleine wijzigingen is Hoofdstuk 2 overgenomen uit [56].

Hoofdstuk 3 – Het construeren van roosteroppervlakken met voorgeschreven Veech groepen: een algoritme

In Hoofdstuk 3 beschouwen we de dynamica van een $\mathrm{SL}_2\mathbb{R}$-actie op families van *translatieoppervlakken*, d.w.z. oppervlakken die zijn geconstrueerd door het identificeren van paren randen van vlakke polygonen via translatie. Er is een natuurlijke, eindige maat ν op elke stratum van eenheidsoppervlakte translatieoppervlakken die dezelfde topologische gegevens delen, en de $\mathrm{SL}_2\mathbb{R}$-actie behoudt ν en is ergodisch op verbonden componenten van de stratum ([132, 186]). Ergodiciteit impliceert dat ν–bijna elk translatieoppervlak een dichte $\mathrm{SL}_2\mathbb{R}$-baan heeft in de verbonden component van de stratum waartoe het behoort. Tot de uitzonderingen behoren *roosteroppervlakken* waarvan de stabilisatoren (of *Veech groepen*) onder deze actie bijzonder groot zijn: het zijn roostersubgroepen van $\mathrm{SL}_2\mathbb{R}$.

We ontwikkelen een nieuw, expliciet algoritme dat alle translatieoppervlakken construeert met een voorgeschreven rooster Veech groep in een gegeven stratum. Bij het ontwikkelen van het algoritme presenteren we een nieuw bewijs van een resultaat van Smillie en Weiss ([179]) dat er ten hoogste eindig veel roosteroppervlakken met een gegeven Veech groep in een gegeven stratum zijn. Bovendien kunnen onze methoden worden gebruikt om obstakels te verkrijgen

voor roostersubgroepen van $\mathrm{SL}_2\mathbb{R}$ die worden gerealiseerd als Veech groepen: we tonen aan dat de vierkante torus het enige translatieoppervlak is met precies één singulariteit waarvan de Veech groep $\mathrm{SL}_2\mathbb{Z}$ is.

Hoofdstuk 3 is—op kleine wijzigingen na—overgenomen uit [170].

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Thanks for the dance...
One, two, three, one, two, three, one

Leonard Cohen

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Curriculum Vitae

Slade Sanderson was born on May 8, 1996, in Dallas, Texas. He enjoyed a happy childhood in the suburbs of Dallas and graduated from Wylie East High School in 2014. Slade attended Pepperdine University in Southern California, graduating *summa cum laude* in 2018 with a Bachelor of Science in mathematics, minor in religion. He began his graduate studies at Oregon State University, where he obtained a Master of Science in mathematics under the supervision of Dr. Thomas A. Schmidt in 2020. In the autumn of 2021, Slade moved to Utrecht, the Netherlands, where he met his wife, Mafe (married June 22, 2024, in Sesquilé, Colombia). From 2021 to 2025, he was a Ph.D. candidate supervised by Dr. Karma Dajani at Utrecht University.

